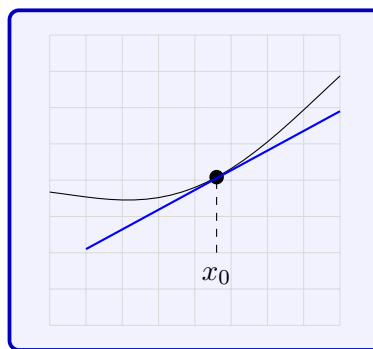
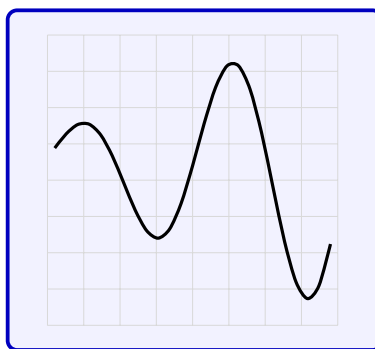
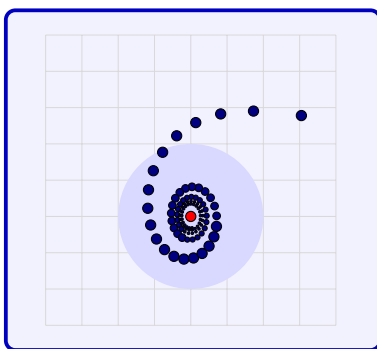


Analysis 1 & 2

A two-semester course for engineering students.

Marko Lindner



Limits, continuity, derivatives, and all the rest of it.

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Contents

I	Analysis 1	1
1	Foundations	3
2	Sequences	5
2.1	Convergent sequences	7
2.2	Convergence and boundedness	10
2.3	Working with null sequences	13
2.4	$\lim$ commutes with $+$, $-$, $\cdot$ and $:$	14
2.5	The relation $\leq$ extends to the limit	16
2.6	The sandwich lemma and some of its applications	17
2.7	Cauchy sequences	20
2.8	Monotonicity and convergence	21
2.9	Subsequences and their limits	25
	Summary	28
3	Series	29
3.1	Some first examples and criteria	30
3.2	Absolute convergence	36
3.3	Criteria by comparison with other series	38
	Summary	42
4	Topology of the real line	45
4.1	Some important kinds of points	45
4.2	Open and closed sets	48
4.3	Bounded sets	52

4.4	Compact sets	53
	Summary	55
5	Continuity	57
5.1	Definition: Limit and continuity	57
5.2	ε - δ characterisation and continuity	65
5.3	Combinations of continuous functions	68
5.4	Continuity on compact sets	70
5.5	The intermediate value theorem	71
5.6	Continuity and the inverse function	73
	Summary	74
6	Elementary functions	77
6.1	Monomials and roots	77
6.2	Polynomials and rational functions	79
6.3	Power series	84
6.4	Exponential function and logarithm	91
6.5	Trigonometric functions and their inverses	98
	Summary	101
7	Differentiation in one variable	105
7.1	Differentiation and derivative	106
7.2	Rules for differentiation	109
7.3	Local extrema	114
7.4	The mean value theorem and the monotonicity rule	116
7.5	De l'Hospitals rule	120
7.6	The second derivative	122
7.7	Taylor expansion	127
7.8	Curve discussion	133
	Summary	135

II	Analysis 2	137
8	Integration in one variable	139
8.1	The indefinite integral: antiderivatives	140
8.1.1	Definitions	140
8.1.2	Integration rules and basic integrals	143
8.1.3	Rational functions	149
8.2	The definite integral: Riemann integration	154
8.2.1	Definitions and examples	154
8.2.2	Criteria and properties	159
8.3	The fundamental theorem of calculus	164
8.4	Some applications	168
8.4.1	Areas of sectors	168
8.4.2	Length, mass and center of gravity of a curve	170
8.4.3	Solids of revolution	175
8.5	Improper integral: unbounded interval or unbounded function	177
8.5.1	Unbounded interval	177
8.5.2	Unbounded function	181
8.5.3	Both: interval and function are unbounded	184
	Summary	184
9	Sequences and series of functions	187
9.1	Pointwise vs. uniform convergence	188
9.2	Conservation laws for uniform convergence	192
9.3	Series of functions	196
	Summary	204
10	Differentiation in several variables	205
10.1	Distance and convergence in $\mathbb{R}^n$	206
10.2	Scalar fields and vector fields	210
10.3	Differentiability and derivative	214
10.4	Partial differentiability and partial derivatives	217
10.4.1	Jacobian matrix of a scalar field	217

10.4.2	Jacobian matrix of a vector field	223
10.5	Partial vs. total differentiability	223
10.6	Differentiation rules	226
10.7	Directional derivatives and level sets	231
10.8	Higher order derivatives and Taylors formula	236
10.8.1	Higher order derivatives and Schwarz's theorem	236
10.8.2	Taylor's theorem in several variables	241
10.9	Local extrema	246
10.9.1	Necessary condition: Fermat's condition	246
10.9.2	A sufficient condition and the Hessian matrix	247
10.9.3	Examples of extrema in domains with or without boundary points	254
10.10	Divergence and curl of a vector field	257
	Summary	261
11	Outlook: Ordinary Differential Equations (ODEs)	265
11.1	First order ODEs	266
11.2	Systems of first-order differential equations	270
11.3	Linear differential equations of order n	271
	Index	273

Part I

Analysis 1

Chapter 1

Foundations

For this we refer to the first chapter of the lecture notes of your “Linear Algebra I” course.

Chapter 2

Sequences

Germany started recording daily temperatures in 1881. For Hamburg this record might start like this:

time	day 1	day 2	day 3	day 4	day 5	day 6	...
temperature in °C	14.3	12.6	10.8	10.9	17.5	15.3	...

Written down consecutively, we have this sequence of numbers:

$$(x_1, x_2, x_3, x_4, x_5, x_6, \dots) = (14.3, 12.6, 10.8, 10.9, 17.5, 15.3, \dots).$$

Instead of saying a number sequence is a function on the day-based time axis since 1881, we rather say the following:

Definition 2.1. Sequence or number sequence

A sequence (or number sequence) is a function $x : \mathbb{N} \rightarrow \mathbb{R}$.

For $n \in \mathbb{N}$ we write x_n instead of $x(n)$ for the n th member of the sequence.

Instead of n th member, we will also say n th term or n th entry of the sequence.

When we mean the whole sequence, we write $(x_n)_{n \in \mathbb{N}}$ or $(x_n)_{n=1}^{\infty}$ or just $(x_n)_n$.

We know that it is unimportant which name we give a function. Still, we prefer calling sequences x , y or z .

An essential feature of sequences is that they never end.

Instead of writing down all(!) the numbers x_n explicitly, we try to give a formula for x_n whenever we can.

Example 2.2. some first simple sequences

a) Let x_n be given by $\frac{1}{n}$ for each $n \in \mathbb{N}$, so that the sequence starts $x_1 = \frac{1}{1}$, $x_2 = \frac{1}{2}$, $x_3 = \frac{1}{3}$, $x_4 = \frac{1}{4}$, $\dots$. We abbreviate this by saying $(x_n)_{n \in \mathbb{N}} = \left(\frac{1}{n}\right)_{n \in \mathbb{N}}$.

b) Look at $(y_n)_{n \in \mathbb{N}} = ((-1)^n)_{n \in \mathbb{N}} = (-1, 1, -1, 1, \dots)$. This time we have

$$y_n = \begin{cases} -1, & n \text{ odd}, \\ 1, & n \text{ even}, \end{cases} \quad n \in \mathbb{N}.$$

c) Let $z_n = z_{n-1} + 2n - 1$ for $n = 2, 3, \dots$ and let $z_1 = 1$.

🧐 Nerd box 2.3. explicit vs recursive description

In example c) we speak of a recursive description (the members of $(z_n)_n$ are given in terms of other members of $(z_n)_n$), while the description in a) and b) is referred to as explicit. (x_n and y_n can be computed directly in terms of n .)

Can you give an explicit formula for z_n from c)?

2.1 Convergent sequences

We speak of convergence of a sequence if the members x_n get closer to the so-called limit as n gets larger. The formally correct definition comes below.

Since we stop writing $x : \mathbb{N} \rightarrow \mathbb{R}$ for the sequence $(x_n)_{n \in \mathbb{N}}$ after Definition 2.1, the notation x as a single letter is free again. We will use it for the limit, which is a number $x \in \mathbb{R}$.

Definition 2.4. Convergence and limit of a sequence

We say that a sequence $(x_n)_{n \in \mathbb{N}}$ converges to a number $x \in \mathbb{R}$ if, for every $\varepsilon > 0$, there is an $n_0 \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon \quad \text{for all } n \geq n_0$$

In this case x is called the limit of the sequence $(x_n)_{n \in \mathbb{N}}$, and we write

$$x = \lim_{n \rightarrow \infty} x_n \quad \text{or} \quad x_n \rightarrow x \quad \text{for } n \rightarrow \infty.$$

A sequence with a limit is called convergent; otherwise it is called divergent.

The essence of the definition again, now with all the quantors:

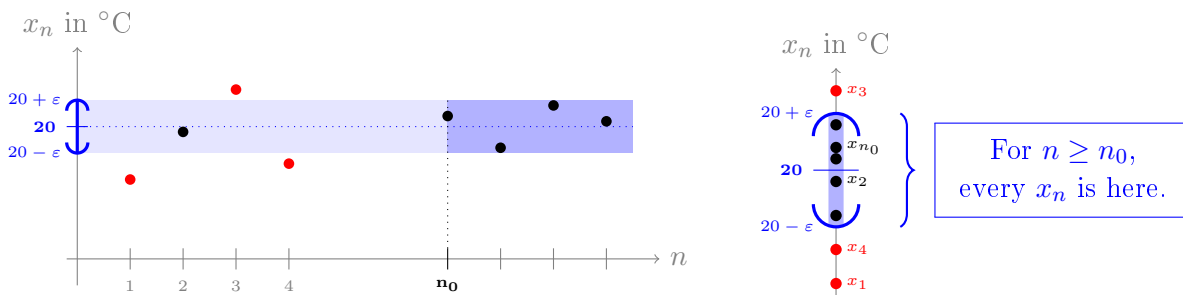
Convergence without words

$$\forall \varepsilon > 0 \quad \exists n_0 \in \mathbb{N} : \quad \forall n \geq n_0 : \quad |x_n - x| < \varepsilon$$

This sounds more complicated than it is. Some more attempts (with many more words):

- $|x_n - x| < \varepsilon$ means $x_n \in (x - \varepsilon, x + \varepsilon)$,
- this holds for all x_n , except perhaps $x_1, x_2, \dots, x_{n_0-1}$,
- i.e., for all sequence members x_n up to finitely many exceptions,
- the whole thing has to be true for arbitrarily small precisions $\varepsilon > 0$,
- n_0 (or the number of exceptions) may depend on ε .

So if someone claims the temperature in Hamburg converges to 20°C then, after a certain day n_0 , the temperature would have to enter the interval $(19, 21)$ – and never leave that interval again. A little later it would even enter the interval $(19.9, 20.1)$, and so on.



Example 2.5. some convergent and some divergent sequences

- a) $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$: For $\varepsilon = 0.1$, $n_0 = 11$ is enough, since $|\frac{1}{n} - 0| < \frac{1}{10} = 0.1$ for all $n \geq 11$. For $\varepsilon = 0.01$ we already need $n_0 = 101$, and in general, n_0 has to be chosen a bit larger than $\frac{1}{\varepsilon}$.
- b) $(-1)^n \not\rightarrow 1$ since already for $\varepsilon = \frac{1}{2}$ the sequence keeps leaving the interval $(1 - \varepsilon, 1 + \varepsilon) = (0.5, 1.5)$. This sequence doesn't converge to any other real number either. But this is something we will show a little bit further down.
- c) Constant sequences are clearly convergent: If $x_n = 5$ for all $n \in \mathbb{N}$ then $x_n \in (5 - \varepsilon, 5 + \varepsilon)$ for all $n \geq 1$, no matter how small the $\varepsilon > 0$. Hence, $x_n \rightarrow 5$.

For a more fluent language and argument, let us introduce the following handy notation:

Definition 2.6. short notation $x \overset{\varepsilon}{\approx} y$

For two numbers $x, y \in \mathbb{R}$ and an $\varepsilon > 0$, let us write

$$x \overset{\varepsilon}{\approx} y \quad \text{if} \quad |x - y| < \varepsilon.$$

The following important rule holds

$$x \overset{\varepsilon}{\approx} y \overset{\delta}{\approx} z \quad \implies \quad x \overset{\varepsilon+\delta}{\approx} z \quad (2.1)$$

for all $x, y, z \in \mathbb{R}$ and $\varepsilon, \delta > 0$, which corresponds to the so-called

triangle inequality

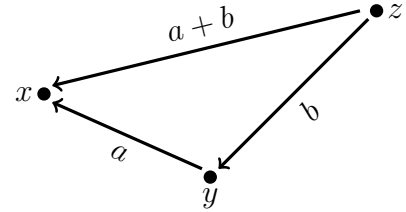
$$|x - z| \leq |x - y| + |y - z|, \quad x, y, z \in \mathbb{R},$$

in other words,

$$|a + b| \leq |a| + |b|, \quad a, b \in \mathbb{R}. \quad (2.2)$$

(Put $a = x - y$ and $b = y - z$, so that $a + b = x - y + y - z = x - z$.)

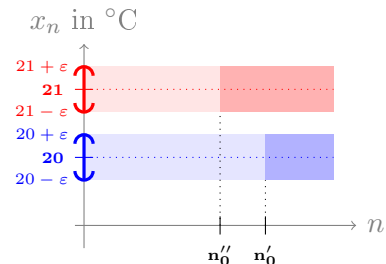
It is named like this since the inequality extends to vectors x, y, z in the plane (and beyond), where $|x - y|$, $|y - z|$ and $|x - z|$, resp. $|a|$, $|b|$ and $|a + b|$, are the lengths of the sides of a triangle with corners x, y, z . In that situation it is obvious that the sum of two side lengths cannot be smaller than the length of the third side.



With our short notation from Definition 2.6, $x_n \rightarrow x$ translates into $x_n \overset{\varepsilon}{\approx} x$ holding sooner or later (meaning: for all $n \geq \text{some } n_0$), for any prescribed precision $\varepsilon > 0$.

When a sequence converges to 20, it cannot also converge to 21, because, if it did, then x_n were contained in both $(20 - \varepsilon, 20 + \varepsilon)$ and $(21 - \varepsilon, 21 + \varepsilon)$ for sufficiently large n . But already for $\varepsilon = \frac{1}{2}$ there is no number in the intersection of both intervals. Or, arguing even swifter:

$$20 \overset{\varepsilon}{\approx} x_n \overset{\varepsilon}{\approx} 21 \implies 20 \overset{2\varepsilon}{\approx} 21, \text{ which is only possible for } \varepsilon > \frac{1}{2}.$$



Proposition 2.7. The limit is unique

A convergent sequence can only have one limit.

To be precise, it is only now that the notation “ $\lim x_n$ ” is legitimate.

Proof: Let's just copy the argument from above: Suppose $x_n \rightarrow x$ and $x_n \rightarrow y$, while $x \neq y$.

■

2.2 Convergence and boundedness

We will now get to know a minimum requirement for convergence.

Definition 2.8. bounded sequence

A sequence $(x_n)_{n \in \mathbb{N}}$ is called

- bounded from below, if there is an $a \in \mathbb{R}$ with $a \leq x_n$ for all $n \in \mathbb{N}$,

- bounded from above, if there is a $b \in \mathbb{R}$ with $x_n \leq b$ for all $n \in \mathbb{N}$,

- bounded, if it is bounded from below and above, i.e., if there exist $a, b \in \mathbb{R}$ with $x_n \in [a, b]$ for all $n \in \mathbb{N}$. In particular, then holds $|x_n| \leq c$ for all $n \in \mathbb{N}$, where $c = \max(|a|, |b|)$,

- unbounded if it is not bounded.

Such numbers a and b are then called lower bounds resp. upper bounds on $(x_n)_{n \in \mathbb{N}}$.

The enclosing interval for bounded sequences can always be chosen symmetric to zero since $[a, b] \subset [-c, c]$ with c as above. The condition $|x_n| \leq c$ then means $x_n \in [-c, c]$.

Example 2.9. bounded and unbounded sequences

a) The sequence $(x_n)_{n \in \mathbb{N}}$ with $x_n = (-1)^n$ for $n \in \mathbb{N}$ is bounded since

$$|x_n| = |(-1)^n| = 1 \leq 1, \quad \text{i.e. } x_n \in [-1, 1] \quad \text{for all } n \in \mathbb{N}.$$

b) The sequence $(y_n)_{n \in \mathbb{N}}$ with $y_n = n$ for $n \in \mathbb{N}$ is bounded from below (by $a = 1$) but not from above since every number $b \in \mathbb{R}$ will be exceeded by y_n once $n > b$. The sequence is hence unbounded.

We will now see that boundedness is necessary (it is a minimal requirement) for convergence. Unbounded sequences like $(y_n)_{n \in \mathbb{N}}$ from b) cannot be convergent.

Proposition 2.10. convergence $\implies$ boundedness

Every convergent sequence is bounded.

Proof: Let $x_n \rightarrow x$ and $\varepsilon > 0$ be arbitrary.

■

⚠ Attention! The implication goes only one way!

The reverse implication of Proposition 2.10 does not hold. Not every bounded sequence converges, see e.g. $((-1)^n)_{n \in \mathbb{N}}$ from Example 2.9 a) and Example 2.5 b).

Next we look at sequences that eventually exceed every number $M \in \mathbb{R}$ without ever getting below again – or the other way round.

Definition 2.11. $\lim x_n = \infty$

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence. If, for every $M \in \mathbb{R}$, there exists a $n_0 \in \mathbb{N}$ such that $x_n \geq M$ for all $n \geq n_0$, then we write

$$x_n \rightarrow +\infty \quad \text{or} \quad \lim_{n \rightarrow \infty} x_n = +\infty.$$

If, for every $M \in \mathbb{R}$, there exists a $n_0 \in \mathbb{N}$ such that $x_n \leq M$ for all $n \geq n_0$, then we write

$$x_n \rightarrow -\infty \quad \text{or} \quad \lim_{n \rightarrow \infty} x_n = -\infty.$$

Such sequences are unbounded and hence divergent by Proposition 2.10! That's why we refrain from talking about “convergence to ∞ ” or similar. We will however use the handy notations from Definition 2.11 – “by abuse of notation”, so to say.

 Nerd box 2.12. Neighbourhoods of $+\infty$ and $-\infty$

If we really wanted to accept $x_n \rightarrow -\infty$ or $x_n \rightarrow +\infty$ as convergence, we would first have to add the two desired limits as elements to the real numbers and then extend the notion of an ε -neighbourhood to the new members:

What the interval $(x - \varepsilon, x + \varepsilon)$ is for good old real numbers, $x \in \mathbb{R}$, namely, the set where convergent sequences eventually go and never leave again, has to be replaced by something like $(\frac{1}{\varepsilon}, +\infty]$ in case $x = +\infty$. Similarly for $x = -\infty$. Since these new ε -neighbourhoods are unbounded (in the usual sense), we lose Proposition 2.10 – as was desired.

The concept of boundedness can actually be dropped in that new sense since the new numbers $-\infty$ and $+\infty$ will always work as lower resp. upper bounds on any sequence. Next, one had to extend the usual operations $+, -, \cdot, :$ to the new number set $\mathbb{R} \cup \{-\infty, +\infty\}$ and thereby tackle lots of other problems. We decide that this is not worth the hassle.

Proposition 2.13. $x_n \rightarrow \infty \implies \frac{1}{x_n} \rightarrow 0$

- a) If $x_n \rightarrow +\infty$ or $x_n \rightarrow -\infty$ as in Definition 2.11 then $\frac{1}{x_n} \rightarrow 0$.
- b) If x_n does not change its sign then also the reverse implication holds.

2.3 Working with null sequences

Definition 2.14. null sequence

A convergent sequence with limit zero is called a null sequence.

The next section will simplify quite a bit when we know some rules about null sequences:

Lemma 2.15. null sequence + null sequence = null sequence

If $x_n \rightarrow 0$ and $y_n \rightarrow 0$ then $x_n + y_n \rightarrow 0$.

A lemma is, in our way of using the word, a useful little helper of the bigger propositions and theorems that are often a bit more center stage.

Proof: In order to show that $x_n + y_n \rightarrow 0$, we take an arbitrary $\varepsilon > 0$ and then we have to find a $n_0 \in \mathbb{N}$ such that $x_n + y_n \in (0 - \varepsilon, 0 + \varepsilon)$ for all $n \geq n_0$.

■

Lemma 2.16. convergent sequence = constant + null sequence

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence and let $x \in \mathbb{R}$ be a real number. Then

$$x_n \rightarrow x \iff x_n = x + z_n \text{ for all } n \in \mathbb{N} \text{ with a null sequence } (z_n)_{n \in \mathbb{N}}.$$

Proof: For $\varepsilon > 0$ and $n \in \mathbb{N}$, it holds that $x_n \overset{\varepsilon}{\approx} x$ if and only if $x_n - x \overset{\varepsilon}{\approx} 0$. If the first holds for all $n \geq n_0$ then so does the second. Hence: $x_n \rightarrow x \iff z_n := x_n - x \rightarrow 0$. ■

Lemma 2.17. bounded sequence $\cdot$ null sequence = null sequence

If $(x_n)_n$ is bounded and $(z_n)_n$ is a null sequence then $(x_n \cdot z_n)_n$ is again a null sequence.

Proof: Let $\varepsilon > 0$ be arbitrary. We have to show that there is a $n_0 \in \mathbb{N}$ such that $|x_n z_n - 0| < \varepsilon$ for all $n \geq n_0$.

Fix $c > 0$ such that $|x_n| \leq c$ for all $n \in \mathbb{N}$, and let $n_0 \in \mathbb{N}$ be such that $|z_n - 0| < \frac{\varepsilon}{c}$ for all $n \geq n_0$. Then it holds that $|x_n z_n - 0| = |x_n z_n| = |x_n| |z_n| < c \cdot \frac{\varepsilon}{c} = \varepsilon$ for all $n \geq n_0$. ■

Together with Proposition 2.10 we conclude:

Corollary 2.18. convergent sequence $\cdot$ null sequence = null sequence

If $(x_n)_n$ converges and $(z_n)_n$ is a null sequence then $(x_n \cdot z_n)_n$ is a null sequence.

Proof: By Proposition 2.10, $(x_n)_n$ is bounded, and it remains to apply Lemma 2.17. ■

2.4 $\lim$ commutes with $+$, $-$, $\cdot$ and $:$

We connect two sequences with the basic operations, $+$, $-$, $\cdot$, $:$, by applying them to the members of the sequences (which are real numbers):

$$(x_n)_{n \in \mathbb{N}} \circledast (y_n)_{n \in \mathbb{N}} := (x_n \circledast y_n)_{n \in \mathbb{N}} \quad \text{for} \quad \circledast \in \{+, -, \cdot, :\}.$$

The “new” operation on the left hand side is defined in terms of the “old” operation between the real numbers on the right.

The next proposition says that passing to the limit commutes with e.g. addition: Whether we first add two sequences and then pass to the limit or we first pass to the limits and add them afterwards – the result is the same. The same is true for $-$, $\cdot$ und $:$.

Proposition 2.19. $\lim$ commutes with $+$, $-$, $\cdot$ and $:$

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be convergent sequences with $x_n \rightarrow x$ and $y_n \rightarrow y$.

a) Then also the sequences $(x_n)_n + (y_n)_n$ and $(x_n)_n - (y_n)_n$ converge, where

$$x_n \pm y_n \rightarrow x \pm y, \quad \text{i.e.} \quad \lim_{n \rightarrow \infty} (x_n \pm y_n) = \left(\lim_{n \rightarrow \infty} x_n \right) \pm \left(\lim_{n \rightarrow \infty} y_n \right).$$

b) For every constant $c \in \mathbb{R}$, also $c \cdot (x_n)_n := (cx_n)_n$ converges, where

$$cx_n \rightarrow cx, \quad \text{i.e.} \quad \lim_{n \rightarrow \infty} (cx_n) = c \lim_{n \rightarrow \infty} x_n.$$

c) Also the sequence $(x_n)_n \cdot (y_n)_n$ is convergent, and it holds that

$$x_n \cdot y_n \rightarrow x \cdot y, \quad \text{i.e.} \quad \lim_{n \rightarrow \infty} (x_n \cdot y_n) = \left(\lim_{n \rightarrow \infty} x_n \right) \cdot \left(\lim_{n \rightarrow \infty} y_n \right).$$

d) If $y \neq 0$ then also the sequence $(\frac{x_n}{y_n})_n$ is convergent, and

$$\frac{x_n}{y_n} \rightarrow \frac{x}{y}, \quad \text{i.e.} \quad \lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \frac{\lim_{n \rightarrow \infty} x_n}{\lim_{n \rightarrow \infty} y_n}.$$

Proof: This is looking like a lot of work. Let's try to be economic here:

We start by showing the $+$ part of **a)** and then **c)**. With this shown, we also have **b)** and the $-$ part of **a)** sorted since **b)** is a special case of **c)** and since $x_n - y_n = x_n + (-1) \cdot y_n$. At the end, we still have to show **d)**.

Take home message:

*Try to reduce new problems to previously solved (or soon to be solved) problems.
Whenever you can!*

a) In accordance with Lemma 2.16 we write

c) By Lemma 2.16, we again write

d) By $\frac{x_n}{y_n} = x_n \cdot \frac{1}{y_n}$ and c), it is enough to show $\frac{1}{y_n} \rightarrow \frac{1}{y}$. Therefore, let $y \neq 0$ and $\varepsilon = \frac{|y|}{2} > 0$, such that 0 is surely (with distance ε) outside of the interval $I = (y - \varepsilon, y + \varepsilon)$. For a certain $n_0 \in \mathbb{N}$, all y_n with $n \geq n_0$ are located in I and are hence nonzero. It holds that

$$\frac{1}{y_n} - \frac{1}{y} = \frac{y - y_n}{y_n y} = (y - y_n) \cdot \frac{1}{y} \cdot \frac{1}{y_n}. \quad (2.3)$$

By $y_n \in I$, it holds that $|y_n| \geq \varepsilon = \frac{|y|}{2}$ and hence $|\frac{1}{y_n}| \leq \frac{2}{|y|}$. Consequently, and by Lemma 2.16, the right-hand side of (2.3) is of the form “null sequence $\cdot$ bounded sequence”, so that Lemma 2.17 and 2.16 finish the proof: $\frac{1}{y_n} \rightarrow \frac{1}{y}$ and hence $\frac{x_n}{y_n} = x_n \cdot \frac{1}{y_n} \rightarrow x \cdot \frac{1}{y} = \frac{x}{y}$. ■

Example 2.20. limits – now a bit more complicated

a) We start with

$$x_n = \frac{1}{n^3} =$$

b) Consequently,

$$y_n = \frac{n^2 - 7n + 4}{2n^3 + n - 5} =$$

2.5 The relation $\leq$ extends to the limit

Proposition 2.21. $\leq$ extends to the limit

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be convergent sequences with $x_n \rightarrow x$ and $y_n \rightarrow y$. If $x_n \leq y_n$ for all $n \in \mathbb{N}$ then also $x \leq y$ holds. In short:

$$x_n \leq y_n \text{ for all } n \in \mathbb{N} \implies \lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} y_n.$$

Since changes to finitely many members do not change the limit, the claim, $\lim x_n \leq \lim y_n$, still holds when $x_n \leq y_n$ is only true for all $n \geq n_0$.

Proof: By Proposition 2.19 a), it suffices to show that, for every convergent sequence $(z_n)_n$, the implication

$$z_n \geq 0 \quad \text{for all } n \in \mathbb{N} \quad \implies \quad \lim_{n \rightarrow \infty} z_n \geq 0 \quad (2.4)$$

holds and then to apply it to $z_n = y_n - x_n$.

■

⚠ Attention! The relation $<$ does not extend to the limit!

Example: Although $0 < \frac{1}{n}$ for all $n \in \mathbb{N}$, we have $\lim 0 \not< \lim \frac{1}{n}$ for the limits.

2.6 The sandwich lemma and some of its applications

The following lemma runs under many names: sandwich lemma, squeeze lemma and police lemma are just some of them. It is probably our most valuable tool in the hunt for convergence and limits.

Lemma 2.22. sandwich lemma

Let $x \in \mathbb{R}$ and let $(x_n)_{n \in \mathbb{N}}$ and $(z_n)_{n \in \mathbb{N}}$ be convergent sequences with $x_n \rightarrow x$ and $z_n \rightarrow x$. If $x_n \leq y_n \leq z_n$ for all $n \in \mathbb{N}$ then also $(y_n)_{n \in \mathbb{N}}$ converges, where $y_n \rightarrow x$.

Proof: Let $\varepsilon > 0$ be arbitrary. By $x_n \rightarrow x$ and $z_n \rightarrow x$, there are $n'_0, n''_0 \in \mathbb{N}$, such that

■

Let's get to applications right away:

Example 2.23. Juggling with n -th roots

- a) $\sqrt[n]{5} \rightarrow 1$, *precisely*: $\sqrt[n]{c} \rightarrow 1$ for all $c > 0$.
- b) *Even* $\sqrt[n]{n} \rightarrow 1$ and $\sqrt[n]{n^3} \rightarrow 1$.
- c) $\sqrt[n]{7^n + 4^n} \rightarrow 7$, *precisely*: $\sqrt[n]{a^n + b^n} \rightarrow \max(a, b)$ for all $a, b \geq 0$.

Proof: a) Let $x_n := \sqrt[n]{5} - 1$. We show that $x_n \rightarrow 0$. By the binomial formula,

$$5 = (1 + x_n)^n =$$

since $x_n > 0$. It follows that $0 < x_n \leq \frac{5}{n}$, and, by the sandwich lemma, $x_n \rightarrow 0$.

b) In analogy, let $y_n := \sqrt[n]{n} - 1$. We show that $y_n \rightarrow 0$. By the binomial formula, one has

$$n = (1 + y_n)^n = \binom{n}{0} + \binom{n}{1}y_n + \binom{n}{2}y_n^2 + \cdots + \binom{n}{n}y_n^n \geq \frac{n(n-1)}{2}y_n^2$$

since $y_n > 0$. It follows that $0 < y_n^2 \leq \frac{2}{n-1}$, so that $y_n^2 \rightarrow 0$ (sandwich lemma) and $y_n \rightarrow 0$. It follows that $\sqrt[n]{n} \rightarrow 1$ and also $\sqrt[n]{n^3} = \sqrt[n]{n} \cdot \sqrt[n]{n} \cdot \sqrt[n]{n} \rightarrow$

c)

$$\sqrt[n]{7^n + 4^n} = \sqrt[n]{7^n \left(1 + \frac{4^n}{7^n}\right)} = 7 \cdot \sqrt[n]{1 + \left(\frac{4}{7}\right)^n} \rightarrow 7$$

by the sandwich lemma since

■

Next, we look at three sequences that all grow, nonstop. Precisely, we mean the sequences $(x_n)_n$, $(y_n)_n$ and $(z_n)_n$ with their members equal to

$$x_n = n^a, \quad y_n = b^n, \quad z_n = n!, \quad n \in \mathbb{N},$$

where $a, b \in \mathbb{N}$ are fixed, with $b \geq 2$, and $n! = 1 \cdot 2 \cdot \dots \cdot n$ is the factorial function.

We all agree that (in the sense of Definition 2.11):

$$x_n \rightarrow +\infty, \quad y_n \rightarrow +\infty \quad \text{and} \quad z_n \rightarrow +\infty,$$

while we are wondering which of the three sequences is fastest in their race to $+\infty$.

To find out, we look at pairwise quotients, arguing that, e.g. n^5 grows faster than n^2 since even their ratio, $\frac{n^5}{n^2} = n^3 \rightarrow +\infty$ or, equivalently, since $\frac{n^2}{n^5} = \frac{1}{n^3} \rightarrow 0$.

Proposition 2.24. Exponential function b^n beats power function n^a

Let $a, b \in \mathbb{N}$ with $b \geq 2$. Then

$$\frac{n^a}{b^n} \rightarrow 0 \quad \text{i.e.} \quad \frac{b^n}{n^a} \rightarrow +\infty \quad \text{for} \quad n \rightarrow \infty.$$

Proposition 2.25. Factorial $n!$ beats exponential function b^n

Let $b \geq 2$ be integer. Then it holds that

$$\frac{b^n}{n!} \rightarrow 0 \quad \text{i.e.} \quad \frac{n!}{b^n} \rightarrow +\infty \quad \text{for} \quad n \rightarrow \infty.$$

Proof: For $n > 2b$, we argue as follows:

$$\frac{b^n}{n!} = \frac{b \cdot b \cdot \dots \cdot b}{1 \cdot 2 \cdot \dots \cdot n} = \frac{b^{2b}}{(2b)!} \cdot \underbrace{\frac{b}{2b+1}}_{\leq \frac{1}{2}} \cdot \dots \cdot \underbrace{\frac{b}{n}}_{\leq \frac{1}{2}} \leq \frac{b^{2b}}{(2b)!} \cdot \frac{1}{2^{n-2b}} \rightarrow 0. \quad \blacksquare$$

 Nerd box 2.26. And by how much is $n!$ faster than b^n ?

The so-called Stirling formula shows that $n!$ grows as fast as $\sqrt{n} \cdot \left(\frac{n}{e}\right)^n$. Precisely, the quotient of the two converges to $\frac{1}{\sqrt{2\pi}}$, where $e \approx 2.71828\dots$ denotes the basis of the natural exponential function that we will meet in Section 6.4.

It seems pretty redundant to still check $x_n = n^a$ versus $z_n = n!$. Indeed,

2.7 Cauchy sequences

That $(x_n)_n$ with $x_n = (-1)^n$ is a divergent sequence, was mentioned in Example 2.5 b) already but so far we postponed the proof. And this is understandable because: Who wants to check that $x_n \not\rightarrow x$, for every single $x \in \mathbb{R}$? We will now find a clever argument.

For unbounded sequences we already have such a clever argument:

Lesson learnt: The members of a convergent sequence do not only get increasingly close to a (so far unknown) limit x but also to each other. Sequences with the latter property will get their own name:

Definition 2.27. Cauchy sequence (Augustin-Louis Cauchy, 1789–1857)

A sequence $(x_n)_{n \in \mathbb{N}}$ is called a Cauchy sequence if, for every $\varepsilon > 0$, there is a $n_0 \in \mathbb{N}$ such that $|x_m - x_n| < \varepsilon$ for all $m, n \geq n_0$.

With quantors instead of words:

Cauchy sequence without words

$$\forall \varepsilon > 0 \quad \exists n_0 \in \mathbb{N} : \quad \forall m, n \geq n_0 : \quad |x_m - x_n| < \varepsilon$$

A popular short notation is

$$x_m - x_n \rightarrow 0 \quad \text{for} \quad m, n \rightarrow \infty.$$

Right before Definition 2.27, we learnt this (noting that 2ε is just another ε):

Proposition 2.28. convergent sequence $\implies$ Cauchy sequence

Every convergent sequence is a Cauchy sequence.

Remarkably, in the setting of real numbers, also the reverse implication holds:

Proposition 2.29. In $\mathbb{R}$, we also have the reverse implication.

Every Cauchy sequence of real numbers is convergent.

Of course, when we say “sequence” in this course, we *always* mean a sequence of real numbers. Still, we purposely make that distinction between Propositions 2.28 and 2.29 to emphasize which implication holds in arbitrary so-called metric spaces, while the other implication fails in number spaces like the rational numbers, $\mathbb{Q}$.

Number spaces, where the reverse implication holds (i.e., where all Cauchy sequences converge), are called complete. The completeness of $\mathbb{R}$ is hardwired into the definition of $\mathbb{R}$, which is one of the most difficult topics in each Analysis course for maths students. Let’s skip that issue here.

Corollary 2.30. In $\mathbb{R}$, it holds that: convergent sequence = Cauchy sequence

A sequence of real numbers is convergent if and only if it is a Cauchy sequence.

Let us close the full circle: The sequence $(x_n)_n$ with $x_n = (-1)^n$ is obviously not a Cauchy sequence (and hence not convergent) since $x_m - x_n \not\rightarrow 0$ for $m, n \rightarrow \infty$. In particular, $|x_{n+1} - x_n| = 2$ for all $n \in \mathbb{N}$.

2.8 Monotonicity and convergence

Definition 2.31. monotonicity

A sequence $(x_n)_n$ is called

- monotonically increasing, if $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$,

- strictly increasing, if $x_n < x_{n+1}$ for all $n \in \mathbb{N}$,

- monotonically decreasing, if $x_n \geq x_{n+1}$ for all $n \in \mathbb{N}$,
- strictly decreasing, if $x_n > x_{n+1}$ for all $n \in \mathbb{N}$,
- monotonic, if it is monotonically increasing or decreasing, and
- strictly monotonic, if it is strictly increasing or decreasing.

Instead of “increasing” and “decreasing” we also say “growing” and “falling”.

As a second ingredient, recall the notions of boundedness from Definition 2.8. Bringing the two together, we get one of the two super tools (besides the sandwich lemma) of the current chapter:

Proposition 2.32. monotonicity & boundedness $\implies$ convergence

- a) *A monotonically growing sequence that is bounded from above is convergent.*
- b) *A monotonically falling sequence that is bounded from below is convergent.*
- c) *A monotonic and bounded sequence is convergent.*

Example 2.33. Our Proposition 2.32 in action

- a) Let $x_n = q^n$ with $q \in (0, 1)$. Then $x_{n+1} = q \cdot x_n < x_n$ and $x_n > 0$ for all $n \in \mathbb{N}$. By Proposition 2.32 b), $(x_n)_n$ converges. Let $x := \lim x_n$. A little trick:

so that

By $1 - q \neq 0$, only $x = 0$ remains.

- b) Let $y_n = \sqrt[n]{5}$. We know from Example 2.23 a), that $y_n \rightarrow 1$. Another proof: By $y_n^n = 5 > 1$, we have $y_n > 1$. If y_{n+1} were $\geq y_n$ then $5 = y_{n+1}^{n+1} \geq y_n^{n+1} = y_n \cdot y_n^n = y_n \cdot 5 > 5$. Contradiction! ⚡ Hence, $y_{n+1} < y_n$ for all $n \in \mathbb{N}$. By Proposition 2.32 b), $(y_n)_n$ is convergent. Let $y := \lim y_n$. A little trick:

PS: The trick in b) uses a so-called subsequence and Proposition 2.40 from below.

The limit $\lim x_n$ from Proposition 2.32 plays another special role for the sequence (x_n) :

Proposition 2.34. smallest upper and largest lower bound

- a) Let $(x_n)_n$ be monotonically growing and bounded from above. The limit $x = \lim_{n \rightarrow \infty} x_n$ from Proposition 2.32 is then the smallest upper bound on the sequence $(x_n)_n$.
- b) Let $(x_n)_n$ be monotonically falling and bounded from below. The limit $x = \lim_{n \rightarrow \infty} x_n$ from Proposition 2.32 is then the largest lower bound on the sequence $(x_n)_n$.

From here it is not far to showing that every bounded sequence (also if non-monotonic) has a smallest upper and a largest lower bound.

Definition 2.35. infimum, supremum, minimum and maximum of a sequence

Let $(x_n)_n$ be a bounded sequence.

- The largest lower bound of the sequence is called its infimum.
- If the infimum is equal to a sequence member x_n then it is also called minimum.
- The smallest upper bound of the sequence is called its supremum.
- If the supremum is equal to a sequence member x_n then it is also called maximum.

The corresponding notations are $\inf x_n$, $\min x_n$, $\sup x_n$ and $\max x_n$.

For unbounded sequences $(x_n)_n$ with no lower bound, we put $\inf x_n := -\infty$, and if it has no upper bound then let $\sup x_n := +\infty$.

Time for examples:

Example 2.36. infimum, supremum, minimum and maximum

- The sequence $(x_n)_{n \in \mathbb{N}}$ with $x_n = \frac{n-1}{n} = 1 - \frac{1}{n}$ is bounded. The infimum is $x_1 = 0$ (and hence also the minimum), and the supremum of this monotonically growing sequence is $\lim x_n = 1$. Since 1 is not a sequence member x_n , the supremum is not a maximum.
- The sequence $(y_n)_{n \in \mathbb{N}}$ with $y_n = \sqrt{n}$ is just bounded from below. Infimum (and at the same time minimum) is $y_1 = 1$. Since there is no upper bound, we have $\sup y_n = +\infty$, which is clearly not a maximum.

Also in this section we got very close to the topic of completeness of $\mathbb{R}$:

 Nerd box 2.37. Here we go again: completeness of $\mathbb{R}$

Via Proposition 2.29, we've been using the completeness of $\mathbb{R}$ in the proof of Proposition 2.32. The existence of the supremum and the infimum of a bounded sequence is indeed equivalent to the completeness of $\mathbb{R}$. In $\mathbb{Q}$, the sequence $x_n = \lfloor 10^n \pi \rfloor 10^{-n}$ with the integer round-down function $\lfloor \cdot \rfloor$ is bounded from above but there is no smallest upper bound: The set of all upper bounds in $\mathbb{Q}$ is $\{q \in \mathbb{Q} : q \geq \pi\}$, which indeed has no smallest element.

An alternative proof of Proposition 2.32 a), when Proposition 2.29 is not available but instead the existence of supremum and infimum are guaranteed:

Let $x := \sup x_n$ and take some $\varepsilon > 0$. Since $x - \varepsilon$ is not an upper bound (x is the smallest), there is a $n_0 \in \mathbb{N}$ such that $x_{n_0} \in (x - \varepsilon, x]$. By monotonicity, all x_n with $n \geq n_0$ are located in $[x_{n_0}, x] \subset (x - \varepsilon, x] \subset (x - \varepsilon, x + \varepsilon)$. ■

2.9 Subsequences and their limits

Definition 2.38. subsequence

The sequence $(x_{n_k})_{k \in \mathbb{N}}$ is called a subsequence of the sequence $(x_n)_{n \in \mathbb{N}}$ if $(n_1, n_2, n_3, \dots)$ is a strictly growing sequence of natural numbers, $n_1 < n_2 < n_3 < \dots$.

If, for example, $(n_k)_{k \in \mathbb{N}}$ is the sequence of prime numbers, $(2, 3, 5, 7, 11, \dots)$, then, out of the original sequence $(x_1, x_2, x_3, \dots)$ of real numbers, we are extracting the subsequence $(x_2, x_3, x_5, x_7, x_{11}, \dots)$, e.g., the Hamburg air temperatures on the prime number days.

Important to keep in mind:

- subsequences are still infinitely long,
- they are derived by omitting members from the original sequence,
- as long as there are still infinitely many left.
- Reordering is not allowed.
- Also the original sequence is a subsequence of itself. (No omissions.)

Example 2.39. subsequences

Let the sequence $(x_n)_n = ((-1)^n)_{n \in \mathbb{N}} = (-1, +1, -1, +1, \dots)$ be given.

- a) The subsequence $(x_{n_k})_k$ with $n_k = 2k$ is the constant sequence $(+1, +1, +1, \dots)$.
- b) The subsequence $(x_{n_k})_k$ with $n_k = 2k+1$ is the constant sequence $(-1, -1, -1, \dots)$.
- c) The subsequence $(x_{n_k})_k$ with $n_k = k$ is the original sequence.
- d) The subsequence $(x_{n_k})_k$ with $n_k = 3k$ is equal to the original sequence.

The following statement is hardly surprising:

Proposition 2.40. Subsequences of convergent sequences are convergent

If $x_n \rightarrow x$ then also every subsequence of $(x_n)_n$ converges to x .

Proof: Let $\varepsilon > 0$ be arbitrary. By $x_n \rightarrow x$ it follows that $x_n \overset{\varepsilon}{\approx} x$ for all $n \in \mathbb{N}$ up to finitely many exceptions. This property is preserved if we omit sequence members. ■

We know from Section 2.2 that convergent sequences are bounded but that bounded sequences need not converge. They have, however, always a convergent subsequence:

Proposition 2.41. of Bolzano-Weierstraß

Bernard Bolzano 1781–1848
Karl Weierstraß 1815–1897

Every bounded sequence has a convergent subsequence.

Proof: Let $(x_n)_n$ be a bounded sequence and $[a_1, b_1]$ an interval containing all x_n .

■

Looking back, this seems actually pretty obvious: Infinitely many points sitting in a bounded interval – they have to accumulate somewhere.

The limit of a convergent subsequence of $(x_n)_n$ is called a partial limit (or sometimes an accumulation point) of $(x_n)_n$.

a) *If $(x_n)_n$ is convergent with $x_n \rightarrow x$ then it has just one partial limit: x .*

b) *Our sequence $((-1)^n)_n$ does have the two partial limits -1 and $+1$.*

c) The sequence $(\sin(n\frac{\pi}{2}))_n$ does have the three partial limits -1 , 0 and $+1$.

It is easy to see that also the reverse implication holds in Example a).

 Nerd box 2.44. limes superior and limes inferior

The set S of all partial limits of a bounded sequence $(x_n)_n$ is

- nonempty (Proposition 2.41, Bolzano-Weierstraß),
- bounded (Proposition 2.21) and
- closed (see Definition 4.8 below).

That's why the set S has a largest and a smallest element, that we denote by $\limsup x_n$ and $\liminf x_n$, respectively. By Proposition 2.40 and the reverse direction of Example 2.43 a), it follows that $x_n \rightarrow x \Leftrightarrow \liminf x_n = \limsup x_n = x$.

Summary¹

- Sequences are functions $\mathbb{N} \rightarrow \mathbb{R}$, thought of as infinite vectors $(x_1, x_2, \dots)$ of real numbers. We write $(x_n)_{n \in \mathbb{N}}$ or just $(x_n)_n$.
- A sequence $(x_n)_n$ is said to converge to $x \in \mathbb{R}$, written as $x_n \rightarrow x$, if, for every precision $\varepsilon > 0$, it holds that $x_n \overset{\varepsilon}{\approx} x$ for all $n \in \mathbb{N}$ up to finitely many exceptions.
- Convergent sequences are bounded – but not the other way round.
- $\lim$ commutes with $+$, $-$, $\cdot$ and $\cdot$.
- If $x_n \leq y_n$ then $\lim x_n \leq \lim y_n$.
- From $x_n \leq y_n \leq z_n$ and $x_n \rightarrow y \leftarrow z_n$ it follows that $y_n \rightarrow y$.
- Convergent sequences are Cauchy sequences, i.e., $x_m - x_n \rightarrow 0$ as $m, n \rightarrow \infty$.
- Monotonic and bounded sequences are convergent.
 - In this case, the limit is either the infimum or the supremum of the sequence.
- Every bounded sequence has convergent subsequences.
 - If every one of them has the same limit then the original sequence converges.
 - And vice versa.

¹Disclaimer: At the end of each chapter, we have a very short summary like this. Things that are missing here are however important. (Otherwise they would not be in the notes, and these would be just one page per chapter.)

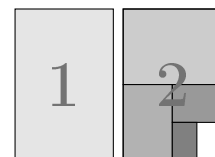
Chapter 3

Series

Granny is not feeling great so I'll go and visit her. If I bring two pieces of cake, that will make her feel better. She looks happy but she doesn't even touch her piece of cake. (She must feel really bad today.) My piece was delicious. It's long finished by now. I could have more... I ask Granny if she's sure about not having any bit of cake. And then, being polite, I have half of her piece. And a bit later half of what's left. Then again half of the rest. And so on, until nothing is left but crumbs.

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \stackrel{?}{=} 2$$

It looks like I had both pieces, after all.
Then I guess I'll have all the wine, too.



This infinite process of summation seems to lead to a finite result. Is it really 2?

- More than two pieces have never been on the table.
- But I didn't have much less than two.
- The amount of crumbs on the plate converges to zero as I keep eating.

In general: Take a sequence $(x_n)_n$ and start adding up:

$$\underbrace{x_1}_{s_1} + x_2 + x_3 + x_4 + \dots$$

$$\underbrace{\hspace{1.5cm}}_{s_2}$$

$$\underbrace{\hspace{2.5cm}}_{s_3}$$

The interim results, $s_1, s_2, s_3, \dots$, form a sequence that is to be studied for convergence. In our cake example, apparently, $s_m \rightarrow 2$ as $m \rightarrow \infty$.

Definition 3.1. series, partial sums

For a given sequence, $(x_n)_{n \in \mathbb{N}}$, the number

$$s_m := \sum_{n=1}^m x_n, \quad m \in \mathbb{N}$$

is called the m -th partial sum of $(x_n)_{n \in \mathbb{N}}$.

The sequence $(s_m)_{m \in \mathbb{N}}$ is called the series with the members x_n .

We say that the series converges to $s \in \mathbb{R}$ if $s_m \rightarrow s$, and we write

$$\sum_{n=1}^{\infty} x_n = s.$$

Remarks:

- Instead of “interim results” we say partial sums. The rest was quite right.
- The symbol $\sum_{n=1}^{\infty} x_n$ is used for the series, $(s_m)_{m \in \mathbb{N}}$, and for its limit.
- In the sense of Definition 2.11, we write $\sum_{n=1}^{\infty} x_n = +\infty$ if $s_m \rightarrow +\infty$.
- Sometimes it makes life easier to start summation at $n = 0$, not 1. Or possibly at 7.

⚠ Attention! series $\neq$ sequence

The words “series” and “sequence” often mean the same in everyday life. But, careful: not in mathematics!

3.1 Some first examples and criteria

Example 3.2. the geometric series

a) Our cake example:

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = \sum_{n=0}^{\infty} \frac{1}{2^n} = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = 2,$$

as we will show in a minute.

- b) A so-called geometric series arises, when we fix some $q \in (-1, 1)$ and form the series $\sum_{n=0}^{\infty} q^n$. For $m \in \mathbb{N}$, let s_m denote the m -th partial sum:

$$s_m = 1 + q + q^2 + \cdots + q^m.$$

Multiplying this equality by q , we get

By Example 2.33 a), $|q^m| = |q|^m \rightarrow 0$ and therefore $q^m \rightarrow 0$, which is why

$$s_m \rightarrow \frac{1}{1-q} = \sum_{n=0}^{\infty} q^n.$$

In our cake example **a)** with $q = \frac{1}{2}$, we thus confirm that $s_m \rightarrow \frac{1}{1-\frac{1}{2}} = 2$.

- c) For $q = 1$, we get $s_m = 1 + \cdots + 1 = m + 1 \rightarrow +\infty$. Divergence!

- d) For $q = -1$, we get $1 - 1 + 1 - 1 + \cdots$. Where is this converging to?

- Answer 1:
- Answer 2:
- Answer 3:

Solution: All wrong. The series does not converge at all, by Proposition 3.3.

⚠ Attention! Don't do this at home!

When convergence of the series is not yet clear, manipulations as in Example 3.2 d) can go arbitrarily wrong.

If someone had guaranteed us the convergence of b) beforehand, we could have argued in b) as sporty as in d):

$$s = \sum_{n=0}^{\infty} q^n \implies qs = \sum_{n=0}^{\infty} q^{n+1} \implies s - qs = q^0 = 1 \implies s = \frac{1}{1-q}.$$

With this kind of argument, you can, e.g., see really fast why $0.999\dots = 1$ since

But first, the convergence of the series (in this case: $s = 9 \cdot \sum_{n=1}^{\infty} 0.1^n$) has to be ensured.

Proposition 3.3. necessary convergence condition for series

If the series $\sum_{n=1}^{\infty} x_n$ converges then $x_n \rightarrow 0$.

Proof:

■

This proposition is almost always used via contraposition:

If $x_n \not\rightarrow 0$ then $\sum_{n=1}^{\infty} x_n$ just cannot converge.

And this is also the solution for Examples 3.2 c) and d) as well as for all $q \in \mathbb{R}$ with $|q| \geq 1$. Is the condition $x_n \rightarrow 0$, besides being necessary, also sufficient? No:

Example 3.4. the harmonic series

This time we study the so-called harmonic series, that is

$$\sum_{n=1}^{\infty} \frac{1}{n} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$$

The members, $\frac{1}{n}$, form a null sequence, so that convergence and divergence are still both possible. By the following neat arrangement,

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \dots + \frac{1}{16} + \dots$$

we can see that the partial sums s_m will eventually exceed every $M \in \mathbb{R}$ and grow monotonically, so that $s_m \rightarrow +\infty$.

For null sequences $(x_n)_n$, it all depends on how fast they decay. $x_n = \frac{1}{2^n}$ is fast enough, while $\frac{1}{n}$ isn't. Where is the transition happening? Let us look at two more examples:

Example 3.5. denominator of second order

a) We switch from $x_n = \frac{1}{n}$ to a term with denominator of second order:

$$x_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

By the clever transformation in the last step, we conclude, for every $m \in \mathbb{N}$:

$$s_m = x_1 + \dots + x_m =$$

b) Also for $x_n = \frac{1}{n^2}$, we get convergence since

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \sum_{n=2}^{\infty} \frac{1}{n^2} = 1 + \sum_{k=1}^{\infty} \frac{1}{(k+1)^2}$$

and by $0 \leq \frac{1}{(k+1)^2} < \frac{1}{k(k+1)}$. Consequently, s_m grows monotonically and is bounded from above by $1 +$ the limit from a). By Proposition 2.32, it converges.

🧐 Nerd box 3.6. And where exactly do we switch from divergence to convergence?

For a series of the form $\sum_{n=1}^{\infty} \frac{1}{n^s}$, the transition happens at $s = 1$: For $s \leq 1$, one has divergence, for $s > 1$, it's already convergence. The proof of this statement uses the same grouping technique as in Example 3.4, also see Nerd box 3.19 below.

PS: The limit in Example 3.5 b), by the way, is $\frac{\pi^2}{6}$. (↗ Basler problem)

The final argument in Example 3.5 b) is often useful. Let us isolate it here:

Proposition 3.7. convergence = boundedness, if $x_n \geq 0$

If $x_n \geq 0$ for all $n \in \mathbb{N}$, then:

$$\sum_{n=1}^{\infty} x_n \text{ converges} \iff (s_m)_{m \in \mathbb{N}} \text{ is bounded.}$$

Proof:

■

In contrast to the harmonic series, $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$, that diverges, as seen in Example 3.4, the so-called alternating harmonic series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + - \dots$$

is convergent (👁️: to $\ln 2$), by the following result:

Proposition 3.8. Leibniz rule

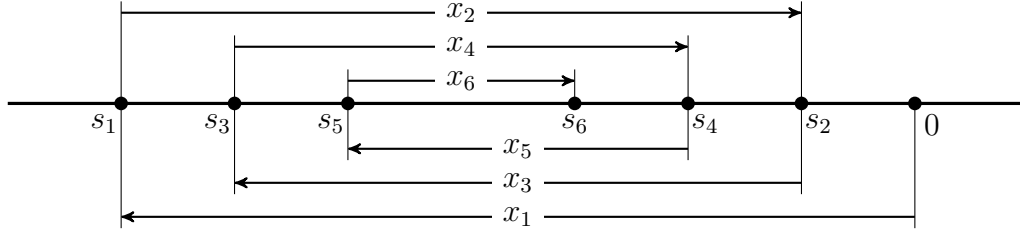
(Gottfried Wilhelm Leibniz 1646–1716)

Let $(x_n)_{n \in \mathbb{N}}$ be a monotonically falling null sequence. Then the alternating series

$$\sum_{n=1}^{\infty} (-1)^n x_n = -x_1 + x_2 - x_3 + x_4 - + \dots$$

converges.

Proof: Look at the partial sums $s_1, s_2, s_3, \dots$:



$\Rightarrow$ nested intervals $\underbrace{[s_1, s_2]}_{x_2} \supset \underbrace{[s_3, s_4]}_{x_4} \supset \underbrace{[s_5, s_6]}_{x_6} \supset \dots$
 with lengths $x_2, x_4, x_6, \dots \rightarrow 0$,

By the so-called principle of nested intervals (👁: yet another equivalent way of imposing the completeness of $\mathbb{R}$), there is exactly one point $s \in \mathbb{R}$ in the intersection of all these intervals, and it holds that $s_m \rightarrow s$. ■

The sketch even shows the error bound, $|s_m - s| \leq x_{m+1}$, at any point in time, $m \in \mathbb{N}$.

Also for series, Corollary 2.30 about the convergence of Cauchy sequences can be used:

$$\begin{aligned}
 \sum_{n=1}^{\infty} x_n \text{ converges} &\iff (s_m)_{m \in \mathbb{N}} \text{ converges} \xrightarrow{\text{F. 2.30}} (s_m)_{m \in \mathbb{N}} \text{ is a Cauchy sequence} \\
 &\iff \forall \varepsilon > 0 \exists n_0 \in \mathbb{N} : \forall m, m' \geq n_0 : |s_{m'} - s_m| < \varepsilon, \\
 &\iff \forall \varepsilon > 0 \exists n_0 \in \mathbb{N} : \forall m, m' \geq n_0 : \left| \sum_{n=m+1}^{m'} x_n \right| < \varepsilon. \quad (3.1)
 \end{aligned}$$

The latter is called the Cauchy criterion for series.

3.2 Absolute convergence

Definition 3.9. absolute and conditional convergence of a series

A series $\sum_{n=1}^{\infty} x_n$ is called absolutely convergent if even the series $\sum_{n=1}^{\infty} |x_n|$ converges.

A convergent but not absolutely convergent series is called conditionally convergent.

Example 3.10. absolute and conditional convergence

- a) The geometric series (our cake example), $\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$, is absolutely convergent.
- b) The alternating harmonic series, $\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$, is convergent, not absolutely convergent though, – hence conditionally convergent.

By Proposition 3.7, a series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent, if the series (i.e. the sequence of partial sums) $\sum_{n=1}^{\infty} |x_n|$ is bounded.

As the wording “absolutely convergent” already suggests, one has:

Proposition 3.11. absolute convergence $\implies$ convergence

Every absolutely convergent series is convergent.

Proof: Let $\varepsilon > 0$ be arbitrary. By the Cauchy criterion, (3.1), for $\sum_{n=1}^{\infty} |x_n|$, one has

$$\begin{aligned} \exists n_0 \in \mathbb{N} : \forall m, m' \geq n_0 : \varepsilon > \left| \sum_{n=m+1}^{m'} |x_n| \right| &= \sum_{n=m+1}^{m'} |x_n| \geq \left| \sum_{n=m+1}^{m'} x_n \right| \\ \implies \sum_{n=1}^{\infty} x_n &\text{ converges by the Cauchy criterion. } \blacksquare \end{aligned}$$

Hereby, the inequality

$$\sum_{n=1}^m |x_n| \geq \left| \sum_{n=1}^m x_n \right|$$

transfers to the limits as $m \rightarrow \infty$.

Example 3.10 b) shows that the reverse implication in Proposition 3.11 need not hold – so conditionally convergent series do exist.

Proposition 3.12. sums and multiples of (absolutely) convergent series

Let $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ be two (absolutely) convergent series, and let $c \in \mathbb{R}$ be constant.

Then also $\sum_{n=1}^{\infty} (x_n \pm y_n)$ and $\sum_{n=1}^{\infty} (cx_n)$ are (absolutely) convergent, and it holds that

$$\begin{aligned} \sum_{n=1}^{\infty} (x_n \pm y_n) &= \sum_{n=1}^{\infty} x_n \pm \sum_{n=1}^{\infty} y_n & \text{as well as} & & \sum_{n=1}^{\infty} (cx_n) &= c \sum_{n=1}^{\infty} x_n, & \text{and} \\ \sum_{n=1}^{\infty} |x_n \pm y_n| &\leq \sum_{n=1}^{\infty} |x_n| + \sum_{n=1}^{\infty} |y_n| & \text{as well as} & & \sum_{n=1}^{\infty} |cx_n| &= |c| \sum_{n=1}^{\infty} |x_n|. \end{aligned}$$

Proof: For the corresponding partial sums, $\sum_{n=1}^m \dots$, we have the formulas as claimed. By assumption, the limits, as $m \rightarrow \infty$, exist on the right-hand side, so that also the left-hand sides converge. For $\sum_{n=1}^{\infty} |x_n \pm y_n|$, we additionally involve Proposition 3.7. ■

3.3 Criteria by comparison with other series

In Examples 3.4 and 3.5, we already used bounds of unknown partial sums by known ones in order to prove convergence or divergence of the series. Here is the proper statement:

Proposition 3.13. direct comparison tests

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences. If $|x_n| \leq |y_n|$ for all $n \in \mathbb{N}$, then

- a) $\sum_{n=1}^{\infty} |y_n|$ converges $\implies \sum_{n=1}^{\infty} |x_n|$ converges $\implies \sum_{n=1}^{\infty} x_n$ converges.
- b) $\sum_{n=1}^{\infty} x_n$ diverges $\implies \sum_{n=1}^{\infty} |x_n|$ diverges $\implies \sum_{n=1}^{\infty} |y_n|$ diverges.

Again, we allow finitely many exceptions from $|x_n| \leq |y_n|$, i.e., $\forall n \geq n_0$ will suffice.

Proof: b) is the contraposition of a).

a) The second implication $\implies$ holds by Proposition 3.11. It remains to prove the first: Let $\varepsilon > 0$ be arbitrary. By the Cauchy criterion for $\sum |y_n|$, there is a $n_0 \in \mathbb{N}$ such that

$$\begin{aligned} \forall m, m' \geq n_0 : \quad \varepsilon &> \left| \sum_{n=m+1}^{m'} |y_n| \right| = \sum_{n=m+1}^{m'} |y_n| \geq \sum_{n=m+1}^{m'} |x_n| = \left| \sum_{n=m+1}^{m'} x_n \right| \\ \implies \sum_{n=1}^{\infty} |x_n| &\text{converges, again by the Cauchy criterion.} \quad \blacksquare \end{aligned}$$

Example 3.14. dominated convergence and its contraposition

a) The series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ converges since

$$\frac{n!}{n^n} = \frac{1}{n} \cdot \frac{(n-1)!}{n^{n-1}} \leq \frac{2}{n^2} \quad \text{for all } n \in \mathbb{N}$$

and since, by Example 3.5 b), $\sum_{n=1}^{\infty} \frac{2}{n^2}$ converges (dominated convergence).

b) The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$ diverges since

$$\frac{1}{\sqrt{n(n+1)}} \geq \frac{1}{n+1} \quad \text{for all } n \in \mathbb{N}$$

and since, by Example 3.4, $\sum_{n=1}^{\infty} \frac{1}{n+1} = \sum_{n=2}^{\infty} \frac{1}{n}$ diverges.

In the case of the series

$$400 + 200 + 100 + 50 + 25 + \dots,$$

we can almost directly roll back to our cake example:

Obviously, $x_n = 400 \cdot \left(\frac{1}{2}\right)^n$ with $n = 0, 1, 2, \dots$, whence the series converges. The hint comes by observing that $\frac{x_{n+1}}{x_n} = \frac{1}{2}$. Alternatively, we could have argued via $\sqrt[n]{x_n} = \sqrt[n]{400} \frac{1}{2} \rightarrow \frac{1}{2}$. Both are standard tests:

Proposition 3.15. the ratio test

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence with at most finitely many members of the kind $x_n = 0$.

a) If $\exists n_0 \in \mathbb{N}, q < 1 : \forall n \geq n_0 : \left| \frac{x_{n+1}}{x_n} \right| \leq q$, then $\sum_{n=1}^{\infty} x_n$ is absolutely convergent.

b) If $\exists n_0 \in \mathbb{N} : \forall n \geq n_0 : \left| \frac{x_{n+1}}{x_n} \right| \geq 1$, then $\sum_{n=1}^{\infty} x_n$ diverges.

Proof: a) From $\forall n \geq n_0 : \left| \frac{x_{n+1}}{x_n} \right| \leq q < 1$, it follows that

$$\forall n \geq n_0 : |x_n| = \underbrace{\left| \frac{x_n}{x_{n-1}} \right|}_{\leq q} \cdot \underbrace{\left| \frac{x_{n-1}}{x_{n-2}} \right|}_{\leq q} \cdot \dots \cdot \underbrace{\left| \frac{x_{n_0+1}}{x_{n_0}} \right|}_{\leq q} \cdot |x_{n_0}| \leq q^{n-n_0} \cdot |x_{n_0}|,$$

which implies the convergence of $\sum_{n=n_0}^{\infty} |x_n|$ via domination by the convergent series

$$\sum_{n=n_0}^{\infty} q^{n-n_0} |x_{n_0}| = \frac{|x_{n_0}|}{q^{n_0}} \cdot \sum_{n=n_0}^{\infty} q^n \quad (\text{geometric series with } q \in [0, 1)).$$

b) By $|x_{n+1}| \geq |x_n|$ for $n \geq n_0$, we have $x_n \not\rightarrow 0$, whence $\sum_{n=1}^{\infty} x_n$ diverges. ■

Proposition 3.16. the root test

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence.

a) If $\exists n_0 \in \mathbb{N}, q < 1 : \forall n \geq n_0 : \sqrt[n]{|x_n|} \leq q$, then $\sum_{n=1}^{\infty} x_n$ is absolutely convergent.

b) If $\exists n_0 \in \mathbb{N} : \forall n \geq n_0 : \sqrt[n]{|x_n|} \geq 1$, then $\sum_{n=1}^{\infty} x_n$ diverges.

Proof: a) By $|x_n| \leq q^n$ for $n \geq n_0$, the series $\sum_{n=n_0}^{\infty} q^n$ dominates our series $\sum_{n=n_0}^{\infty} |x_n|$.

b) By $|x_n| \geq 1$ for $n \geq n_0$, $(x_n)_n$ is not a null sequence, whence $\sum_{n=1}^{\infty} x_n$ diverges. ■

Example 3.17. Both tests have their pro's and con's

- a) Let $a \in \mathbb{R}$ be arbitrary. $\sum_{n=0}^{\infty} \frac{a^n}{n!}$ is absolutely convergent (where we put $0! := 1$) since, by the ratio test,

$$\left| \frac{x_{n+1}}{x_n} \right| = \frac{|a|}{n+1} \rightarrow 0 \quad \text{for } n \rightarrow \infty,$$

$$\text{i.e. } \exists n_0 : \forall n \geq n_0 : \left| \frac{x_{n+1}}{x_n} \right| \leq \frac{1}{2} < 1.$$

Note that the root test is a lot clumsier to use here.

- b) Let $x_n = \begin{cases} \frac{1}{2^n} & , n \text{ even,} \\ \frac{1}{3^n} & , n \text{ odd.} \end{cases} \implies \sqrt[n]{|x_n|} \leq \frac{1}{2} < 1, \text{ (root test)}$

$$\implies \sum_{n=1}^{\infty} x_n = \frac{1}{3^1} + \frac{1}{2^2} + \frac{1}{3^3} + \frac{1}{2^4} + \frac{1}{3^5} + \dots \quad \text{converges (absolutely).}$$

Note that

$$\left| \frac{x_{n+1}}{x_n} \right| = \begin{cases} \left(\frac{3}{2}\right)^n \cdot \frac{1}{2} > 1, & n \text{ odd,} \\ \left(\frac{2}{3}\right)^n \cdot \frac{1}{3} < 1, & n \text{ even,} \end{cases}$$

so that the ratio test is unable to conclude anything!

 Nerd box 3.18. ratio and root test via lim inf & lim sup

When we use $\liminf$ and $\limsup$ from Nerd box 2.44, we can reformulate both the ratio and root test in an almost equivalent way:

÷ a) If $\limsup \left| \frac{x_{n+1}}{x_n} \right| < 1$, then $\sum_{n=1}^{\infty} x_n$ converges absolutely.

÷ b) If $\liminf \left| \frac{x_{n+1}}{x_n} \right| > 1$, then $\sum_{n=1}^{\infty} x_n$ diverges.

$\sqrt[n]{\cdot}$ a) If $\limsup \sqrt[n]{|x_n|} < 1$, then $\sum_{n=1}^{\infty} x_n$ converges absolutely.

$\sqrt[n]{\cdot}$ b) If $\limsup \sqrt[n]{|x_n|} > 1$, then $\sum_{n=1}^{\infty} x_n$ diverges.

- The asymmetry in $\liminf$ vs $\limsup$ is not a typo; it's like it is written here.
- Nothing can be said if $\liminf \left| \frac{x_{n+1}}{x_n} \right| \leq 1 \leq \limsup \left| \frac{x_{n+1}}{x_n} \right|$,
- neither if $\limsup \sqrt[n]{|x_n|} = 1$.

One can show that the root test is more powerful than the ratio test, in the following sense:

- When the ratio test signals convergence, then the root test does that, too.
- This is not true the other way round, as Example 3.17 b) shows.
- However, the ratio test is often easier to apply (if applicable).
- In the example $\sum_{n=1}^{\infty} \frac{1}{n^s}$ however, they are both clueless.

 Nerd box 3.19. Cauchy's condensation test

The grouping strategy from Example 3.4 along the powers of two, $n = 2^k$, is easily generalised: If $(x_n)_n$ is a monotonically falling null sequence then

$$\sum_{n=1}^{\infty} x_n \text{ converges} \iff \sum_{k=0}^{\infty} 2^k \cdot x_{2^k} \text{ converges.}$$

And this is how to crack the nut from Nerd box 3.6, where all our other tests fail.

Summary

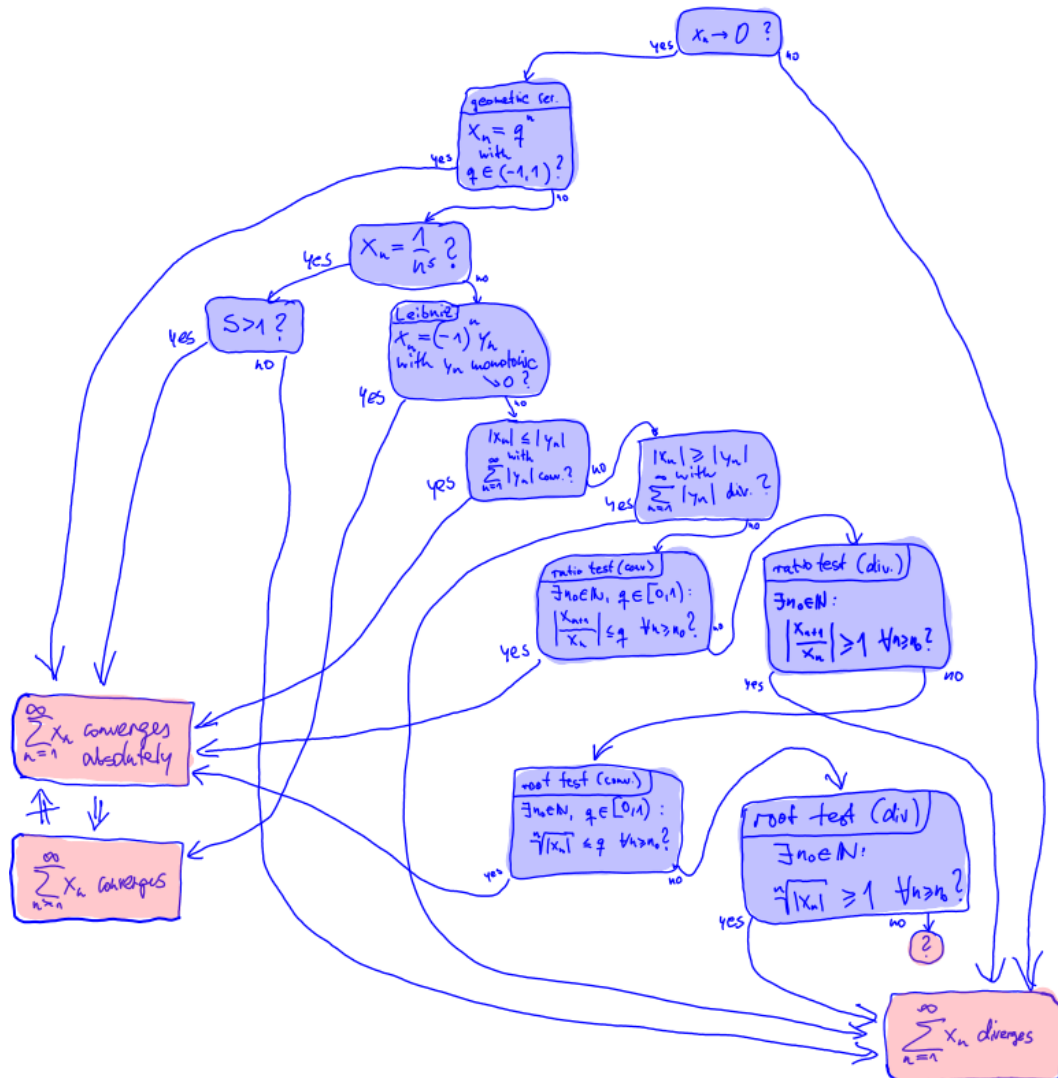
- One takes a sequence $(x_1, x_2, x_3, \dots)$ and starts adding its members:

$$\underbrace{x_1}_{s_1} + x_2 + x_3 + x_4 + \dots$$

$$\underbrace{\hspace{1.5cm}}_{s_2}$$

$$\underbrace{\hspace{2.5cm}}_{s_3}$$

- A series is the sequence $(s_m)_{m=1}^{\infty}$ of partial sums, $s_m = \sum_{n=1}^m x_n$. It is denoted by the symbol $\sum_{n=1}^{\infty} x_n$.
- The geometric series with $|q| < 1$ is a standard example of convergent series.
- For the convergence of the series $\sum_{n=1}^{\infty} x_n$ it is necessary that its members x_n form a null sequence. Otherwise there is no chance for convergence.
 - And is $x_n \rightarrow 0$ also enough (sufficient) for the convergence of the series $\sum_{n=1}^{\infty} x_n$? No, for this, one needs x_n to converge fast enough to 0.
 - The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, for example, while $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.
 - Precisely: In the special case where $x_n = \frac{1}{n^s}$ for some $s \in \mathbb{R}$, one has convergence of the series $\sum_{n=1}^{\infty} x_n$ if $s > 1$ and divergence otherwise.
 - In general cases, we suggest the following tests:



very short roundup

The topic of series essentially comes with two questions:

a) *Is my series convergent?*

We now know a number of tricks for this one.

b) *If yes, what's the limit?*

This question is much tougher, and we know much less.

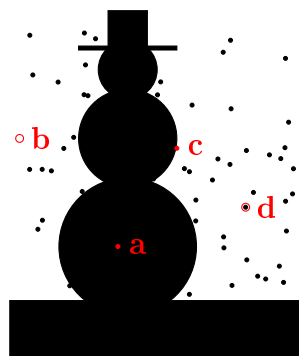
Chapter 4

Topology of the real line

Having discussed sequences and series, we now come to so-called topological properties of sets. Although all our sets will be subsets of $\mathbb{R}$, for the following illustration, let's look at a set in $\mathbb{R}^2$:

In the figure on the right, the points of a particular set $A \subset \mathbb{R}^2$ are shown in black. Then

- a ... interior point of A ,
- b ... exterior point of A ,
- c ... boundary point of A ,
- d ... isolated point of A .



Next we explain these and further kinds of points in more details.

4.1 Some important kinds of points

For our classification of points we still need one more notion:

Definition 4.1. ε -neighbourhood

For $x \in \mathbb{R}$ and $\varepsilon > 0$, we refer to the set

$$U_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |x - y| < \varepsilon\}$$

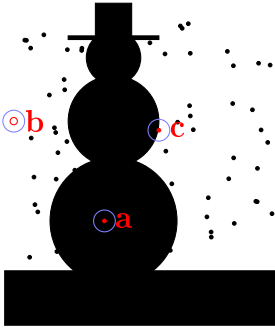
as the ε -neighbourhood of x .

On different length scales $\varepsilon > 0$, we now look at the ε -neighbourhood $U_\varepsilon(x)$ of a point x and decide about the kind of x , depending on how much of $U_\varepsilon(x)$ belongs to A or to its complement, A^c :

Definition 4.2. interior, exterior and boundary points

Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$. The point x is called

- an interior point of A , if $U_\varepsilon(x) \cap A = U_\varepsilon(x)$ for some $\varepsilon > 0$,
i.e., there is an $\varepsilon > 0$, such that $U_\varepsilon(x)$ is entirely in A ,
in particular, we then have $x \in A$,
- exterior point of A , if $U_\varepsilon(x) \cap A = \emptyset$ for some $\varepsilon > 0$,
i.e., there is an $\varepsilon > 0$, such that $U_\varepsilon(x)$ is entirely outside of A ,
i.e., x is an interior point (and hence an element) of the complement, A^c ,
- boundary point of A , if none of the two cases above applies,
that is, when in every $U_\varepsilon(x)$ with $\varepsilon > 0$, there are both points from A and from A^c .



Back to our snowman:

- The point a has an entirely black ε -neighbourhood and is therefore an interior point of A .
- The point b has an entirely white ε -neighbourhood and is therefore an exterior point of A .
- For point c , every ε -neighbourhood is black/white mixed. Therefore, c is a boundary point of A .

Definition 4.3. interior, exterior and closure of A

Let $A \subset \mathbb{R}$ be a set.

- $\text{int } A$ be the set of all interior points of A , and it is called the interior of A ,
- $\text{ext } A$ be the set of all exterior point of A , and it is called the exterior of A ,
- ∂A be the set of all boundary points of A , and it is called the boundary of A .
- The set $\text{clos } A := \text{int } A \cup \partial A$ is called the closure of A .

$$\begin{aligned} \text{I.e., } x \in \text{clos } A &\iff x \notin \text{ext } A \iff \neg(\exists \varepsilon > 0 : U_\varepsilon(x) \cap A = \emptyset) \\ &\iff \forall \varepsilon > 0 : U_\varepsilon(x) \cap A \neq \emptyset \iff \exists x_1, x_2, \dots \in A : x_n \rightarrow x. \end{aligned}$$

Here $B \cup C$ (disjoint union) stands for $B \cup C$ with the extra info that $B \cap C = \emptyset$.

Example 4.4. some simple examples on the real axis

For $A = [0, 1)$ and $B = [1, 2] \cup \{3\}$, we have that

a) $\text{int } A = (0, 1)$, $\text{ext } A = (-\infty, 0) \cup (1, +\infty)$, $\partial A = \{0, 1\}$ and $\text{clos } A = [0, 1]$.

b) $\text{int } B = (1, 2)$, $\text{ext } B = (-\infty, 1) \cup (2, 3) \cup (3, +\infty)$, $\partial B = \{1, 2, 3\}$, $\text{clos } B = B$.

Immediately by Definitions 4.2 and 4.3, we conclude:

Corollary 4.5. connections between interior, exterior and boundary

One has

$$\text{int } A \cup \text{ext } A \cup \partial A = \mathbb{R}$$

as well as

$$\text{int } A = A \setminus \partial A, \quad \text{ext } A = \text{int}(A^c), \quad \partial A = \partial(A^c).$$

We immediately conclude some further formulas for the closure:

Corollary 4.6. Formulas for the closure

$$\begin{aligned} \text{clos } A &= \text{int } A \cup \partial A = A \cup \partial A, \\ \text{clos } A &= (\text{ext } A)^c = (\text{int}(A^c))^c. \end{aligned}$$

We could study further kinds of points but those introduced so far are enough for us.

🧐 **Nerd box 4.7. Isolated points and limit points**

A point $x \in A \subset \mathbb{R}$ is called

- an isolated point of A , if $U_\varepsilon(x) \cap A = \{x\}$ for some $\varepsilon > 0$,
- a limit point of A , if $U_\varepsilon(x) \cap A$ has infinitely many points for all $\varepsilon > 0$.

Isolated points (like d in the snow man) are special boundary points.

x is a limit point of A if and only if $x \in \text{clos}(A \setminus \{x\})$.

The limit points of A are the closure of A without the isolated points.

⚠️ **Attention! accumulation point $\neq$ limit point**

The sequence $(1, 1, 1, \dots)$ has the accumulation point (a.k.a. partial limit) 1.

The set $\{1, 1, 1, \dots\} = \{1\}$ however, has no accumulation points at all.

We prefer “partial limit” over “accumulation point” to avoid confusion.

4.2 Open and closed sets

Definition 4.8. open set, closed set

A set $A \subset \mathbb{R}$ is called

- an open set if every point $x \in A$ is an interior point of A ,
i.e., one has $A = \text{int } A$, so that $A \cap \partial A = \emptyset$,
i.e., the set A does not contain any of its boundary points $x \in \partial A$,
- a closed set if its complement, A^c , is open,
i.e., $A^c = \text{int}(A^c) = \text{ext } A$,
and hence $A = (\text{ext } A)^c = \text{clos } A$, so that $\partial A \subset A$,
i.e., the set A contains all of its boundary points, $x \in \partial A$.
By $\text{clos } A \subset A$, from $x_1, x_2, \dots \in A$ and $x_n \rightarrow x$ it always follows that $x \in A$.

⚠ Attention! Open and closed are not the opposites of each other!

*Both sets $A = \mathbb{R}$ and $B = \emptyset$ are open and closed at the same time.
In contrast, many sets, like $C = [0, 1)$ or $D = (2, 3]$, are neither open nor closed.*

We continue with examples, including proofs of the last claims.

Example 4.9. open and/or closed sets

- a) So-called “open intervals”, $A = (a, b)$, are indeed open sets: Every point $x \in A$ still has an ε -neighbourhood with $U_\varepsilon(x) \subset A$. E.g., take $\varepsilon := \min(x - a, b - x)$.

Shorter argument: The boundary, $\partial A = \{a, b\}$, is entirely outside of A .

- b) So-called “closed intervals”, $A = [a, b]$, are indeed closed sets:
The complement, $A^c = (-\infty, a) \cup (b, +\infty)$, is open (arguing as in a).

Or in short: The boundary, $\partial A = \{a, b\}$, is entirely contained in A .

- c) So-called “half-open intervals”, $A = (a, b]$, are neither open nor closed: They contain their boundary, $\partial A = \{a, b\}$, neither completely nor not at all.
Argued differently: A is not open since $x = a$ has no neighbourhood $U_\varepsilon(x) \subset A$;
 A is not closed since $x_n = a + \frac{1}{n} \in A$ converges to $a \notin A$.

d) The set $A = \mathbb{R}$ is open since, for every $x \in \mathbb{R}$, one has $U_\varepsilon(x) \subset \mathbb{R}$ with $\varepsilon := 1$.

e) The set $A = \emptyset$ is open. Who contradicts, has to find an $x \in \emptyset$ (⚡) that is not an interior point of A .

f) The set $A = \mathbb{R}$ is closed since its complement, $A^c = \emptyset$, is open.

g) The set $A = \emptyset$ is closed since its complement, $A^c = \mathbb{R}$, is open.

It is not difficult to show that

- $\text{int } A$ is open, precisely: the largest open set B with $B \subset A$,
- $\text{clos } A$ is closed, precisely: the smallest closed set C with $A \subset C$
- and that ∂A is closed.

In Example 4.9 b), we claimed that the union of two open intervals is open. The following proposition finalizes this observation and more:

Proposition 4.10. Union of open sets and intersection of closed sets

- a) The union of arbitrarily many open sets is open.
 b) The intersection of arbitrarily many closed sets is closed.

Proof: a)

b) Let A_i be closed for all $i \in I$. Then all A_i^c are open and, by a), also $\cup_{i \in I} A_i^c = (\cap_{i \in I} A_i)^c$ is open (De Morgan). Consequently, $\cap_{i \in I} A_i$ is closed. ■

⚠ Attention! Do not confuse!

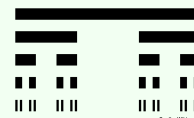
The intersection of infinitely many open sets need not be open. E.g., $\bigcap_{i \in \mathbb{N}} (-\frac{1}{i}, \frac{1}{i}) = \{0\}$ is not open.

For finitely many open sets, also the intersection is open. Similarly for the union of closed sets.

Example 4.11. Cantor set

(Georg Cantor, 1845–1918)

Starting from $A_1 = [0, 1]$, we construct A_{n+1} in a set sequence $A_1, A_2, \dots$ by deleting the open middle third from every interval in A_n . The Cantor set is then the intersection $A := \bigcap_{n \in \mathbb{N}} A_n$.



We show that A is closed. This is best seen via the complement:

A_{n+1}^c arises from A_n^c via union with certain open intervals (the open middle thirds of all intervals in A_n). Since $A_1^c = (-\infty, 0) \cup (1, +\infty)$ is open, it follows, inductively, by Proposition 4.10 a), that every A_n^c is open – and so is their union, $\bigcup_{n \in \mathbb{N}} A_n^c$. But the latter is A^c (De Morgan). Hence, A is closed.

4.3 Bounded sets

In analogy to bounded sequences and Definition 2.8, we now continue with sets:

Definition 4.12. bounded sets and their bounds

A set $A \subset \mathbb{R}$ is called

- bounded from below if there is an $a \in \mathbb{R}$ with $a \leq x$ for all $x \in A$,
- bounded from above if there is a $b \in \mathbb{R}$ with $x \leq b$ for all $x \in A$,
- bounded if it is bounded from below and above, i.e., if there are numbers $a, b \in \mathbb{R}$ such that $A \subset [a, b]$. In particular, one then has $|x| \leq c$ for all $x \in A$ with $c = \max(|a|, |b|)$,
- unbounded if it is not bounded.

Such numbers a and b are then called lower bound, resp. upper bound, on A .

Due to the completeness of $\mathbb{R}$ (see Nerd box 2.37), every bounded set $A \subset \mathbb{R}$ has a largest lower bounds and a smallest upper bound in $\mathbb{R}$:

Definition 4.13. infimum, supremum, minimum and maximum of sets

Let $A \subset \mathbb{R}$ be a set that is bounded from below or above (or both).

- The largest lower bound of the set is its infimum.
- If the infimum is itself an element $x \in A$, then we also call it the minimum.
- The smallest upper bound of the set is its supremum.
- If the supremum is itself an element $x \in A$, then we also call it the maximum.

The corresponding notations are $\inf A$, $\min A$, $\sup A$ and $\max A$.

For unbounded sets A without a lower bound, we set $\inf A := -\infty$, and if it is without upper bound, then put $\sup A := +\infty$.

Example 4.14. inf, sup, min and max

- a) If $A = (2, 5]$ then $\inf A = 2$ and $\sup A = 5 = \max A$. $\min A$ does not exist.
- b) The set $B = \{x \in \mathbb{R} : x^2 < 2\} = (-\sqrt{2}, \sqrt{2})$ is bounded. One has $\inf B = -\sqrt{2}$ and $\sup B = \sqrt{2}$. Minimum and maximum do not exist.
- c) The set $C = \{x \in \mathbb{Q} : x^2 < 2\} = (-\sqrt{2}, \sqrt{2}) \cap \mathbb{Q}$ is bounded. It has, however, no largest lower bound and no smallest upper bound in $\mathbb{Q}$. Their existence is a special feature (equivalent to completeness) of $\mathbb{R}$ (Nerd box 2.37).

4.4 Compact sets

Finite sets always have a minimum and a maximum, and they also have other quantities that make working with them just a bit more convenient. Compact sets behave, in many aspects, like finite sets.

Definition 4.15. compact set

A set $A \subset \mathbb{R}$ is called compact if every sequence from A has a convergent subsequence whose limit is again in A .

Example 4.16. compact sets

- a) $A = \{1, 2, 3\}$, as well as any other finite set, is compact. Indeed: Every sequence from A contains infinitely often the 1 or the 2 or the 3. The corresponding subsequence is constant, hence convergent, and its limit is in A .
- b) The interval $A = [3, 4]$ is compact. Indeed: Every sequence from A is bounded, hence, by Proposition 2.41 (Bolzano-Weierstraß), it has a convergent subsequence. Since A is closed, the limit of that subsequence is still in A .
- c) The argument from b) actually proves a lot more:
If $A \subset \mathbb{R}$ is bounded and closed, then A is compact.
- d) The empty set is compact. Who contradicts, has to find a sequence $(x_n)_n$ with $x_n \in \emptyset$ ($\nexists$), that is without a convergent subsequence with limit in $\emptyset$.
- e) $\mathbb{N}$ is not compact: $(n)_n$ has no convergent subsequence with limit in $\mathbb{N}$.
- f) $(0, 1]$ is not compact: $(\frac{1}{n})_n$ has no convergent subsequence with limit in $(0, 1]$.
- g) The Cantor set from Example 4.11 is closed (see there) and bounded (it is contained in $[0, 1]$), whence it is compact, by c).

Also the reverse direction holds in the implication of Example 4.16 c):

Proposition 4.17. by Heine-Borel

Eduard Heine 1821–1881
Emile Borel 1871–1956

For $A \subset \mathbb{R}$, one has: A is compact $\iff A$ is bounded and closed.

Proof: $\boxed{\Leftarrow}$ holds by Example 4.16 c). $\boxed{\Rightarrow}$: Let A be compact and let $(x_n)_n$ be a convergent sequence from A . By compactness, $(x_n)_n$ has a convergent subsequence with its limit $x \in A$. Consequently, also $\lim x_n = x \in A$, i.e., A is closed. If A were not bounded, there were a sequence $(x_n)_n$ in A with $|x_n| > n$. But this sequence has no bounded, and hence no convergent, subsequence – contradicting the compactness of A . ■

Now compactness of a set $A \subset \mathbb{R}$ is much easier checked than by Definition 4.15.

🧐 **Nerd box 4.18.** Here we go again: completeness of $\mathbb{R}$

The direction $\boxed{\Rightarrow}$ holds in arbitrary so-called metric spaces. The direction $\boxed{\Leftarrow}$, i.e. Example 4.16 c), relies on Proposition 2.41 and hence on Propositions 2.32 and 2.29, bringinng us back to the completeness of $\mathbb{R}$. This direction can be extended to $\mathbb{R}^n$.

Our start into this section was that compact sets in $\mathbb{R}$, just like finite sets, should have a minimum and a maximum. Let us go the full circle now:

Proposition 4.19. Compact sets possess a minimum and a maximum

Nonempty compact sets $A \subset \mathbb{R}$ possess a minimum and a maximum.

Proof: Let $A \subset \mathbb{R}$ be compact. By Proposition 4.17, A is bounded and closed.

■

Summary

- With respect to a given set $A \subset \mathbb{R}$, the number line $\mathbb{R}$ splits into three disjoint subsets: the interior, the exterior and the boundary of A .
- When we switch from A to A^c (e.g. black $\leftrightarrow$ white for the snowman), then interior and exterior swap places, and the boundary stays the same.

- Put together, interior and boundary make up the closure of A .
- A set is called open if it is equal to its interior, i.e., if it contains none of its boundary points.
- A set is called closed if it is equal to its closure, i.e., if it contains all of its boundary points.
- Sets that contain part but not all of their boundary are neither open nor closed.
- $\emptyset$ and $\mathbb{R}$ are both open and closed at the same time.
- A is open $\iff A^c$ is closed.
- The union of arbitrarily many and the intersection of finitely many open sets is open.
- The intersection of arbitrarily many and the union of finitely many closed sets is closed.
- A set is called bounded if it fits into an interval $[a, b]$ of finite length.
- A set A is called compact if every sequence from A has a convergent subsequence with limit in A .
- Finite sets are compact.
- $A \subset \mathbb{R}$ is compact $\iff A$ is bounded and closed.

Chapter 5

Continuity

With exactly 100 g filling, the sausage reaches the desired length of 13 cm. We may deviate a little from this ideal case:

- Desired outcome: 13 ± 0.5 cm length
- Allowed input: $100 \pm \boxed{?}$ g filling

Continuity of f at $x_0 = 100$ means that $f(x) \approx f(100)$ if $x \approx 100$. More precisely: Any desired approximation quality of the result, $f(x) \approx f(100)$, is achieved if the approximation of the input variable, $x \approx 100$, is sufficiently good.



5.1 Definition: Limit and continuity

In this chapter we analyse functions $f : D \rightarrow \mathbb{R}$ with domain $D \subset \mathbb{R}$.

Reminder: $x_0 \in \mathbb{R}$ lies in the closure of D , in short: $x_0 \in \text{clos } D$, if there are points from D in every neighbourhood of x_0 , i. e. there is a sequence $x_1, x_2, \dots \in D$ with $x_n \rightarrow x_0$.

So let $f : D \rightarrow \mathbb{R}$ be a function with domain $D \subset \mathbb{R}$.

Definition 5.1. limit of a function at a point

$y \in \mathbb{R}$ is the limit of f in $x_0 \in \text{clos } D$, if

$$f(x_n) \rightarrow y \text{ for all sequences } (x_n)_n \text{ in } D \text{ with } x_n \rightarrow x_0. \quad (5.1)$$

Notation: $y = \lim_{\substack{x \rightarrow x_0 \\ x \in D}} f(x)$ or $y = \lim_{x \rightarrow x_0} f(x)$ or $f(x) \rightarrow y$ as $x \rightarrow x_0$

It holds that $\text{clos } D \supset D$. If $x_0 \in D$, the constant sequence $x_n = x_0 \in D$ can also be selected for the approximation $x_n \rightarrow x_0$ by points $x_1, x_2, \dots \in D$.

Proposition 5.2. limit at $x_0 \in D$

If $x_0 \in D$ and $\lim_{x \rightarrow x_0} f(x)$ exist, then $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Proof: Assume $y := \lim_{x \rightarrow x_0} f(x)$. $f(x_n) \rightarrow y$ holds for all sequences $(x_n)_n$ in D with $x_n \rightarrow x_0$, especially for the constant sequence $x_n = x_0$. Therefore, it is also true that $f(x_0) \rightarrow y$, that is, $y = f(x_0)$.

■

 Nerd box 5.3. $\lim$ commutes with f

In the case of $x_0 \in D$ if the limit of f exists in x_0 , then you have

$$x_n \rightarrow x_0 \implies f(x_n) \rightarrow f(x_0) \quad \text{resp.} \quad \lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n).$$

Similar to Proposition 2.19 read this as: $\lim$ commutes with f (at least nearby x_0).

To bring a little order into the chaos, all possible approximation sequences $(x_n)_n$ in D to $x_0 \in \mathbb{R}$, we distinguish at least between approximation from the left and from the right:

$$\begin{array}{c} \longrightarrow x_0 \longleftarrow \\ | \\ \hline \end{array}$$

For each of the two directions, the approximation of x_0 is of course still possible along various sequences $(x_n)_n$. For $x_0 \in \mathbb{R}$ and $D \subset \mathbb{R}$ we write

$$D_{<x_0} := D \cap (-\infty, x_0) \quad \text{and} \quad D_{>x_0} := D \cap (x_0, +\infty).$$

Definition 5.4. one-sided limits

for $f : D \rightarrow \mathbb{R}$ and $x_0 \in \text{clos } D$ as well as $y \in \mathbb{R}$ we write

$$y = \lim_{x \nearrow x_0} f(x) \quad \text{or} \quad y = \lim_{x \rightarrow x_0 - 0} f(x) \quad \text{or} \quad y = f(x_0 - 0),$$

if $x_0 \in \text{clos } D_{<x_0}$ and $f(x_n) \rightarrow y$ for all sequences $(x_n)_n$ in $D_{<x_0}$ with $x_n \rightarrow x_0$,

as well as

$$y = \lim_{x \searrow x_0} f(x) \quad \text{or} \quad y = \lim_{x \rightarrow x_0 + 0} f(x) \quad \text{or} \quad y = f(x_0 + 0),$$

if $x_0 \in \text{clos } D_{>x_0}$ and $f(x_n) \rightarrow y$ for all sequences $(x_n)_n$ in $D_{>x_0}$ with $x_n \rightarrow x_0$.

y is then called left-sided limit or right-sided limit of f in x_0 .

Note that this time x_0 itself is not allowed to be part of the approximating sequence $(x_n)_n$.

Example 5.5. limits of functions in points

a) Let $f(x) = 5$ for all $x \in D = \mathbb{R}$ (constant function) and $x_0 \in \mathbb{R}$. From $x_n \rightarrow x_0$ it follows that $f(x_n) = 5 \rightarrow 5$, which means $\lim_{x \rightarrow x_0} f(x)$ exists and is equal to 5.

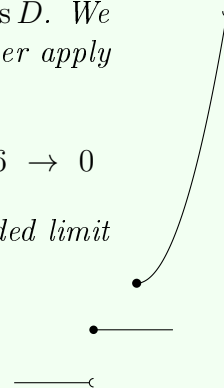
b) Let $f(x) = 2x + 1$ for all $x \in D = \mathbb{R}$ and choose an arbitrary $x_0 \in \mathbb{R}$. From $x_n \rightarrow x_0$ follows $f(x_n) = 2x_n + 1 \rightarrow 2x_0 + 1 = f(x_0)$, which means $\lim_{x \rightarrow x_0} f(x)$ exists and is equal to $f(x_0)$.

c) Let $f(x) = x^2$ for all $x \in D = [0, 3) \subset \mathbb{R}$ and let $x_0 = 3 \in \text{clos } D$. We show that the limit of f exists in x_0 and is equal to 9. Either apply Proposition 2.19 c) or recalculate:

$$|f(x_n) - 9| = |x_n^2 - 9| = |x_n - 3| \cdot |x_n + 3| \leq |x_n - 3| \cdot 6 \rightarrow 0$$

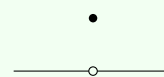
and therefore $f(x_n) \rightarrow 9$ as $x_n \rightarrow 3$. In $x_0 = 3$ no right-sided limit value can be formed since $D_{>3} = \emptyset$.

d) Set $f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$ with $D = \mathbb{R}$ and set $x_0 = 0$.



Then the one-sided limit values $\lim_{x \nearrow 0} f(x) = 0$ and $\lim_{x \searrow 0} f(x) = 1$ exist. However, the limit $\lim_{x \rightarrow 0} f(x)$ does not exist because, for example the sequences $(x_n)_n = (\frac{1}{n})_n$ and $(x'_n)_n = (-\frac{1}{n})_n$ both converge to $x_0 = 0$ but $f(x_n) \rightarrow 1$, while $f(x'_n) \rightarrow 0$, i.e. (5.1) does not apply.

e) Set $f(x) = \begin{cases} 0, & x \neq 0 \\ 1, & x = 0 \end{cases}$ with $D = \mathbb{R}$ and set $x_0 = 0$.



Then the one-sided limits $\lim_{x \nearrow 0} f(x) = 0$ and $\lim_{x \searrow 0} f(x) = 0$ exist.

However, the limit $\lim_{x \rightarrow 0} f(x)$ does not exist, since $f(0) = 1$, i.e. the constant sequence $(x_n)_n = (0)_n$ leads to $f(x_n) = 1 \rightarrow 1$. (5.1) therefore does not apply.

f) Set $f(x) = \begin{cases} x^2, & x \in [-1, 1) \\ \frac{1}{2}, & x = 2 \end{cases}$ with $D = [-1, 1) \cup \{2\}$.



For $x_0 \in [-1, 1)$, the limit $\lim_{x \rightarrow x_0} f(x)$ exists and is equal to $f(x_0) = x_0^2$.

At the point $x_0 = 1$, the limit and the left-sided limit exist, but not the right-sided limit (see c).

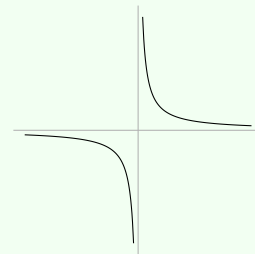
At the (isolated 😊) point $x_0 = 2$ there is neither a left-sided nor a right-sided limit, but the usual limit. Due to the lack of other sequences with $x_n \rightarrow 2$, (5.1) is reduced to the constant sequence $(x_n)_n = (2)_n$, which is why $\lim_{x \rightarrow 2} f(x) = f(2) = \frac{1}{2}$.

g) Let $f(x) = \frac{1}{x}$ be for all $x \in D = \mathbb{R} \setminus \{0\}$.

The limit of f exists in every $x_0 \in D$, because

$$x_n \rightarrow x_0 \implies f(x_n) = \frac{1}{x_n} \rightarrow \frac{1}{x_0} = f(x_0)$$

with respect to Proposition 2.19 d). In $x_0 = 0 \in \text{clos } D$ neither the one-sided nor the usual limits exist.



We extend the statements $f(x) \rightarrow y$ as $x \rightarrow x_0$ and $\lim_{x \rightarrow x_0} f(x) = y$ as well as their one-sided variants $\lim_{x \nearrow x_0} f(x) = y$ and $\lim_{x \searrow x_0} f(x) = y$ to the cases $x_0 = \pm\infty$ and $y = \pm\infty$ by interpreting $x_n \rightarrow x_0$ and $f(x_n) \rightarrow y$ in the sense of Definition 2.11 .

Example 5.6. infinite limit or limit at infinity

Proposition 2.13 provides:

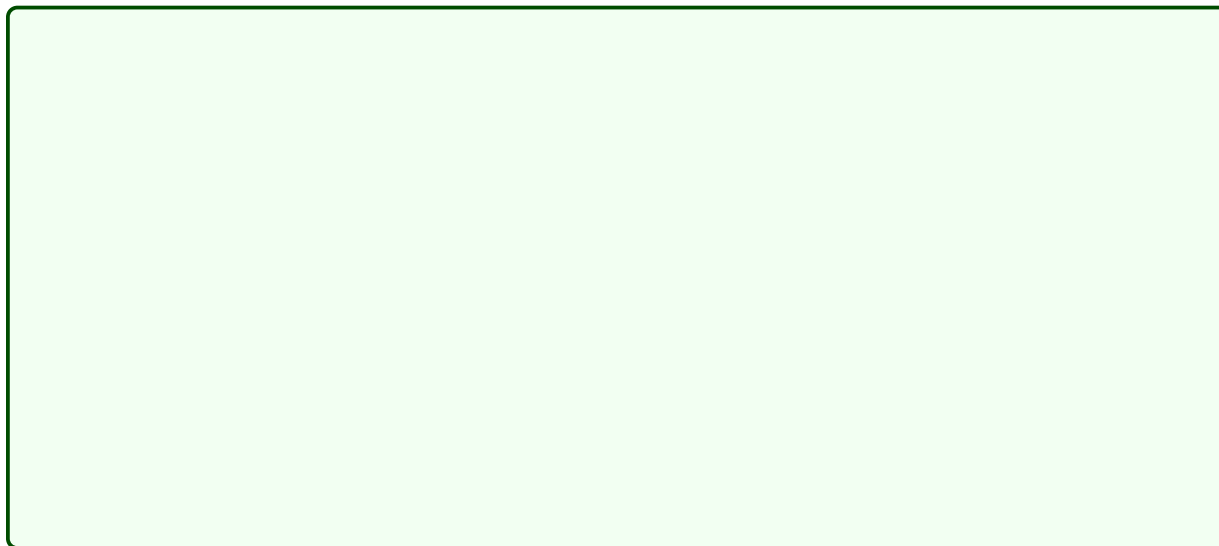
a) $\frac{1}{x} \rightarrow 0$ as $x \rightarrow +\infty$, because $\frac{1}{x_n} \rightarrow 0$ for every sequence $(x_n)_n$ with $x_n \rightarrow +\infty$,

b) $\frac{1}{x} \rightarrow +\infty$ as $x \searrow 0$, because $\frac{1}{x_n} \rightarrow +\infty$ for every sequence $(x_n)_n$ with $x_n > 0$ and $x_n \rightarrow 0$,

c) $\frac{1}{x} \rightarrow -\infty$ as $x \nearrow 0$, because $\frac{1}{x_n} \rightarrow -\infty$ for every sequence $(x_n)_n$ with $x_n < 0$ and $x_n \rightarrow 0$, $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist, because $\lim_{x \searrow 0} \frac{1}{x} \neq \lim_{x \nearrow 0} \frac{1}{x}$.

Both are also possible at the same time:

d) $\sqrt{x} \rightarrow +\infty$ as $x \rightarrow +\infty$, because $\sqrt{x} > M > 0$ for all $x > M^2$.



For $f : D \subset \mathbb{R} \rightarrow \mathbb{R}$ and points x_0 that can be approximated from both sides, left and right, by points from D , i.e. $x_0 \in \text{clos } D_{<x_0} \cap \text{clos } D_{>x_0}$, the following proposition applies:

Proposition 5.7. limit and one-sided limits

- a) If the limit $y = \lim_{x \rightarrow x_0} f(x)$ of f exists in x_0 , then the left-sided and right-sided limits also exist and are the same, both equal to y .
- b) In the case of $x_0 \notin D$, the reverse also applies.
- c) In the case of $x_0 \in D$, the inverse applies if, in addition

$$\lim_{x \nearrow x_0} f(x) = \lim_{x \searrow x_0} f(x) = f(x_0).$$

Proof: Part a) and b) follow directly from Definitions 5.1 and 5.4.

c) Since Definition 5.1 includes the case $x_n = x_0 \in D$ but Definition 5.4 does not, the constant sequence $(x_n)_n = (x_0)_n$, must also be considered here (cf. Example 5.5 e). ■

Definition 5.8. continuity

Let $x_0 \in \text{clos } D$ and let $\lim_{x \rightarrow x_0} f(x) =: y$ be existent.

- If $x_0 \in D$, then $y = f(x_0)$ and f is continuous in x_0 .
- If $x_0 \notin D$, then we call f continuously extendable to x_0 , and the function

$$g : D \cup \{x_0\} \rightarrow \mathbb{R} \quad \text{with} \quad g(x) = \begin{cases} f(x), & x \in D, \\ y, & x = x_0 \end{cases}$$

is called continuous extension of f to x_0 .

If $f : D \rightarrow \mathbb{R}$ is continuous at every point $x_0 \in D$ then we call f continuous (on D).

In Example 5.5 we therefore have continuity for a) and b) and a continuous extension for c), while for d) there is only so-called right-sided continuity and for e) not even this:

Definition 5.9. one-sided continuity

Let $x_0 \in D$. We call a function $f : D \rightarrow \mathbb{R}$

- left-continuous at x_0 , if $\lim_{x \nearrow x_0} f(x) = f(x_0)$,
- right-continuous at x_0 , if $\lim_{x \searrow x_0} f(x) = f(x_0)$,

provided the one-sided limits can be applied.

The functions from Example 5.5 f) and g) are continuous at every point of D and therefore continuous functions – although we cannot draw their graphs in one uninterrupted line.

5.2 ε - δ characterisation and continuity

Definition 5.1 is often impractical for verifying continuity or the existence of $\lim_{x \rightarrow x_0} f(x)$ as it is impossible to examine all sequences $(x_n)_n$ with $x_n \rightarrow x_0$. Rescue is close:

Proposition 5.10. ε - δ - criterion for continuity

Let $f : D \subset \mathbb{R} \rightarrow \mathbb{R}$ and $x_0 \in \text{clos } D$.

Then $y = \lim_{x \rightarrow x_0} f(x)$ (this is (5.1)) holds if and only if

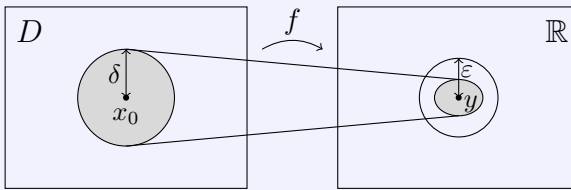
$$\forall \varepsilon > 0 \exists \delta > 0 : \underbrace{\forall x \in U_\delta(x_0) \cap D : f(x) \in U_\varepsilon(y)}_{\text{Kurz: } f(U_\delta(x_0) \cap D) \subset U_\varepsilon(y)}. \quad (5.2)$$

If $x_0 \in D$, then both statements are equivalent to: f is continuous in x_0 .

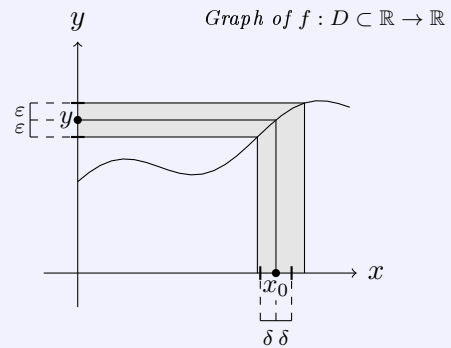
Proof: (5.2) $\Rightarrow$ (5.1):

$\neg(5.2) \Rightarrow \neg(5.1)$: $\exists \varepsilon > 0 \forall \delta > 0 \exists x \in U_\delta(x_0) \cap D : f(x) \notin U_\varepsilon(y)$. So for $\delta = \frac{1}{1}, \frac{1}{2}, \dots$ there are $x_1, x_2, \dots$ with $x_n \in U_{\frac{1}{n}}(x_0) \cap D$ and $f(x_n) \notin U_\varepsilon(y)$. Hence, $x_n \rightarrow x_0$ but $f(x_n) \not\rightarrow y$. ■

Graphical illustration of the ε - δ criterion



The permissible tolerance ε for the output $f(x)$ can be guaranteed by limiting the input to a suitable δ -environment of x_0 .



If $x_0 \in D$, where $y = f(x_0)$, this pretty much brings us back to the sausage machine:

$$\forall \varepsilon > 0 \exists \delta > 0 : x \overset{\delta}{\approx} x_0 \implies f(x) \overset{\varepsilon}{\approx} f(x_0) \quad (5.3)$$

Often $D = \mathbb{R}$ will apply. Then every $x_0 \in \mathbb{R}$ is an element of D and the question about the existence of $\lim_{x \rightarrow x_0} f(x)$ is the question about the continuity of f in x_0 , i.e. (5.1) $\Leftrightarrow$ (5.2) $\Leftrightarrow$ (5.3). Here are two more important characterisations of continuity on $D = \mathbb{R}$:

Proposition 5.11. further characterisations of continuity

The following statements are equivalent for $f : \mathbb{R} \rightarrow \mathbb{R}$.

- (i) f is continuous (on all of $\mathbb{R}$).
- (ii) $\forall x_0 \in \mathbb{R} \forall$ sequences $(x_n)_n$ in $\mathbb{R}$ with $x_n \rightarrow x_0 : f(x_n) \rightarrow f(x_0)$.
- (iii) $\forall x_0 \in \mathbb{R} \forall \varepsilon > 0 \exists \delta > 0 : f(U_\delta(x_0)) \subset U_\varepsilon(f(x_0))$.
- (iv) For every open set $V \subset \mathbb{R}$, the preimage $f^{-1}(V)$ is open.
- (v) For every closed set $V \subset \mathbb{R}$, the preimage $f^{-1}(V)$ is closed.

Proof: The equivalence of (i), (ii) and (iii) are already shown.

(iii) $\Rightarrow$ (iv): Let $V \subset \mathbb{R}$ be open and let $U := f^{-1}(V)$. To be shown: U is open. Let $x_0 \in U$ be arbitrary, i.e., $f(x_0) \in V$. V is open $\Rightarrow \exists \varepsilon > 0 : U_\varepsilon(f(x_0)) \subset V$. (iii) $\Rightarrow \exists \delta > 0 : f(U_\delta(x_0)) \subset U_\varepsilon(f(x_0)) \subset V$. $\Rightarrow U_\delta(x_0) \subset f^{-1}(V) = U \Rightarrow x_0 \in \text{int}(U) \Rightarrow U$ is open.

(iv) $\Rightarrow$ (iii):

(iv) $\Leftrightarrow$ (v): We use that $f^{-1}(V^c) = [f^{-1}(V)]^c$ and that V is closed $\Leftrightarrow V^c$ is open. ■

⚠ Attention! preimage, not image

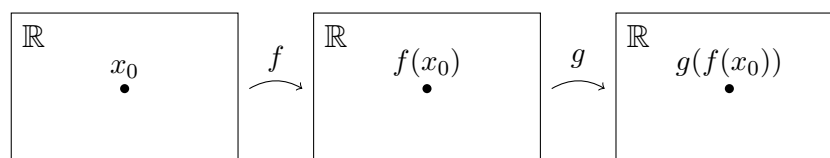
The image of an open set with respect to a continuous function does not have to be open. For the constant function $f(x) = 5$, for example, $f((0, 1)) = \{5\}$ is not open.

🧐 Nerd box 5.12. ε - δ -characterisation for infinite limits

When $x_0 = \pm\infty$, $y = \pm\infty$ or both, the statement $\lim_{x \rightarrow x_0} f(x) = y$ (as well as its one-sided variants with $x \nearrow x_0$ and $x \searrow x_0$) again corresponds to our ε - δ -characterisation with the environments of $\pm\infty$ from our Nerd-Box 2.12.

5.3 Combinations of continuous functions

The first kind of combination that we have in mind is the composition of functions.



Proposition 5.13. continuous followed by continuous stays continuous

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous in x_0 and $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous in $f(x_0)$, then $g \circ f$ is continuous in x_0 .

Proof:

■

Because that went so quickly, here is an alternative proof (for continuity on all $\mathbb{R}$):

Proof:

■

Example 5.14. Composition of continuous functions

The functions f and g with $f(x) = 2x + 1$ and $g(x) = \frac{1}{x}$ are continuous on their respective domains, $\mathbb{R}$ resp. $\mathbb{R} \setminus \{0\}$, see Example 5.5. Therefore, $f \circ g$ and $g \circ f$, precisely,

are also continuous where they are defined.

We also think of the combination of two functions $f, g : D \subset \mathbb{R} \rightarrow \mathbb{R}$ in terms of the basic arithmetic operations $+$, $-$, $\cdot$ and $:$. So if $\odot$ stands for one of these four operations, $\odot \in \{+, -, \cdot, :\}$, then let the function $f \odot g : \mathbb{R} \rightarrow \mathbb{R}$ be pointwise defined via

$$(f \odot g)(x) := f(x) \odot g(x), \quad x \in D,$$

where the known arithmetic operations between two real numbers are meant on the right, and, for division, $g(x) \neq 0$ must apply.

Proposition 5.15. Sum, product & quotient of continuous functions are continuous.

For continuous $f, g : \mathbb{R} \rightarrow \mathbb{R}$ in $x_0 \in \mathbb{R}$, also $f + g$ and $f \cdot g$ are continuous in x_0 . If furthermore $g(x_0) \neq 0$, then $\frac{f}{g}$ is defined nearby x_0 and is also continuous in x_0 .

Proof: Let $(x_n)_n$ be an arbitrary sequence with $x_n \rightarrow x_0$. Then $f(x_n) \rightarrow f(x_0)$ and $g(x_n) \rightarrow g(x_0)$ apply. By the rules for limits of sequences (Proposition 2.19),

$$(f + g)(x_n) = f(x_n) + g(x_n) \rightarrow f(x_0) + g(x_0) = (f + g)(x_0) \quad \text{and similarly for } f \cdot g.$$

If every neighbourhood $U_{\frac{1}{n}}(x_0)$ contained a zero z_n of g , then $g(x_0) = \lim g(z_n) = 0$ would apply. ⚡ So there is a neighbourhood $U_\varepsilon(x_0)$, where g has no zero.

For all $n \geq n_0$, $x_n \in U_\varepsilon(x_0)$ applies, so that $g(x_n) \neq 0$.

With Proposition 2.19 it follows that $\left(\frac{f}{g}\right)(x_n) = \frac{f(x_n)}{g(x_n)} \rightarrow \frac{f(x_0)}{g(x_0)} = \left(\frac{f}{g}\right)(x_0)$. ■

🧐 **Nerd box 5.16.** Actually, that was to be expected.

We had already established in Nerd-Box 5.3 that continuity of f is equivalent to f commuting with $\lim$. Since, according to Proposition 2.19, $\lim$ also commutes with $+$, $-$, $\cdot$, $:$, it is clear that $\lim$ commutes with $f \odot g$ if it commutes with f and g and if $\odot \in \{+, -, \cdot, :\}$

Example 5.17. Composition of continuous functions via $+$, $-$, $\cdot$, $:$

- Like $f : x \mapsto 5$ from Example 5.5 a), $f : x \mapsto c$ is continuous for every $c \in \mathbb{R}$.
- As for $g : x \mapsto 2x + 1$ from Example 5.5 b), we see that $g : x \mapsto x$ is continuous.
- So-called polynomials, i.e. sums and products of $f : x \mapsto c$ and $g : x \mapsto x$ from a) and b), such as $p : x \mapsto 3x^2 - 5x + 2$, are therefore continuous on $\mathbb{R}$.
- Quotients of two polynomials, e.g. $q : x \mapsto \frac{3x^2 - 5x + 2}{7x^4 + x^2 - 2x + 6}$, are continuous wherever they are defined, i.e., where the polynomial in the denominator is nonzero.

5.4 Continuity on compact sets

We start with a nice preservation rule:

Proposition 5.18. Continuous functions map compact sets to compact sets

If $f : D \rightarrow \mathbb{R}$ is continuous and $D \subset \mathbb{R}$ is compact, then $f(D) \subset \mathbb{R}$ is compact.

Proof:

■

We mainly talk about compact sets because of the following proposition:

Proposition 5.19. Existence of minimum and maximum

Let $D \subset \mathbb{R}$ be compact and non-empty. A continuous function $f : D \rightarrow \mathbb{R}$ attains both its maximum and its minimum in D .

Proof: By Proposition 5.18, $f(D)$ is compact and, by Proposition 4.19, has a minimum α and a maximum β . Thus, there exist $x_\alpha, x_\beta \in D$ such that $\alpha = f(x_\alpha) \leq f(x) \leq f(x_\beta) = \beta \quad \forall x \in D$. ■

A typical case in the above theorem is $D = [a, b]$ with $a, b \in \mathbb{R}$.

🧐 **Nerd box 5.20.** continuous, more continuous, most continuous

Continuous functions on compact sets are actually uniformly continuous; this means

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x_0 \in \mathbb{R} : f(U_\delta(x_0)) \subset U_\varepsilon(f(x_0)).$$

In contrast to (iii) from Proposition 5.11 (i.e., continuity), a single δ works for all x_0 . Even better: f is called Lipschitz continuous with $L > 0$, if $\delta := \frac{\varepsilon}{L}$ can be chosen.

5.5 The intermediate value theorem

Proposition 5.21. Intermediate value theorem (IVT)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous.

a) If $f(a), f(b)$ have different signs, there exists an $x \in (a, b)$ such that $f(x) = 0$.

b) Every y between $f(a)$ and $f(b)$ is assumed as $y = f(x)$ with $x \in [a, b]$

The proof is constructive. The described algorithm is called bisection.

Proof: **a)**

b) The function g with $g(x) := f(x) - y$ is continuous, with $g(a)$ and $g(b)$ having different signs. By part **a)**, there exists an $x \in [a, b]$ with $g(x) = 0$. Thus, $y = f(x)$. ■

In what follows we refer to "interval" as any open, closed, or half-open interval – bounded or unbounded. Here is our next preservation rule:

Proposition 5.22. Continuous functions map intervals to intervals

If $D \subset \mathbb{R}$ is an interval and $f : D \rightarrow \mathbb{R}$ is continuous then $f(D)$ is also an interval.

Proof:

■

Example 5.23. on the intermediate value theorem

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^2$ is continuous. For $x \in [-1, 2]$, $f(x)$ continuously covers the interval $[f(-1), f(2)] = [1, 4]$ – actually even $[0, 4]$.

For the latter, one must segment $[-1, 2]$ into the so-called monotonicity intervals of f : decreasing on $[-1, 0]$, then increasing on $[0, 2]$, and apply the IVT separately.

Monotonicity is also one of the key concepts for the next section.

5.6 Continuity and the inverse function

For continuous functions, invertibility is closely related to monotonicity. Therefore:

Definition 5.24. Monotonicity of functions

A function $f : D \rightarrow \mathbb{R}$ with $D \subset \mathbb{R}$ is called

- monotonically increasing, if $\forall x_1, x_2 \in D : x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$

- strictly monotonically increasing, if $\forall x_1, x_2 \in D : x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$,
- monotonically decreasing, if $\forall x_1, x_2 \in D : x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)$,
- strictly monotonically decreasing, if $\forall x_1, x_2 \in D : x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$,
- (strictly) monotonic, if it is (strictly) monotonically increasing or decreasing.

And here is the announced theorem:

Proposition 5.25. Monotonicity vs. Invertibility

Let $D \subset \mathbb{R}$ be an interval and $f : D \rightarrow \mathbb{R}$ be continuous.

- a) f is injective $\iff f$ is strictly monotonic.
- b) f strictly monotonic $\implies f^{-1} : f(D) \rightarrow D$ strictly monotonic and continuous.

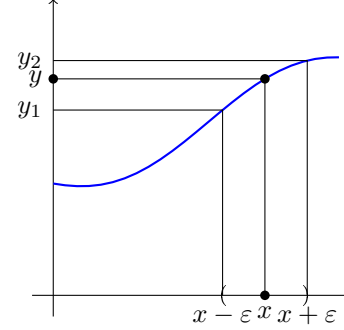
Proof: a) $\Leftarrow$ With strict monotonicity, we have: $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

$\Rightarrow$ Suppose f is not strictly monotonic. Then $\exists x_1 < x_2 < x_3$ in D such that $f(x_1) \geq f(x_2) \leq f(x_3)$ or $f(x_1) \leq f(x_2) \geq f(x_3)$.



- b) Let f be strictly monotonically increasing (analogously for decreasing) and let $Y := f(D)$. Due to **a)**, $f : D \rightarrow Y$ is even bijective. To be shown: f^{-1} is strictly monotonically increasing. Let $y_1 < y_2 \in Y$, i.e., $y_1 = f(x_1)$, $y_2 = f(x_2)$ with $x_1, x_2 \in D$. If $x_1 \geq x_2$, then $y_1 = f(x_1) \geq f(x_2) = y_2$ ∇ . Thus, $x_1 < x_2$, i.e., $f^{-1}(y_1) < f^{-1}(y_2)$.

It remains to show: f^{-1} is continuous: Let $y \in Y$. By Proposition 5.22, $Y = f(D)$ is also an interval. Suppose y is an interior point. Due to strict monotonicity, then $x = f^{-1}(y)$ is also an interior point of the interval D . Let $\varepsilon > 0$ be arbitrary. It suffices to consider the ε with $x \pm \varepsilon \in D$. We set $y_1 := f(x - \varepsilon) < y$ and $y_2 := f(x + \varepsilon) > y$, as well as $\delta := \min(y - y_1, y_2 - y)$.



Then the following holds (again due to the monotonicity of f^{-1})

$$f^{-1}((y - \delta, y + \delta)) \subset f^{-1}((y_1, y_2)) = (x - \varepsilon, x + \varepsilon),$$

hence, f^{-1} is continuous (ε - δ criterion). For boundary points $y \in Y$, the argument is analogous. ■

We will see plenty of application examples in the next chapter.

Summary

- We write $f(x) \rightarrow y$ as $x \rightarrow x_0$ or $y = \lim_{x \rightarrow x_0} f(x)$ if $f(x_n) \rightarrow y$ for every sequence $(x_n)_n$ in the domain D of f with $x_n \rightarrow x_0$.
- Equivalently:

$$\forall \varepsilon > 0 \exists \delta > 0 : x \overset{\delta}{\approx} x_0 \implies f(x) \overset{\varepsilon}{\approx} y$$

- If $x_0 \in D$, i.e., if $f(x_0)$ is defined, then $\lim_{x \rightarrow x_0} f(x)$ is equal to $f(x_0)$, provided the limit exists. f is then called continuous at x_0 .
- f is called continuous in a set $D \subset \mathbb{R}$ if it is continuous at every point $x_0 \in D$.
- This is equivalent to saying that the preimage of every open set is open.
- ...and also that the preimage of every closed set is closed.
- We also examine left-sided and right-sided limits and continuity.
- The combinations of continuous functions using $+$, $-$, $\cdot$, $:$ and $\circ$ are continuous.
- Continuous functions map compact sets onto compact sets.
- If D is compact, then any continuous function $f : D \rightarrow \mathbb{R}$ assumes its minimum and maximum in D .

- Continuous functions $f : [a, b] \rightarrow \mathbb{R}$ assume every value between $f(a)$ and $f(b)$.
- For continuous functions on intervals, injectivity is the same as strict monotonicity.
- The inverse function f^{-1} of such a function is also continuous.

Chapter 6

Elementary functions

With the available tools that we encountered, we can now explore many old and new functions together with their inverses:

6.1 Monomials and roots

Definition 6.1. Monomial

Let $n \in \mathbb{N}_0$. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^n$ is called (the n -th) monomial.

For $n \in \mathbb{N}$ it is easy to verify that

$$\lim_{x \rightarrow -\infty} x^n = \begin{cases} -\infty, & n \text{ odd,} \\ +\infty, & n \text{ even,} \end{cases} \quad \text{and} \quad \lim_{x \rightarrow +\infty} x^n = +\infty. \quad (6.1)$$

Proposition 6.2. about monomials

Let $n \in \mathbb{N}$ and $f(x) = x^n$ the n -th monomial. Then f is

- a) continuous on $\mathbb{R}$,
- b) strictly monotonically increasing on the interval $[0, \infty)$, and
- c) bijective as a restricted mapping $f : [0, \infty) \rightarrow [0, \infty)$.

Proof: a) f is the n -fold product of the continuous function $g(x) = x$ with itself.
b)

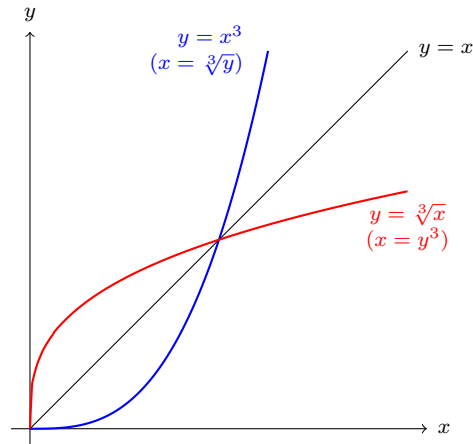
c) Injectivity follows from a), b) and Proposition 5.25 a). Surjectivity follows from a), $f(0) = 0$, $\lim_{x \rightarrow +\infty} f(x) = +\infty$ and Proposition 5.22.

■

By Proposition 6.2 c), there exists the inverse function $f^{-1} : [0, +\infty) \rightarrow [0, +\infty)$ of the n -th monomial $f(x) = x^n$, i.e.

$$f^{-1} : x^n \mapsto x \quad \text{in short,} \quad f^{-1}(y) = \sqrt[n]{y},$$

which we encountered earlier in the root test. According to Proposition 5.25 b), the n -th root is a strictly monotonically increasing and continuous function from $[0, +\infty)$ to $[0, +\infty)$. Its graph is the reflection of the graph of the corresponding monomial f across the line $y = x$.



6.2 Polynomials and rational functions

Definition 6.3. Polynomial, Degree

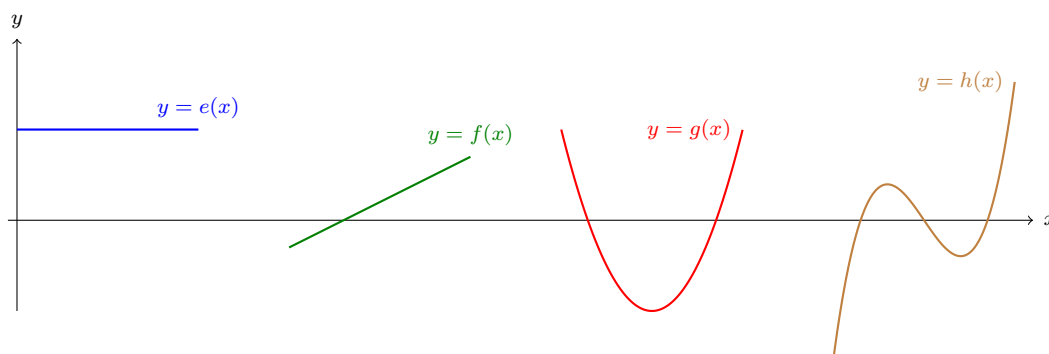
A polynomial is a (finite) linear combination of monomials, i.e., a sum

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x^1 + a_0$$

with fixed coefficients $a_0, a_1, \dots, a_n \in \mathbb{R}$, understood as a function $p: \mathbb{R} \rightarrow \mathbb{R}$.

The highest power of x occurring (above: n , if $a_n \neq 0$) is called the degree of p .

Polynomials are continuous on the entire real line $\mathbb{R}$, as Example 5.17 c) illustrates.



from left to right: Graphs of four polynomials, e, f, g and h , of degree 0, 1, 2, respectively, 3.

The four polynomials above each have as many zeros as their degrees. They could have fewer zeros (consider $g(x) + 5$ or $h(x) + 3$); however, the following proposition always holds:

Proposition 6.4. Fundamental Theorem of Algebra (light)

Let p be a polynomial of degree $n \in \mathbb{N}$. Then one has

- a) p has at most n (real) zeros.
- b) If n is odd, then p has at least one (real) zero.

Proof: a) The actual Fundamental Theorem of Algebra (by C. F. Gauß) states that p has exactly n zeros – in the domain of the *complex* numbers $\mathbb{C}$, which you will soon encounter in Linear Algebra. However, $\mathbb{R} \subset \mathbb{C}$.

b) Let $a_n > 0$ ($a_n < 0$: similar). It is easy to see that $\lim_{x \rightarrow \pm\infty} p(x) = \lim_{\pm\infty} a_n x^n = \pm\infty$ by (6.1) since n is odd. By the IVT (Proposition 5.21), $p(c) = 0$ for some $c \in \mathbb{R}$.

■

For polynomial degrees $n = 1$ and $n = 2$, we know the formulas

$$x_0 = -\frac{a_0}{a_1} \quad \text{and} \quad x_{1,2} = -\frac{b_1}{2b_2} \pm \sqrt{\left(\frac{b_1}{2b_2}\right)^2 - \frac{b_0}{b_2}}$$

for the polynomial zeros, i.e., the solutions of

$$0 = a_1x + a_0 \quad \text{and} \quad 0 = b_2x^2 + b_1x + b_0 = b_2 \cdot \left(x^2 + \frac{b_1}{b_2}x + \frac{b_0}{b_2}\right).$$

For polynomial degrees $n = 3$ and $n = 4$, there are also solution formulas (N. Tartaglia, G. Cardano) using $+$, $-$, $\cdot$, $:$ and roots.



Nerd box 6.5. Birth of complex numbers and then also that of algebra

- You obtain all the n solutions of the fundamental theorem by also allowing roots of negative numbers (thus using complex numbers).
- For $n \geq 5$, there are generally no solution formulas for $p(x) = 0$ using $+$, $-$, $\cdot$, $:$ and roots anymore (N.H. Abel, P. Ruffini)! This theorem and its emergence are considered by many as the birth of Algebra.

Even without a solution formula, the zeros can be determined one by one. We start with a very simple example:

Example 6.6. Zeros vs. linear factors

The equation

$$(x - 2) \cdot (x - 3) = 0 \quad (6.2)$$

is a quadratic equation, as can be seen by expanding the product

$$(x - 2) \cdot (x - 3) = x^2 - 3x - 2x + (-2) \cdot (-3) = x^2 - 5x + 6.$$

Hence, (6.2) is in fact the same equations as

$$x^2 - 5x + 6 = 0. \quad (6.3)$$

Looking at (6.3), the standard pq -formula yields

$$x_{1,2} = -\frac{-5}{2} \pm \sqrt{\left(\frac{-5}{2}\right)^2 - 6} = \frac{5}{2} \pm \sqrt{\frac{1}{4}} = \frac{5 \pm 1}{2} \in \{2, 3\}.$$

But in the form (6.2), it was a much simpler problem since the product of $(x - 2)$ and $(x - 3)$ is zero if and only if one of the factors is zero, that is, when $x = 2$ or $x = 3$.

We see that it would be actually smart if we could turn equations of the form (6.3) into equations of the form (6.2) since that would show us their solutions directly, without a pq -formula or others.

The transition from (6.3) into (6.2) is called factorisation into linear factors. In hindsight, we confirm¹ that 6 is the product of the two solutions and 5 is their sum, which turns out helpful for the factorisation (6.3) $\rightarrow$ (6.2).

Indeed, the ability to factorise a polynomial into linear factors is just as good as the ability of computing its zeros:

Proposition 6.7. Factorization into linear factors

A polynomial p has a number $x_0 \in \mathbb{R}$ as a zero if and only if it contains the factor $x - x_0$, i.e., $p(x) = (x - x_0) \cdot q(x)$ for another polynomial q .

Proof: If you divide $p(x)$ by, e.g., $x - 5$, a constant c will be the remainder, i.e., $p(x) = (x - 5) \cdot q(x) + c$ with a polynomial q and a $c \in \mathbb{R}$. But then $p(5) = 0 \cdot q(5) + c$ and, hence, $c = p(5)$: Thus, the division is exact if and only if 5 is a zero of p . The general proof is analogous, with x_0 instead of 5. ■

Knowing that 2 is a zero of $p(x) = x^2 - 5x + 6$

¹This observation goes back to F. Vieta.

Strategy: divide out the linear factor

Once a zero, x_0 , of p is found, $p(x) : (x - x_0)$ is a polynomial of degree one lower. Its zeros are the remaining zeros of p .

This approach provides another proof by induction for Proposition 6.4 a).

But how do we divide a polynomial p by $(x - x_0)$?

Example 6.8. Polynomial division and zeros

Let, for example, $p(x) = 2x^3 - 6x^2 - 18x - 10$. Computation shows that $x_0 = 5$ is a zero of p , meaning $p(5) = 0$.

Division by $x - 5$ (so-called polynomial division) works like written division from school (grade 4):

$$(2x^3 - 6x^2 - 18x - 10) : (x - 5) =$$

remainder:

🧐 Nerd box 6.9. It practically is the same

Our division problem $5238 : 27$ from grade 4 literally corresponds to the polynomial division $(5x^3 + 2x^2 + 3x + 8) : (2x + 7)$ at the point $x = 10$.

The division was exact, which, by Proposition 6.7, has to do with $p(5) = 0$. The result, $q(x) = 2x^2 + 4x + 2$, is now of degree 2 (quadratic); this is something we are more familiar with. In this case, the 1st binomial formula helps directly:

$$q(x) = 2x^2 + 4x + 2 =$$

Thus,

$$\begin{aligned} p(x) &= (x - 5) \cdot q(x) = \\ &= \end{aligned}$$

The zeros of p are thus $x_0 = 5$, $x_1 = -1$ and $x_2 = -1$. The factor 2 is the same that was already in front of the highest monomial, x^3 , in p .

Multiplying everything out, we end up, as a check, back at $p(x) = 2x^3 - 6x^2 - 18x - 10$.

In particular, $2 \cdot 5 \cdot (-1) \cdot (-1) = 10$, thus $x_0 \cdot x_1 \cdot x_2 = \pm \frac{a_0}{2}$ with $a_0 = -10$ being the coefficient of the 0-th monomial of p (Vieta).

Assuming the exercise creators only use integer zeros, the initially guessed zero $x_0 = 5$ wasn't such a spectacular guess after all.

Definition 6.10. rational function

If p and q are polynomials, the function

$$f = \frac{p}{q}, \quad \text{meaning} \quad f(x) = \frac{p(x)}{q(x)},$$

is called a rational function. The domain of f is $D = \mathbb{R} \setminus N$, where N is the set of zeros of q .

By Example 5.17 d), rational functions are continuous on their domain. The gaps x_0 in the domain of $\frac{p}{q}$, where $q(x_0) = 0$ but $p(x_0) \neq 0$, are so-called poles. Precisely:

Definition 6.11. pole

Let $f : D \subset \mathbb{R} \rightarrow \mathbb{R}$ and $x_0 \in \text{clos}(D) \setminus D$. The point x_0 is called a pole of f , if

$$\lim_{x \nearrow x_0} f(x) = -\infty \text{ or } \lim_{x \nearrow x_0} f(x) = +\infty \text{ or } \lim_{x \searrow x_0} f(x) = -\infty \text{ or } \lim_{x \searrow x_0} f(x) = +\infty.$$

Example 6.12. Pole or not?

The function $f : x \mapsto \frac{x-3}{x^2-9} = \frac{x-3}{(x-3)(x+3)}$ has a pole at $x_0 = -3$, since even both

$$\lim_{x \nearrow -3} f(x) = -\infty \quad \text{and} \quad \lim_{x \searrow -3} f(x) = +\infty.$$

The gap in the domain at $x_0 = 3$ is however not a pole since $\lim_{x \rightarrow 3} f(x) = \frac{1}{6}$.

6.3 Power series

In sloppy terms: power series are polynomials of degree ∞ .

More precisely:

Definition 6.13. power series

For a sequence $(a_n)_n$ in $\mathbb{R}$ and $x \in \mathbb{R}$, a series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

is called a power series. The set of all $x \in \mathbb{R}$, for which the above series converges, is its region of convergence.

Example 6.14. power series

a) geometric series, i.e., $a_n \equiv 1$: $\sum_{n=0}^{\infty} 1 \cdot x^n$ converges for $x \in (-1, 1)$.

b) Exponential series, $a_n = \frac{1}{n!}$: $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$ converges for all $x \in \mathbb{R}$,
since $x_n := a_n x^n$ implies $|\frac{x_{n+1}}{x_n}| = \frac{|x^{n+1}| n!}{(n+1)! |x^n|} = \frac{|x|}{n+1} \rightarrow 0$ for all $x \in \mathbb{R}$.

c) $a_n = n!$: $\sum_{n=0}^{\infty} n! x^n$ only converges when $x = 0$, because for $x_n := a_n x^n$, it holds that $|\frac{x_{n+1}}{x_n}| = \frac{(n+1)! |x^{n+1}|}{n! |x^n|} = (n+1)|x| \rightarrow \infty \geq 1$, as $x \neq 0$.

The following holds for the region of convergence of a power series:

Lemma 6.15. power series and absolute convergence

If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges at a point $x_0 \neq 0$, then it converges absolutely for every $x \in \mathbb{R}$ with $|x| < |x_0|$.

Proof: $\sum_{n=0}^{\infty} a_n x_0^n$ converges $\implies (a_n x_0^n)_n$ is a null sequence $\implies (a_n x_0^n)_n$ is bounded.
Let $M := \sup_{n \in \mathbb{N}_0} |a_n x_0^n|$ and let $q = |\frac{x}{x_0}| < 1$ for fixed x with $|x| < |x_0|$. Because

$$|a_n x^n| = |a_n x_0^n| \cdot \left| \frac{x}{x_0} \right|^n \leq M \cdot q^n,$$

$M \cdot \sum_{n=0}^{\infty} q^n$ is a convergent majorant for $\sum_{n=0}^{\infty} |a_n x^n| \Rightarrow \sum_{n=0}^{\infty} a_n x^n$ converges absolutely. ■

Let us therefore seek to maximize $|x|$ with x in the region of convergence:

Definition 6.16. radius of convergence

The number $R \in \mathbb{R} \cup \{+\infty\}$ with

$$R = \sup \left\{ |x| : x \in \mathbb{R}, \sum_{n=0}^{\infty} a_n x^n \text{ converges} \right\}$$

is called the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n x^n$.

It holds that

Proposition 6.17. location and convergence

The power series $\sum_{n=0}^{\infty} a_n x^n$ with radius of convergence R is

- a) absolutely convergent for all $x \in (-R, R)$ and
- b) divergent for all $x \notin [-R, R]$.

No statement can be made for $x \in \{-R, R\}$.

Proof: a) Let $x \in (-R, R)$ and let $|x_0|$ be such that $|x| < |x_0| < R$ and $\sum_n a_n x_0^n$ converges ($\nearrow$ Definition 6.16). Then $\sum_n a_n x^n$ is absolutely convergent by Lemma 6.15.

b) If $\sum_n a_n x^n$ converges for an $x \in \mathbb{R}$ with $|x| > R$, this would contradict Def. 6.16. ■

Thus, the region of convergence D is always an interval of the form $(-R, R)$, $(-R, R]$, $[-R, R)$, or $[-R, R]$ with $R \geq 0$, including the case $D = \mathbb{R}$ for $R = +\infty$.

- In the interior, $\text{int } D = (-R, R)$, there is even absolute convergence,
- in the exterior, $\text{ext } D = [-R, R]^c$, there is divergence,

- and on the boundary, $\partial D = \{-R, R\}$, anything can happen ($\nearrow$ Example 6.19).

The ratio test and the root test provide formulas for the radius of convergence R :

Proposition 6.18. formulas for the radius of convergence

For the convergence radius R of the power series $\sum_{n=0}^{\infty} a_n x^n$, the following applies:

- a) If $q := \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ exists, then $R = \frac{1}{q}$.
- b) If $L := \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ exists, then $R = \frac{1}{L}$.
- c) In any case, $\ell := \limsup \sqrt[n]{|a_n|}$ exists (see Nerd-Box 2.44 for $\limsup$), and $R = \frac{1}{\ell}$ (the so-called Cauchy-Hadamard formula) holds.

Here we interpret $\frac{1}{0}$ as $+\infty$ and $\frac{1}{+\infty}$ as 0.

Proof: a)

c) According to the root test (in the version in Nerd-Box 3.18), there is absolute convergence of $\sum_n a_n x^n$ if $1 > \limsup \sqrt[n]{|a_n x^n|} = \limsup \sqrt[n]{|a_n|} \cdot |x| = \ell \cdot |x|$, i.e., $|x| < \frac{1}{\ell}$, and divergence if $1 < \limsup \sqrt[n]{|a_n x^n|} = \ell \cdot |x|$, i.e., $|x| > \frac{1}{\ell}$. Hence, $R = \frac{1}{\ell}$.

b) If $L = \lim \sqrt[n]{|a_n|}$ exists, then $L = \ell$ and according to c), $R = \frac{1}{\ell} = \frac{1}{L}$. ■

Example 6.19. possible behavior on the boundary of the convergence region

- a) The geometric series $\sum_n x^n$ has a region of convergence equal to $(-1, 1)$, i.e., $R = 1$ according to Example 6.14a) or $L = \lim \sqrt[n]{1} = 1$ or $q = \lim \frac{1}{1} = 1$. For neither boundary point, $x = \pm 1$, does convergence occur, as then $x^n \not\rightarrow 0$.

- b) Let $a_n = \frac{1}{n}$, i.e., the series is $\sum_{n=1}^{\infty} \frac{x^n}{n}$. The boundary points are again $x = \pm 1$:
- $$\left. \begin{array}{ll} x = -1 : & \text{Convergence (Leibniz)} \\ x = 1 : & \text{Divergence (harmonic series)} \end{array} \right\} \begin{array}{l} \Rightarrow R \geq 1, \\ \Rightarrow R \leq 1, \end{array} \Rightarrow R = 1.$$

The latter is confirmed by $q = \lim_{n \rightarrow \infty} \frac{1/(n+1)}{1/n} = 1$ and $L = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{n}} = 1$.

- c) Lastly, let $a_n = \frac{1}{n^2}$, i.e., the series is $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$. Again, $R = 1$, because $q = \lim_{n \rightarrow \infty} \frac{1/(n+1)^2}{1/n^2} = 1$ and $L = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{n^2}} = 1$. This time, there is convergence (even absolute) for both boundary points, $x = \pm 1$: $\sum_{n=1}^{\infty} \left| \frac{x^n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{|x|^n}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$.

Nerd box 6.20. Disc of convergence?

The concept of power series also comes to full bloom when one admits complex instead of real numbers, x . The above propositions then continue to hold – only that $|x| < R$ describes a disc around the origin and not just the interval $(-R, R) \subset \mathbb{R}$.

The region of convergence is then also called the disc of convergence. The term radius of convergence fits even better.

Hence, within the region of convergence $D \subset \mathbb{R}$ of the series,

$$f(x) := \sum_{n=0}^{\infty} a_n x^n$$

is computable and can be considered as a function $f : D \rightarrow \mathbb{R}$.

In fact, we have so far only considered particular power series:

$$\sum_{n=0}^{\infty} a_n x^n \quad \Rightarrow \quad \begin{array}{c} \text{region of convergence} \\ U_R(0) = (-R, R) \\ \text{(possibly plus boundary points)} \end{array} \quad \text{---} \left(\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right) \begin{array}{c} R \\ \text{ } \\ 0 \end{array}$$

Shifting by $x_0 \in \mathbb{R}$, i.e. switching from x to $y := x - x_0$, results in

$$\sum_{n=0}^{\infty} a_n \underbrace{(x - x_0)}_y^n \quad \Rightarrow \quad \begin{array}{c} \text{region of convergence} \\ U_R(x_0) = (x_0 - R, x_0 + R) \\ \text{(possibly plus boundary points)} \end{array} \quad \text{---} \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \begin{array}{c} x_0 \\ \text{ } \\ 0 \end{array}$$

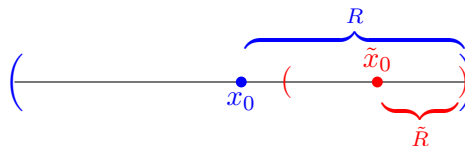
Example 6.21. shifted geometric series

$$\sum_{n=0}^{\infty} (x - 2)^n = \frac{1}{1 - (x - 2)} = \frac{1}{3 - x}, \quad \text{where } |x - 2| < 1, \text{ i.e., } x \in U_1(2) = (1, 3).$$

Definition 6.22. center of a power series

A series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$ is referred to as a power series centered at x_0 , and x_0 is called the center (or the point of expansion) of that power series.

Within the convergence interval $(x_0 - R, x_0 + R)$, one may also choose any new center, $\tilde{x}_0$. When moving the center, new coefficients, $\tilde{a}_0, \tilde{a}_1, \tilde{a}_2, \dots$, and a new convergence radius $\tilde{R}$ arise:

**Example 6.23. moving the center of a geometric series**

$$\sum_{n=0}^{\infty} \frac{1}{3^n} x^n =$$

Proposition 6.24. continuity of a power series

A power series $f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$ with radius of convergence R is continuous on the interval $U_R(x_0) = (x_0 - R, x_0 + R)$.

Proof: We will see later in Chapter 7 that f is even differentiable on the interval $(x_0 - R, x_0 + R)$, and that continuity follows from it. ■

6.4 Exponential function and logarithm

The concept of power series opens the door to entirely new functions:

Definition 6.25. the exponential function

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

is called the exponential function and is denoted by $\exp$.

According to Example 3.17 a) and 6.14 b), the convergence domain of $\exp$ is indeed all of $\mathbb{R}$. In this context, x^0 and $0!$ are both to be understood as 1. The constant

$$e := \exp(1) = \sum_{n=0}^{\infty} \frac{1}{n!} = 2.718281\dots$$

is the famous Euler's number.

Proposition 6.26. $\exp$ turns $+$ into $\cdot$

For all $x, y \in \mathbb{R}$, we have:

- a) $\exp(x + y) = (\exp x) \cdot (\exp y)$.
- b) $\exp(-x) = \frac{1}{\exp x}$, in particular, $\exp x \neq 0$ for all $x \in \mathbb{R}$.

Proof: **a)** We have not yet considered products of series. Here we have

$$(\exp x) \cdot (\exp y) = \sum_{i=0}^{\infty} \frac{x^i}{i!} \cdot \sum_{j=0}^{\infty} \frac{y^j}{j!} = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{x^i}{i!} \frac{y^j}{j!}.$$

By enumerating the region $(i, j) \in \{0, 1, 2, \dots\}^2$ along “diagonals” with $i + j = k$ for $k = 0, 1, 2, \dots$, one arrives at

$$(\exp x) \cdot (\exp y) = \sum_{k=0}^{\infty} \sum_{i=0}^k \frac{x^i}{i!} \frac{y^{k-i}}{(k-i)!} = \sum_{k=0}^{\infty} \frac{1}{k!} \underbrace{\sum_{i=0}^k \binom{k}{i} x^i y^{k-i}}_{(x+y)^k} = \exp(x+y).$$

In almost every step, we use the *absolute* convergence of the exponential series.

b) Because of part **a)**, for all $x \in \mathbb{R}$, we have

$$1 = \exp(0) =$$

■

Proposition 6.27. monotonicity and bijectivity of the function $\exp : \mathbb{R} \rightarrow (0, +\infty)$

For the exponential function $\exp : \mathbb{R} \rightarrow \mathbb{R}$, we have:

- a)** $\forall x \in \mathbb{R} : \exp x > 0$.
- b)** $\lim_{x \rightarrow +\infty} \exp x = +\infty$.
- c)** $\lim_{x \rightarrow -\infty} \exp x = 0$.
- d)** $\exp$ is strictly monotonically increasing.

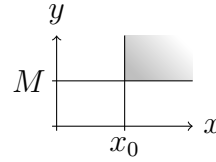
- e) $\exp$ is continuous.
 f) $\exp : \mathbb{R} \rightarrow (0, +\infty)$ is bijective.

Proof: a) For $x = 0$, $\exp x = 1 > 0$. Now let $x > 0$ be arbitrary. Then we have

$$\exp x = 1 + \frac{x}{1!} + \underbrace{\frac{x^2}{2!} + \frac{x^3}{3!} + \dots}_{>0} > 1 + x \quad (*)$$

and hence, $\exp x > 1 + x > 1 > 0$. For $x < 0$, we have $\exp x = \frac{1}{\exp(-x)} > 0$, since $-x > 0$.

- b) We must show that
 $\forall M \in \mathbb{R} \quad \exists x_0 \in \mathbb{R} :$
 $\forall x > x_0 : \exp x > M.$



Let $M \in \mathbb{R}$. According to (*), we have $\forall x > x_0 := M : \exp x > 1 + x > 1 + M > M$.

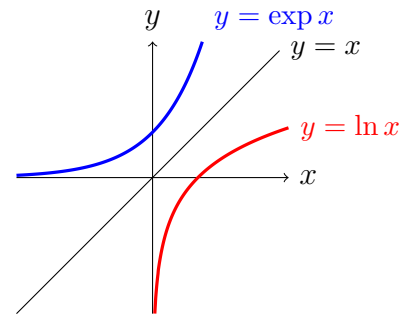
- c) For $x \rightarrow -\infty$, $\exp(-x) \rightarrow +\infty$ and hence $\exp x = \frac{1}{\exp(-x)} \rightarrow 0$ by Proposition 2.13.
 d) For $\delta > 0$, we have $\exp(x + \delta) = (\exp x)(\exp \delta) > (\exp x)(1 + \delta) > \exp x$ by (*).
 e) Continuity follows directly from Proposition 6.24.
 f) Injectivity follows from d), e), and Proposition 5.25 a). For surjectivity: According to Proposition 5.22, $I := \exp(\mathbb{R})$ is an interval. By a), $I \subset (0, +\infty)$. Due to b) and c), $\forall M, \varepsilon > 0 : (\varepsilon, M) \subset I$. Thus, $I = (0, +\infty)$, i.e., $\exp(\mathbb{R}) = (0, +\infty)$. ■

Definition 6.28. the natural logarithm

The inverse function $\exp^{-1} : (0, +\infty) \rightarrow \mathbb{R}$ is called the natural logarithm and is denoted by $\ln$.

According to Proposition 5.25, the natural logarithm, as a function $\ln : (0, +\infty) \rightarrow \mathbb{R}$, is strictly monotonically increasing and continuous.

Since $\exp$ turns addition into multiplication, it is the other way round with $\ln$:



Corollary 6.29. $\ln$ turns $\cdot$ into $+$

$$\ln(x \cdot y) = \ln x + \ln y \quad \text{for all } x, y > 0.$$

By repeated application of Corollary 6.29 (or from $\exp(ny) = (\exp y)^n$), the following formula is obtained for all $x > 0$ and $n \in \mathbb{N}$:

$$\ln(x^n) = n \cdot \ln x. \quad (6.4)$$

Definition 6.30. exponential function $\exp_a$ with any $a > 0$

Let $a > 0$. We define $\exp_a : \mathbb{R} \rightarrow (0, +\infty)$ as

$$\exp_a(x) := \exp(x \cdot \ln a).$$

Between us: Actually, we would like to introduce $\exp_a(x)$ as a^x .

If we already knew about potency laws and also knew that $\exp x = e^x$ holds, we could write $a = \exp(\ln a) = e^{\ln a}$ and then conclude that

$$a^x =$$

But since we don't know any of these yet, we start the story from the other end.

The more general exponential function $\exp_a$ inherits the following property from $\exp$:

Proposition 6.31. $\exp_a$ turns $+$ into $\cdot$

For all $x, y \in \mathbb{R}$, the following holds: $\exp_a(x + y) = (\exp_a x) \cdot (\exp_a y)$.

Proof: This follows from the corresponding rule for $\exp$:

$$\begin{aligned} \exp_a(x + y) &= \exp((x + y) \ln a) = \exp(x \ln a + y \ln a) \\ &= \exp(x \ln a) \cdot \exp(y \ln a) = (\exp_a x) \cdot (\exp_a y) \blacksquare \end{aligned}$$

Proposition 6.32.

Let $a > 0$. The formula $\exp_a(x) = a^x$ holds

- a) for all $x \in \mathbb{N}$,
- b) even for all $x \in \mathbb{Z}$,
- c) and even for all $x \in \mathbb{Q}$.

Proof: a) For all $n \in \mathbb{N}$, we have $\exp_a(n) =$

b) Moreover, $\exp_a(-n) = \exp(-n \cdot \ln a) = \frac{1}{\exp(n \ln a)} = \frac{1}{a^n} = a^{-n}$ for all $n \in \mathbb{N}$ and $\exp_a(0) = \exp(0 \cdot \ln a) = \exp 0 = 1 = a^0$.

c) Now let $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Then $\frac{m}{n} \in \mathbb{Q}$, and we have

$$\left(\exp_a\left(\frac{m}{n}\right)\right)^n =$$

It remains to apply $\sqrt[n]{}$ on both sides. ■

The above proposition motivates the following notation

$$\exp_a(x) =: a^x \quad \text{for all } x \in \mathbb{R}.$$

In particular, $e^x = \exp(x \cdot \underbrace{\ln e}_1) = \exp x$ with Euler's number $e = \exp 1$.

Naturally, this also defines the power function

$$f(x) := x^b = \exp_x(b) = \exp(b \cdot \ln x), \quad x > 0$$

for any $b \in \mathbb{R}$ as a function $f : (0, +\infty) \rightarrow (0, +\infty)$. Now, the statements of Proposition 2.24 regarding the race of the functions $n \mapsto n^a$ and $n \mapsto b^n$ towards $+\infty$ translate from integer to real parameters $a > 0$ and $b > 1$.

Proposition 6.33. potency laws

Let $a, b > 0$ and $x, y \in \mathbb{R}$. Then the potency laws are:

$$a^{x+y} = a^x \cdot a^y, \quad (a^x)^y = a^{x \cdot y}, \quad a^x \cdot b^x = (a \cdot b)^x, \quad \left(\frac{1}{a}\right)^x = a^{-x}.$$

Proof: The first formula is Proposition 6.31. The other formulas are seen as follows:

$$(a^x)^y = \exp(y \cdot \ln(a^x)) = \exp(xy \ln a) = a^{xy}$$

$$a^x b^x = \exp(x \ln a) \cdot \exp(x \ln b) = \exp(x \cdot (\ln a + \ln b)) = \exp(x \cdot \ln(ab)) = (ab)^x$$

$$\left(\frac{1}{a}\right)^x = \exp\left(x \cdot \ln \frac{1}{a}\right) \stackrel{\frac{1}{a} = a^{-1}}{=} \exp(x \cdot (-1) \cdot \ln a) = \exp(-x \cdot \ln a) = a^{-x} \quad \blacksquare$$

Proposition 6.34. monotonicity and invertibility of $\exp_a$

Let $a > 0$. For the function $\exp_a : \mathbb{R} \rightarrow (0, +\infty)$, the following holds:

- a) $\exp_a$ is continuous.
- b) $\exp_a$ is $\begin{cases} \text{strictly increasing} & \text{for } a > 1, \\ \text{strictly decreasing} & \text{for } 0 < a < 1, \\ \text{constantly equal to } 1 & \text{for } a = 1. \end{cases}$
- c) $\exp_a(\mathbb{R}) = (0, +\infty)$, if $a \neq 1$.
- d) $\exp_a : \mathbb{R} \rightarrow (0, +\infty)$ is bijective if $a \neq 1$.

Proof: a) The product and composition of continuous functions are continuous.
 b) $\ln a$ is > 0 for $a > 1$, < 0 for $0 < a < 1$, and $= 0$ for $a = 1$. The outer function $\exp$ is also strictly increasing.
 c) Since $\ln a \neq 0$, $x \mapsto x \cdot \ln a$ is a bijection $\mathbb{R} \rightarrow \mathbb{R}$ and $\exp_a(\mathbb{R}) = \exp(\mathbb{R}) = (0, +\infty)$.
 d) Injectivity follows from a), b), and Proposition 5.25 a), while surjectivity follows from c). ■

Definition 6.35. logarithm to the base $a > 0$

Let $a > 0$ but $a \neq 1$. The inverse function $\exp_a^{-1} : (0, +\infty) \rightarrow \mathbb{R}$ is called the logarithm to the base a and is denoted by $\log_a$.

According to Proposition 5.25, $\log_a$ is strictly increasing and continuous.

Corollary 6.36. Also $\log_a$ turns $\cdot$ into $+$

Let $a > 0$. Then we have

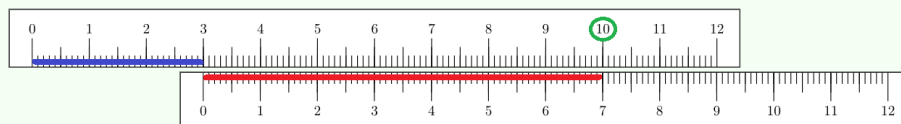
$$\log_a(x \cdot y) = \log_a x + \log_a y \quad \text{for all } x, y > 0.$$

Proof: This follows directly from Proposition 6.31 and $\log_a = \exp_a^{-1}$. ■

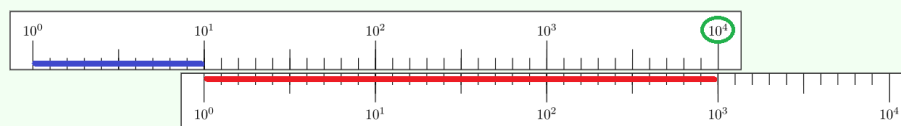
Here is an immediate application:

Example 6.37. Slide rulers: The pocket calculator from 1650 to 1970.

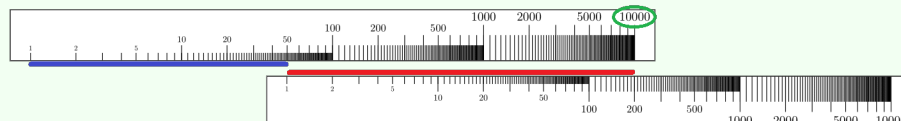
Addition can easily be performed with two rulers. $3 + 7 = 10$:



If one changes the scale on both rulers from linear to logarithmic, then the above alignment results in adding the logarithms, and thus multiplying the displayed numbers. In this example, $1 + 3 = 4$ becomes $10^1 \cdot 10^3 = 10^4$:



On the “proper” slide ruler, the scales are labeled with consecutive values instead of powers of ten, but otherwise, the two rulers remain the same. The rest works as before. In the next example, you can see that $50 \cdot 200 = 10.000$:



If one can then, for example, by placing a linear scale beside it, double logarithmic distances or divide them by five, one accordingly obtains the square or the fifth root of the initial number.

Speaking of pocket calculators: The following Proposition 6.38 explains why it’s sufficient to have only one log button on calculators. Up to a constant factor, there is essentially only one logarithm function:

Proposition 6.38. $\log_a x$ is just $c \cdot \ln x$ with the right constant c

Let $a > 0$ but $a \neq 1$. Then, for all $x \in \mathbb{R}$, we have

$$\log_a x = \frac{\ln x}{\ln a}.$$

Proof: $x = \exp_a(y) = a^y$ holds if and only if $y = \exp_a^{-1}(x) = \log_a x$. It follows that $x = a^y = \exp(y \cdot \ln a)$, hence $\ln x = \ln \exp(y \cdot \ln a) = y \cdot \ln a$, proving the claim. ■

In short: The rescaling $\frac{1}{\ln a}$ cancels the factor $\ln a$ from Definition 6.30 of $\exp_a(x)$.

6.5 Trigonometric functions and their inverses

Sine and cosine are also defined using power series:

Definition 6.39. sine and cosine

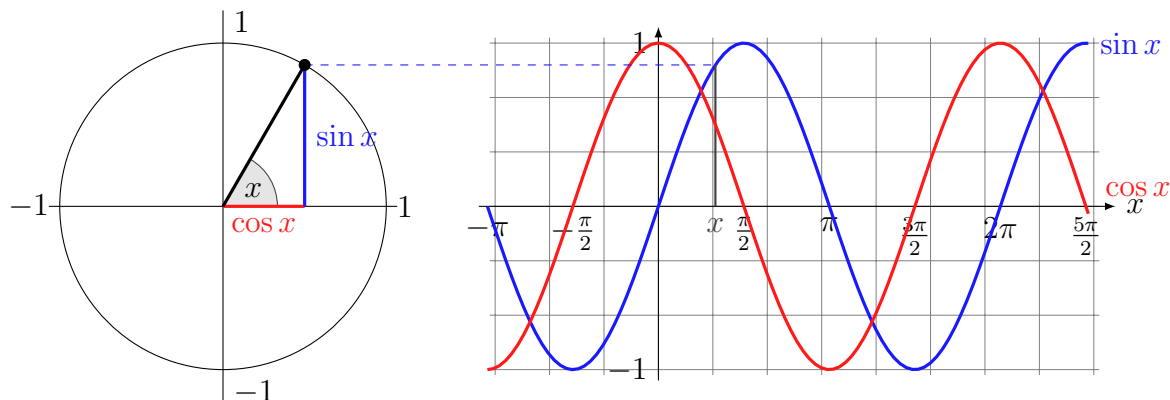
For $x \in \mathbb{R}$, we define

$$\cos x := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - + \dots$$

and

$$\sin x := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - + \dots$$

As with $\exp$, one can see that in both cases the radius of convergence is indeed $R = +\infty$. Being power series, $\sin$ and $\cos$ are again continuous functions. These result in the same functions $\sin$ and $\cos$, which were formerly defined geometrically:



🧐 Nerd box 6.40. Complex numbers would greatly simplify the following

With the complex exponential function $z \mapsto e^z$ and $i^2 = -1$, we have

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad \text{and} \quad e^{ix} = \cos x + i \cdot \sin x,$$

and the following proposition can easily be proven. (Thus, we omit the proof.)

Proposition 6.41. trigonometric identities, symmetry and pythagorean identity

For all $x, y \in \mathbb{R}$, the following formulas hold:

- a) $\cos(x + y) = \cos x \cdot \cos y - \sin x \cdot \sin y$,
 $\sin(x + y) = \sin x \cdot \cos y + \cos x \cdot \sin y$,
- b) $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$,
- c) $(\cos x)^2 + (\sin x)^2 = 1$.

With $a := x + y$ and $b := x - y$, we get $x = \frac{a+b}{2}$ and $y = \frac{a-b}{2}$ as well as

$$\left. \begin{aligned} \cos a - \cos b &= -2 \sin x \sin y = -2 \sin \frac{a+b}{2} \sin \frac{a-b}{2}, \\ \sin a - \sin b &= 2 \cos x \sin y = 2 \cos \frac{a+b}{2} \sin \frac{a-b}{2}. \end{aligned} \right\} \quad (6.5)$$

If we can introduce $\sin$ and $\cos$ without geometry, then we can also introduce π :

Proposition & Definition 6.42. This is how we define π

The function $\cos$ has exactly one zero in $[0, 2]$, let's call it p .

Now define $\pi := 2p$, i.e. $p = \frac{\pi}{2}$. We then have $\cos \frac{\pi}{2} = 0$ and $\sin \frac{\pi}{2} = 1$.

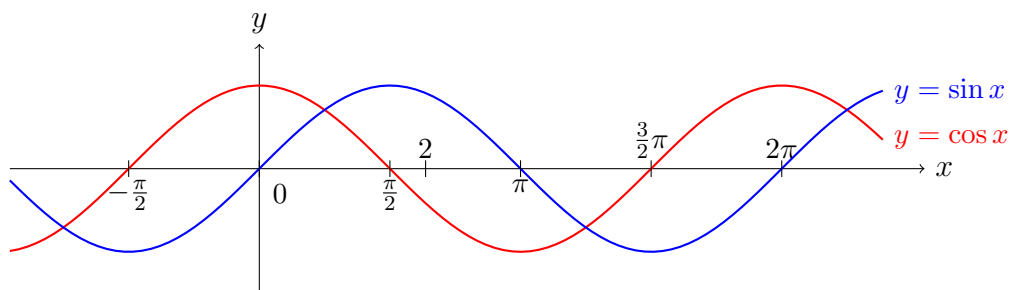
Proof: With the proof technique of the Leibniz rule, one shows $\cos 0 > 0 > \cos 2$. By the intermediate value theorem, $\cos$ has at least one zero in $[0, 2]$. With (6.5), one shows that $\cos$ is monotonically decreasing on $[0, 2]$ and thus has at most one zero in $[0, 2]$. ■

Along with the addition theorems for $\cos$ and $\sin$, it follows:

Corollary 6.43. $\sin$ and $\cos$ under shifts by multiples of $\frac{\pi}{2}$

For all $x \in \mathbb{R}$, the following hold:

$$\begin{aligned} \cos\left(x + \frac{\pi}{2}\right) &= -\sin x, & \cos(x + \pi) &= -\cos x, & \cos(x + 2\pi) &= \cos x, \\ \sin\left(x + \frac{\pi}{2}\right) &= \cos x, & \sin(x + \pi) &= -\sin x, & \sin(x + 2\pi) &= \sin x. \end{aligned}$$



After a shift by 2π , $\sin$ and $\cos$ start again from the beginning. The more technical expression is the following: $\sin$ and $\cos$ are periodic functions with a period of 2π .

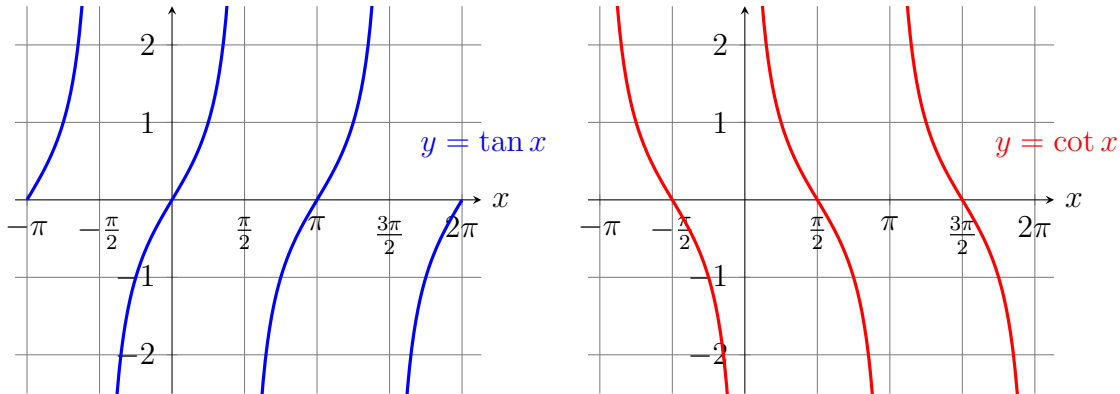
Definition 6.44. periodic function

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called periodic with period $P \in \mathbb{R}$, if for all $x \in \mathbb{R}$ it holds that $f(x + P) = f(x)$.

Whenever possible (i.e. denominator $\neq 0$), we define

$$\tan x := \frac{\sin x}{\cos x} \quad \text{and} \quad \cot x := \frac{\cos x}{\sin x}, \quad x \in \mathbb{R}.$$

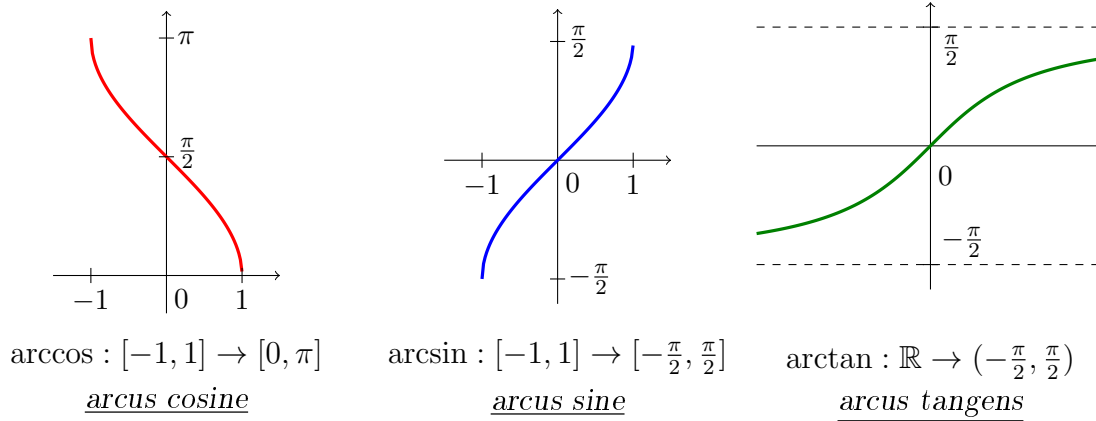
By definition, $\tan$ and $\cot$ are also 2π -periodic, indeed only π -periodic.



From (6.5) and Corollary 6.43, we obtain:

- $\cos$ is strictly decreasing on $[0, \pi] \rightarrow [-1, 1]$,
- $\sin$ is strictly increasing on $[-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$,
- $\tan x = \frac{\sin x}{\cos x}$ is strictly increasing on $(-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$.

Moreover, the functions with the above interval pairs are surjective. Injectivity again follows from strict monotonicity, continuity, and Proposition 5.25 a). Therefore, bijectivity exists in these intervals, i.e., there are strictly monotonic continuous inverse functions:



With the chosen monotonicity intervals of $\cos, \sin, \tan$ (i.e., the ranges of $\arccos, \arcsin, \arctan$), one speaks of the so-called principal branches of $\arccos, \arcsin$, and $\arctan$.

Summary

- Monomials $x \mapsto x^n$ with $n \in \mathbb{N}$ are continuous, strictly increasing on $[0, +\infty)$ and bijective $[0, +\infty) \rightarrow [0, +\infty)$.
- The corresponding inverse function $[0, +\infty) \rightarrow [0, +\infty)$ with $x \mapsto \sqrt[n]{x}$ is hence also continuous, strictly increasing and bijective.

- Polynomials
 - are linear combinations of monomials.
 - They are continuous on all of $\mathbb{R}$,
 - they have at most as many real zeros as their degree n is.
 - The zeros are easy to find if $n \in \{1, 2\}$,
 - for $n \geq 3$, one first has to guess a zero x_0 and then divide by $(x - x_0)$.
- Power series
 - are of the form $\sum_{n=0}^{\infty} a_n(x - x_0)^n$,
 - they converge absolutely in $U_R(x_0) = (x_0 - R, x_0 + R)$,
 - where they are also continuous as a function of x ,
 - they may or may not converge at $x_0 \pm R$,
 - they diverge at any other point.
 - We have formulas for the convergence radius R .
- The exponential function $\exp : \mathbb{R} \rightarrow (0, +\infty)$
 - is given by the power series, $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$, that converges on all of $\mathbb{R}$,
 - turns $+$ into $\cdot$,
 - is continuous, strictly increasing and bijective,
- The corresponding inverse function $\ln : (0, +\infty) \rightarrow \mathbb{R}$
 - is called the natural logarithm,
 - is continuous, strictly increasing and bijective,
 - turns $\cdot$ into $+$,
- Now we can also define $a^b := \exp(b \cdot \ln a)$ for $a > 0$ and $b \in \mathbb{R}$.
- The well-known potency laws hold.
- Also for the logarithm, one can pass to an arbitrary basis $a > 0$ ($a \neq 1$), leading to nothing but a scaled version of the $\ln$ function with scaling factor $\frac{1}{\ln a}$.
- $\sin$ and $\cos$
 - are also given by a power series with $R = +\infty$ and are hence continuous,
 - are equal to themselves when shifted by 2π
 - and equal to one another when shifted by $\frac{\pi}{2}$,
 - where π is defined as two times the smallest positive zero of the $\cos$ function.
 - Their quotients are $\tan$ and $\cot$.
 - On appropriate pairs of intervals, all four functions are invertible.

Overview: powers, roots and logarithms

We have $2^5 = 32$. Often two of the values are given and the third is to be determined:

<i>operation</i>	<i>problem</i>	<i>solution</i>	<i>functions</i>	
powers	$2^5 = ?$	$? = 2^5 = 32$	$x \mapsto x^5$	$x \mapsto 2^x$
roots	$?^5 = 32$	$? = \sqrt[5]{32} = 2$	$x \mapsto \sqrt[5]{x}$	$x \mapsto \sqrt[x]{32}$
logarithms	$2^? = 32$	$? = \log_2 32 = 5$	$x \mapsto \log_2 x$	$x \mapsto \log_x 32$

Roots are special cases of powers: $\sqrt[5]{x} = x^{\frac{1}{5}}$.

Since $\log_x 32 = \frac{1}{\log_{32} x}$, the latter is not really a new function either.

Thus, the three function classes remaining are

$x \mapsto x^a$ (power function),

$x \mapsto b^x$ (exponential function) and

$x \mapsto \log_c x$ (logarithmic function).

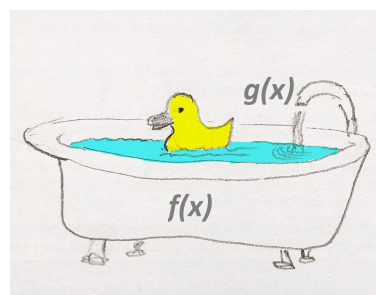
Chapter 7

Differentiation in one variable

GER vs. ENG – this has a long tradition; not only in football or battles over sun loungers by the hotel pool. Around 1670, Leibniz and Newton independently discovered differential and integral calculus and disputed till death (of the english man) over glory and fame. What is it about? A brief example:

We fill a bathtub: Plug in, turn the tap on, water flows. The water level in the tub (in liters) rises with time x and is denoted by $f(x)$. The flow rate at the tap (in liters per second) is $g(x)$. We might open the tap more or less at times, changing the flow rate $g(x)$ with time x . How are the functions f and g related? As follows:

- The filling level $f(x)$ is the accumulated inflow $g(x)$ over time;
- The flow $g(x)$ is the current rate of change of the fill level $f(x)$.



Such pairs f and g are found everywhere: for example, $f(x)$ could be the distance traveled so far and $g(x)$ the current speed. Or $f(x)$ could be the account balance and $g(x)$ the current inflow and outflow.

f is always a kind of accumulator for g , and g is the current rate of change of f . We will soon write $g = f'$, say "derivative" instead of "rate of change", and "integral" instead of "accumulator". And the fact that the two processes are essentially inverses of each other will be called the "fundamental theorem of calculus". But let's start from the beginning:

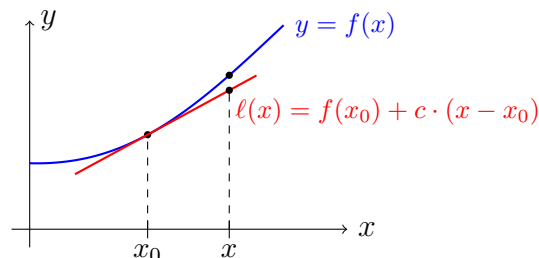
7.1 Differentiation and derivative

Let $D \subset \mathbb{R}$ be an open set and $f : D \rightarrow \mathbb{R}$. We fix a point $x_0 \in D$.

Fundamental idea of differentiation:

We approximate f locally (nearby x_0) by an (affine-)linear function:

$$\ell(x) = f(x_0) + c \cdot (x - x_0)$$



Thus, $\ell(x_0) = f(x_0)$, i.e., at the point x_0 the approximation is exact.

The optimal choice for c (slope of ℓ) is the slope/derivative of f at x_0 :

Definition 7.1. Differentiability, derivative

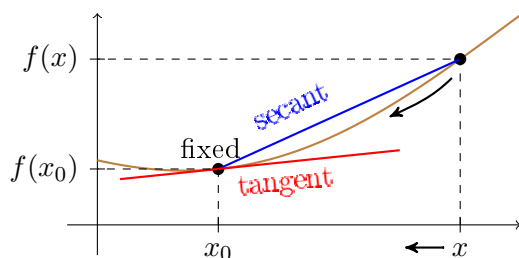
The function $f : D \rightarrow \mathbb{R}$ is called differentiable at $x_0 \in D$ if the following limit exists:

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{often shortened as:} \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}(x_0). \quad (7.1)$$

The limit is then called the derivative of f at x_0 . Notations: $f'(x_0)$, $\frac{df}{dx}(x_0)$, $\frac{d}{dx}f(x_0)$.
The function f is called differentiable if it is differentiable at every $x_0 \in D$.

The quotient in (7.1) is called the difference quotient, and it indicates the change in output $f(x)$ relative to the change in input x when $x \approx x_0$.

Geometrically, it describes the slope of the so-called secant through the points $(x_0, f(x_0))$ and $(x, f(x))$:



$$\text{„tangent}(x_0) = \lim_{x \rightarrow x_0} \text{secant}(x_0, x)\text{“}$$

Proposition 7.2. Differentiability reformulated

For $f : D \rightarrow \mathbb{R}$, $x_0 \in D$ and $c \in \mathbb{R}$, the following statements are equivalent:

- (i) f is differentiable at x_0 and $f'(x_0) = c$.
- (ii) The function $\varphi : x \mapsto \frac{f(x)-f(x_0)}{x-x_0}$ is continuously extended to x_0 via $\varphi(x_0) := c$.
- (iii) It holds that $f(x) = f(x_0) + c \cdot (x - x_0) + r(x)$, where $\frac{r(x)}{x-x_0} \rightarrow 0$ as $x \rightarrow x_0$.

Proof: Simple algebra. ■

Example 7.3. Some elementary functions and their derivatives

- a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = ax + b$ for fixed, $b \in \mathbb{R}$. Then for every $x_0 \in \mathbb{R}$

$$\frac{f(x) - f(x_0)}{x - x_0} =$$

so that the limit exists for $x \rightarrow x_0$ and is equal to a , i.e., $f'(x_0) = a$.

- b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^2$. For every $x_0 \in \mathbb{R}$, it holds that $f'(x_0) = 2x_0$:

$$\frac{f(x) - f(x_0)}{x - x_0} =$$

- c) A power series $f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n$ is differentiable at the expansion point x_0 and $f'(x_0) = a_1$ holds. Initially holds $f(x) = a_0 + a_1(x-x_0) + r(x)$ with

$$r(x) = \sum_{n=2}^{\infty} a_n(x-x_0)^n = (x-x_0)^2 \sum_{n=2}^{\infty} a_n(x-x_0)^{n-2},$$

so that

$$|r(x)| = |x-x_0|^2 \left| \sum_{n=2}^{\infty} a_n(x-x_0)^{n-2} \right| \leq |x-x_0|^2 \sum_{n=2}^{\infty} |a_n| |x-x_0|^{n-2}$$

and hence

$$\left| \frac{r(x)}{x-x_0} \right| = \frac{|r(x)|}{|x-x_0|} \leq |x-x_0| \underbrace{\sum_{n=2}^{\infty} |a_n| |x-x_0|^{n-2}}_{< \infty} \rightarrow 0 \quad \text{for } x \rightarrow x_0,$$

since the radius of convergence of the power series $\sum_n |a_n|(x-x_0)^n$ is the same according to Cauchy-Hadamard. It remains to apply (i) $\Leftrightarrow$ (iii) from Proposition 7.2.

- d) A power series is differentiable at every interior point of its region of convergence D , because by moving the center, the new center can be anywhere in $\text{int } D$ and then part c) can be applied.
- e) Let $f = \exp$, i.e., $f(x) = e^x$ and let $x_0 \in D = \mathbb{R}$ be arbitrary as well as $x = x_0 + h$.

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{e^{x_0+h} - e^{x_0}}{h} = \quad .$$

- f) Let $f = \sin$ and $x_0 \in D = \mathbb{R}$. By (6.5) it holds $f'(x_0) = \cos x_0$, because

$$\frac{\sin x - \sin x_0}{x - x_0} = \frac{2 \cos \frac{x+x_0}{2} \sin \frac{x-x_0}{2}}{x - x_0} \xrightarrow{x \rightarrow x_0}$$

Proposition 7.4. differentiable $\implies$ continuous

If $f : D \rightarrow \mathbb{R}$ is differentiable at $x_0 \in D$, then f is also continuous at x_0 .

Proof:

■

The converse of Proposition 7.4 does not hold. An example is $f : x \mapsto |x|$ at $x_0 = 0$.

7.2 Rules for differentiation

Proposition 7.5. The derivative commutes with $+$ and $-$ but not with $\cdot$ and $:$

Let f and g be differentiable in $x_0 \in \mathbb{R}$ and let $\alpha \in \mathbb{R}$. Then $\alpha f, f + g, f - g, f \cdot g$ and in case $g(x_0) \neq 0$ also $\frac{f}{g}$ are differentiable at x_0 and:

- a) $(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$
- b) $(\alpha f)'(x_0) = \alpha \cdot f'(x_0)$
- c) $(f \cdot g)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$ *(product rule)*
- d) $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}$ *(quotient rule)*

Proof: Taking the limit $h \rightarrow 0$ in each case leads to the assertion:

a)

$$\frac{(f+g)(x_0+h) - (f+g)(x_0)}{h} = \frac{f(x_0+h) - f(x_0)}{h} + \frac{g(x_0+h) - g(x_0)}{h}$$

b)

$$\frac{(\alpha f)(x_0+h) - (\alpha f)(x_0)}{h} = \alpha \cdot \frac{f(x_0+h) - f(x_0)}{h}$$

c)

$$\frac{(f \cdot g)(x_0+h) - (f \cdot g)(x_0)}{h} = \frac{f(x_0+h) - f(x_0)}{h} g(x_0+h) + \frac{g(x_0+h) - g(x_0)}{h} f(x_0)$$

d) Let $g(x_0) \neq 0$. Due to the continuity of g (Proposition 7.4), $g(x_0+h) \neq 0$ holds for sufficiently small $|h|$. For such h it holds

$$\frac{\left(\frac{f}{g}\right)(x_0+h) - \left(\frac{f}{g}\right)(x_0)}{h} = \frac{\frac{f(x_0+h)-f(x_0)}{h}g(x_0) - f(x_0)\frac{g(x_0+h)-g(x_0)}{h}}{g(x_0+h)g(x_0)}.$$

Example 7.6. Derivative of tangent

As shown in Example 7.3 f), analog to $\sin' = \cos$, we have $\cos' = -\sin$. From the quotient rule it follows:

$$\tan' x = \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) =$$

The next rule concerns the derivative of the composition $f \circ g$:

Proposition 7.7. Chain rule

In the situation $D \xrightarrow{g} D' \xrightarrow{f} \mathbb{R}$ let g in $x_0 \in D$ and f in $g(x_0) \in D'$ be differentiable. Then $f \circ g$ is differentiable in x_0 as well, and the following holds

$$(f \circ g)'(x_0) = f'(g(x_0)) \cdot g'(x_0). \quad (\text{chain rule})$$

Proof: Roughly speaking, to prove the chain rule, you have to use

$$\frac{f(g(x)) - f(g(x_0))}{x - x_0} = \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} \cdot \frac{g(x) - g(x_0)}{x - x_0}, \quad \text{i.e.} \quad \frac{\Delta f}{\Delta x} = \frac{\Delta f}{\Delta g} \cdot \frac{\Delta g}{\Delta x},$$

and to pass to the continuous extension of the quotients, see Proposition 7.2 (ii). ■

Example 7.8. ... on the chain rule

a) Let $f(x) = \exp(2x) = \exp(g(x))$ with $g(x) = 2x$.

$$\Rightarrow f'(x) =$$

Further $f(x) = \exp(2x) = (\exp x)^2 = h(\exp x)$ with $h(y) = y^2$.

$$\Rightarrow f'(x) =$$

b) Consider $\ln$ on $(0, +\infty)$. Let $x_0, x \in (0, +\infty)$.

- 1. Path: Using the definition and a few transformations:

$$\ln' x_0 = \lim_{x \rightarrow x_0} \frac{\ln x - \ln x_0}{x - x_0} =$$

- 2. Path:

- 3. Path:

 Nerd box 7.9. Our first differential equation

$f(x) = e^{2x}$ from example 7.8 a) solves the so-called differential equation $f' = 2f$.

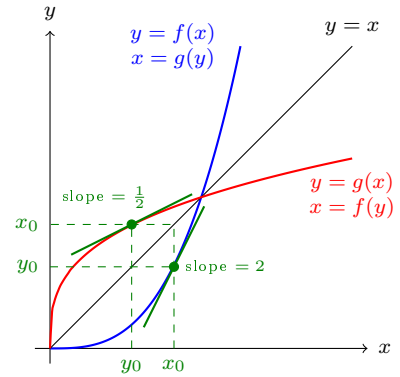
Result and method from Example 7.8 b) are not limited to $\ln$ and $\exp$:

Proposition 7.10. Derivative of the inverse function

Let $f : D \rightarrow f(D)$ be continuous and strictly monotonic, and $g : f(D) \rightarrow D$ be the inverse function of f .

If f is differentiable in $x_0 \in D$ with $f'(x_0) \neq 0$, then g is differentiable in $y_0 = f(x_0) \in f(D)$ and the following holds

$$g'(y_0) = \frac{1}{f'(x_0)} = \frac{1}{f'(g(y_0))}.$$



Proof: f is injective by Proposition 5.25 and surjective by construction. So the inverse function exists, $g : f(D) \rightarrow D$. The differentiability of g in $y_0 = f(x_0)$ can be seen by continuous extension of the quotients

$$\frac{g(y) - g(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}}, \quad \text{i.e.} \quad \frac{\Delta g}{\Delta y} = \frac{\Delta x}{\Delta f} = \frac{1}{\frac{\Delta f}{\Delta x}},$$

and the formula for $g'(y_0)$ follows from the above formula or as in Example 7.8 b). ■

Example 7.11. ...on the derivative of the inverse function

a) $g = \arctan : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$, as inverse function of $f = \tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$.

Example 7.6 $\Rightarrow \tan$ is differentiable on $(-\frac{\pi}{2}, \frac{\pi}{2})$ and $\tan' x = 1 + \tan^2 x$

Proposition 7.10 $\Rightarrow \forall y \in \mathbb{R} : \arctan$ is differentiable in y and

$$\arctan' y =$$

b) $g = \arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$, as inverse function of $f = \sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$.

$$\sin' x = \cos x = \sqrt{1 - \sin^2 x} \quad (\nearrow \text{Example 7.3f}))$$

Proposition 7.10 $\Rightarrow \arcsin$ is differentiable with

$$\arcsin' y =$$

With the help of $\exp'$ and $\ln'$ we are able to derive the derivative of $x \mapsto x^a$ and $x \mapsto a^x$:

Example 7.12. Derivative of power functions and exponential function

a) Let $a \in \mathbb{R}$ and $f(x) = x^a = \exp(a \cdot \ln x)$ for $x \in D = (0, +\infty)$.

$$\Rightarrow f'(x) =$$

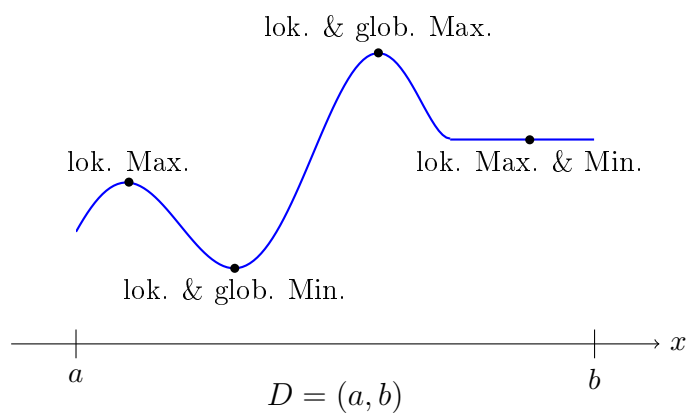
Of course, we have known this formula for a long time. Now finally with proof.

b) Let $a > 0$ and $f(x) = a^x = \exp(x \cdot \ln a)$ for $x \in D = \mathbb{R}$.

$$\Rightarrow f'(x) =$$

7.3 Local extrema

One of the main applications of differential calculus is to find local extrema (i.e. peaks and valleys in the graph) of f .



Let $D \subset \mathbb{R}$ be an open set and $f : D \rightarrow \mathbb{R}$ a function.

Definition 7.13. local extrema

f has a local minimum in $x_0 \in D$ if a neighborhood $U_\varepsilon(x_0) \subset D$ exists with

$$\forall x \in U_\varepsilon(x_0) : f(x_0) \leq f(x).$$

If $f(x_0) \geq f(x)$ in the last term then f is said to have a local maximum at x_0 .

In both cases we say that f has a local extremum in x_0 .

Proposition 7.14. Fermats proposition

(Pierre de Fermat 1607 - 1665)

If f is differentiable and has a local extremum in $x_0 \in D$, then

$$f'(x_0) = 0.$$

Proof:

■

In short:

Necessary criterion for local extrema: For differentiable functions f ,

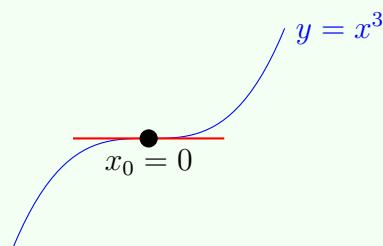
$$f \text{ has a local extremum in } x_0 \implies f'(x_0) = 0.$$

Example 7.15. The opposite direction, „ $\Leftarrow$ “ does not hold.

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = x^3$$

has the derivative $f'(x_0) = 3x_0^2 = 0$ at $x_0 = 0$,
but it has no local extremum there.



A point $x_0 \in D$ with $f'(x_0) = 0$ is called a stationary point or an extreme value suspect.

Example 7.16. another third grade polynomial

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ with

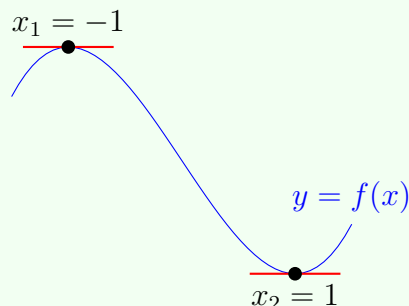
$$f(x) = x^3 - 3x$$

has the following derivative

$$f'(x) =$$

with roots at $x_1 = -1$ and $x_2 = 1$.

This time we have two local extrema.



Whether there is actually a local extremum at a stationary point (as in Example 7.16) or not (as in Example 7.15), can be found out after further investigations:

7.4 The mean value theorem and the monotonicity rule

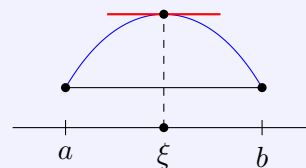
We start with a small auxiliary result:

Proposition 7.17. Rolle's proposition

(Michel Rolle 1652 - 1719)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable in (a, b) . If $f(a) = f(b)$, then there is a $\xi \in (a, b)$ with

$$f'(\xi) = 0.$$



Proof:

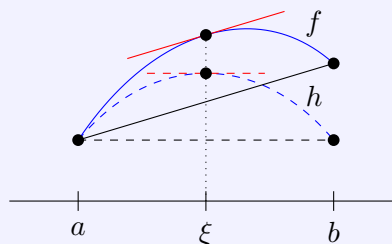
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Here is a straightforward consequence:

Proposition 7.18. Mean value theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable in (a, b) . Then there is a $\xi \in (a, b)$ with

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$



Proof: We „straighten“ the picture: Let

$$h(x) := f(x) - \underbrace{(x - a) \cdot \frac{f(b) - f(a)}{b - a}}_{\text{„slope“}}.$$

Then

■

Example 7.19. The mean value theorem and quadratic polynomials

For $f : [a, b] \rightarrow \mathbb{R}$ with $f : x \mapsto x^2$, the mean value ξ from the mean value theorem lies exactly at the middle of the interval:

$$\frac{f(b) - f(a)}{b - a} = \frac{b^2 - a^2}{b - a} = b + a = 2 \cdot \frac{a + b}{2} = f' \left(\frac{a + b}{2} \right)$$

This is therefore the case for all quadratic polynomials. (👁️: And only for those!)

Between us: speed control!

If your vehicle was flashed in town at a speed of $60 \frac{\text{km}}{\text{h}}$, then approximately the following has happened: You only took $b - a = 0.6$ seconds for a distance of $f(b) - f(a) = 10$ metres. The difference quotient

$$\frac{f(b) - f(a)}{b - a} = \frac{10 \text{ m}}{0.6 \text{ s}} \approx 16.66 \frac{\text{m}}{\text{s}} \approx 60 \frac{\text{km}}{\text{h}} \quad (\text{minus } 3 \frac{\text{km}}{\text{h}} \text{ tolerance})$$

is what you receive by post. Strictly speaking, this was not an instantaneous speed. (An 0.6 s long moment?) No one actually caught you with over $50 \frac{\text{km}}{\text{h}}$!

However, if the letter quotes the mean value theorem correctly, you have to pay...

A simple conclusion of the mean value theorem is:

Corollary 7.20. monotonicity rule

Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then the following holds

- a) $f' > 0$ on $(a, b) \implies f$ strictly increasing on (a, b)
- b) $f' < 0$ on $(a, b) \implies f$ strictly decreasing on (a, b)
- c) $f' \geq 0$ on $(a, b) \iff f$ monotonically increasing on (a, b)
- d) $f' \leq 0$ on $(a, b) \iff f$ monotonically decreasing on (a, b)
- e) $f' = 0$ on $(a, b) \iff f$ is constant on (a, b)

If f is continuous on $[a, b]$, all the statements on the right even apply to $[a, b]$.

Proof: All „ $\Leftarrow$ “ implications follow from Definition 7.1.

All „ $\Rightarrow$ “ implications follow from the mean value theorem, i.e. from

$$f(x_2) - f(x_1) = f'(\xi) \cdot (x_2 - x_1)$$

with $x_1, x_2 \in (a, b)$ resp. $[a, b]$ and with corresponding $\xi \in (x_1, x_2) \subset (a, b)$. ■

Note that in a) and b) „ $\Leftarrow$ “ does not apply (example $f : x \mapsto \pm x^3$).

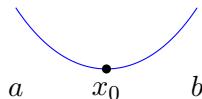
The monotonicity criterion now provides an initial decision-making aid for stationary points:

Corollary 7.21. first sufficient criterion for local extrema

Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable and $x_0 \in (a, b)$ a stationary point, i.e., $f'(x_0) = 0$.

- a) *If f' changes its sign at x_0 from $-$ to $+$,
i.e., $f' \leq 0$ in (a, x_0) and $f' \geq 0$ on (x_0, b) , then f has a local minimum at x_0 .*
- b) *If f' changes its sign at x_0 from $+$ to $-$,
i.e., $f' \geq 0$ on (a, x_0) and $f' \leq 0$ on (x_0, b) , then f has a local maximum at x_0 .*

Proof: a) decreasing on (a, x_0) , increasing on $(x_0, b) \Rightarrow$ local minimum



b) analogously

■

Example 7.22. back to our two polynomials of degree three

a) Looking at $f : x \mapsto x^3$ from Example 7.15, $x_0 = 0$ is a stationary point, since $f'(x) = 3x^2 = 0$ at $x_0 = 0$. However, since f' has no sign change at x_0 , Corollary 7.21 cannot make any statement here.

b) Looking at $f : x \mapsto x^3 - 3x$ from Example 7.16 on the other side, $f'(x) = 3x^2 - 3$ changes its sign at $x_1 = -1$ from $+$ to $-$ and at $x_2 = 1$ from $-$ to $+$. Corollary 7.21 shows: local maximum at x_1 , local minimum at x_2 .

7.5 De l'Hospitals rule

First, we need a new and improved version of the mean value theorem:

Proposition 7.23. generalized mean value theorem

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous and on (a, b) differentiable. Let furthermore $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists a $\xi \in (a, b)$ with

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

⚠ Attention!

- a) To prove this, it is not sufficient to apply the mean value theorem to f and g and divide the formulae. The result on the right-hand side would be $\frac{f'(\xi_1)}{g'(\xi_2)}$, where ξ_1 and ξ_2 do not need to be equal.
- b) The case $g(b) - g(a) = 0$ (denominator of the left-hand side) is impossible, because then there would be (Rolle's proposition) a $\xi \in (a, b)$ with $g'(\xi) = 0$, which has been excluded.

Proof: We refer back to Rolle's proposition with a similar construction as in the proof of the mean value theorem: Be this time

$$h(x) := f(x) - (g(x) - g(a)) \cdot \frac{f(b) - f(a)}{g(b) - g(a)}, \quad x \in (a, b).$$

Then h satisfies the conditions of Rolle's proposition, and thus

$$\exists \xi \in (a, b) : 0 = h'(\xi) = f'(\xi) - g'(\xi) \frac{f(b) - f(a)}{g(b) - g(a)}. \quad \blacksquare$$

Proposition 7.24. de l'Hospitals rule

(Marquis de l'Hospital 1661 - 1704)

Let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for all $x \in (a, b)$.

If $\lim_{x \searrow a} f(x) = 0$ and $\lim_{x \searrow a} g(x) = 0$ as well as $\lim_{x \searrow a} \frac{f'(x)}{g'(x)}$ exists, then $\lim_{x \searrow a} \frac{f(x)}{g(x)}$ exists and

$$\lim_{x \searrow a} \frac{f(x)}{g(x)} = \lim_{x \searrow a} \frac{f'(x)}{g'(x)}.$$

All versions of de l'Hospital's rule summarised:

The following rule

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

holds in the setting of

- one-sided limits $x \searrow a$ of the form $\frac{0}{0}$ (Proposition 7.24) and
- one-sided limits $x \nearrow b$ of the form $\frac{0}{0}$ (proof analogously).

With a little trick, the rule can be transferred to

- one-sided limits $x \searrow a$ and $x \nearrow b$ of the form $\frac{\infty}{\infty}$,
- but also to limits $x \rightarrow +\infty$ and $x \rightarrow -\infty$ of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We accept all these versions as proven and can now start to apply de l'Hospital's rule in different situations:

Example 7.25. ...on the rule of de l'Hospital

a) $\lim_{x \rightarrow 0} \frac{\sin x}{x} =$

(of the form $\frac{0}{0}$)

b) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} =$

(of the form $\frac{0}{0}$)

c) $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^a} =$

for $a > 0$ (of the form $\frac{\infty}{\infty}$)

d) $\lim_{x \rightarrow +\infty} \frac{x^n}{e^x} =$

for $n \in \mathbb{N}$

e) $\lim_{x \rightarrow 0} (x \cdot \cot x) =$

(of the form $0 \cdot \infty \sim \frac{0}{0}$)

$$\text{f) } \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) =$$

(of the form $\frac{1}{0} - \frac{1}{0} \sim \frac{0-0}{0-0}$)

7.6 The second derivative

Let $D \subset \mathbb{R}$ be open and $f : D \rightarrow \mathbb{R}$ differentiable. Then by $f' : D \rightarrow \mathbb{R}$ with $x \mapsto f'(x)$ again a function is given. If this function is as well differentiable, you can consider its derivative $(f')'$ and denote it by f'' . This is then again a function $f'' : D \rightarrow \mathbb{R}$ and is called the second derivative of f .

As long as differentiability holds on, you can continue this process and obtain ever higher derivatives: $f''', f''', f''', \dots$. At the latest from $n \geq 4$ dashes you write $f^{(n)}$.

Definition 7.26. multiple (continuous) differentiability

$f : D \rightarrow \mathbb{R}$ is called n times differentiable for some $n \in \mathbb{N}$, if $f^{(n)}$ exists. If $f^{(n)}$ is still continuous, we call f n times continuously differentiable; we write $f \in C^n(D)$.

Example 7.27. an exactly n times differentiable function

For $n \in \mathbb{N}_0$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f_n(x) = x^n \cdot |x| = \begin{cases} x^{n+1}, & x \geq 0, \\ -x^{n+1}, & x < 0. \end{cases}$$

Then f_n is n times differentiable but not $n+1$ times. This can be seen inductively: f_0 is continuous but not differentiable and $f'_{n+1} = (n+1)f_n$ holds.

For the rest of this section, we restrict ourselves to $n = 2$, i.e., to the 2nd derivative, f'' .

Proposition 7.28. second sufficient criterion for local extremas

Let $f : (a, b) \rightarrow \mathbb{R}$ be 2-times differentiable and $x_0 \in (a, b)$ be a stationary point.

- a) If $f''(x_0) > 0$, then f has a local minimum in x_0 .
- b) If $f''(x_0) < 0$, then f has a local maximum in x_0 .

Proof: a) With $f''(x_0) > 0$, also the difference quotient of f' is positive nearby x_0 , let's say in $U_\varepsilon(x_0)$:

$$0 < \frac{f'(x) - f'(x_0)}{x - x_0} = \frac{f'(x) - 0}{x - x_0} = \frac{f'(x)}{x - x_0}, \quad x \in U_\varepsilon(x_0)$$

Since $x - x_0$ changes its sign at x_0 from $-$ to $+$, $f'(x)$ does the same. It remains to apply Corollary 7.21 a). Part b) is proven in the same way. ■

Example 7.29. and again: back to our two polynomials of degree three

- a) Looking back to $f : x \mapsto x^3$ from Example 7.15, $x_0 = 0$ is a stationary point, because $f'(x) = 3x^2 = 0$ at $x_0 = 0$. By $f''(x) = 6x = 0$ in $x_0 = 0$, there is nothing more that Proposition 7.28 could say.
- b) However, looking back to $f : x \mapsto x^3 - 3x$ from Example 7.16 on the other side, $f''(x) = 6x$ is negative at $x_1 = -1$ and positive at $x_2 = 1$, so that Proposition 7.28 comes to the following conclusion: local maximum at x_1 , local minimum at x_2 .

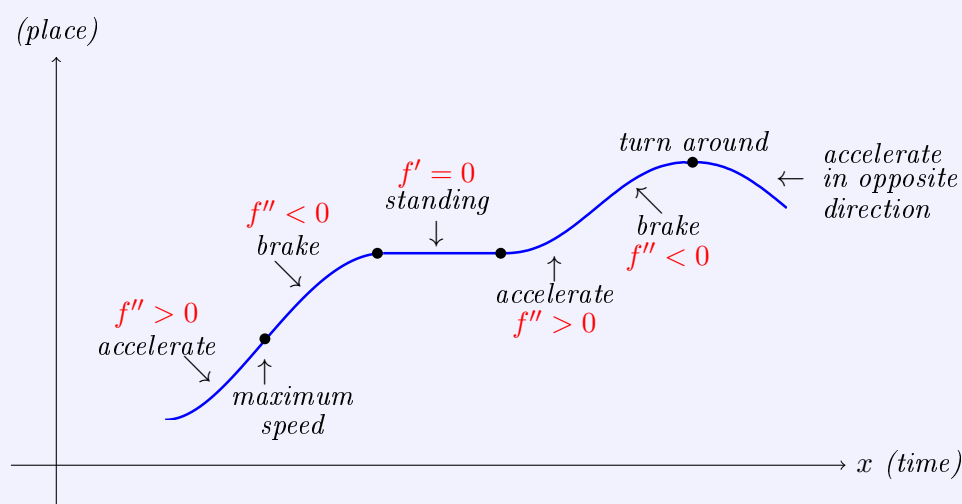
If $f'(x_0) = 0$ and $f''(x_0) = 0$, higher derivatives may have to be considered.

Interaction of f , f' and f'' : An object moves along a line

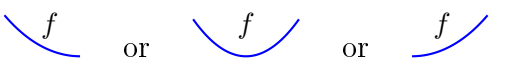
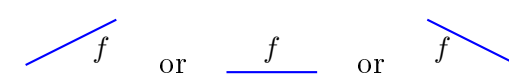
$f(x)$... location at which the object is located at the time x ,

$f'(x)$... speed at time x ,

$f''(x)$... acceleration at time x



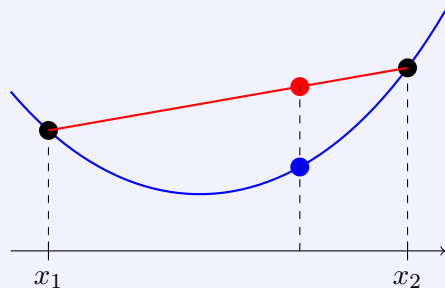
The sign of f'' determines the direction in which the graph of f curves, and the greater the magnitude of f'' , the more the graph of f is curved:

- $f'' > 0 \implies f'$ increases, which means 
- $f'' < 0 \implies f'$ decreases, which means 
- $f'' = 0 \implies f'$ is constant, which means 

The two directions of curvature are called convex and concave:

Definition 7.30. convex, concave

Let $D \subset \mathbb{R}$ be an interval. A function $f : D \rightarrow \mathbb{R}$ is called convex if for all $x_1, x_2 \in D$ with $x_1 < x_2$ the graph of f on (x_1, x_2) lies below the secant to f through x_1 and x_2 :



In formulas: For all $x_1, x_2 \in D$ with $x_1 < x_2$ the following applies

$$\forall \lambda \in (0, 1) : \quad f\left((1 - \lambda)x_1 + \lambda x_2\right) \leq (1 - \lambda)f(x_1) + \lambda f(x_2). \quad (7.2)$$

f is called concave if $-f$ is convex, which means if in (7.2) the relation „ $\geq$ “ applies. f is called strictly convex (strictly concave), if in (7.2) „ $<$ “ (or „ $>$ “) holds.

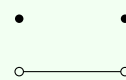
Example 7.31. convex functions

a) $f : x \mapsto x^2$ is strictly convex on $\mathbb{R}$.

b) Every affine-linear function $f : x \mapsto ax + b$, with $a, b \in \mathbb{R}$ fixed, is convex and concave at the same time, but in both not strictly.

c) The absolute value function $x \mapsto |x|$ is convex (but not strictly convex) on $\mathbb{R}$.

d) The function $f : [0, 1] \rightarrow \mathbb{R}$ with $f(x) = \begin{cases} 3, & x = 0, \\ 3, & x = 1, \\ 2, & x \in (0, 1), \end{cases}$ is convex (but not strictly convex) – and discontinuous at the boundary points.



As already indicated, convexity can be characterised by the monotonicity of the first or the sign of the second derivative:

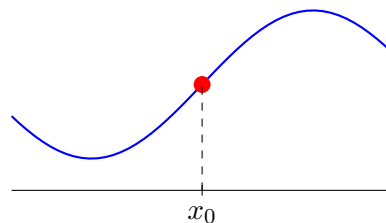
Proposition 7.32. convexity vs. f' and f''

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If f is one or two times differentiable on (a, b) then the corresponding parts of the following statements apply (analogously for concave):

$$\begin{array}{llll} f \text{ is convex} & \Longleftrightarrow & f' \text{ is monotonically increasing} & \Longleftrightarrow & f'' \geq 0, \\ f \text{ is strictly convex} & \Longleftrightarrow & f' \text{ is strictly increasing} & \Longleftarrow & f'' > 0. \end{array}$$

A point $x_0 \in (a, b)$, in which the behaviour of f changes from convex to concave, or vice versa, is called inflection point of f .

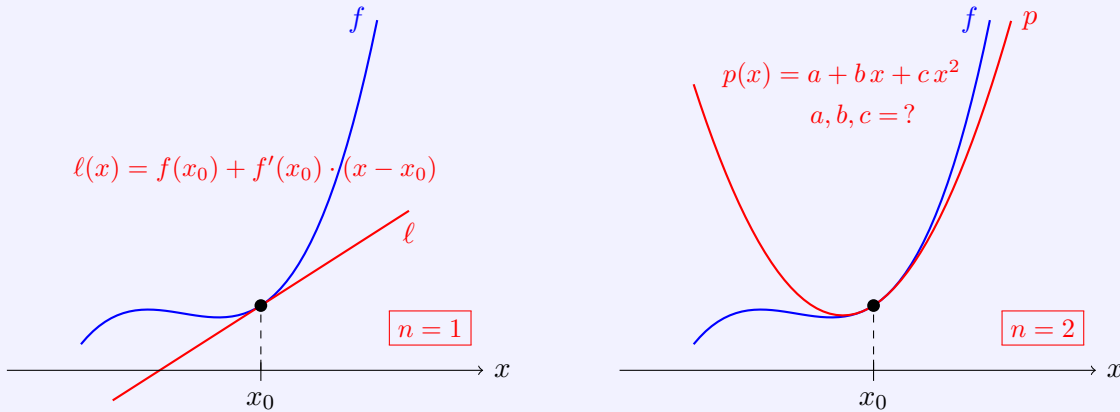
For 2-times continuously differentiable f , this is equivalent to f'' having a change of sign in x_0 . It is then necessary that $f''(x_0) = 0$.



7.7 Taylor expansion

Taylor Polynomial. Since the beginning of the previous section, we now have higher derivatives $f^{(n)}$ available, and we ask ourselves the following question:

If one can use $f(x_0)$ and $f'(x_0)$ to approximate the function f in $U_\varepsilon(x_0)$ beautifully in an affine-linear manner, can one perhaps use $f(x_0), f'(x_0), \dots, f^{(n)}(x_0)$ to approximate f even better by a polynomial of degree n ?



Answer: Yes.

Taylor polynomial: problem description

given: $f(x_0), f'(x_0), \dots, f^{(n)}(x_0)$

desired: polynomial of degree n , p , that behaves in x_0 exactly (value, slope, curvature, ...) as f , i.e., it should hold that:

$$\left. \begin{aligned} p(x_0) &= f(x_0), \\ p'(x_0) &= f'(x_0), \\ &\vdots \\ p^{(n-1)}(x_0) &= f^{(n-1)}(x_0), \\ p^{(n)}(x_0) &= f^{(n)}(x_0). \end{aligned} \right\} \quad (7.3)$$

Approach: Put

$$p(x) := a_0 + a_1(x - x_0)^1 + a_2(x - x_0)^2 + \dots + a_n(x - x_0)^n$$

with initially unknown coefficients $a_0, a_1, \dots, a_n \in \mathbb{R}$.

We now compute the left hand sides of (7.3) and then construct $a_0, a_1, \dots, a_n$ so, that the equations (7.3) fit. ($n + 1$ equations for $n + 1$ unknowns: sounds fair.)

$$\begin{aligned} p(x) &= a_0 + a_1(x - x_0)^1 + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \dots + a_n(x - x_0)^n \\ \Rightarrow p(x_0) &= \end{aligned}$$

$$\begin{aligned} p'(x) &= 1 \cdot a_1 + 2 \cdot a_2(x - x_0)^1 + 3 \cdot a_3(x - x_0)^2 + \dots + n \cdot a_n(x - x_0)^{n-1} \\ \Rightarrow p'(x_0) &= \end{aligned}$$

$$\begin{aligned} p''(x) &= 1 \cdot 2 \cdot a_2 + 2 \cdot 3 \cdot a_3(x - x_0)^1 + \dots + (n - 1) \cdot n \cdot a_n(x - x_0)^{n-2} \\ \Rightarrow p''(x_0) &= \end{aligned}$$

$$\vdots$$

$$p^{(n)}(x) = 1 \cdot 2 \cdot \dots \cdot n \cdot a_n.$$

Solving for a_k , we get the formula for the so-called Taylor coefficients:

$$a_k = \frac{p^{(k)}(x_0)}{k!} \stackrel{(7.3)}{=} \frac{f^{(k)}(x_0)}{k!}, \quad k = 0, \dots, n \quad (\text{mit } p^{(0)} = p \text{ und } 0! = 1).$$

Plugging a_k into the polynomial p above, we get:

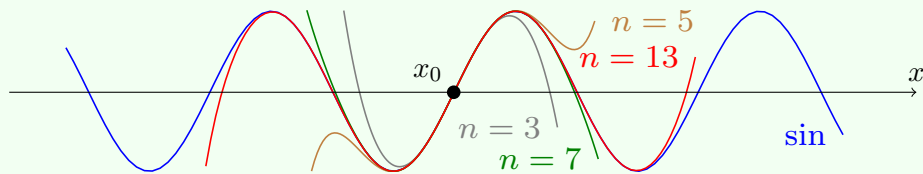
Taylor polynomial of order n

(Brook Taylor, 1685 – 1731)

$$p(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad (7.4)$$

Example 7.33. Taylor polynomial of $f = \sin$ at $x_0 = 0$

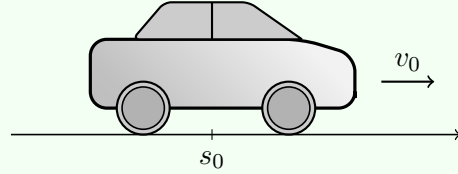
$$p(x) = \sum_{k=0}^n \frac{\sin^{(k)}(0)}{k!} x^k = \frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - + \dots (\pm) \frac{x^n}{n!}, \quad n \in \mathbb{N} \text{ odd}$$



Example 7.34. Second order Taylor polynomial in driving

$f(x)$... position at time x ,
 $f'(x)$... speed at time x ,
 $f''(x)$... acceleration at time x .

Let $x_0 = 0$ (for simplicity).



$n = 0$: All I know: Lately, my car was at point $s_0 \in \mathbb{R}$.

$$f(0) = s_0 \quad (\text{last known location})$$

$\implies$ Best possible prediction:

$$p(x) = s_0 \quad \forall x$$

$n = 1$: All I know:

$$f(0) = s_0 \quad (\text{last known location})$$

$$f'(0) = v_0 \quad (\text{last known speed})$$

$\implies$ Prediction:

$$p(x) = s_0 + v_0 x \implies \begin{cases} p(0) = s_0 = f(0) \\ p'(0) = v_0 = f'(0) \end{cases}$$

$n = 2$: I know:

$$f(0) = s_0 \quad (\text{last known location})$$

$$f'(0) = v_0 \quad (\text{last known speed})$$

$$f''(0) = a_0 \quad (\text{last known acceleration})$$

$\implies$ Prediction:

$$p(x) = s_0 + v_0 x + \frac{a_0}{2} x^2 \implies \begin{cases} p(0) = s_0 = f(0) \\ p'(0) = v_0 = f'(0) \\ p''(0) = a_0 = f''(0) \end{cases}$$

Next, we address the following questions:

- 1.) How well does the Taylor polynomial p approximate the original function f ?
- 2.) What happens to p as we let $n \rightarrow \infty$? Convergence? To what limit? f ?

1.) Error estimation for the approximation $p \approx f$. First, we estimate the approximation error of the Taylor polynomial p to the original function f :

Proposition 7.35. Taylor with Lagrange remainder term (J.-L. Lagrange 1736 - 1813)

Let $D \subset \mathbb{R}$ be an open interval, $n \in \mathbb{N}$, and $f \in C^{n+1}(D)$.

For all $x_0, x \in D$, there exists a ξ between x_0 and x such that

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{p(x)} + \underbrace{\frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}}_{\text{Lagrange remainder}}.$$

Proof: Let $x_0, x \in D$ and let M be such that $f(x) = p(x) + \frac{M}{(n+1)!} (x - x_0)^{n+1}$. Through $(n+1)$ applications of Rolle's theorem, one shows: $\exists \xi : f^{(n+1)}(\xi) = M$. ■

Example 7.36. Error estimate for the exponential function when $n = 5$

Let $f = \exp$ and $x_0 = 0$. Then $f^{(n)}(x_0) = \exp(0) = 1$ for all n .

$$e^x = \underbrace{1 + x + \frac{1}{2!}x^2 + \dots + \frac{1}{n!}x^n}_{p(x)} + \underbrace{\quad}_{=: R(x)} \quad \text{with } \xi \in [0, x].$$

For $n = 5$, we conclude at $x = 1$ that $\xi \leq 1$ and $|R(1)| \leq \frac{e^1}{6!} 1^6 = \frac{e}{720} < 0.0038$.

Taylor polynomial with Lagrange remainder (alternative notation)

With $x = x_0 + h$ and ξ between x_0 and x from Proposition 7.35, we get:

$$f(x_0 + h) = f(x_0) + \frac{f'(x_0)}{1!}h + \frac{f''(x_0)}{2!}h^2 + \dots + \frac{f^{(n)}(x_0)}{n!}h^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}h^{n+1}$$

Based on upper bounds for $|f^{(n+1)}(\xi)|$ and $|x - x_0|$, the Lagrange remainder and thus the worst-case approximation error can be estimated:

Corollary 7.37. Taylor polynomial: error estimate

Let $D = (a, b)$, $n \in \mathbb{N}$, and let $f \in C^{n+1}(D)$ and $M \geq \sup_{\xi \in D} |f^{(n+1)}(\xi)|$.

Then for all $x, x_0 \in D$, it holds that

$$|f(x) - p(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$$

and thus for all $x_0 \in D$

$$\sup_{x \in D} |f(x) - p(x)| \leq \frac{M}{(n+1)!} \sup_{x \in D} |x - x_0|^{n+1} = \frac{M}{(n+1)!} d^{n+1},$$

where $d := \max(x_0 - a, b - x_0)$ is the distance from x_0 to the further end point.

2.) Now the Taylor series. Now let us send $n \rightarrow \infty$ in the Taylor polynomial.

To ensure that all Taylor coefficients $a_n = \frac{f^{(n)}(x_0)}{n!}$ are defined, let $f \in C^\infty(D) := \bigcap_{n \in \mathbb{N}} C^n(D)$.

Definition 7.38. Taylor series

Let $D \subset \mathbb{R}$ be an open interval. For $f \in C^\infty(D)$ and $x_0 \in D$, let Tf be the following power series,

$$(Tf)(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n,$$

which we call the Taylor series of f expanded / centered at x_0 .

Let $f \in C^\infty(D)$ already be given by a power series with radius of convergence $R > 0$,

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad x \in D = U_R(x_0). \quad (7.5)$$

In this case, f is also called analytic at x_0 .

Then f is differentiable in $D = U_R(x_0)$ (Example 7.3 d), and one has

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x - x_0)^{n-1}, \quad x \in D.$$

In particular, the power series can be differentiated term by term and the radius of convergence remains the same, $R > 0$, because

$$\sqrt[n]{|n a_n|} = \sqrt[n]{n} \cdot \sqrt[n]{|a_n|} \quad \text{and} \quad \sqrt[n]{n} \rightarrow 1.$$

Just like in the derivation of (7.4), we get $f(x_0) = a_0$, $f'(x_0) = 1 \cdot a_1$ as well as $f''(x_0) = 1 \cdot 2 \cdot a_2$, etc.: The a_n in (7.5) are automatically the Taylor coefficients of f .

$$f \text{ analytic at } x_0 \implies f = Tf \text{ in } U_R(x_0). \quad (7.6)$$

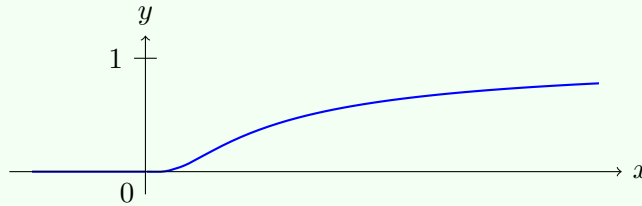
Thus, the known power series of $\exp, \sin, \cos, \dots$ are indeed their Taylor series.

The radius of convergence of Tf can also be $R = 0$. And even if $R > 0$, Tf does not necessarily converge to f ! (In which case f is of course not analytic, see (7.6).)

Example 7.39. Taylor series Tf of $f \in C^\infty(\mathbb{R})$ converges, but $Tf \neq f$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & x > 0, \\ 0 & x \leq 0. \end{cases}$$



It can be shown that $f \in C^\infty(\mathbb{R})$ with $f^{(n)}(0) = 0$, $n = 0, 1, 2, \dots$, so that $Tf \equiv 0 \neq f$ holds with Taylor expansion at $x_0 = 0$.

7.8 Curve discussion

Let a function $f : D \rightarrow \mathbb{R}$ be given.

Typical steps of a curve discussion

- 1.) Find the domain $D \subset \mathbb{R}$ (if not given)
- 2.) Symmetries (for example $f(x) = f(-x)$ or $f(x) = -f(-x)$ for all $x \in D$)
- 3.) Continuity
- 4.) Poles and behavior at gaps in the domain
- 5.) Asymptotic behavior for $x \rightarrow \pm\infty$
- 6.) Zeros
- 7.) Monotonicity intervals and local extrema
- 8.) Convexity and concavity as well as inflection points

A quick example:

Example 7.40. Curve discussion

Let the rule $f : x \mapsto \frac{2x^2-12x+10}{x^2}$ be given. We simplify it a bit:

$$f(x) = \frac{2x^2 - 12x + 10}{x^2} =$$

- 1.) Due to the division by x^2 , $x = 0$ must be excluded, whence $D = \mathbb{R} \setminus \{0\}$.
- 2.) Obviously, $f(-x) =$
- 3.) According to Example 5.17d), f is continuous on the entire domain $D = \mathbb{R} \setminus \{0\}$.
- 4.) The only gap in the domain is at $x_0 = 0$.

5.) For $x \rightarrow +\infty$ and for $x \rightarrow -\infty$,

6.) It holds that $f(x) = 0 \iff 2x^2 - 12x + 10 = 0 \iff$

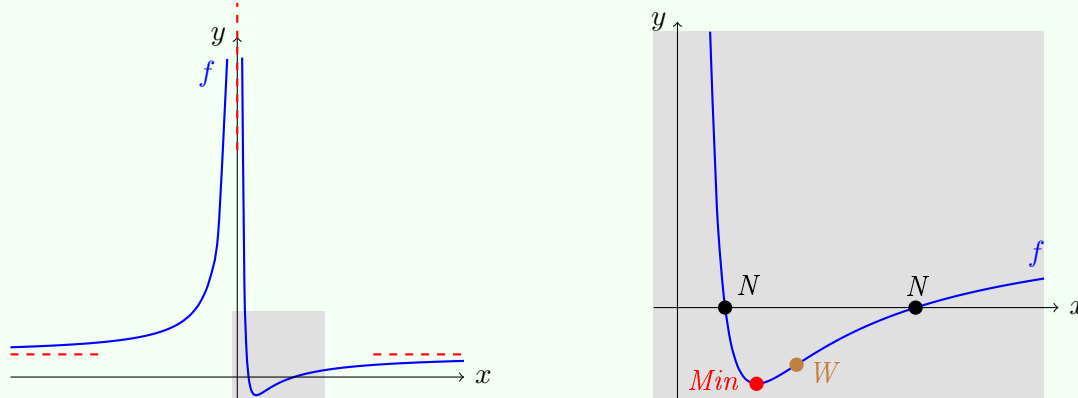
7.) Here we investigate when the derivative f' has which sign. It holds that

$$f'(x) =$$

8.) Finally, let's examine when the second derivative f'' has which sign. It holds that

$$f''(x) =$$

We conclude the example with a graph sketch, showing the asymptotics at the pole and as $x \rightarrow \pm\infty$, and a zoom into the gray area showing the two zeros, the minimum, and the inflection point:



Summary

- $f : D \rightarrow \mathbb{R}$ is differentiable at $x_0 \in \text{int } D$ if $\frac{f(x)-f(x_0)}{x-x_0}$ has a limit as $x \rightarrow x_0$.

- The limit value is then the derivative $f'(x_0)$.
- Continuity follows from differentiability – but not vice versa.
- With f and g , also $f + g, f - g, f \cdot g, f : g, f \circ g$ and f^{-1} are all differentiable, and we know a formula for the derivative in each case.
- For f to have a local extremum in x_0 , it is necessary that $f'(x_0) = 0$ applies. (assuming $x_0 \in \text{int } D$ and differentiability of f in x_0)
- However, $f'(x_0) = 0$ alone is not sufficient to conclude a local extremum.
 - This is then only referred to as an extreme value suspect or stationary point.
 - However, if f' also changes its sign at x_0 , we can conclude a local extremum,
 - If $f''(x_0) \neq 0$, then also.
 - Minimum or maximum? Depends on the sign change of f' or the sign of $f''(x_0)$.
- Monotonicity of f can be characterized equivalently via $f' \geq 0$ or $f' \leq 0$.
- Strict monotonicity follows from $f' > 0$ or $f' < 0$ – but not vice versa.
- If $\frac{f(x)}{g(x)}$ for $x \rightarrow x_0$ or $x \rightarrow \pm\infty$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the limit value of $\frac{f(x)}{g(x)}$ can be calculated by the limit value of $\frac{f'(x)}{g'(x)}$, if the latter exists.
- The second derivative f'' has something to do with the curvature of the graph of f :
 - For $f'' = 0$ on the whole of D , the graph is a straight line,
 - for $f'' > 0$ the graph is curved like a “U”; f is then strictly convex,
 - for $f'' < 0$ the curve goes the other way round and f is called strictly concave,
 - the magnitude $|f''|$ is related to the intensity of the curvature.
- Just as $f(x_0)$ and $f'(x_0)$ are used to determine the tangent to f in x_0 , all of $f(x_0), f'(x_0), \dots, f^{(n)}(x_0)$ can be used to determine a polynomial of n th degree that optimally matches f at x_0 , the so-called Taylor polynomial.
 - The polynomial coefficient in front of x^k is calculated via $a_k = \frac{f^{(k)}(x_0)}{k!}$,
 - the approximation error is estimated using the Lagrange remainder,
 - if we send $n \rightarrow \infty$, we get the Taylor series.
- In a curve discussion, you determine the domain of definition, symmetries, any discontinuities, in particular poles, the asymptotics at $\pm\infty$, zeros, intervals with monotonicity, local extrema, intervals with convexity, concavity and inflection points.

Part II

Analysis 2

Chapter 8

Integration in one variable

We return to the interplay between the level f and the rate of change $g = f'$, or vice versa: inflow g and reservoir f . This time it might be a fuel tank whose level f we cannot see.

To determine the current level $f(x)$, we have to continuously sum the inflow $g(t)$ over the period of time from $t = 0$ to x . This “continuous way of summing” is called integrating, and we write

$$f(x) = \int_0^x g(t) dt, \quad \text{also} \quad f(x) = \int_0^x f'(t) dt. \quad (8.1)$$

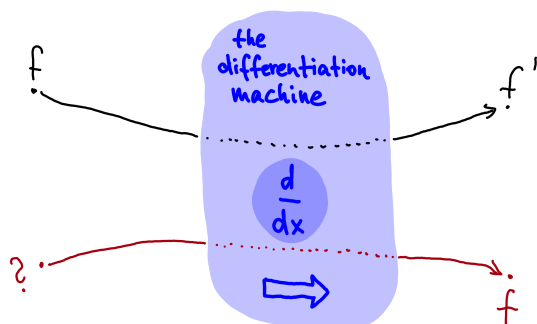
How much is a drop of gasoline?

Nothing at all!

Then please drip my tank full.

8.1 The indefinite integral: antiderivatives

8.1.1 Definitions



After understanding in Chapter 7 how to determine the rate of change / derivative f' of a function f , we now want to go in the reverse direction:

Which function(s) F has (or have) the derivative f , i.e., when does $F' = f$ hold?

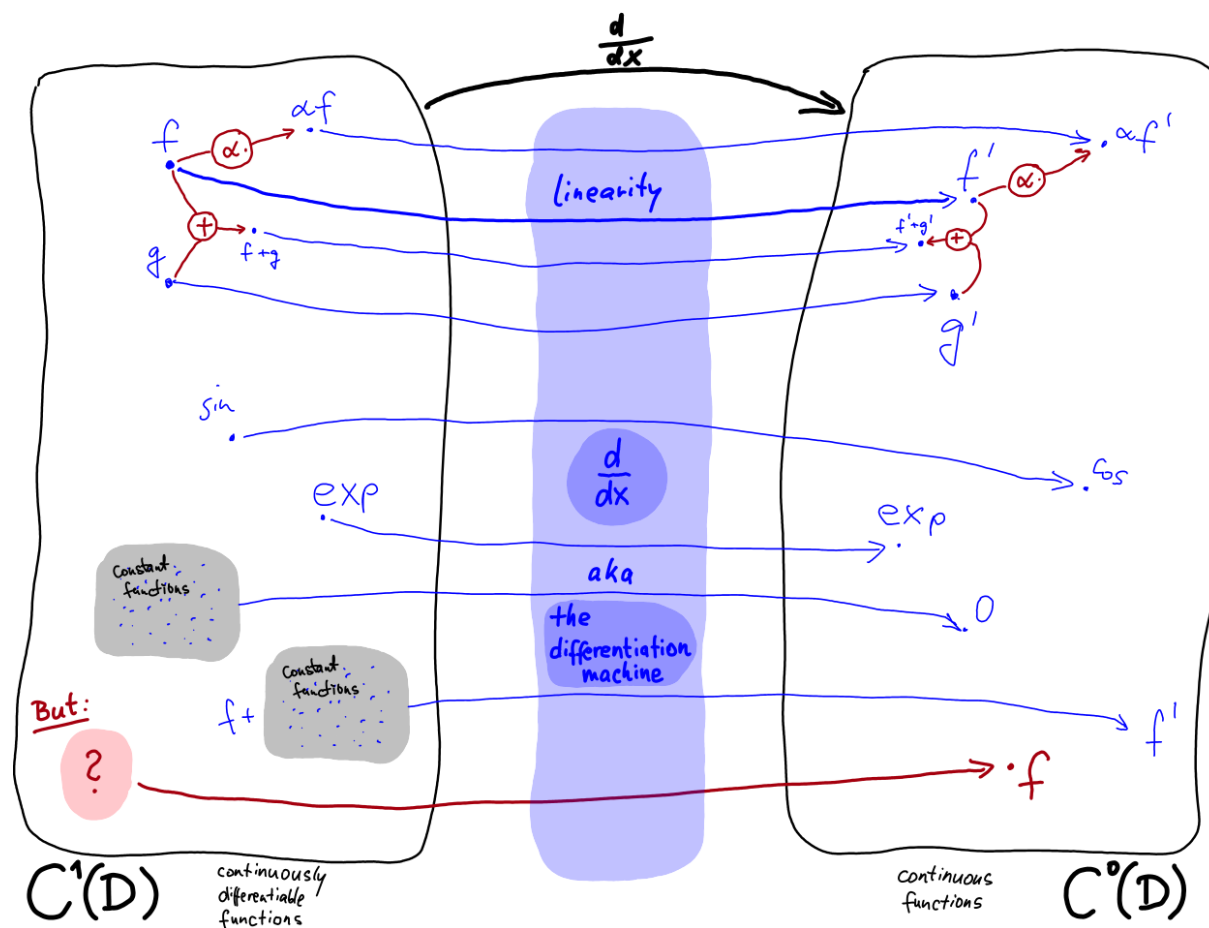
Definition 8.1. antiderivative

Let D be an open interval in $\mathbb{R}$ and $f : D \rightarrow \mathbb{R}$. A function $F : D \rightarrow \mathbb{R}$ is called an antiderivative of f if F is differentiable on D and $F' = f$ holds.

Example 8.2. two antiderivatives

Both $F_1 : \mathbb{R} \rightarrow \mathbb{R}$ and $F_2 : \mathbb{R} \rightarrow \mathbb{R}$ given by $F_1(x) = \frac{x^3}{3}$ and $F_2(x) = \frac{x^3}{3} + 5$ are both antiderivatives $\mathbb{R} \rightarrow \mathbb{R}$ of $f(x) = x^2$.

We must therefore learn to run the differentiation mechanism $\frac{d}{dx}$ "backwards". To do this, we use as many properties of the mapping $\frac{d}{dx}$ as possible:



Proposition 8.3. the antiderivative is unique up to "+C"

If F is an antiderivative of f on D , then...

- ...every function G on D of the form $G(x) = F(x) + C$ with a constant $C \in \mathbb{R}$ is also an antiderivative of f
- and every antiderivative of f is of this form.

Proof: a) is clear:

b) Assume F, G are both antiderivatives of f , i.e.,

■

Because of this, the set of all antiderivatives of f is of the form:

$$\{ \text{some antiderivative } F \text{ of } f + \text{all possible constants } C \}.$$

The common notation for this is $F + C$, and it describes the set $\{F + C : C \in \mathbb{R}\}$.

 **Nerd box 8.4.** Don't we already know this pattern from linear algebra?

a) The set of all solutions of an inhomogeneous linear equation $\mathbf{Ax} = \mathbf{b}$ is

$$\begin{aligned} & \{ \text{some ("particular") solution of } \mathbf{Ax} = \mathbf{b} \\ & \quad + \text{all possible solutions of the (homogeneous) equation } \mathbf{Ax} = \mathbf{0} \} \\ &= \text{some ("particular") solution of } \mathbf{Ax} = \mathbf{b} + \text{Kern}(\mathbf{A}). \end{aligned}$$

This is not a coincidence. Because the differential operator $\frac{d}{dx}$ is linear, we are basically doing linear algebra right now. Our “ $+C$ ” is the shorthand for $+\text{Kern}(\frac{d}{dx})$.

b) In case D is a union of multiple intervals, the “constant function” C can have a different value on each interval, because $C' = 0$ still holds. In this case, we speak of a locally constant function.

For the **set of all** antiderivatives of f , we introduce a new term:

Definition 8.5. the indefinite integral

By the phrase indefinite integral of f , written as

$$\int f(x) dx,$$

we mean the set of all antiderivatives of f .

So it holds $\int f(x) dx = F(x) + C$ with some antiderivative F of f .

The interval D , on which the function is defined, is omitted in this notation most of the time (it is clear from the context).

8.1.2 Integration rules and basic integrals

Let $F' = f$ and $G' = g$. From the fundamental rules of differentiation, there follow the...

rules of integration

- *sum:* $(F + G)' = F' + G'$

$$\begin{aligned}\Rightarrow \int (f + g) dx &= \int (F' + G') dx = \int (F + G)' dx = F + G + C \\ &= \int f dx + \int g dx\end{aligned}$$

- *scaling:* $(\alpha F)' = \alpha \cdot F'$

$$\Rightarrow \int \alpha \cdot f dx = \int \alpha \cdot F' dx = \int (\alpha F)' dx = \alpha \cdot F + C = \alpha \cdot \int f dx$$

- *product rule:* $(uv)' = u'v + uv'$

$$(uv)' - uv' = u'v$$

$$\Rightarrow \int u'v dx = \int ((uv)' - uv') dx = uv - \int uv' dx$$

integration by parts or partial integration

- *chain rule:* $\frac{d}{dx}(F(g(x))) = F'(g(x)) \cdot g'(x)$

$$\begin{aligned}\Rightarrow \int f(g(x)) \cdot g'(x) dx &= \int F'(g(x)) \cdot g'(x) dx = \int \frac{d}{dx}(F(g(x))) dx \\ &= F(g(x)) + C\end{aligned}$$

substitution

In contrast to differentiation, we see that:

Attention!

There are no concrete formulas for:

$$\int f(x) \cdot g(x) dx, \quad \int \frac{f(x)}{g(x)} dx \quad \text{or} \quad \int f(g(x)) dx$$

This makes things significantly more tedious. The antiderivatives of, for example,

$$x \mapsto e^{-x^2}, \quad x \mapsto \frac{\sin x}{x}, \quad x \mapsto \cos(x^2), \quad x \mapsto \frac{e^x}{x}, \quad x \mapsto \frac{1}{\ln x}$$

can (provably!) not be represented using a finite number of applications of $+$, $-$, $\cdot$, $:$ and $\circ$ from polynomials, roots, exponential, logarithmic, and trigonometric functions.

With the four integration rules above, however, you can still get quite far. In contrast to

$$\int \cos(x^2) dx \quad \text{we can, for example, compute:} \quad \int \cos(x^2) \cdot x dx$$

Example 8.6. substitution rule in action

The trick is to work with the substitution rule here because we observe that the factor x is approximately (upto a factor of 2) the derivative of the "inner function" $g(x) = x^2$:

$$\int \cos(x^2) \cdot x dx =$$

In the end, we still had to transition from $f = \cos$ to its antiderivative $F = \sin$.

As a check, it is always advisable to differentiate the result,

$$\frac{d}{dx} \left(\frac{1}{2} \sin(x^2) + C \right) = \frac{1}{2} \sin'(\underbrace{x^2}_{g(x)}) \cdot \underbrace{2x}_{g'(x)} = \cos(x^2) \cdot x,$$

in the hope that we retrieve the original function. Check passed.

Such derivative examples like $\sin' = \cos$ should always be kept in mind in order to be used here in the reverse direction: from $\cos$ to its antiderivative $\sin$. Let's collect everything we already know in this direction:

basic integrals

- $\int x^a dx =$
 - for $a \in \{0, 1, 2, \dots\}$ and $D = \mathbb{R}$,
 - for $a \in \{-2, -3, -4, \dots\}$ and $D = (-\infty, 0)$ as well as $D = (0, +\infty)$,
 - for $a \in \mathbb{R} \setminus \{-1\}$ and $D = (0, +\infty)$,
 - however not for $a = -1$. In that case:
- $\int x^{-1} dx =$ for $D = (-\infty, 0)$ and $D = (0, +\infty)$. Or like this:

$$\int x^{-1} dx = \begin{cases} \ln x + C & \text{for } D = (0, +\infty), \\ \ln(-x) + C & \text{for } D = (-\infty, 0). \end{cases}$$
- $\int e^{ax} dx =$ for $a \neq 0$ and $D = \mathbb{R}$.
- $\int a^x dx =$ for $a > 0$ with $a \neq 1$ and $D = \mathbb{R}$.
- $\int \sin x dx =$ for $D = \mathbb{R}$.
- $\int \cos x dx =$ for $D = \mathbb{R}$.
- $\int \frac{1}{1+x^2} dx =$ for $D = \mathbb{R}$.
- $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$
 $= \frac{\pi}{2} - \arccos x + C = -\arccos x + \tilde{C}$ for $D = (-1, 1)$.

Time for examples¹:

¹People used to say that to learn integration properly, one must have solved 500 integrals. Let's start with eight of them.

Example 8.7. some indefinite integrals

a) Using the sum and scaling rules, as well as two basic integrals, we solve:

$$\begin{aligned}\int x^3 + \sqrt{x} - 5 \sin x \, dx &= \\ &= \\ &= \frac{1}{4}x^4 + \frac{2}{3}\sqrt{x^3} + 5 \cos x + C\end{aligned}$$

b) Using the substitution rule, we have already seen (example 8.6):

$$\int \cos(x^2) \cdot x \, dx = \frac{1}{2} \sin(x^2) + C$$

c) Now let's change from x^2 to x in the cosine:

$$\int \cos(x) \cdot x \, dx$$

Now the second factor, x , is no longer the derivative of the inner function, so substitution no longer works. What's left is integration by parts:

$$\int u'v \, dx = uv - \int uv' \, dx.$$

$$\int \underbrace{\cos x} \cdot \underbrace{x} \, dx = \sin x \cdot x + \cos x + C$$

The price for getting rid of (differentiating) the factor x is that we had to integrate the other factor, $\cos x$. But we know how to do that.

d) For the following integral, substitution is again out of discussion. So, integration by parts it is:

$$\begin{aligned}\int \cos^2 x \, dx &= \\ &= \sin x \cdot \cos x + \int \sin x \cdot \sin x \, dx\end{aligned}$$

If one integrates the new problem, $\int \sin x \cdot \sin x \, dx$, again by parts, one goes back the above way and ends up with $\int \cos^2 x \, dx = \int \cos^2 x \, dx$. Bummer.

Better idea: We use that $\sin^2 x + \cos^2 x = 1$ and rewrite $\int \sin^2 x \, dx$ accordingly:

$$\int \cos^2 x \, dx = \sin x \cdot \cos x + \int \underbrace{(1 - \cos^2 x)}_{\sin^2 x} \, dx =$$

Now we got back to $\int \cos^2 x \, dx$ with our right hand side – but with a minus sign in front! Let's bring it to the other side and divide by two:

- e) *Coming from the previous task, let's bring in another factor, $\sin x$, under the integral:*

$$\int \cos^2 x \cdot \sin x \, dx = \int (\cos x)^2 \cdot \sin x \, dx$$

This simplifies things considerably, because the new factor is (except for a factor of -1) the derivative of the inner function $g(x) = \cos x$:

$$\int (\cos x)^2 \cdot \sin x \, dx = \quad \quad \quad = - \int f(g(x)) \cdot g'(x) \, dx,$$

where $f(y) = y^2$. According to the substitution rule, the result is $F(g(x)) + C$, where F is an antiderivative of f , e.g., $F(y) = \frac{1}{3}y^3$. We summarize:

$$\int \cos^2 x \cdot \sin x \, dx = \dots = -\frac{1}{3}(\underbrace{\cos x}_{g(x)})^3 + C = -\frac{1}{3}\cos^3 x + C$$

Between us: substitutions are often written this way:

Let $t := g(x)$. Then $\frac{dt}{dx} = g'(x)$ and thus

$$\int f(g(x)) \cdot g'(x) dx = \int f(t) \cdot \frac{dt}{dx} dx$$

$$\text{and } F(g(x)) + C = F(t) + C = \int f(t) dt.$$

According to the substitution rule, both left sides are equal – thus the right sides are so, too. Thus, one may write:

$$\frac{dt}{dx} dx = dt$$

That the whole thing looks like simple fraction arithmetic is thanks to Leibniz's ingenious and suggestive notation for differentiation and integration. (While Newton was having apples drop on his head.)

The last example again with $t := \cos x$, such that $\frac{dt}{dx} = -\sin x$:

f) In the next example, we substitute $t := \sin x$, such that $\frac{dt}{dx} = \cos x$:

$$\int \cot x dx =$$

=

g) For the next integral, initially nothing comes to mind. Then sometimes the following trick helps along with integration by parts:

$$\int \ln x \, dx =$$

=

h) Finally, we substitute $t := 3x - 7$, so that $\frac{dt}{dx} = 3$.

$$\int \frac{1}{3x - 7} \, dx =$$

The derivative $\frac{dt}{dx} = 3$ was not initially in the desired place in the original integral, but such a constant can be easily produced and cancelled elsewhere. This is the typical trick with such linear substitutions.

8.1.3 Rational functions

In fact, we can also integrate any rational functions, i.e., functions of the form

$$f = \frac{p}{q} = \frac{\text{polynomial}}{\text{polynomial}}.$$

As a small preview, we begin with this example:

Example 8.8. Integral of a simple rational function

$$\int \frac{2x + 1}{x^2 + x} \, dx = \int \frac{1}{x} + \frac{1}{x + 1} \, dx = \ln |x| + \ln |x + 1| + C$$

The most challenging part was the decomposition

$$\frac{2x + 1}{x^2 + x} =$$

a so-called *partial fraction decomposition*.

The roots of the denominator polynomial play an important role in this decomposition. Additionally, it was important that the degree of the numerator polynomial was smaller than that of the denominator. In more general examples, much more needs to be done. We illustrate this by the next example:

Example 8.9. Integral of a more complicated rational function

Let $p(x) = 2x^4 + x^3 - 2x^2 - 8x + 13$ and $q(x) = x^3 - 3x + 2$. We are looking for

$$\int \frac{p(x)}{q(x)} dx = \int \frac{2x^4 + x^3 - 2x^2 - 8x + 13}{x^3 - 3x + 2} dx.$$

1.) Since the degree of p is greater than (or equal to) that of q , we begin with polynomial division and obtain

$$(2x^4 + x^3 - 2x^2 - 8x + 13) : (x^3 - 3x + 2) =$$

$$p(x) : q(x) = 2x + 1, \quad \text{remainder: } r(x) = 4x^2 - 9x + 11,$$

so we have

$$\frac{p(x)}{q(x)} = 2x + 1 + \frac{r(x)}{q(x)} \quad \text{with} \quad \text{Grad}(r) = 2 < 3 = \text{Grad}(q). \quad (8.2)$$

Since $\int 2x + 1 dx$ is straightforward, we only focus on $\int \frac{r(x)}{q(x)} dx$.

2.) The three roots of q are 1, 1 and -2 . (For example, $q(1) = 0$ by trying, and the rest by polynomial division $q(x) : (x - 1)$.) Thus, $q(x) = (x - 1)^2 \cdot (x + 2)$.

3.) According to a theorem on the so-called partial fraction decomposition (see literature), $\frac{r(x)}{q(x)}$ can be written as

$$\frac{r(x)}{q(x)} = \frac{4x^2 - 9x + 11}{(x-1)^2 \cdot (x+2)} = \frac{A_1}{x-1} + \frac{A_2}{(x-1)^2} + \frac{B_1}{x+2} \quad (8.3)$$

with appropriate constants $A_1, A_2, B_1 \in \mathbb{R}$.

4.) The constants A_1, A_2, B_1 can be found by multiplying (8.3) by $q(x)$,

$$(8.4)$$

and plugging in the roots of q , here $x = 1$ and $x = -2$, into (8.4) :

$$\begin{aligned} r(1) &= 6 = \\ r(-2) &= 45 = \end{aligned}$$

From this, we immediately obtain $A_2 = 2$ and $B_1 = 5$. For A_1 we need another equation, e.g. (8.4) with $x = 0$:

$$r(0) = 11 =$$

This results in $A_1 = -1$. Therefore, all the ingredients of (8.3) are known.

5.) With (8.2) and (8.3) we are almost there: $\int \frac{p(x)}{q(x)} dx$

$$\begin{aligned} &= \int 2x + 1 \quad + \quad \frac{-1}{x-1} \quad + \quad \frac{2}{(x-1)^2} \quad + \quad \frac{5}{x+2} \quad dx \\ &= \int 2x + 1 \, dx - \int \frac{1}{x-1} \, dx + 2 \int (x-1)^{-2} \, dx + 5 \int \frac{1}{x+2} \, dx \\ &= \frac{x^2 + x}{x^2 + x} - \ln |x-1| - \frac{2}{2(x-1)^{-1}} + 5 \ln |x+2| + C \end{aligned}$$

This example already covers the general case quite well. A few more comments:

- In step 4, one could expand the polynomial on the right side of (8.4) first, then sort by monomials and then compare the coefficients in front of x^2, x^1, x^0 on both sides, to get a system of equations to determine A_1, A_2, B_1 .
- In step 3, for each real root x_0 of q we have the terms

$$\frac{A_1}{(x-x_0)^1} + \frac{A_2}{(x-x_0)^2} + \cdots + \frac{A_k}{(x-x_0)^k} \quad \text{with} \quad A_1, A_2, \dots, A_k \in \mathbb{R} \quad (8.5)$$

(partial fractions of the 1st kind), where k is the multiplicity of x_0 in q .

- In step 2, it may happen that q has fewer real roots than its degree. The missing roots are then complex. You can either continue with the partial fraction decomposition using complex numbers, or argue as follows: Complex roots always come in pairs, say z_1 and z_2 , and it holds

$$(x - z_1) \cdot (x - z_2) = x^2 + bx + c \quad \text{with certain numbers} \quad b, c \in \mathbb{R},$$

where $\frac{b^2}{4} - c < 0$, because $x \mapsto x^2 + bx + c$ has no real root. Then the corresponding terms in the partial fraction decomposition in step 3, instead of being of the form (8.5), are of the following form

$$\frac{B_1x + C_1}{(x^2 + bx + c)^1} + \frac{B_2x + C_2}{(x^2 + bx + c)^2} + \dots + \frac{B_kx + C_k}{(x^2 + bx + c)^k} \quad (8.6)$$

with $B_1, \dots, B_k, C_1, \dots, C_k \in \mathbb{R}$, where k is again the multiplicity of the pair of roots (z_1, z_2) . One talks about partial fractions of the 2nd kind.

For the first term in (8.6) we still dare to perform the integration (step 5):

Example 8.10. partial fractions of the 2nd kind

a) *The topic is*

$$\int \frac{B_1x + C_1}{x^2 + bx + c} dx \quad \text{with} \quad B_1, C_1, b, c \in \mathbb{R} \quad \text{and} \quad \frac{b^2}{4} - c < 0.$$

We substitute the denominator, $t := x^2 + bx + c$, and now try to express the numerator, $B_1x + C_1$, as closely as possible as the derivative $\frac{dt}{dx}$:

$$B_1x + C_1 =$$

Thus we have

$$\begin{aligned} \int \frac{B_1x + C_1}{x^2 + bx + c} dx &= \int \frac{\frac{B_1}{2} \frac{dt}{dx} \cancel{dx}}{t} + \int \frac{-\frac{B_1b}{2} + C_1}{x^2 + bx + c} dx \\ &= \frac{B_1}{2} \ln |x^2 + bx + c| + \left(-\frac{B_1b}{2} + C_1 \right) \int \frac{1}{x^2 + bx + c} dx. \end{aligned}$$

b) Part a) still owes us the following integral:

$$\int \frac{1}{x^2 + bx + c} dx = \int \frac{1}{\left(x + \frac{b}{2}\right)^2 + \left(-\frac{b^2}{4} + c\right)} dx = \int \frac{1}{t^2 + a^2} dt,$$

where we substitute $t := x + \frac{b}{2}$ ($\frac{dt}{dx} = 1$) and set $a := \sqrt{-\frac{b^2}{4} + c}$.

c) Time to solve it:

$$\int \frac{1}{t^2 + a^2} dt = \int \frac{1}{a^2 \left(\left(\frac{t}{a} \right)^2 + 1 \right)} dt = \frac{1}{a} \int \frac{\frac{1}{a}}{\left(\frac{t}{a} \right)^2 + 1} dt \quad \text{with} \quad a > 0.$$

If we substitute $u := \frac{t}{a}$, such that $\frac{du}{dt} = \frac{1}{a}$, then we get

$$\int \frac{1}{t^2 + a^2} dt = \frac{1}{a} \int \frac{\frac{du}{\cancel{dt}}}{u^2 + 1} \cancel{dt} = \frac{1}{a} \arctan u + C = \frac{1}{a} \arctan \left(\frac{t}{a} \right) + C.$$

It remains to insert the integrals from bottom to top.

The following terms in (8.6) are no fun to solve anymore. The formulas for these can be derived by partial integration. (They can be verified by differentiation.)

🧐 Nerd box 8.11. for the sake of completeness...

The next integrals in (8.6) turn out to be even more extravagant:

$$\int \frac{B_j x + C_j}{(x^2 + bx + c)^j} dx = \frac{-B_j}{2(j-1)(x^2 + bx + c)^{j-1}} + \left(C_j - \frac{bB_j}{2} \right) \underbrace{\int \frac{1}{(x^2 + bx + c)^j} dx}_{\text{see below}}$$

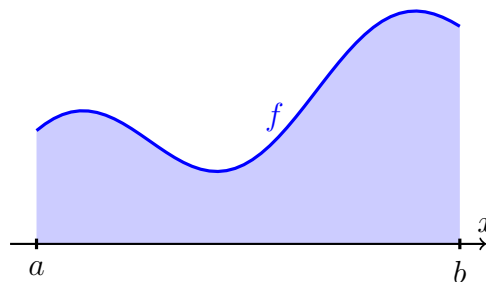
$$\int \frac{1}{(x^2 + bx + c)^j} dx = \frac{2x + b}{(j-1)(4c - b^2)(x^2 + bx + c)^{j-1}} + \frac{2(2j-3)}{(j-1)(4c - b^2)} \underbrace{\int \frac{1}{(x^2 + bx + c)^{j-1}} dx}_{\text{analogously, until } j=1}$$

8.2 The definite integral: Riemann integration

The purpose of the definite integral, written as

$$\int_a^b f(x) dx,$$

is to calculate the (signed) area between the graph of a function $f : [a, b] \rightarrow \mathbb{R}$ and the x -axis.



The Riemann integral is a common way to solve this problem. The close connection between the definite and the indefinite integral (not only in the name, but also in the notation) will be resolved in the next chapter.

8.2.1 Definitions and examples

Let $[a, b]$ be a bounded and closed interval with $a < b$. To calculate the area we are looking for, we divide the interval $[a, b]$ into many small pieces and replace each piece of the area by a rectangle.

Definition 8.12. partition of an interval

A set $P = \{x_0, x_1, \dots, x_n\}$ is called a partition of $[a, b]$ if

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b.$$

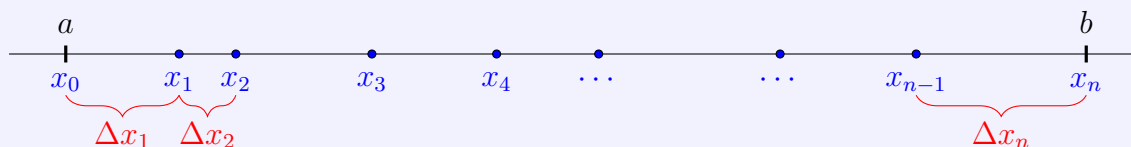
By

$$\Delta x_i := x_i - x_{i-1}, \quad i = 1, 2, \dots, n$$

we denote the lengths of the subintervals, and let

$$\mu(P) := \max_{i=1}^n \Delta x_i$$

refer to the so-called fineness of the partition P .



⚠ Attention!

We give ourselves the freedom to choose the length of each interval Δx_i independently.

The fineness $\mu(P)$ is in some sense the “biggest gap” in the partition – comparable to the weakest link in a chain. From $\mu(P) < \delta$ we conclude $\Delta x_i < \delta$ for all $i \in \{1, \dots, n\}$.

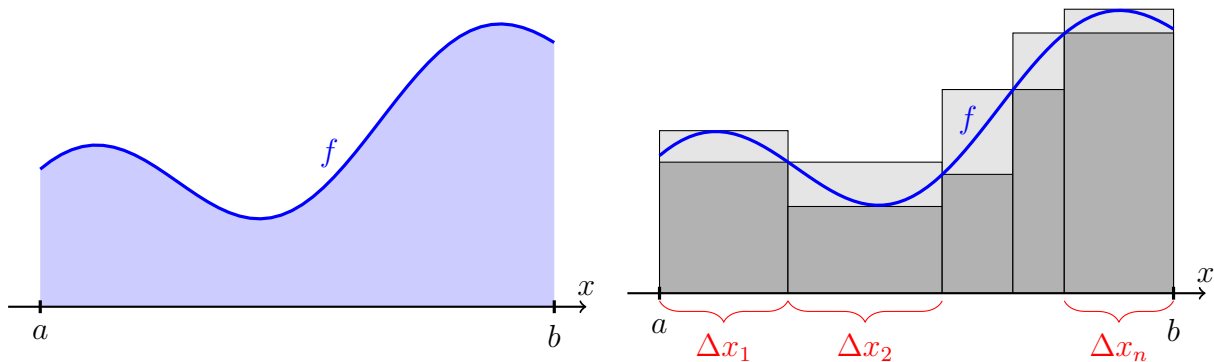
Definition 8.13. lower and upper sum

For a given partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ and $f : [a, b] \rightarrow \mathbb{R}$, let

$$\underline{\Sigma}(f, P) := \sum_{i=1}^n \left(\inf_{x \in [x_{i-1}, x_i]} f(x) \right) \Delta x_i \quad \text{and} \quad \overline{\Sigma}(f, P) := \sum_{i=1}^n \left(\sup_{x \in [x_{i-1}, x_i]} f(x) \right) \Delta x_i$$

be, respectively, the so-called lower sum and upper sum of f with respect to P .

In the left image, we see the **area we want to calculate** and on the right the **lower** and **upper sum** for a given function f and a partition P . The upper sum is represented by the whole grey column (**light grey** + **dark grey**):

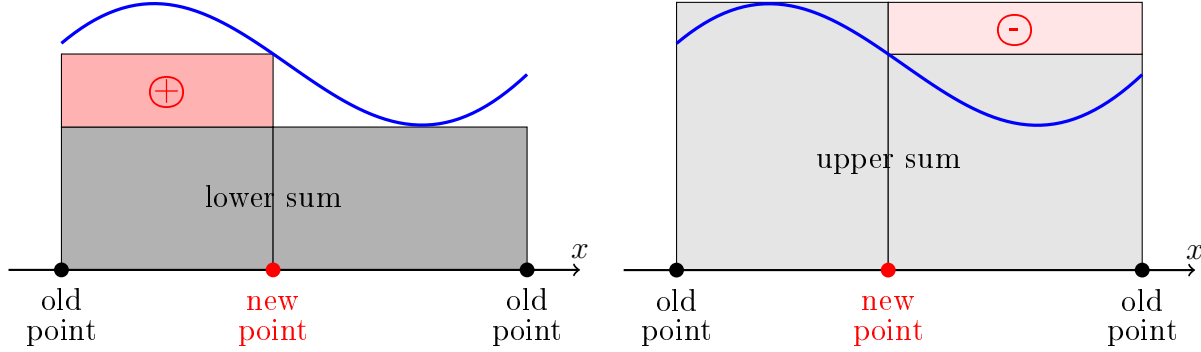


For the areas depicted left and right, we can see that

$$\text{lower sum} \leq \text{area we want to calculate} \leq \text{upper sum.} \quad (8.7)$$

For “nice” functions f , we can make this sandwich (8.7) of the **area we want to calculate** arbitrarily tight via letting the fineness of the partition P go to zero, $\mu(P) \rightarrow 0$. Instead of “nice” we will soon say “Riemann integrable”. But first:

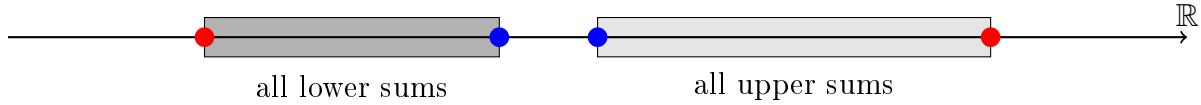
If we add a point to an existing partition P , the lower sum increases (or does not change) and the upper sum decreases (or does not change):



Therefore, for two arbitrary partitions P_1 and P_2 of $[a, b]$:

$$\underline{\Sigma}(f, P_1) \leq \underline{\Sigma}(f, P_2) \quad \text{and} \quad \bar{\Sigma}(f, P_1) \geq \bar{\Sigma}(f, P_2)$$

For a fixed function f , the set of all upper and lower sums looks a bit like this:



The two red dots are

$$\inf_{x \in [a, b]} f(x) \cdot (b - a) \quad \text{and} \quad \sup_{x \in [a, b]} f(x) \cdot (b - a),$$

while the blue dots are the so called lower integral as well as the so called upper integral of f on $[a, b]$, that is,

$$\int_a^b f(x) dx := \sup_P \underline{\Sigma}(f, P) \quad \text{and} \quad \int_a^b f(x) dx := \inf_P \bar{\Sigma}(f, P).$$

Proposition 8.14. Equality of the upper and lower sums

For a bounded function $f : [a, b] \rightarrow \mathbb{R}$, the following are equivalent:

- (i) $\int_a^b f(x) dx = \int_a^b f(x) dx$
- (ii) For every $\varepsilon > 0$ there is a partition P of $[a, b]$, such that $\bar{\Sigma}(f, P) - \underline{\Sigma}(f, P) < \varepsilon$.
- (iii) $\bar{\Sigma}(f, P) - \underline{\Sigma}(f, P) \rightarrow 0$ as $\mu(P) \rightarrow 0$.

Proof: See literature. ■

Definition 8.15. Riemann integrability and the Riemann integral

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is called *Riemann integrable*, if one (and therefore all) of the statements in Proposition 8.14 hold. In this case we call

$$\int_a^b f(x) dx := \int_a^b f(x) dx = \int_a^b f(x) dx$$

its Riemann integral. Let now $\mathcal{R}([a, b])$ denote the set of all Riemann integrable functions on $[a, b]$.

The name of the integration variable is not important for this. We have

$$\int_a^b f(x) dx = \int_a^b f(y) dy = \int_a^b f(t) dt = \dots$$

We start with an example where the lower and upper integrals are different.

Example 8.16. The Dirichlet function is not Riemann integrable

Let $[a, b] = [0, 1]$ and

$$f(x) = \begin{cases} 1, & x \text{ rational,} \\ 0, & x \text{ irrational,} \end{cases}$$

be the so called Dirichlet function.

For each partition P of $[0, 1]$, we have

$$\underline{\Sigma}(f, P) = \quad \text{and} \quad \overline{\Sigma}(f, P) =$$

since each subinterval $[x_{i-1}, x_i]$ contains rational as well as irrational numbers x . Therefore we have

$$\int_a^b f(x) dx = \sup_P 0 = 0, \quad \text{while} \quad \int_a^b f(x) dx = \inf_P 1 = 1,$$

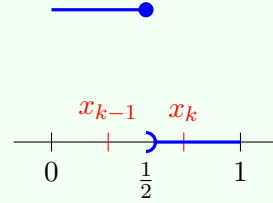
i.e., the statement (i) in Proposition 8.14 does not hold, so $f \notin \mathcal{R}([0, 1])$.

The function in the last example had too many jumps between 0 and 1. For a function with only one (or maybe finitely many) jumps, we still have Riemann integrability:

Example 8.17. A piecewise constant function

Let $[a, b] = [0, 1]$ again and this time let

$$f(x) = \begin{cases} 1, & x \in [0, \frac{1}{2}], \\ 0, & x \in (\frac{1}{2}, 1]. \end{cases}$$



If P is a partition of $[0, 1]$ and k is such that $\frac{1}{2} \in [x_{k-1}, x_k]$ then we have

$$\underline{\Sigma}(f, P) = \underbrace{\sum_{i=1}^{k-1} \inf_{x \in [x_{i-1}, x_i]} f(x) \Delta x_i}_{\text{first part}} + \underbrace{\inf_{x \in [x_{k-1}, x_k]} f(x) \Delta x_k}_{\text{middle part}} + \underbrace{\sum_{i=k+1}^n \inf_{x \in [x_{i-1}, x_i]} f(x) \Delta x_i}_{\text{last part}},$$

$$\overline{\Sigma}(f, P) = \underbrace{\sum_{i=1}^{k-1} \sup_{x \in [x_{i-1}, x_i]} f(x) \Delta x_i}_{\text{first part}} + \underbrace{\sup_{x \in [x_{k-1}, x_k]} f(x) \Delta x_k}_{\text{middle part}} + \underbrace{\sum_{i=k+1}^n \sup_{x \in [x_{i-1}, x_i]} f(x) \Delta x_i}_{\text{last part}}.$$

Because of

$$0 \leq \overline{\Sigma}(f, P) - \underline{\Sigma}(f, P) = \Delta x_k \leq \mu(P),$$

$\mu(P) \rightarrow 0$ implies $\overline{\Sigma}(f, P) - \underline{\Sigma}(f, P) \rightarrow 0$. So statement (iii) in Proposition 8.14 holds, i.e. $f \in \mathcal{R}([0, 1])$. To be precise: For $\mu(P) \rightarrow 0$ we have

$$\underline{\Sigma}(f, P) = x_{k-1} \rightarrow \frac{1}{2} \leftarrow x_{k-1} + \Delta x_k = \overline{\Sigma}(f, P), \quad \text{so that} \quad \int_0^1 f(x) dx = \frac{1}{2}.$$

8.2.2 Criteria and properties

Checking whether or not a function is Riemann integrable is already tedious for a function like the one in Example 8.17. Because of this we are happy about the following proposition:

Proposition 8.18. Monotonic and continuous functions are Riemann integrable

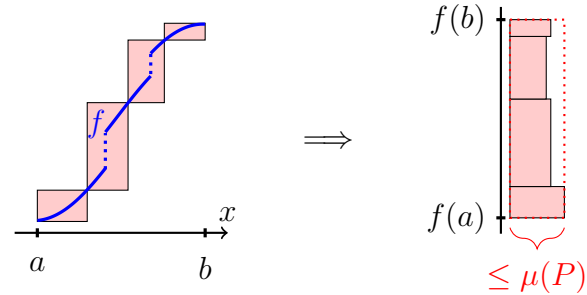
- a) Bounded monotonic functions are Riemann integrable.
- b) Continuous functions are Riemann integrable.

Proof: We will show in either case that $\overline{\Sigma}(f, P) - \underline{\Sigma}(f, P) \rightarrow 0$ if $\mu(P) \rightarrow 0$. The difference between the upper sums and the lower sums is now marked in red.

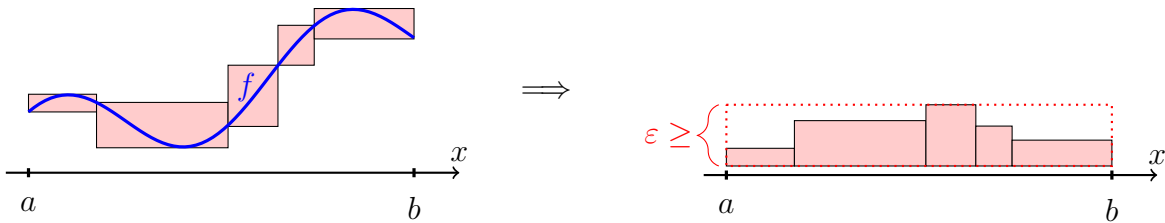
a) Let f be monotonic and bounded. We will move the red boxes to the y -axis. Now we see, that

$$\overline{\Sigma}(f, P) - \underline{\Sigma}(f, P) \leq |f(a) - f(b)| \cdot \mu(P),$$

so that $\overline{\Sigma}(f, P) - \underline{\Sigma}(f, P) \rightarrow 0$ as $\mu(P) \rightarrow 0$ and thus $f \in \mathcal{R}([a, b])$.



b) Being continuous on $[a, b]$, f is already uniformly continuous (see Nerd box 5.20). So we know that on subintervals of length δ , f cannot vary by more than ε , where ε is dependent on δ (but independent of the position of the δ -interval) and $\varepsilon \rightarrow 0$ if $\delta \rightarrow 0$. This time we move the red boxes to the x -axis:



Obviously, $\overline{\Sigma}(f, P) - \underline{\Sigma}(f, P) \leq (b - a) \cdot \varepsilon$ holds, while $\varepsilon \rightarrow 0$ as $\mu(P) = \delta \rightarrow 0$. ■

So continuity is sufficient for Riemann integrability – but not necessary. It may be violated up to a certain degree. To be precise:

 **Nerd box 8.19.** Precise characterization of Riemann integrability

A function is Riemann integrable if and only if M , being the set of all its points of discontinuity, has measure zero, i.e., for all $\varepsilon > 0$, we can cover the set M with several intervals with overall length ε .

For the Riemann integral, we again have the following rules:

Proposition 8.20. Composition of Riemann integrable functions

Let $f, g \in \mathcal{R}([a, b])$ and $\alpha \in \mathbb{R}$. Then we have:

a) $f \pm g \in \mathcal{R}([a, b])$ and $\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$

b) $\alpha f \in \mathcal{R}([a, b])$ and $\int_a^b \alpha \cdot f(x) dx = \alpha \cdot \int_a^b f(x) dx$

c) If also $f \in \mathcal{R}([b, c])$, then $f \in \mathcal{R}([a, c])$ and it holds that

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx.$$

d) If $f \leq g$, i.e. $f(x) \leq g(x)$ for all $x \in [a, b]$, then also

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

e) If $W \subset \mathbb{R}$ is the range of f and $h : W \rightarrow \mathbb{R}$ is continuous then $h \circ f \in \mathcal{R}([a, b])$.

Proof: a) to d) follow from definition 8.15. Part e) is more difficult. See literature. ■

Because of Proposition 8.20 e), also $f^2, f^3, \dots$ are Riemann integrable as soon as f is. You can see this by setting $h(y) = y^2$ and y^3 and so on.

Corollary 8.21. It also works with products.

With $f, g \in \mathcal{R}([a, b])$ also their product, $f \cdot g$, is Riemann integrable.

Proof: The trick is to combine the first and second binomial formula, like this:

The terms on the right hand side are covered by Proposition 8.20 with $h(y) = y^2$. ■

After 1, 2, $\dots$ and even infinitely many summations, we can now also recover the triangle inequality for integrals:

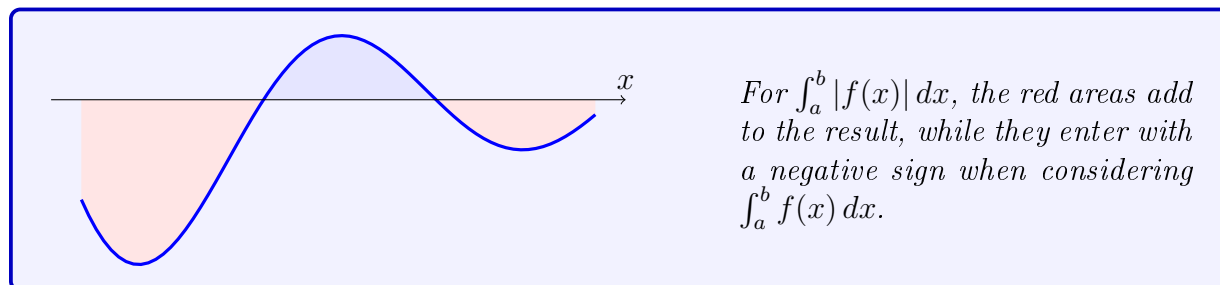
Corollary 8.22. triangle inequality for integrals

With $f \in \mathcal{R}([a, b])$ also $|f|$ is Riemann integrable and we have

$$\int_a^b |f(x)| dx \geq \left| \int_a^b f(x) dx \right|.$$

Proof: $|f| \in \mathcal{R}([a, b])$ by Proposition 8.20 e) with $h(y) = |y|$. By Proposition 8.20 d),

$$-|f(x)| \leq f(x) \leq |f(x)| \implies -\int_a^b |f(x)| dx \leq \int_a^b f(x) dx \leq \int_a^b |f(x)| dx. \quad \blacksquare$$



Moreover, put

$$\int_b^a f(x) dx := - \int_a^b f(x) dx, \quad \text{so that} \quad \int_a^a f(x) dx = - \int_a^a f(x) dx = 0.$$

We conclude this section with a close relative of the mean value theorem of differential calculus:

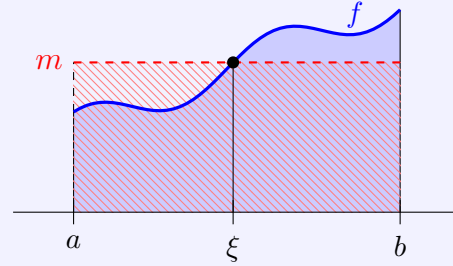
Proposition 8.23. mean value theorem for integrals (MVT.I)

Let $f, g \in \mathcal{R}([a, b])$ and $g \geq 0$. Then there is an $m \in \left[\inf_{x \in [a, b]} f(x), \sup_{x \in [a, b]} f(x) \right]$ with

$$\int_a^b f(x)g(x) dx = m \cdot \int_a^b g(x) dx.$$

In the special case of $g \equiv 1$, we have:

$$\int_a^b f(x) dx = m \cdot (b - a).$$



If f is even continuous, $m = f(\xi)$ for some point $\xi \in [a, b]$.

The numbers

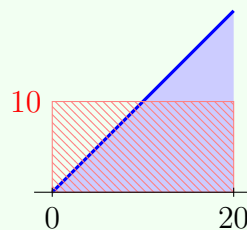
$$\frac{\int_a^b f(x) dx}{b - a} \quad \text{and} \quad \frac{\int_a^b f(x)g(x) dx}{\int_a^b g(x) dx}$$

are called average of f over $[a, b]$ and weighted average of f over $[a, b]$ with respect to the weight function g , respectively.

Example 8.24. average and weighted average

Let $f : [0, 20] \rightarrow \mathbb{R}$ with $f(x) = x$.

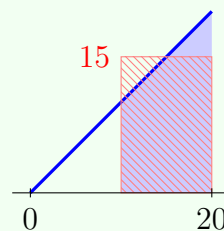
From our understanding (linear increase of values from 0 to 20) and geometrically (red area = blue area) it is clear that the average (without weight function, i.e. with weight $g \equiv 1$) is $m = 10$.



If we choose the weight function

$$g(x) = \begin{cases} 0, & x \in [0, 10), \\ 1, & x \in [10, 20], \end{cases}$$

then all $x \in [0, 10)$ have no weight in the ‘averaging’ process, while all $x \in [10, 20]$ have the same weight, 1 in this case, whence the weighted average is $m = 15$.



8.3 The fundamental theorem of calculus

In this section, we bring together the concepts from the previous two sections. Suppose we know two math superheroes:



- **$\int$ -woman:**
Knows how to calculate all the antiderivatives, i.e., indefinite integrals.
- **$\int_a^b$ -man:**
Knows how to calculate areas under curves, i.e., Riemann integrals, like no one else in the galaxy.



At first, it sounds like they don't have much in common (aside from the funny names). But then the following happens...

- Monday: $\int_a^b$ -man is sick. While he's not around, we teach $\int$ -woman how she can simply sort out his super-job as a by-product of hers. (fundamental theorem part 1)
- Tuesday: the other way round (fundamental theorem part 2)
- From Wednesday on, the two share an office as integration officers.

Theorem 8.25. Fundamental theorem of calculus

Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable, i.e., $f \in \mathcal{R}([a, b])$.

- **Part 1:** If F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) \, dx = F(b) - F(a) =: [F(x)]_a^b. \quad (8.8)$$

- **Part 2:** For all $x \in [a, b]$, we define

$$F(x) := \int_a^x f(t) \, dt. \quad (8.9)$$

Then F is continuous on $[a, b]$. At points $x \in [a, b]$, where f is continuous, F is even differentiable and $F'(x) = f(x)$ holds.

If f is continuous on the whole interval $[a, b]$, then F is differentiable on $[a, b]$ and $F' = f$, i.e., F is an antiderivative of f .

Remarks:

- The fundamental theorem builds the announced bridge between the indefinite integral (antiderivatives) and the definite integral (Riemann integral)...
- ...and thus also between differentiation and integration: part 1 deals with the integral of the derivative (f of F) and part 2 with the derivative of the integral.
- The lower endpoint of the integral $\int_a^x$ in (8.9) could also be another point $c \in [a, b]$ instead of a . The new antiderivative would then differ from the previous one only by the additive constant $C = \int_a^c f(x) dx$.
- Thanks to part 1, we no longer have to calculate Riemann integrals by Definition 8.15.
- Part 2 shows that, in particular, every continuous function has an antiderivative, namely (8.9).
- Now we can close the circle for reconstructing the fill level of the bathtub or gas tank from the cumulative inflow and outflow (rates of change), see (8.1).
- The concept of an antiderivative on $[a, b]$ requires the concept of a derivative on $[a, b]$ (which we had previously only considered on open sets D). In the boundary points, $x_0 = a$ or $x_0 = b$, $F'(x_0)$ is defined by the existing one-sided limit of the difference quotient $\frac{F(x) - F(x_0)}{x - x_0}$.
- Our superhero story and the fundamental theorem show that neither of the two jobs is harder than the other. Each task can be traced back to the other one. This is similarly the case, for example, with the determination of polynomial roots and eigenvalues. The field of complexity theory (in theoretical computer science) revolves around numerous computational problems and which one can be reduced to the other.

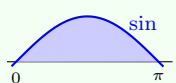
We should not forget to prove the fundamental theorem:

Proof: We prove the theorem for the special case where f is continuous on $[a, b]$. The general case, $f \in \mathcal{R}([a, b])$, is slightly (but not much) more difficult to show - see literature.

- **Part 2:** Let f be continuous, F given by (8.9), and let $x \in [a, b]$ be arbitrary. Then

- **Part 1:** Let f be continuous. With the antiderivative F from (8.9), we have

■

Example 8.26. The area under a sine bump

$$\int_0^{\pi} \sin x \, dx =$$

Due to part 1 of the fundamental theorem, the following rules apply to definite integrals:

- integration by parts:

$$\int_a^b u'(x)v(x)dx = [u(x)v(x)]_a^b - \int_a^b u(x)v'(x)dx$$

- integration by substitution:

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let $g : [\alpha, \beta] \rightarrow [a, b]$ be continuously differentiable. Then

$$\int_{\alpha}^{\beta} f(g(t)) \cdot g'(t) \, dt = \int_{g(\alpha)}^{g(\beta)} f(x) \, dx.$$

We will see both formulas in action in examples:

Example 8.27. for integration by parts

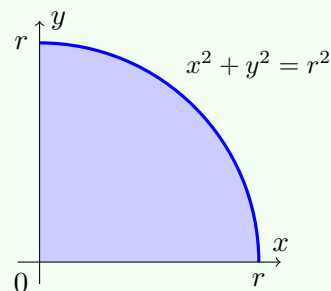
$$\begin{aligned} \int_0^{\pi} \underbrace{x}_{v(x)} \cdot \underbrace{\sin x}_{u'(x)} \, dx &= \left[\underbrace{(-\cos x)}_{u(x)} \cdot \underbrace{x}_{v(x)} \right]_0^{\pi} - \int_0^{\pi} \underbrace{(-\cos x)}_{u(x)} \cdot \underbrace{1}_{v'(x)} \, dx \\ &= \\ &= \end{aligned}$$

Example 8.28. for substitution: the area of a quarter disk

We are looking for the area of a quarter disk with radius $r > 0$. We place an x - y -coordinate system as in the image to the right.

The points (x, y) on the circle satisfy the equation $x^2 + y^2 = r^2$ (pythagorean theorem) and thus

$$y = \quad \quad \quad =: f(x) \in [0, r], \quad x \in [0, r].$$



The area A that we are looking for is then given by

$$A = \int_0^r f(x) dx = \int_0^r \sqrt{r^2 - x^2} dx.$$

Inspired by the geometry of the disk, we substitute

$$x = r \sin t =: g(t) \quad \text{with} \quad t \in \left[0, \frac{\pi}{2}\right] \quad \text{and thus} \quad g'(t) = r \cos t :$$

$$A =$$

$$=$$

$$\stackrel{*}{=} r^2 \frac{1}{2} \left[\sin t \cos t + t \right]_0^{\frac{\pi}{2}}$$

$$=$$

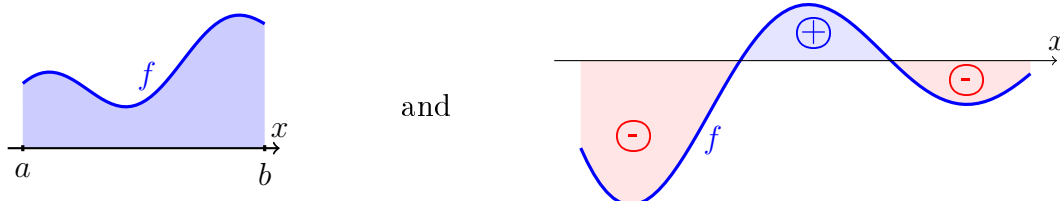
where $\stackrel{S}{=}$ uses substitution and $\stackrel{*}{=}$ uses the result of Example 8.7 d) .

Now we can also reconstruct the well-known formula stating that the area of a disk with radius r is equal to πr^2 .

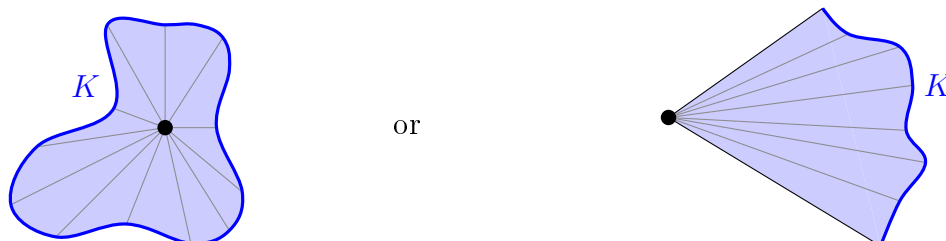
8.4 Some applications

8.4.1 Areas of sectors

After looking at areas like



between the graph and the x -axis, we will now learn about more interesting areas like



between a so called curve K and a point.

Definition 8.29. path, curve

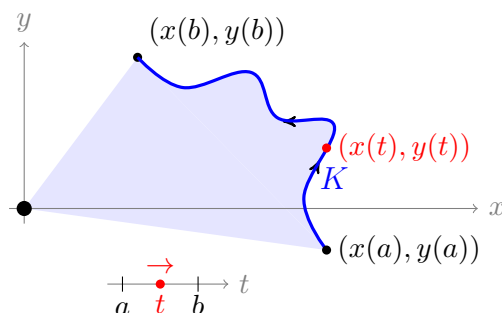
Let $x, y : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable and let $f(t) := \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ for all $t \in [a, b]$.

- The function $f : [a, b] \rightarrow \mathbb{R}^2$ is called a path or a parametrization,
- its range $K = f([a, b]) \subset \mathbb{R}^2$ is called a curve in $\mathbb{R}^2$.

Between us: path vs. curve

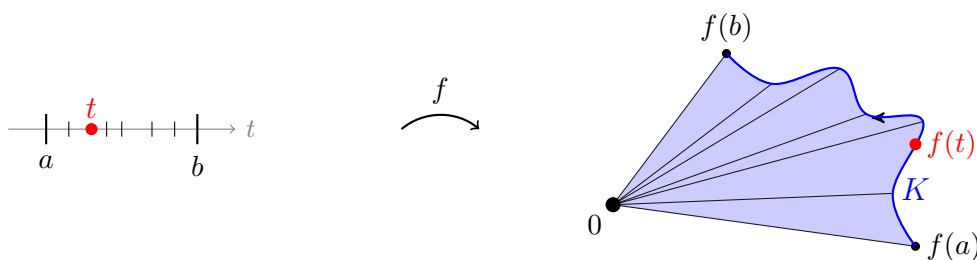
For the path / parametrization f , think about a bus timetable that tells you for each time $t \in [a, b]$ where the bus $f(t)$ is right now. Then the curve K is the trace the bus leaves on the street, if it went thorough a paddle of paint before.

Let now f be a path with a curve K , such that each ray originating in the origin intersects K at most once.



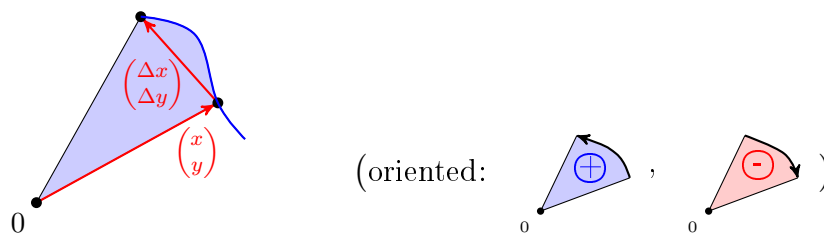
We are now looking for the **area** of the sector between the origin and the curve K .

Idea: partition P of the interval $[a, b] \implies$ partition of the sector into many $\approx$ triangles



For each of the small segments we calculate the area of the according triangle:

$$\text{area}(\text{triangle}) = \frac{1}{2} \det \begin{pmatrix} x & \Delta x \\ y & \Delta y \end{pmatrix} = \frac{1}{2} (x \cdot \Delta y - \Delta x \cdot y)$$



Summing over all triangles, we get the following approximation of the area of the sector:

$$\sum_{i=1}^n \quad = \quad \frac{1}{2} \sum_{i=1}^n \quad \rightarrow \quad \frac{1}{2} \int_a^b (x(t)y'(t) - x'(t)y(t)) dt$$

for $\mu(P) \rightarrow 0$. The last term is called the sector formula.

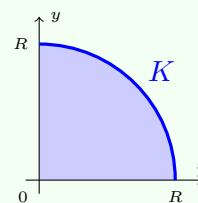
Example 8.30. the area of the quarter disk again

We parametrize the circle K with polar coordinates:

$$\left. \begin{aligned} x(t) &= r \cos t \\ y(t) &= r \sin t \end{aligned} \right\} \quad f(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \quad t \in [0, \frac{\pi}{2}].$$

Using the sector formula, we have

$$\begin{aligned} A &= \\ &= . \end{aligned}$$



This time we could also have used $t \in [0, 2\pi]$ and would have gotten the full circle area instead.

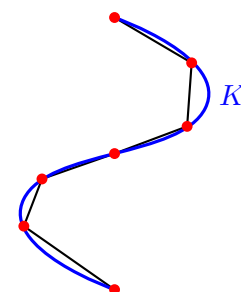
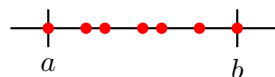
8.4.2 Length, mass and center of gravity of a curve

Let again $f = \begin{pmatrix} x \\ y \end{pmatrix} : [a, b] \rightarrow \mathbb{R}^2$ be a curve with $x, y \in C^1([a, b])$.

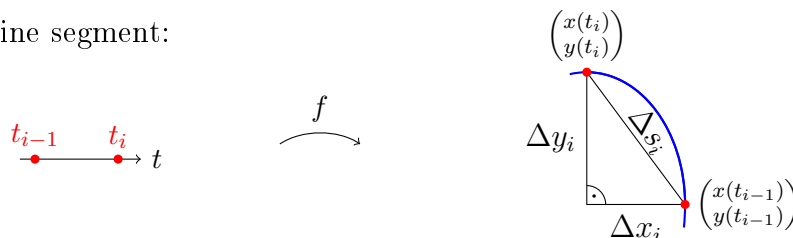
This time we are looking for the **length** of the curve $K = f([a, b]) \subset \mathbb{R}^2$.

Idea:

partition of $[a, b]$ $\implies$ approximation of K with a polygonal chain



An individual line segment:



Length of the i -th line segment:

$$\Delta s_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} =$$

with $\xi_i, \eta_i \in (t_{i-1}, t_i)$ using the mean value theorem.

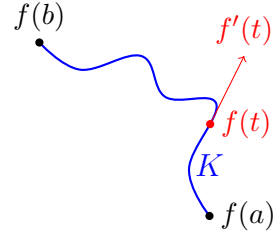
Now we add up the lengths of all line segments:

$$\begin{aligned}
 \text{length}(\text{polygon chain}) &= \sum_{i=1}^n \Delta s_i \\
 &= \sum_{i=1}^n \sqrt{(x'(\xi_i))^2 + (y'(\eta_i))^2} \cdot \Delta t_i \\
 &\rightarrow \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt
 \end{aligned}$$

as $\mu(P) \rightarrow 0$. The integrand is apparently the length of the following vector:

$$\begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix} =: \begin{pmatrix} x \\ y \end{pmatrix}'(t) = f'(t).$$

We call this vector the *velocity vector* of f (the bus, if you want) at time t . The direction of $f'(t)$ is tangential to the curve K in $f(t)$, and the length of the vector corresponds to the velocity at that time.



The length s of the curve K is thus calculated as follows:

$$s = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt = \int_a^b \|f'(t)\| dt \quad (\text{arc length}),$$

where $\|\cdot\|$ denotes the Euclidean norm (in $\mathbb{R}^2$).

Between us: physicist-speak

To calculate the distance s that was traveled, we integrate the velocity over time t . At constant velocity $v(t) := \|f'(t)\|$ we get the distance-time law we know from school $s = v \cdot t$, so $v = \frac{s}{t}$.

The distance traveled up to a time $T \in [a, b]$ can thus be calculated like this:

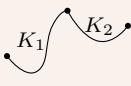
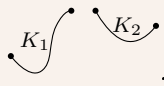
$$s(T) := \int_a^T \|f'(t)\| dt, \quad T \in [a, b].$$

The fundamental theorem of calculus tells us that s then is an antiderivative of $\|f'(\cdot)\|$, i.e., we have $s'(t) = \frac{ds}{dt} = \|f'(t)\|$, which motivates the following abbreviation and notation:

$$ds = \|f'(t)\| dt \quad \text{and} \quad \int_K ds := \int_a^b \|f'(t)\| dt$$

We use the same formula to calculate the arc length of C^1 -paths $f = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : [a, b] \rightarrow \mathbb{R}^n$.

🧐 Nerd box 8.31. Possible is also also a piecewise C^1 -path instead of C^1 .

...for example a curve K like  or even .

Then we have

$$\int_{K_1 \cup K_2} ds = \int_{K_1} ds + \int_{K_2} ds.$$

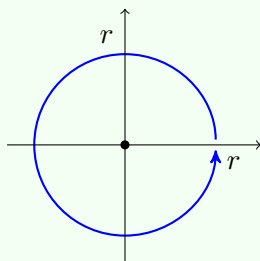
Back to C^1 -smooth curves in 2D:

Example 8.32. circumference of a circle

We again use the parametrization

$$f(t) = \begin{pmatrix} r \cos t \\ r \sin t \end{pmatrix}, \quad t \in [0, 2\pi].$$

It follows that



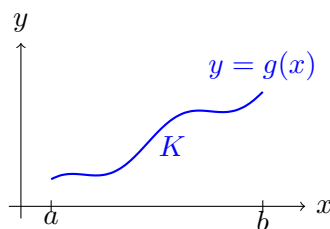
$$\|f'(t)\| =$$

The “bus” drives with constant velocity r (only the direction changes all the time), so the calculation for the traveled distance is simple:

$$s = \int_0^{2\pi} \|f'(t)\| dt =$$

And thus we also have the well-known formula for the circumference.

Special case of the curve K : Let the curve be the graph of a function $g \in C^1([a, b])$.



$$\left. \begin{array}{l} x =: t \\ y = g(x) = g(t) \end{array} \right\} \implies f(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} t \\ g(t) \end{pmatrix}$$

$$s = \int_a^b \|f'(t)\| dt = \int_a^b \quad dt$$

We now talk about the **mass and center of gravity** of a curve K in $\mathbb{R}^n$. For this, let the curve be equipped with a density function $\rho : K \rightarrow [0, +\infty)$, i.e. each point $x \in K \subset \mathbb{R}^n$ has the mass $\rho(x) \geq 0$.

Again, the idea is: Choose a path $f : [a, b] \rightarrow \mathbb{R}^n$ with $f([a, b]) = K$, partition of $[a, b] \Rightarrow$ finitely many mass points; then $\mu(P) \rightarrow 0$

Result:

$$\begin{aligned} \text{mass: } m &= \int_K \rho \, ds = \int_a^b \rho(f(t)) \underbrace{\|f'(t)\|}_{ds} dt, \\ \text{center of gravity: } p &= \begin{pmatrix} p_1 \\ \vdots \\ p_n \end{pmatrix} \quad \text{with} \quad p_j = \frac{\int_K x_j \rho \, ds}{m} = \frac{\int_a^b x_j(t) \cdot \rho(f(t)) \|f'(t)\| dt}{m} \end{aligned}$$

for $j = 1, \dots, n$. The formula for the coordinates $p_1, \dots, p_n$ of the center of gravity p reminds us of the weighted average in the mean value theorem for integration.

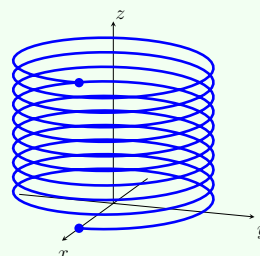
Example 8.33. screw line

The path

$$f(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} 3 \cos t \\ 3 \sin t \\ 4t \end{pmatrix}, \quad t \in [0, 20\pi]$$

describes ten windings on a screw.

Let K be the curve of f in $\mathbb{R}^3$. From



$$f'(t) = \begin{pmatrix} \quad \quad \quad \end{pmatrix} \quad \text{we get} \quad \|f'(t)\| =$$

So the “bus” is again traveling with constant velocity (but is changing directions all the time). Thus the arc length of the curve K is

$$s = \int_K ds = \int_0^{20\pi} \underbrace{\|f'(t)\|}_{ds} dt = \int_0^{20\pi} \quad dt = \left[\quad \right]_0^{20\pi} = \quad .$$

Now let the curve $K \subset \mathbb{R}^3$ be equipped with a density function $\rho : K \rightarrow [0, +\infty)$, $\rho(x, y, z) = \frac{1}{4}z$. Therefore its mass equals

$$m = \int_K \rho \, ds = \int_0^{20\pi} \rho(f(t)) \|f'(t)\| \, dt =$$

and its center of gravity is $p = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}$ with

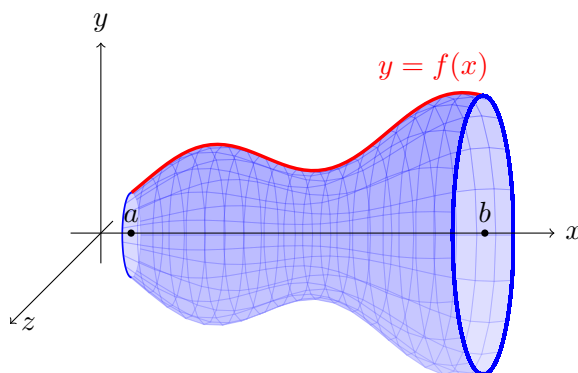
$$p_1 = \frac{\int_K x \cdot \rho \, ds}{m} = \frac{\int_0^{20\pi} 3 \cos t \cdot t \cdot 5 \, dt}{m}, \quad p_3 = \frac{\int_K z \cdot \rho \, ds}{m} = \frac{\int_0^{20\pi} 4t \cdot t \cdot 5 \, dt}{m}$$

and p_2 analogous to p_1 – just with $\sin$ instead of $\cos$. The numerators for p_1 and p_2 are integrated by parts, as in Example 8.27. All in all we have

$$p_1 = 0, \quad p_2 = -\frac{3}{10\pi}, \quad p_3 = \frac{160}{3}\pi.$$

8.4.3 Solids of revolution

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous with $f \geq 0$. We lift the graph of f into $\mathbb{R}^3$ and rotate it around the x -axis. This results in a so called solid of revolution, R .



Partitioning the interval $[a, b]$ leads to an approximation of the volume of R and of the area of its lateral surface:

$$\begin{aligned} \text{volume}(R) &\approx \sum_{i=1}^n \text{volume} \left(\text{disk with radius } f(x_i) \text{ and thickness } \Delta x_i \right) \\ &= \sum_{i=1}^n \underbrace{\pi f(\xi_i)^2}_{\text{area of the disk}} \cdot \underbrace{\Delta x_i}_{\text{thickness}} \\ &\xrightarrow{\mu(P) \rightarrow 0} \pi \int_a^b f(x)^2 dx \end{aligned} \quad (8.10)$$

volume of a solid of revolution

$$\begin{aligned} \text{lat.surface}(R) &\approx \sum_{i=1}^n \text{surface} \left(\text{frustum with radius } f(x_i) \text{ and slanted thickness } \Delta s_i \right) \\ &\approx \sum_{i=1}^n \underbrace{2\pi f(\xi_i)}_{\text{circumference}} \cdot \underbrace{\Delta s_i}_{\text{thickness (slanted!)}} \\ &\xrightarrow{\mu(P) \rightarrow 0} 2\pi \int_a^b f(x) \underbrace{\sqrt{1 + (f'(x))^2}}_{ds} dx \end{aligned} \quad (8.11)$$

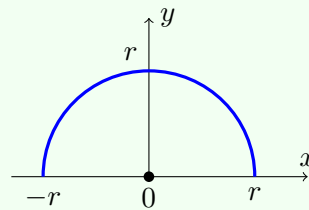
lateral surface of a solid of revolution

Example 8.34. The ball as a solid of revolution

Let $r > 0$. The graph of the function f with

$$y = f(x) = \sqrt{r^2 - x^2}, \quad x \in [-r, r],$$

makes up the upper half of the circle with the equation $x^2 + y^2 = r^2$, i.e. with radius r around the origin.



Rotating this upper half circle around the x -axis of $\mathbb{R}^3$, we get as the solid of revolution R a ball with radius r . Its volume is calculated as follows:

$$\text{volume}(R) \stackrel{(8.10)}{=}$$

$$=$$

Since $f(-r) = 0 = f(r)$, the whole surface is lateral surface, for the computation of which we still need

$$f'(x) = \frac{1}{2\sqrt{r^2 - x^2}} \cdot (-2x) = \frac{-x}{\sqrt{r^2 - x^2}}, \quad \text{so that} \quad (f'(x))^2 = \frac{x^2}{r^2 - x^2}.$$

Then we have

$$\text{surface}(R) \stackrel{(8.11)}{=} 2\pi \int_{-r}^r f(x) \sqrt{1 + (f'(x))^2} dx$$

$$=$$

$$=$$

And like that we recover the next well-known formula from geometry.

🧐 Nerd box 8.35. Did you ever notice, that...

...the surface of the sphere, $4\pi r^2$, is exactly the derivative of the volume, $\frac{4}{3}\pi r^3$, with respect to the radius r ? Similarly, the circumference of the circle, $2\pi r$, is exactly the derivative of the area, πr^2 . Why is this different with a cube and a square?

8.5 Improper integral: unbounded interval or unbounded function

So far: definite integral $\int_a^b f(x) dx$ where $\begin{cases} [a, b] \dots \text{bounded interval,} \\ \text{and } f \dots \text{bounded function.} \end{cases}$

Now: drop one or even both of these boundedness conditions

8.5.1 Unbounded interval

Definition 8.36. improper integral over $[a, +\infty)$

Let $f : [a, +\infty) \rightarrow \mathbb{R}$ be a bounded function and assume $f \in \mathcal{R}([a, b])$ for all $b > a$.
We define

$$\int_a^{+\infty} f(x) dx := \lim_{b \rightarrow +\infty} \int_a^b f(x) dx,$$

if the limit exists. Wording: improper integral of f over $[a, +\infty)$.

We also say that $\int_a^{+\infty} f(x) dx$ converges (otherwise: $\int_a^{+\infty} f(x) dx$ diverges).

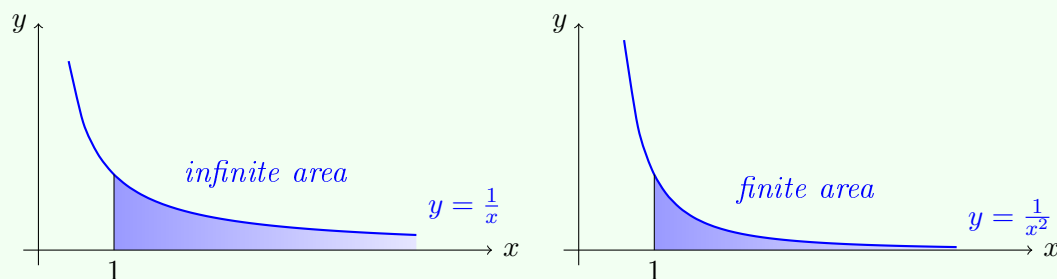
Example 8.37. improper integrals over $[1, +\infty)$

$$\int_1^{+\infty} \frac{1}{x} dx =$$

$$\int_1^{+\infty} \frac{1}{x^\alpha} dx =$$

$$= \begin{cases} & , \quad \alpha > 1 \text{ (convergence)}, \\ & , \quad \alpha < 1 \text{ (divergence)}, \end{cases} \quad (\text{for } \alpha \neq 1),$$

i.e.,



This reminds us of the series $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$, $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$.

Here is a theorem that establishes this connection:

Proposition 8.38. monotonicity: improper integral vs. series

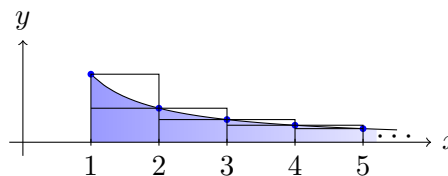
Let $f : [1, +\infty) \rightarrow [0, +\infty)$ be monotonically decreasing. Then

$$\int_1^{+\infty} f(x) dx \text{ converges} \iff \sum_{k=1}^{+\infty} f(k) \text{ converges}$$

and

$$\sum_{k=2}^{+\infty} f(k) \leq \int_1^{+\infty} f(x) dx \leq \sum_{k=1}^{+\infty} f(k).$$

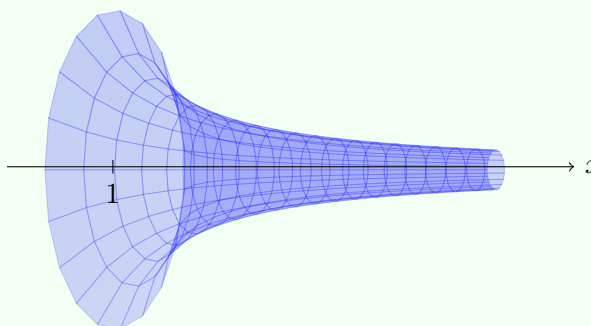
Proof: The Riemann integrability of f follows from its monotonicity. The rest of the proof works without words. ■



Example 8.39. Gabriel's horn

Let R be the solid of revolution of $f(x) = \frac{1}{x}$ on $[1, +\infty)$ rotated around the x -axis.

Assume R is made of glass, hollow on the inside, and is to be painted in blue.



$$\text{volume} \stackrel{(8.10)}{=} \pi \int_1^{+\infty} \frac{1}{x^2} dx = \pi < \infty \quad (\text{Example 8.37})$$

$$\begin{aligned} \text{lateral surface} &\stackrel{(8.11)}{=} 2\pi \int_1^{+\infty} \frac{1}{x} \underbrace{\sqrt{1 + (f'(x))^2}}_{\geq 1} dx \\ &\geq 2\pi \int_1^{+\infty} \frac{1}{x} dx = +\infty \quad (\text{Example 8.37}) \end{aligned}$$

Curiously, painting the outside in blue requires an infinite amount of paint. But when pouring π liters of blue paint inside, the solid of revolution is entirely blue.

In analogy to the case of an infinite upper limit, “ $b = +\infty$ ”, we define

$$\int_{-\infty}^b f(x) dx := \lim_{a \rightarrow -\infty} \int_a^b f(x) dx,$$

if the limit exists, and finally even

$$\int_{-\infty}^{+\infty} f(x) dx := \int_{-\infty}^c f(x) dx + \int_c^{+\infty} f(x) dx = \lim_{a \rightarrow -\infty} \int_a^c f(x) dx + \lim_{b \rightarrow +\infty} \int_c^b f(x) dx,$$

if both limits exist.

The intermediate point $c \in \mathbb{R}$ can be chosen arbitrarily.

⚠ Attention!

For the convergence of $\int_{-\infty}^{+\infty} f(x) dx$ both limits $\lim_{a \rightarrow -\infty} \int_a^c f(x) dx$ and $\lim_{b \rightarrow +\infty} \int_c^b f(x) dx$ have to exist.

If (at least) one of these limits does not exist, it might be the case that at least the following limit exists:

$$\lim_{a \rightarrow +\infty} \int_{-a}^a f(x) dx =: p.v. \int_{-\infty}^{+\infty} f(x) dx,$$

which is then called the Cauchy principal value² of $\int_{-\infty}^{+\infty} f(x) dx$.

$$\int_{-\infty}^{+\infty} f(x) dx \text{ converges} \quad \begin{matrix} \Rightarrow \\ \nRightarrow \end{matrix} \quad p.v. \int_{-\infty}^{+\infty} f(x) dx \text{ converges.}$$

²„p.v.“ = principal value, sometimes also: „v.p.“ = valeur principale [french]

Example 8.40. integral $\int_{-\infty}^{+\infty}$ exists only as Cauchy principal value

$$p.v. \int_{-\infty}^{+\infty} \sin x \, dx = \lim_{a \rightarrow +\infty} \int_{-a}^a \sin x \, dx = \lim_{a \rightarrow +\infty} \underbrace{[-\cos x]_{-a}^a}_0 = 0$$

but

$$\int_{-\infty}^{+\infty} \sin x \, dx = \int_{-\infty}^0 \sin x \, dx + \int_0^{+\infty} \sin x \, dx$$

does not converge.

After discussing bounded functions on unbounded intervals, we now turn to the reverse case.

8.5.2 Unbounded function

Definition 8.41. improper integral

Let $f : [a, b) \rightarrow \mathbb{R}$ be Riemann integrable on every interval $[a, b - \varepsilon]$ with $\varepsilon > 0$. Then we define

$$\int_a^b f(x) \, dx := \lim_{\varepsilon \searrow 0} \int_a^{b-\varepsilon} f(x) \, dx = \lim_{c \nearrow b} \int_a^c f(x) \, dx,$$

if the limit exists. Wording: improper integral of f over $[a, b)$.

One says again: $\int_a^b f(x) \, dx$ converges, respectively diverges.

The point of this construction is that $f(x)$ may be unbounded as $x \nearrow b$. For example, f could have a pole at b . (However, if f is bounded on $[a, b)$, then the limit from Definition 8.41 yields the usual definite integral $\int_a^b f(x) dx$.)

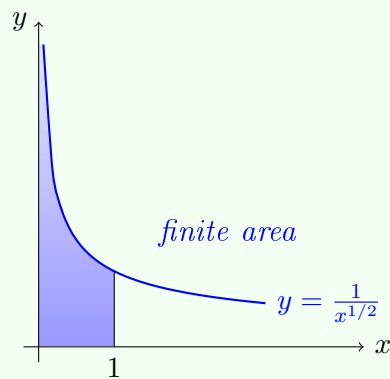
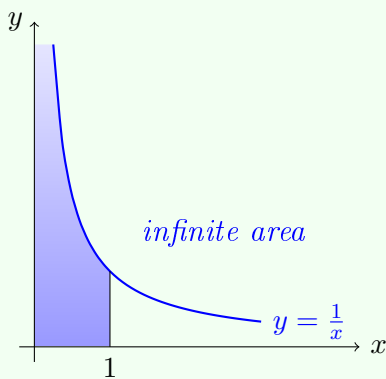
We define the improper integral analogously in case the function is unbounded at the left endpoint a :

$$\int_a^b f(x) dx := \lim_{\varepsilon \searrow 0} \int_{a+\varepsilon}^b f(x) dx = \lim_{c \searrow a} \int_c^b f(x) dx$$

Example 8.42. Unboundedness at the left endpoint

$$\int_0^1 \frac{1}{x} dx = +\infty,$$

$$\int_0^1 \frac{1}{x^\alpha} dx = \begin{cases} +\infty & \alpha > 1 & (\text{divergence}) \\ \frac{1}{1-\alpha} & \alpha < 1 & (\text{convergence}) \end{cases}$$

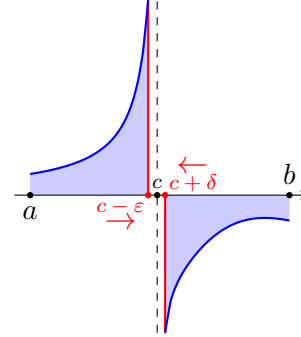


Instead of being unbounded at a or b , f can also be unbounded at an intermediate point $c \in (a, b)$. Then we define

$$\begin{aligned}\int_a^b f(x) dx &:= \int_a^c f(x) dx + \int_c^b f(x) dx \\ &= \lim_{\varepsilon \searrow 0} \int_a^{c-\varepsilon} f(x) dx + \lim_{\delta \searrow 0} \int_{c+\delta}^b f(x) dx,\end{aligned}$$

if both limits exist.

If (at least) one limit does not exist, one can attempt the following single limit. Here we couple ε with δ :



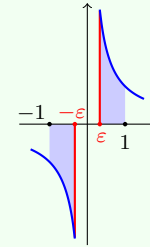
$$p.v. \int_a^b f(x) dx := \lim_{\varepsilon \searrow 0} \left(\int_a^{c-\varepsilon} f(x) dx + \int_{c+\varepsilon}^b f(x) dx \right), \quad (8.12)$$

and call it (again) the Cauchy principal value.

Example 8.43. indefinite integral with an unbounded function

$$\text{a) } p.v. \int_{-1}^1 \frac{1}{x} dx = \lim_{\varepsilon \searrow 0} \underbrace{\left(\int_{-1}^{-\varepsilon} \frac{1}{x} dx + \int_{\varepsilon}^1 \frac{1}{x} dx \right)}_0 = 0,$$

$$\text{but } \int_{-1}^1 \frac{1}{x} dx = \int_{-1}^0 \frac{1}{x} dx + \int_0^1 \frac{1}{x} dx \text{ doesn't exist.}$$

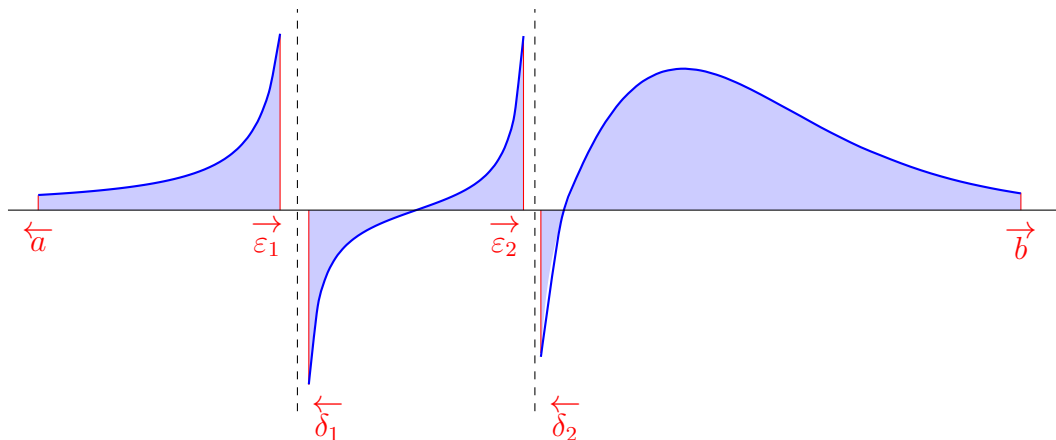


$$\text{b) In contrast, the integral } \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx \text{ converges:}$$

$$\begin{aligned}\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx &= \lim_{a \searrow -1} \int_a^0 \frac{1}{\sqrt{1-x^2}} dx + \lim_{b \nearrow 1} \int_0^b \frac{1}{\sqrt{1-x^2}} dx \\ &= \lim_{a \searrow -1} [\arcsin x]_a^0 + \lim_{b \nearrow 1} [\arcsin x]_0^b \\ &= \underbrace{\arcsin(0) - \arcsin(-1)}_0 + \underbrace{\arcsin(1) - \arcsin(0)}_{\frac{\pi}{2}} = \pi\end{aligned}$$

8.5.3 Both: interval and function are unbounded

When combining unbounded intervals and unbounded functions we also have to combine the techniques from above. Thus we may have to deal with many limits:



$$\int_{-\infty}^{+\infty} f(x) dx = \text{sum of 6 limits (all 6 have to exist)}$$

Summary

- The indefinite integral
 - asks for the set of all antiderivatives F of f (such that $F' = f$),
 - if you know one, you know them all (append “ $+C$ ”),
 - the derivative operation $\frac{d}{dx}$ has to be reversed,
 - rules such as $\frac{d}{dx} : \sin \mapsto \cos$ simply have to be reversed (“basic integrals”),
 - simple rules such as those for sum and scaling can be transferred,
 - unfortunately, this does not work for product and quotient. Here you have to find clever ways to follow the product and chain rule backwards (“partial integration” and “substitution”)
 - one can even integrate all rational functions (polynomial divided by polynomial)
- The definite integral
 - asks for the (signed) surface A between the graph of a function $f : [a, b] \rightarrow \mathbb{R}$ and the x-axis,
 - where $[a, b]$ is a bounded interval and f is a bounded function,
 - the Riemann integral is the most common approach to calculate A (later on, you may also hear about the Lebesgue integral: this is a little more technical but also more widely applicable with even nicer properties)

- you divide $[a, b]$ into subintervals, approximating the area you are looking for with narrow strips that almost look like rectangles
 - if you replace the strips by rectangles through the highest or lowest point of f per subinterval, you will get the so-called upper and lower sums, which always sandwich A in between,
 - if this sandwich becomes arbitrarily tight the finer you break down $[a, b]$, then you have determined A as the limit and f is called Riemann integrable,
 - continuous functions are Riemann integrable,
 - monotonic (and thereby bounded) functions are Riemann integrable,
 - the sum and the product of Riemann integrable functions are also Riemann integrable,
 - mean value theorem: If f is continuous then A is equal to the width of the interval $[a, b]$ times a certain function value $f(\xi)$ of f .
- The fundamental theorem
 - brings together indefinite and definite integral
 - and thus also differential and integral calculus,
 - if the filling level f of a tank is observed over time, the rate of change g of f (due to inflow and outflow) is the derivative of f ; vice versa, f can be reconstructed from g by integration,
 - now theorems and rules for the indefinite integral can be transferred to the definite integral – and vice versa (formulas for partial integration, integrability of continuous functions, etc.)
 - By means of definite integrals we calculate:
 - not only the area underneath the graph of f but also
 - so-called sectorial areas,
 - the length, mass and center of gravity of a curve,
 - as well as the volume and lateral surface of a solid of revolution.
 - The indefinite integral
 - bypasses the boundedness conditions on $[a, b]$ and f ,
 - approaches the problem by forming limits,
 - if not all of the resulting limits exist, Cauchy's principal value provides a possibility to get at a result (in a weaker sense).

Chapter 9

Sequences and series of functions

This chapter is about sequences again – but this time it's about sequences of functions and not sequences of real numbers.

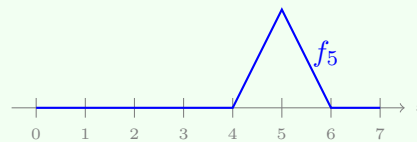
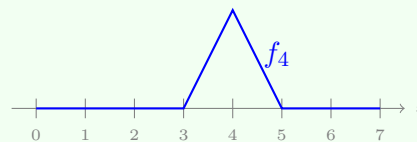
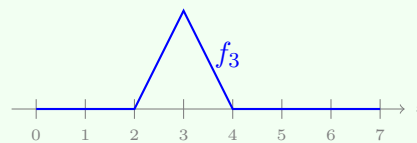
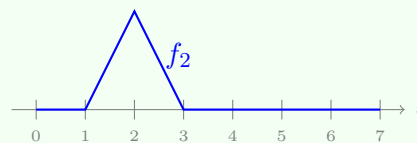
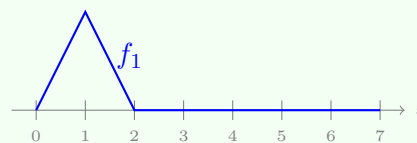
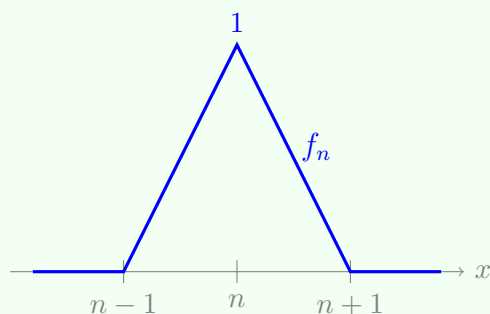
Example 9.1. A little hat walks to $+\infty$

We consider the sequence $(f_1, f_2, \dots)$ of functions on $D = [0, +\infty)$, with their graphs to the right.

For $n \in \mathbb{N}$ let $f_n : D \rightarrow \mathbb{R}$ with

$$f_n(x) := \begin{cases} x - n + 1, & x \in [n - 1, n), \\ -x + n + 1, & x \in [n, n + 1), \\ 0, & \text{otherwise,} \end{cases}$$

i.e.,



⋮

Just like with sequences of real numbers we will mostly talk about convergence again. Does the sequence $(f_n)_{n \in \mathbb{N}}$ in Example 9.1 converge? In which sense? Towards what?

9.1 Pointwise vs. uniform convergence

Firstly, there is the so-called pointwise convergence: You choose an arbitrary point $x \in D$ and check, whether and towards which number $(f_n(x))_{n \in \mathbb{N}}$ converges as $n \rightarrow \infty$.

Definition 9.2. pointwise convergence, $f_n \rightarrow f$

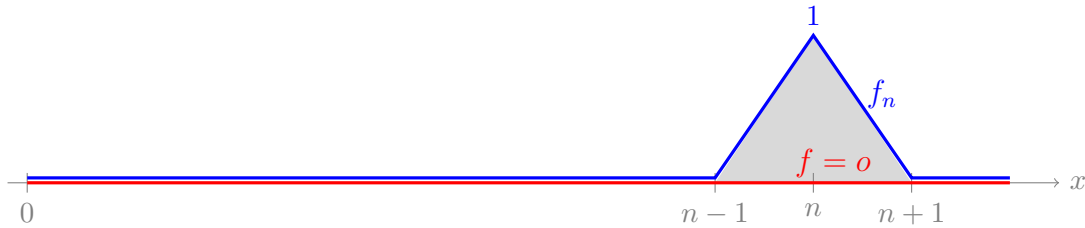
Let $D \subset \mathbb{R}$ and let $f, f_1, f_2, \dots : D \rightarrow \mathbb{R}$ be functions. If, for each $x \in D$, we have

$$f_n(x) \rightarrow f(x) \quad \text{for} \quad n \rightarrow \infty,$$

we say that the sequence $(f_n)_{n \in \mathbb{N}}$ converges pointwise to f , the so-called pointwise limit of $(f_n)_{n \in \mathbb{N}}$, and we write $f_n \rightarrow f$.

In Example 9.1 we saw that for each $x \in D$ the sequence $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$, because for $n \geq x + 1$, the hat has walked past x already, and from there on we have $f_n(x) = 0$. Thus $f_n \rightarrow o$, where o is the zero function on $D = [0, +\infty)$, i.e., $o(x) = 0$ for all $x \in D$.

The pointwise convergence, $f_n \rightarrow f$, is convergence from the *local* perspective of each $x \in D$. Looking at it on a *global* level, there is always exactly “one hat” difference in Example 9.1 between the graph of f_n and the graph of $f = o$:



The biggest difference $|f_n(x) - f(x)|$ of 1 is at¹ $x = n$. That global perspective is watching out for the largest difference between f_n and f on the whole of D ,

$$\sup_{x \in D} |f_n(x) - f(x)|, \tag{9.1}$$

and we would conclude that f_n “converges to f even globally”, if $(9.1) \rightarrow 0$ as $n \rightarrow \infty$.

¹The pointwise convergence does not care, because this point (the location of the n -th hat) changes with n and it walks towards $+\infty$. For each fixed point $x \in D$ we have $f_n(x) \rightarrow o(x) = 0$ as $n \rightarrow \infty$.

Definition 9.3. uniform convergence, $f_n \rightrightarrows f$

Let $D \subset \mathbb{R}$ and let $f, f_1, f_2, \dots : D \rightarrow \mathbb{R}$ be functions. If

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty,$$

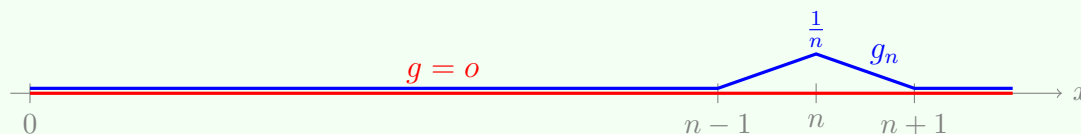
we say that the sequence $(f_n)_{n \in \mathbb{N}}$ converges uniformly to f , the so called uniform limit of $(f_n)_{n \in \mathbb{N}}$, and we write $f_n \rightrightarrows f$.

Example 9.4. The little hat is shrinking as it walks.

If we now put

$$g_n(x) := \frac{1}{n} f_n(x), \quad n \in \mathbb{N}, \quad x \in D = [0, +\infty)$$

with the hat functions f_n from before,



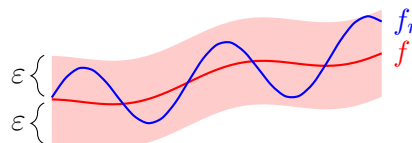
then, clearly, $\sup_{x \in D} |g_n(x) - g(x)| =$ as $n \rightarrow \infty$, showing that $g_n \rightrightarrows 0$.

Written a bit more explicitly, uniform convergence means that for all $\varepsilon > 0$

$$\sup_{x \in D} |f_n(x) - f(x)| < \varepsilon$$

as soon as n is sufficiently large (i.e., $n \geq n_0$ for a suitable n_0).

Then the graph of f_n cannot be further away than ε from the graph of f ; so for large enough n , it lies in the “ ε -strip” around the graph of f , depicted on the right.



In Example 9.1 with $f = o$, the convergence is apparently only pointwise but not uniform, since

$$\sup_{x \in D} |f_n(x) - f(x)| = \quad \text{as} \quad n \rightarrow \infty.$$

In other words, the graph of f_n always lies inside the 1-strip around f , but never in the 0.9-strip, for example – not matter how large you make n .

 Nerd box 9.5. pointwise vs. uniform: swapped quantifiers

If we spell both definitions out using quantifiers, we get

$$\begin{aligned} f_n \rightarrow f &\iff \forall \varepsilon > 0 \quad \forall x \in D \quad \exists n_0 : \forall n \geq n_0 : |f_n(x) - f(x)| < \varepsilon, \\ f_n \rightrightarrows f &\iff \forall \varepsilon > 0 \quad \exists n_0 : \forall x \in D \quad \forall n \geq n_0 : |f_n(x) - f(x)| < \varepsilon. \end{aligned}$$

The existence of one n_0 , that “works” for all $x \in D$ is much stronger than allowing each $x \in D$ its own n_0 and thus its own speed for $f_n(x) \rightarrow f(x)$. So uniform convergence is a stronger property than pointwise convergence.

You can actually see this much more easily: From

$$0 \leq |f_n(x_0) - f(x_0)| \leq \sup_{x \in D} |f_n(x) - f(x)| \quad \text{for all} \quad x_0 \in D$$

and the sandwich lemma we conclude:

$$\text{uniform convergence } f_n \rightrightarrows f \implies \text{pointwise convergence } f_n \rightarrow f.$$

Thus, it is also easy to see that the uniform limit of (f_n) , if it exists, can only be the pointwise limit of (f_n) , i.e. $f : D \rightarrow \mathbb{R}$ with

$$f(x) := \lim_{n \rightarrow \infty} f_n(x), \quad x \in D,$$

in case the pointwise limit f exists.

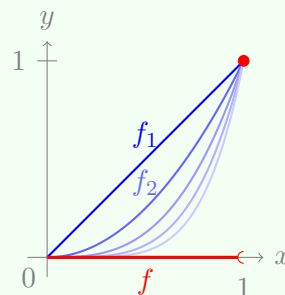
Example 9.6. three more sequences of functions

a) Let $D = [0, 1]$ and $f_1, f_2, \dots : D \rightarrow \mathbb{R}$ with

$$f_n(x) = x^n, \quad x \in D$$

for all $n \in \mathbb{N}$. Then we have pointwise convergence

$$f_n \rightarrow f \quad \text{with} \quad f(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$



But we do not have uniform convergence:

In this case the limit would have to be the same f . But instead, we have

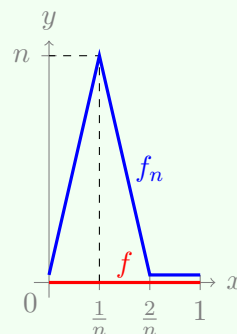
$$\sup_{x \in D} |f_n(x) - f(x)| = \sup_{x \in [0,1)} |x^n - 0| = \left(\sup_{x \in [0,1)} x \right)^n = 1^n = 1 \not\rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

The story is different if we change from $D = [0, 1]$ to $D = [0, 0.99]$. Then we have $f_n \Rightarrow o$, since

$$\sup_{x \in D} |f_n(x) - o(x)| =$$

b) We try little hats $f_n : D \rightarrow \mathbb{R}$ again, this time on $D = [0, 1]$. To be precise, for $n = 1, 2, \dots$,

$$f_n(x) = \begin{cases} n^2 x, & x \in [0, \frac{1}{n}], \\ 2n - n^2 x, & x \in (\frac{1}{n}, \frac{2}{n}), \\ 0, & x \in [\frac{2}{n}, 1], \end{cases} \quad x \in D.$$



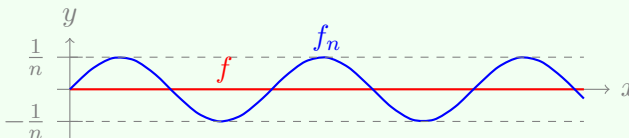
Again, we have $f_n \rightarrow o$ pointwise since, for each $x \in D \setminus \{0\}$, we have $f_n(x) = 0$ as soon as $n \geq n_0 := \frac{2}{x}$.

And again, we certainly do not have uniform convergence since

$$\sup_{x \in D} |f_n(x) - \underbrace{f(x)}_0| =$$

c) Last but not least, let $f_n : D \rightarrow \mathbb{R}$ on $D = [0, 4]$ be given by

$$f_n(x) = \frac{1}{n} \sin(nx), \quad x \in D.$$



This time we have $f_n \Rightarrow f = 0$, since

$$\sup_{x \in D} |f_n(x) - f(x)| =$$

Put differently, the graph of f_n is inside the $\frac{1}{n}$ -strip around the graph of f . As $n \rightarrow \infty$, the width of that strip obviously goes to zero, so $f_n \Rightarrow f$.

9.2 Conservation laws for uniform convergence

We start with the following question:

Under pointwise convergence, $f_n \rightarrow f$, do

- a) continuity,
- b) integrability and the integral, i.e. $\int_D f_n(x) dx \xrightarrow{?} \int_D f(x) dx$,
- c) differentiability and the derivative, i.e. $f'_n \xrightarrow{?} f'$ (Does f' even exist?),
transfer from the functions f_n to the limit f ?

Answers:

- a) no, see Example 9.6 a): every f_n is continuous, but f is not,
- b) no, see Example 9.6 b): $\int_0^1 f_n(x) dx = 1$ for all n but $\int_0^1 f(x) dx = 0$,
- c) no, see Example 9.6 c): $f'_n(x) = \cos(nx) \not\rightarrow f'(x) = 0$.

Pointwise convergence $\rightarrow$ is apparently not strong enough to transfer these properties and quantities to the limit function f . We therefore put all our hopes on uniform convergence $\Rightarrow \dots$ and we won't be disappointed:

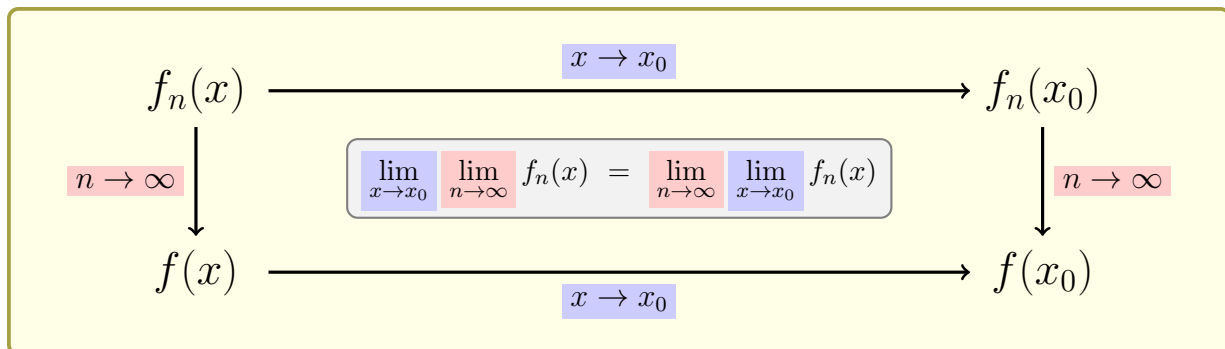
Proposition 9.7. $\Rightarrow$ and continuity

Let $D \subset \mathbb{R}$ and let the functions $f_1, f_2, \dots : D \rightarrow \mathbb{R}$ be continuous.
If $f_n \Rightarrow f$ then also f is continuous.

Proof: Let $x_0 \in D$ and $\varepsilon > 0$ be arbitrary. Choose $n \in \mathbb{N}$ with $\sup_{x \in D} |f_n(x) - f(x)| < \frac{\varepsilon}{3}$. Since f_n is continuous, there is a $\delta > 0$, such that $x \overset{\delta}{\approx} x_0 \implies f_n(x) \overset{\varepsilon/3}{\approx} f_n(x_0)$. So we have

$$x \overset{\delta}{\approx} x_0 \implies f(x) \overset{\varepsilon/3}{\approx} f_n(x) \overset{\varepsilon/3}{\approx} f_n(x_0) \overset{\varepsilon/3}{\approx} f(x_0) \quad \text{and therefore} \quad f(x) \overset{\varepsilon}{\approx} f(x_0). \blacksquare$$

Proposition 9.7 can also be interpreted to mean that in the case of uniform convergence, $f_n \Rightarrow f$, the following limits can be interchanged:



Proposition 9.8. $\Rightarrow$ and the integral

Let $D = [a, b]$ and let $f_1, f_2, \dots : D \rightarrow \mathbb{R}$ be Riemann integrable functions.

In case $f_n \Rightarrow f$ we have $f \in \mathcal{R}([a, b])$ and $\int_a^b f_n(x) dx \rightarrow \int_a^b f(x) dx$ as $n \rightarrow \infty$.

Proof: Let $\varepsilon > 0$ be arbitrary and $n \in \mathbb{N}$, such that $\sup_{x \in D} |f_n(x) - f(x)| < \frac{\varepsilon}{3(b-a)}$. Since $f_n \in \mathcal{R}([a, b])$ and because of Proposition 8.14, there is a partition P of $[a, b]$ with $O(f_n, P) - U(f_n, P) < \frac{\varepsilon}{3}$. From $|U(f, P) - U(f_n, P)| \leq (b-a) \cdot \sup_{x \in D} |f_n(x) - f(x)| < \frac{\varepsilon}{3}$ (and analogously for O), it follows that

$$U(f, P) \overset{\varepsilon/3}{\approx} U(f_n, P) \overset{\varepsilon/3}{\approx} O(f_n, P) \overset{\varepsilon/3}{\approx} O(f, P) \quad \text{and therefore} \quad U(f, P) \overset{\varepsilon}{\approx} O(f, P).$$

According to Proposition 8.14, also $f \in \mathcal{R}([a, b])$.

Additionally, for $n \rightarrow \infty$, we have

$$\left| \int_a^b f_n(x) dx - \int_a^b f(x) dx \right| \leq \int_a^b |f_n(x) - f(x)| dx \leq \sup_{x \in D} |f_n(x) - f(x)| \cdot (b-a) \rightarrow 0. \quad \blacksquare$$

The analogous proposition for derivatives looks slightly different. We already saw in Example 9.6 c) that $f_n \rightrightarrows f$ is not sufficient for $f'_n \rightarrow f'$ (let alone for $f'_n \rightrightarrows f'$).

Proposition 9.9. $\rightrightarrows$ and derivatives

Let $D = [a, b]$ and let $f_1, f_2, \dots : D \rightarrow \mathbb{R}$ be continuously differentiable functions. If $f_n \rightarrow f$ and $f'_n \rightrightarrows g$ hold, then f is differentiable and $g = f'$.

Proof: For each $n \in \mathbb{N}$ and all $x \in D$, we have, by part 1 of the fundamental theorem,

$$f_n(x) = f_n(a) + \int_a^x f'_n(t) dt. \quad (*)$$

Since $f_n \rightarrow f$ and $f'_n \rightrightarrows g$, sending $n \rightarrow \infty$ leads to

$$f(x) = f(a) + \int_a^x g(t) dt \quad (**)$$

for each $x \in D$, by Proposition 9.8. Because of the continuity of each f'_n and Proposition 9.7, g is continuous. By part 2 of the fundamental theorem, we obtain that f is differentiable, and by differentiation of (**) it follows that $f'(x) = g(x)$ for all $x \in D$. ■

For the function $f_n(x) = \frac{1}{n} \sin(nx)$ from Example 9.6 c), the derivative is $f'_n(x) = \cos(nx)$. Although f_n is pointwise (and even uniformly) convergent, $f'_n(\pi)$ assumes the values $-1, 1$ in turns as $n = 1, 2, \dots$. So f'_n is not even pointwise convergent, let alone uniformly. Proposition 9.9 is therefore not applicable in Example 9.6 c).

🧐 Nerd box 9.10. relaxing the conditions of Proposition 9.9

It is possible to relax the condition $f_n \rightarrow f$, i.e. $f_n(x) \rightarrow f(x)$ for **each** $x \in [a, b]$: It is enough if $f_n(x) \rightarrow f(x)$ holds for **one** $x \in [a, b]$, for example $x = a$. For all other x we then get $f_n(x) \rightarrow f(x)$ from (*) and $f'_n \rightrightarrows g$ (plus Proposition 9.8). We again obtain that $(\text{pointwise-lim } f_n)' = g = \text{uniform-lim}(f'_n)$.

We will see applications of this in the next chapter.

9.3 Series of functions

Series are particularly interesting sequences (of the corresponding partial sums). This is also true in the context of series of functions – recall power series and Taylor series, to which we will return at the end of the section.

Definition 9.11. function series

Let $D \subset \mathbb{R}$ and let $f_0, f_1, f_2, \dots : D \rightarrow \mathbb{R}$ be functions.

We consider the sequence of functions $(s_n)_{n=0}^\infty$ with

$$s_n = f_0 + f_1 + \dots + f_n = \sum_{k=0}^n f_k.$$

If $s_n \rightarrow s$ for a function $s : D \rightarrow \mathbb{R}$, i.e., for each $x \in D$ we have

$$\sum_{k=0}^n f_k(x) \rightarrow s(x) \quad \text{as} \quad n \rightarrow \infty,$$

we call the function s a function series. We write $s = \sum_{k=0}^\infty f_k$.

For each point $x \in D$, the function series $s = \sum_{k=0}^\infty f_k$ becomes a series of real numbers,

$$s(x) = \sum_{k=0}^\infty f_k(x) = \lim_{n \rightarrow \infty}$$

Propositions 9.7, 9.8 and 9.9 transfer to series:

Proposition 9.12. continuity, integral and derivative of function series

- a) Let $s_n \Rightarrow s$ and let each f_k be continuous. Then $s = \sum_{k=0}^\infty f_k$ is also continuous.
- b) Let $s_n \Rightarrow s$, $D = [a, b]$ and each $f_k \in \mathcal{R}([a, b])$. Then we have $s \in \mathcal{R}([a, b])$ and

$$\int_a^b \sum_{k=0}^\infty f_k(x) dx = \sum_{k=0}^\infty \int_a^b f_k(x) dx$$

holds.

- c) Let $s_n \rightarrow s$, $D = [a, b]$ and each f_k be continuously differentiable with $s'_n \Rightarrow g$. Then s is differentiable and we have $s' = g$, i.e.

$$\left(\sum_{k=0}^\infty f_k \right)' = \sum_{k=0}^\infty f'_k.$$

Proof: a) With each f_k , also all $s_n = \sum_{k=0}^n f_k$ and hence s are/is continuous by Prop. 9.7.

b) With each f_k , also all $s_n = \sum_{k=0}^n f_k$ are Riemann-integrable and hence also $s \in \mathcal{R}([a, b])$ by Proposition 9.8 with

$$\begin{aligned} \int_a^b s(x) dx &= \int_a^b \lim_{n \rightarrow \infty} s_n(x) dx \stackrel{\text{Prop. 9.8}}{=} \lim_{n \rightarrow \infty} \int_a^b s_n(x) dx = \lim_{n \rightarrow \infty} \int_a^b \sum_{k=0}^n f_k(x) dx \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_a^b f_k(x) dx = \sum_{k=0}^{\infty} \int_a^b f_k(x) dx. \end{aligned}$$

c) Each f_k is differentiable $\implies$ each s_n is differentiable.

$s_n \rightarrow s, s'_n \rightrightarrows g \implies s$ is differentiable and $g = s'$ (Propositions 9.9), so

$$s'(x) = g(x) = \lim_{n \rightarrow \infty} s'_n(x) = \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n f_k(x) \right)' = \lim_{n \rightarrow \infty} \sum_{k=0}^n f'_k(x) = \sum_{k=0}^{\infty} f'_k(x). \blacksquare$$

We want to apply Theorem 9.12 particularly to power series. A brief preparation:

A function series $s = \sum_{k=0}^{\infty} f_k$ is called

- uniformly convergent, in case $s_n \rightrightarrows s$,
- absolutely convergent, if $\sum_{k=0}^{\infty} |f_k(x)|$ converges (i.e., it is bounded) for each $x \in D$.

Lemma 9.13. a sufficient condition

If $\sum_{k=0}^{\infty} \sup_{x \in D} |f_k(x)| < \infty$ then $s = \sum_{k=0}^{\infty} f_k$ converges absolutely and uniformly.

Proof: The series $\sum_{k=0}^{\infty} |f_k(x)|$ converges for each $x \in D$ by dominated convergence. The uniform convergence of the series of functions s follows, since

$$\sup_{x \in D} |s(x) - s_n(x)| = \sup_{x \in D} \left| \sum_{k=n+1}^{\infty} f_k(x) \right| \leq \sup_{x \in D} \sum_{k=n+1}^{\infty} |f_k(x)| \leq \sum_{k=n+1}^{\infty} \sup_{x \in D} |f_k(x)| \rightarrow 0$$

and therefore $s_n \rightrightarrows s$ for $n \rightarrow \infty$. $\blacksquare$

We now come to the case of power series

$$s(x) = \sum_{k=0}^{\infty} \underbrace{a_k \cdot (x - x_0)^k}_{f_k(x)}, \quad x \in \underbrace{(x_0 - R, x_0 + R)}_D$$

with $x_0, a_0, a_1, a_2, \dots \in \mathbb{R}$ and radius of convergence $R > 0$.

Example 9.14. the geometric series

For $n = 1, 2, \dots$, let

$$s_n(x) = \sum_{k=0}^n x^k.$$

We know from Chapter ??, that $s_n \rightarrow s$ on $D = (-1, 1)$ with $s(x) = \frac{1}{1-x}$. Because

$$|s(x) - s_n(x)| = \left| \sum_{k=n+1}^{\infty} x^k \right| \rightarrow +\infty \quad \text{as } x \rightarrow 1$$

for each $n \in \mathbb{N}$, we have

$$\sup_{x \in (-1, 1)} |s(x) - s_n(x)| = +\infty \not\rightarrow 0,$$

i.e., $s_n \not\rightarrow s$ as $n \rightarrow \infty$.

Restricting D to $[-0.99, 0.99]$ leads to

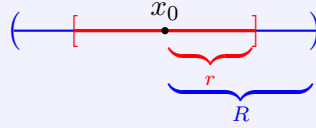
$$\sup_{x \in [-0.99, 0.99]} |s(x) - s_n(x)| = \rightarrow 0$$

and therefore to $s_n \Rightarrow s$ as $n \rightarrow \infty$.

We can see the same phenomenon for power series in general.

Proposition 9.15. convergence of power series

If $r < R$ then $\sum_{k=0}^{\infty} a_k(x - x_0)^k$ converges absolutely and uniformly on $[x_0 - r, x_0 + r]$.



Proof: For $f_k(x) = a_k(x - x_0)^k$ and $x \in [x_0 - r, x_0 + r]$ we have

$$|f_k(x)| =$$

and hence

$$\sup_{x \in [x_0 - r, x_0 + r]} |f_k(x)| \leq |a_k| r^k \quad \text{and therefore} \quad \sum_{k=0}^{\infty} \sup_{x \in [x_0 - r, x_0 + r]} |f_k(x)| \leq \sum_{k=0}^{\infty} |a_k| r^k < \infty,$$

since $x = x_0 + r$ is inside the region of convergence. The rest is by Lemma 9.13. ■

⚠ Attention!

On $(x_0 - R, x_0 + R)$ the convergences is absolute (see Section 6.3) but not necessarily uniform, see Example 9.14.

So, according to Proposition 9.12 and 9.15, we get

Proposition 9.16. continuity, derivatives and integrals of power series

Let $s(x) = \sum_{k=0}^{\infty} a_k(x - x_0)^k$ be a power series with radius of convergence R . Then:

a) s is continuous on $(x_0 - R, x_0 + R)$.

b) s is differentiable on $(x_0 - R, x_0 + R)$ with $s'(x) = \sum_{k=1}^{\infty} k a_k(x - x_0)^{k-1}$.

c) $\int_a^b s(x) dx = \sum_{k=0}^{\infty} \int_a^b a_k(x - x_0)^k dx = \sum_{k=0}^{\infty} \frac{a_k}{k+1} [(x - x_0)^{k+1}]_a^b, \quad a, b \in (x_0 - R, x_0 + R)$.

In short: We can integrate and differentiate power series term by term.

Proof: For $x \in (x_0 - R, x_0 + R)$ there is an $r < R$ with $x \in [x_0 - r, x_0 + r]$. ■

Example 9.17. term by term differentiation of the geometric series

$$s(x) = \frac{1}{1-x} = \sum_{k=0}^{\infty} x^k, \quad x \in (-1, 1),$$

$$= s'(x) =$$

$$= \frac{1}{2} s''(x) =$$

Proposition 9.12 allows to interchange the sum $\sum_{k=0}^{\infty}$ with differentiation and integration in a power series. This is very handy, for example, when we are looking for the power series of f but that of f' is much easier to determine, because

$$\begin{aligned} (\text{power series}(f))' &= \text{power series}(f') \\ \text{i.e., } \text{power series}(f) &= \int \text{power series}(f') \end{aligned}$$

holds.

Example 9.18. power series of arctan

$$\arctan b = \arctan b - \underbrace{\arctan 0}_0 =$$

$$= \frac{b^1}{1} - \frac{b^3}{3} + \frac{b^5}{5} - \frac{b^7}{7} + - \dots$$

 Nerd box 9.19. Three supplements to Example 9.18

- a) We know from Section 6.3 that the power series is automatically the Taylor series of $f = \arctan$ in $x_0 = 0$. This is why the coefficient in front of b^7 , so $-\frac{1}{7}$, is equal to $\frac{f^{(7)}(0)}{7!}$ and therefore

$$f^{(7)}(0) = -\frac{1}{7} \cdot 7! = -6! = -720.$$

Try to calculate this manually by differentiating $\arctan$ seven times...

- b) Because of $\tan(\frac{\pi}{4}) = 1$ we have

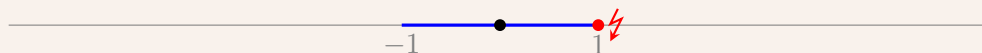
$$\pi = 4 \arctan(1) = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

This is a rapidly converging series for π . The convergence becomes even faster if one uses, for example, Machin's formula,

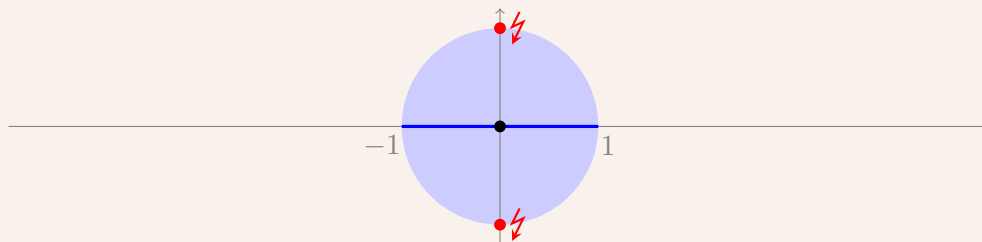
$$\pi = 16 \arctan\left(\frac{1}{5}\right) - 4 \arctan\left(\frac{1}{239}\right)$$

and works with the power series of $\arctan$.

- c) The power series of $f = \arctan$, centered at $x_0 = 0$, has convergence radius $R = 1$. By our previous experience with power series, at the edge of the convergence interval $(x_0 - R, x_0 + R)$, there is usually a pole of f , preventing a bigger area of convergence, see Example 9.14 with $f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$ on $D = (-1, 1)$.



But neither $f = \arctan$ nor $f' : x \mapsto \frac{1}{1+x^2}$ have a pole at the edge of $(-1, 1)$ – And also nowhere else on the real axis! Why isn't the area of convergence bigger? Because $f' : x \mapsto \frac{1}{1+x^2}$ has a pole at $x = \pm i$ each as a function on the complex numbers $\mathbb{C}$.



This is an example of how some effects of real analysis find their explanation in complex analysis and that $\mathbb{C}$ is the natural domain for power series.

Here is another, very similar example:

Example 9.20. power series of the shifted logarithm

$$\ln(1+b) = [\ln(1+x)]_0^b =$$

$$= \frac{b}{1} - \frac{b^2}{2} + \frac{b^3}{3} - + \dots,$$

According to Leibniz' rule, the latter series also converges for $b = 1$. With the so-called Abel's theorem and a continuity argument, we get

$$\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - + \dots = \ln(1+1) = \ln 2. \quad (\textit{alternating harmonic series})$$

Summary

- pointwise convergence, $f_n \rightarrow f$, means: for every $x \in D$ holds $f_n(x) \rightarrow f(x)$, i.e., for all $x \in D$ and all $\varepsilon > 0$ we have $|f_n(x) - f(x)| < \varepsilon$ for $n \geq n_0$ with appropriate n_0 .
- If n_0 depends only on ε but not on x , this is called uniform convergence, $f_n \Rightarrow f$.
- From $f_n \Rightarrow f$ it follows that $f_n \rightarrow f$ (but not vice versa) with the same limit f .
- In $f_n \Rightarrow f$, the continuity of f_n carries over to f ;
- in other words: $\lim_{n \rightarrow \infty}$ commutes with $\lim_{x \rightarrow x_0}$.
- In $f_n \Rightarrow f$, the Riemann integrability of f_n is also carried over to f , and the Riemann integrals of f_n converge to that of f ;
- in other words: $\mathcal{R}([a, b])$ is closed under $\Rightarrow$ and $\int_a^b$ is continuous in this respect;
- yet another way: $\lim_{n \rightarrow \infty}$ commutes with $\int_a^b$.
- Let $f_n \rightarrow f$ and the f_n be continuously differentiable. If f'_n is uniformly convergent, then the limit function is equal to f' . In particular, f is differentiable.
- For uniform convergence of f_n and f'_n , $\lim_{n \rightarrow \infty}$ commutes with differentiation.
- As a consequence, power series can be differentiated or integrated term by term;
- or like this: For power series, $\sum_{k=0}^{\infty}$ commutes with differentiation and integration.
- An application is to obtain the Taylor series of f via that of f' if the latter is simpler.

Chapter 10

Differentiation in several variables

Find the highest and lowest points of the 3D landscape on the right, which is given as the graph

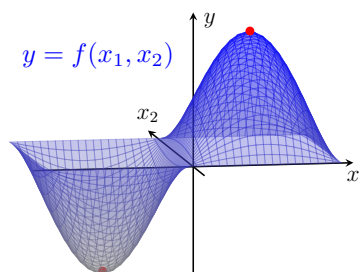
$$\left\{ (x_1, x_2, f(x_1, x_2)) : (x_1, x_2) \in D \subset \mathbb{R}^2 \right\}$$

of a function $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^1$.

Do you set the derivative equal to zero again?

And what is the derivative of f here anyway?

Or: How many noodles should I eat how many hours before the marathon?



$$\text{running performance} = f(\text{amount of noodles, waiting time}) \rightarrow \max$$

This chapter is about differential calculus for functions $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ for $m, n \in \mathbb{N}$ and thus a generalization of differential calculus in the 1D case, $\mathbb{R}^1 \rightarrow \mathbb{R}^1$, of Chapter 7. Again:

The derivative $f'(\mathbf{x}_0)$ of f at the point $\mathbf{x}_0$ maps the change of the input, $\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0$, into a change of the output, $\Delta f = f(\mathbf{x}) - f(\mathbf{x}_0)$, in the best possible **linear** way if $\mathbf{x}$ is close to $\mathbf{x}_0$. More precisely:

$$\Delta f = f'(\mathbf{x}_0)\Delta \mathbf{x} + r(\mathbf{x}) \quad \text{where} \quad \frac{\|r(\mathbf{x})\|}{\|\Delta \mathbf{x}\|} \rightarrow 0 \quad \text{as} \quad \mathbf{x} \rightarrow \mathbf{x}_0. \quad (10.1)$$

In contrast to the 1D-case, the above quantities, $\Delta f, r(\mathbf{x}) \in \mathbb{R}^m$ and $\Delta \mathbf{x} \in \mathbb{R}^n$, are vectors and not numbers. Accordingly,

- $f'(\mathbf{x}_0)$, as a **linear** mapping $\mathbb{R}^n \rightarrow \mathbb{R}^m$, is not a number but an m -by- n matrix,
- $\|r(\mathbf{x})\|$ and $\|\Delta \mathbf{x}\|$ in (10.1) are vector lengths and not absolute values of real numbers,
- we should first clarify what $\mathbf{x} \rightarrow \mathbf{x}_0$ and "close to $\mathbf{x}_0$ " means in $\mathbb{R}^n$.

10.1 Distance and convergence in $\mathbb{R}^n$

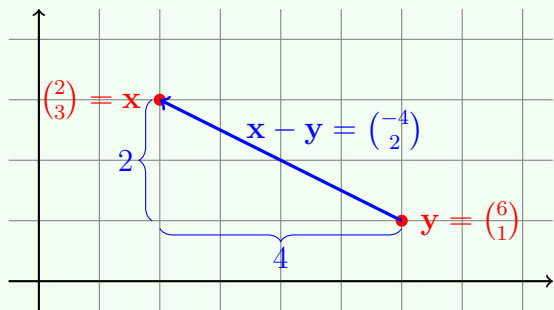
We recall the euclidean norm¹

$$\|\mathbf{v}\| = \left\| \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \right\| = \sqrt{v_1^2 + \cdots + v_n^2}$$

of a vector² $\mathbf{v} \in \mathbb{R}^n$ and, based on this, denote the euclidean distance between two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ as the length (euclidean norm) of their connecting vector, i.e.

$$\|\mathbf{x} - \mathbf{y}\| = \left\| \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} - \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \right\| = \left\| \begin{pmatrix} x_1 - y_1 \\ \vdots \\ x_n - y_n \end{pmatrix} \right\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Example 10.1. euclidean distance in $\mathbb{R}^2$



The euclidean distance of the points

$$\mathbf{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{and} \quad \mathbf{y} = \begin{pmatrix} 6 \\ 1 \end{pmatrix} \quad \text{is}$$

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}\| &= \left\| \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ 1 \end{pmatrix} \right\| = \left\| \begin{pmatrix} -4 \\ 2 \end{pmatrix} \right\| \\ &= \sqrt{(-4)^2 + 2^2} = \sqrt{20}. \end{aligned}$$

¹See Linear Algebra, Definition 2.10. We also continue to use the convention of writing double dashes – also to avoid confusion with the absolute value of numbers in the following.

²Another convention that we adopt from the Linear Algebra course is to write vectors in bold face and numbers (e.g. their components) in normal font.

Let $\mathbf{v} \in \mathbb{R}^n$ and $M := \max\{|v_1|, \dots, |v_n|\}$. Then one component v_i has magnitude M while all others have a magnitude **between 0 and M** .

Accordingly, $\|\mathbf{v}\|$ can be estimated by the two extreme cases

$$\underbrace{\sqrt{M^2 + 0^2 + \dots + 0^2}}_M \leq \underbrace{\sqrt{|v_1|^2 + \dots + |v_n|^2}}_{\|\mathbf{v}\|} \leq \underbrace{\sqrt{M^2 + M^2 + \dots + M^2}}_{\sqrt{nM^2} = \sqrt{n}M}.$$

We obtain the important estimate

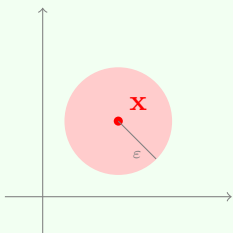
$$\max_{i=1}^n |v_i| \leq \|\mathbf{v}\| \leq \sqrt{n} \cdot \max_{i=1}^n |v_i|. \quad (10.2)$$

Using the euclidean distance, for each $\varepsilon > 0$ and each $\mathbf{x} \in \mathbb{R}^n$, we define an ε -neighbourhood

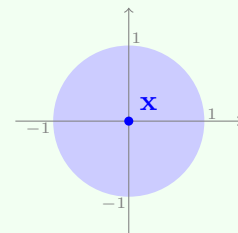
$$U_\varepsilon(\mathbf{x}) := \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{y}\| < \varepsilon\}$$

in $\mathbb{R}^n$.

Example 10.2. ε -neighbourhood in $\mathbb{R}^2$

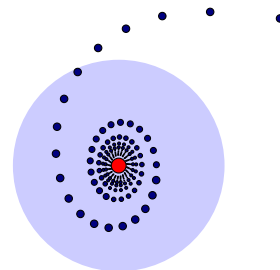


On the left, we see the ε -neighbourhood of an arbitrary point $\mathbf{x} \in \mathbb{R}^2$. The concrete example with $\varepsilon = 1$ and $\mathbf{x} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is shown on the right. The edge of the disc does not belong to $U_\varepsilon(\mathbf{x})$.



With the term ε -neighbourhood we automatically also import the following terms to the context of the vector space $\mathbb{R}^n$:

- convergence, limits: $\mathbf{x}_k \rightarrow \mathbf{x}$
for all $\varepsilon > 0$ all $\mathbf{x}_k$ starting with a certain index k_0 are in $U_\varepsilon(\mathbf{x})$,
in short: $\|\mathbf{x}_k - \mathbf{x}\| \rightarrow 0$ as $k \rightarrow \infty$,
- open/closed sets
 M is open: every point $\mathbf{x} \in M$ has an $\varepsilon > 0$ such that $U_\varepsilon(\mathbf{x}) \subset M$,
 M is closed: the complement of M is open,
- continuity
 $f(\mathbf{x}_n) \rightarrow f(\mathbf{x})$, as $\mathbf{x}_n \rightarrow \mathbf{x}$
or: preimages of open sets are open
or via neighbourhoods: $\forall \varepsilon > 0 \exists \delta > 0 : f(U_\delta(\mathbf{x})) \subset U_\varepsilon(f(\mathbf{x}))$.



The following result takes us back to sequences of numbers from the first semester:

Proposition 10.3. convergence of a sequence of vectors

A sequence $(\mathbf{v}_k)_{k \in \mathbb{N}}$ of vectors in $\mathbb{R}^n$ converges to a vector $\mathbf{v} \in \mathbb{R}^n$, if and only if each component of the vector sequence $(\mathbf{v}_k)$ in $\mathbb{R}$ converges to the corresponding component of $\mathbf{v}$ in $\mathbb{R}$:

$$\mathbf{v}_k = \begin{pmatrix} v_{k,1} \\ \vdots \\ v_{k,n} \end{pmatrix} \xrightarrow{k \rightarrow \infty} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \mathbf{v} \quad \Longleftrightarrow \quad v_{k,j} \xrightarrow{k \rightarrow \infty} v_j \quad \text{for } j = 1, \dots, n$$

To simplify the index notation, we write the statement down for $n = 2$ and $\mathbf{v} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$:

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}, \dots \rightarrow \begin{pmatrix} 3 \\ 5 \end{pmatrix} \quad \Longleftrightarrow \quad x_k \rightarrow 3 \text{ and } y_k \rightarrow 5 \text{ as } k \rightarrow \infty.$$

The x components of the vector sequence converge to the x component of the limit vector $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$. The same applies to the y components. This extends accordingly to vector sequences with more than two components.

Proof: We only prove the statement for the latter special situation with $n = 2$ and $\mathbf{v} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$. The general proof works in the same way. So let $\mathbf{v}_k = \begin{pmatrix} x_k \\ y_k \end{pmatrix} \in \mathbb{R}^2$ for $k = 1, 2, \dots$

■

🧐 Nerd box 10.4. maximum norm

- a) The expression $\max\{|v_1|, \dots, |v_n|\}$ from the inequality in (10.2) is also called the maximum norm^a of the vector $\mathbf{v} \in \mathbb{R}^n$ and is abbreviated by $\|\mathbf{v}\|_\infty$.
- b) The reason for the notation $\|\mathbf{v}\|_\infty$ is as follows: the Euclidean norm can be generalized to the so-called p-norm, which is

$$\|\mathbf{v}\|_p := \sqrt[p]{|v_1|^p + \dots + |v_n|^p}, \quad \mathbf{v} \in \mathbb{R}^n, \quad \text{with } p \in [1, \infty).$$

The Euclidean norm is then obtained in the special case $p = 2$, and the maximum norm is the limit value of $\|\mathbf{v}\|_p$ as $p \rightarrow \infty$, see Example 2.23 c).

- c) The inequality (10.2) then states that $\|\mathbf{v}\|$ and $\|\mathbf{v}\|_\infty$ always differ, multiplicatively, only by values from $[1, \sqrt{n}]$. This is referred to as equivalence of the two norms. In particular, convergence with respect to one norm also leads to convergence with respect to the other one. This is exactly what happened in the proof of Proposition 10.3.

^aThe superordinate concept of a norm is discussed in Chapter 7 of Linear Algebra

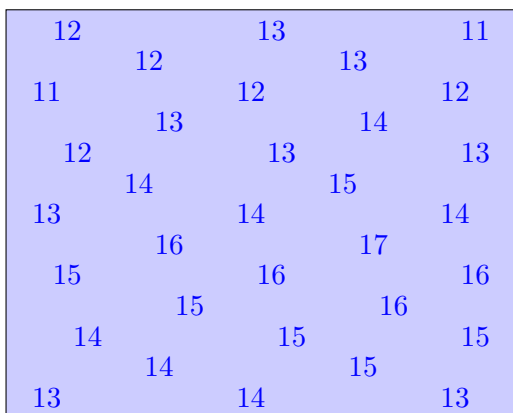
Proposition 10.3 reduces the investigation of vector sequences in $\mathbb{R}^n$ to the investigation of n scalar sequences in $\mathbb{R}$ (by decomposing the vector sequence into n sequences of the corresponding vector components). This strategy of componentwise decomposition is also the key to continuity – and ultimately even to differentiability – of vector-valued functions. We will now take a closer look at that:

10.2 Scalar fields and vector fields

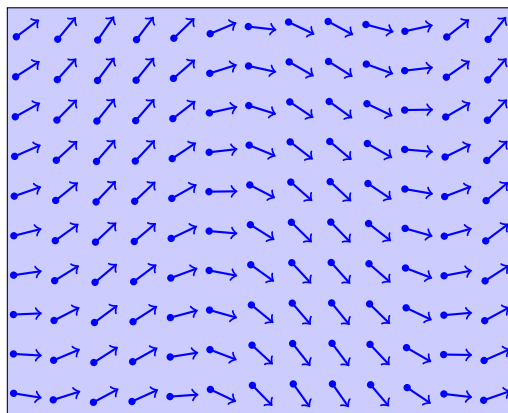
The current chapter, as already mentioned, is about functions $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$.

- If $m = 1$, we call f a scalar-valued function or a scalar field,
- if $m > 1$, we call f a vector-valued function or a vector field.
- The value of n is irrelevant for these terms.

Think about, for example, the weather forecast for a rectangular region $D \subset \mathbb{R}^2$:



The temperature chart: a scalar field $\mathbb{R}^2 \rightarrow \mathbb{R}^1$.
Each point in D gets assigned a number.



The wind chart: a vector field $\mathbb{R}^2 \rightarrow \mathbb{R}^2$.
Each point in D gets a vector in $\mathbb{R}^2$.

If we decompose the wind vector into its x - and y -components, we get a scalar field each. Together, both give the same information as the 2D-vector field of the wind.

In general:

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a vector field. With $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in D \subset \mathbb{R}^n$ we get $f(\mathbf{x}) \in \mathbb{R}^m$,
i.e.

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} =: \begin{pmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{pmatrix}.$$

We call this process the componentwise decomposition of the $\mathbb{R}^m$ -vector field f into its m components, the scalar fields $f_1, \dots, f_m$. We also write, a little shorter,

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{pmatrix} \quad \text{and sometimes even just} \quad f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}.$$

Proposition 10.5. continuity of a vector field

Let $\mathbf{x}_0 \in D \subset \mathbb{R}^n$ and let $f : D \rightarrow \mathbb{R}^m$ be a vector field with componentwise decomposition $f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$ into scalar fields $f_1, \dots, f_m : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$. Then

$$f \text{ is continuous in } \mathbf{x}_0 \iff \text{its components } f_1, \dots, f_m \text{ are continuous in } \mathbf{x}_0.$$

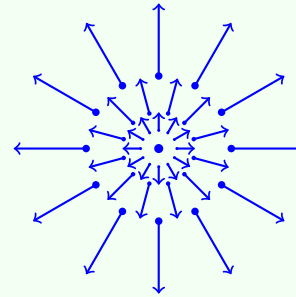
Proof: Let $\mathbf{x}_1, \mathbf{x}_2, \dots \in D$ be arbitrary with $\mathbf{x}_k \rightarrow \mathbf{x}_0$ as $k \rightarrow \infty$. Continuity of f means that $f(\mathbf{x}_k) \rightarrow f(\mathbf{x}_0)$. Continuity of the scalar fields $f_1, \dots, f_m$ means that $f_j(\mathbf{x}_k) \rightarrow f_j(\mathbf{x}_0)$ for $j = 1, \dots, m$. According to Proposition 10.3, both statements are equivalent. ■

Example 10.6. continuous and discontinuous vector fields

a) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $f(\mathbf{x}) = \mathbf{x}$, i.e.,

$$f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

Being the identity function, f is continuous since



continuous

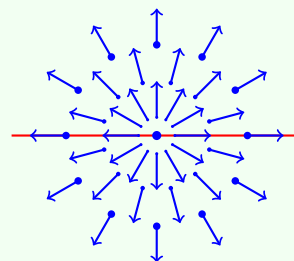
Alternatively, one could check the continuity of its components,

$$f_1(x_1, x_2) = x_1 \quad \text{and} \quad f_2(x_1, x_2) = x_2.$$

- b) This time, let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(\mathbf{o}) = \mathbf{o}$ and $f(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|}$ for $\mathbf{x} \in \mathbb{R}^2 \setminus \{\mathbf{o}\}$, i.e.,

$$f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} \frac{x_1}{\|\mathbf{x}\|} \\ \frac{x_2}{\|\mathbf{x}\|} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 \setminus \{\mathbf{o}\}.$$

In contrast to a), the vectors are scaled to length 1 this time. Since $\|f(\mathbf{o})\| = 0$ and $\|f(\mathbf{x})\| = 1$ for $\mathbf{x} \neq \mathbf{o}$, f can't be continuous at $\mathbf{o}$. We check the first component and detect discontinuity in $(0,0)$ since



discontinuous at $\mathbf{o}$

So f_1 (and hence f) is discontinuous at $\mathbf{o}$.

- c) This time let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by

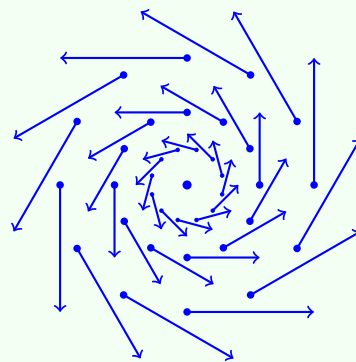
$$f : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}, \quad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

We can easily check that the components

$$f_1(x_1, x_2) = -x_2 \quad \text{und} \quad f_2(x_1, x_2) = x_1$$

are both continuous on the whole of $\mathbb{R}^2$. So f is a continuous vector field.

(Still you don't want to see this vector field as part of your weather forecast.)



continuous

In the next section, as soon as we have properly defined what differentiation and derivatives mean, we will prove the following statement:

Proposition 10.7. differentiability and derivatives of a vector field

Let $\mathbf{x}_0 \in D \subset \mathbb{R}^n$ and let $f : D \rightarrow \mathbb{R}^m$ be a vector field with componentwise decomposition $f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$ into scalar fields $f_1, \dots, f_m : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$. Then

f is differentiable in $\mathbf{x}_0 \iff$ its components $f_1, \dots, f_m$ are differentiable in $\mathbf{x}_0$

and the derivative $f'(\mathbf{x}_0)$ is calculated by

$$f'(\mathbf{x}_0) = \begin{pmatrix} f'_1(\mathbf{x}_0) \\ \vdots \\ f'_m(\mathbf{x}_0) \end{pmatrix}.$$

The last formula describes how a $m \times n$ -matrix is constructed by stacking m matrices of size $1 \times n$ on top of each other.

Between us: a) The message of Proposition 10.7 and b) an outlook

a) If we understand differentiation for scalar fields, we hereby also understand it for vector fields. Because of this, we can often restrict ourselves to the case of $m = 1$.

b) On the other hand, n has to be left as it is. However, we will see that the major points become clear by looking at $n = 2$. Then we are in the case of $\mathbb{R}^2 \rightarrow \mathbb{R}^1$ that we met in the examples at the beginning, where the graph of the function can be visualized in 3D – which is often very helpful.

10.3 Differentiability and derivative

Now all preparations have been made and we can return to (10.1), where we had used the abbreviations $\Delta f = f(\mathbf{x}) - f(\mathbf{x}_0)$ and $\Delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0$.

Avoiding these Δ notations, we arrive at the following definition:

Definition 10.8. (total) differentiability and derivative

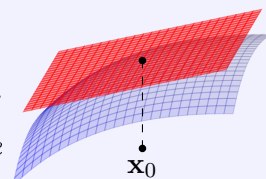
A function $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called (totally) differentiable in $\mathbf{x}_0 \in D$, if there exists a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ and therefore a matrix $J_f(\mathbf{x}_0) \in \mathbb{R}^{m \times n}$ such that

$$f(\mathbf{x}) = f(\mathbf{x}_0) + J_f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + r(\mathbf{x}) \quad \text{where} \quad \frac{\|r(\mathbf{x})\|}{\|\mathbf{x} - \mathbf{x}_0\|} \rightarrow 0 \quad \text{as} \quad \mathbf{x} \rightarrow \mathbf{x}_0. \quad (10.3)$$

The linear mapping $\mathbb{R}^n \rightarrow \mathbb{R}^m$ mentioned above is then uniquely determined, and it is called the derivative of f in $\mathbf{x}_0$. Notation: $f'(\mathbf{x}_0)$ or $Df(\mathbf{x}_0)$ or $df(\mathbf{x}_0)$

If f is differentiable in all $\mathbf{x}_0 \in D$ then f is called (totally) differentiable on D .

The above matrix $J_f(\mathbf{x}_0) \in \mathbb{R}^{m \times n}$ is called Jacobian matrix of f in $\mathbf{x}_0$. From now on, we will only write J_f for $J_f(\mathbf{x}_0)$ if we have already fixed the point $\mathbf{x}_0$.



In the next section, we will study partial differentiability and partial derivatives. To distinguish between these different concepts (partial and not partial), the differentiability in Definition 10.8 is referred to as total differentiability.

As already discussed before in (10.1), $f'(\mathbf{x}_0)$ is the linear part of the best affine-linear approximation of f in the neighbourhood of $\mathbf{x}_0$. In the case of $\mathbb{R}^2 \rightarrow \mathbb{R}^1$, $f'(\mathbf{x}_0)$ describes the slope of the tangent plane of the three-dimensional graph of f . An alternative notation for (10.3) is again

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + J_f \mathbf{h} + r(\mathbf{x}_0 + \mathbf{h}) \quad \text{where} \quad \frac{\|r(\mathbf{x}_0 + \mathbf{h})\|}{\|\mathbf{h}\|} \rightarrow 0 \quad \text{as} \quad \mathbf{h} \rightarrow \mathbf{o}. \quad (10.4)$$

⚠ Attention! linear mapping vs. matrix

In Chapter 8 of Linear Algebra you will learn about the relation between linear mappings $\ell : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $m \times n$ -matrices J – and also that ℓ and J are of course not the same object.

However, if we fix the standard basis $\mathcal{B} = (\mathbf{e}_i)_{i=1}^n$ and $\mathcal{C} = (\mathbf{e}_i)_{i=1}^m$ of the corresponding canonical unit vectors $\mathbf{e}_i$ in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, then the coupling between linear mapping $\ell = f'(\mathbf{x}_0)$ and transformation matrix $J_f(\mathbf{x}_0) = \ell_{\mathcal{C} \leftarrow \mathcal{B}}$ is fixed, and we can switch back and forth between $f'(\mathbf{x}_0)$ and $J_f(\mathbf{x}_0)$ without losing too many words.

As in the 1D case, $f : D \subset \mathbb{R}^1 \rightarrow \mathbb{R}^1$, also for functions $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, we have:

Proposition 10.9. differentiable $\implies$ continuous

If f is totally differentiable in a point $\mathbf{x}_0 \in D$ then f is also continuous in $\mathbf{x}_0$.

Proof: Let f be differentiable in $\mathbf{x}_0$. If $\mathbf{x} \rightarrow \mathbf{x}_0$ we obtain $\|J_f \cdot (\mathbf{x} - \mathbf{x}_0)\| \rightarrow 0$ and furthermore

$$\|r(\mathbf{x})\| = \underbrace{\frac{\|r(\mathbf{x})\|}{\|\mathbf{x} - \mathbf{x}_0\|}}_{\rightarrow 0} \cdot \underbrace{\|\mathbf{x} - \mathbf{x}_0\|}_{\rightarrow 0} \rightarrow 0,$$

and thus our assertion $f(\mathbf{x}) \rightarrow f(\mathbf{x}_0)$ follows with (10.3). ■

Example 10.10. differentiability and derivative

- a) Let $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be defined by $f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$. Then f is differentiable in every point $\mathbf{x}_0 \in \mathbb{R}^n$ and $J_f = A$ holds. The computation:

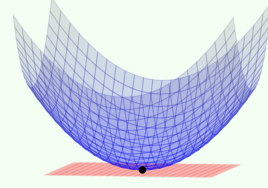
if we use $J_f = A$.

b) We consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ with

$$\mathbf{x} \mapsto \|\mathbf{x}\|^2, \quad \text{i.e.} \quad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto x_1^2 + x_2^2.$$

Then f is differentiable in $\mathbf{x}_0 = \mathbf{o}$ and $J_f = (0 \ 0) \in \mathbb{R}^{1 \times 2}$ holds.

Reasoning: Because of $f(\mathbf{x}) = \|\mathbf{x}\|^2$ and $\mathbf{x}_0 = \mathbf{o}$ we obtain



if one uses the zero matrix $(0 \ 0) \in \mathbb{R}^{1 \times 2}$ for J_f .

(10.3) describes the equality of two vectors in $\mathbb{R}^m$. It holds if and only if equality holds componentwise, i.e. if, for $i = 1, \dots, m$,

$$f_i(\mathbf{x}) = f_i(\mathbf{x}_0) + J_i \cdot (\mathbf{x} - \mathbf{x}_0) + r_i(\mathbf{x}) \quad \text{with} \quad \frac{r_i(\mathbf{x})}{\|\mathbf{x} - \mathbf{x}_0\|} \rightarrow 0 \quad \text{as} \quad \mathbf{x} \rightarrow \mathbf{x}_0 \quad (10.5)$$

holds, where J_i denotes the i -th row of the Jacobian matrix J_f of f in $\mathbf{x}_0$ and $r_i : \mathbb{R}^n \rightarrow \mathbb{R}$ denotes the i -th component function of the vector field $r = \begin{pmatrix} r_1 \\ \vdots \\ r_m \end{pmatrix} : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

The fact that $\frac{r(\mathbf{x})}{\|\mathbf{x} - \mathbf{x}_0\|}$ converges to $\mathbf{o}$ if and only if all components $\frac{r_i(\mathbf{x})}{\|\mathbf{x} - \mathbf{x}_0\|}$ converge to 0 follows again from Proposition 10.3. Equation (10.5) however, describes the differentiability of the i -th component function f_i of f in $\mathbf{x}_0$ with $f'_i(\mathbf{x}_0) \cong J_i$. This proves Proposition 10.7.

10.4 Partial differentiability and partial derivatives

Now we want to find a general method to calculate the entries of the Jacobian matrix

$$J_f = \begin{pmatrix} j_{1,1} & \cdots & j_{1,n} \\ \vdots & & \vdots \\ j_{m,1} & \cdots & j_{m,n} \end{pmatrix} = \begin{pmatrix} - & J_1 & - \\ & \vdots & \\ - & J_m & - \end{pmatrix} \in \mathbb{R}^{m \times n} \quad (10.6)$$

of $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ in $\mathbf{x}_0 \in D$ also beyond simple examples such as Example 10.10.

By Proposition 10.7, we know that the rows $J_1, \dots, J_m$ of J_f are exactly the Jacobian matrices of the components (i.e., the scalar fields) $f_1, \dots, f_m : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ of $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$. So, for now, let us consider the special case of Jacobian matrices of scalar fields.

10.4.1 Jacobian matrix of a scalar field

Let now $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a scalar field itself, and let f be differentiable in $\mathbf{x}_0 \in D$ with Jacobian matrix

$$J_f = (j_1 \ \cdots \ j_n) \in \mathbb{R}^{1 \times n}. \quad (10.7)$$

Let $k \in \{1, \dots, n\}$. We know that we obtain the k -th column of J_f in the following way:

$$j_k = \quad \text{where} \quad \mathbf{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^n$$

is the k -th vector of the standard basis of $\mathbb{R}^n$, so it has the one as its k -th entry.

To calculate $J_f \mathbf{e}_k$, we simply rearrange (10.4) as follows:

$$j_k = J_f \mathbf{e}_k = \frac{J_f t \mathbf{e}_k}{t} = \frac{J_f t \mathbf{e}_k}{\|t \mathbf{e}_k\|} \stackrel{(10.4)}{=} \frac{f(\mathbf{x}_0 + t \mathbf{e}_k) - f(\mathbf{x}_0)}{\|t \mathbf{e}_k\|} = \underbrace{\frac{r(\mathbf{x}_0 + t \mathbf{e}_k)}{\|t \mathbf{e}_k\|}}_{\rightarrow 0 \text{ as } t \rightarrow 0}, \quad (10.8)$$

so that

$$\begin{aligned}
 j_k &= \lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{e}_k) - f(\mathbf{x}_0)}{\|t\mathbf{e}_k\|} = \lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{e}_k) - f(\mathbf{x}_0)}{t} \\
 &= \lim_{t \rightarrow 0} \frac{f(x_1, \dots, x_{k-1}, x_k + t, x_{k+1}, \dots, x_n) - f(x_1, \dots, x_{k-1}, x_k, x_{k+1}, \dots, x_n)}{t},
 \end{aligned} \tag{10.9}$$

where $(x_1, \dots, x_n)$ are the coordinates of $\mathbf{x}_0$. We arrive at the classical 1D-derivative again, the derivative of $g : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ with

$$g : t \mapsto f(\mathbf{x}_0 + t\mathbf{e}_k) =$$

in $t = 0$.

Definition 10.11. partial differentiability and partial derivatives

Let $n \in \mathbb{N}$ and $k \in \{1, \dots, n\}$. A scalar field $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ is called partially differentiable with respect to its k -th variable in $\mathbf{x}_0 \in D$, if the function

$$g : t \mapsto f(\mathbf{x}_0 + t\mathbf{e}_k)$$

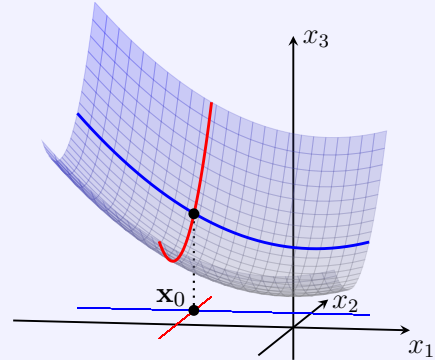
is differentiable in $t = 0$. The number $g'(0)$ is then called the partial derivative of f w.r.t the k -th variable x_k in $\mathbf{x}_0$. It is denoted by

$$\frac{\partial f(\mathbf{x}_0)}{\partial x_k}, \quad f_{x_k}(\mathbf{x}_0) \quad \text{or} \quad D_k f(\mathbf{x}_0).$$

In case all partial derivatives

$$\frac{\partial f(\mathbf{x}_0)}{\partial x_1}, \quad \dots, \quad \frac{\partial f(\mathbf{x}_0)}{\partial x_n}$$

exist, we call f partially differentiable in $\mathbf{x}_0$.



Thus, one of several ($n > 1$) variables is selected and f is differentiated with respect to it, while all other variables are treated as constants. The word "partial" and the notation with ∂ signal this selective kind of differentiation.

Example 10.12. partial derivatives

If there are only $n = 2$ (or $n = 3$) variables, these are often denoted by x, y (and z) instead of x_1, x_2 (and x_3).

- a)** Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ with $f(x, y, z) = 3x + 4y + 5z$. Then f is partially differentiable in $\mathbb{R}^3$ and for each point $(x, y, z) \in \mathbb{R}^3$ we have:

$$\frac{\partial f(x, y, z)}{\partial x} = \quad , \quad \frac{\partial f(x, y, z)}{\partial y} = \quad , \quad \frac{\partial f(x, y, z)}{\partial z} = \quad .$$

- b)** Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ with $g(x, y, z) = x^2 e^y + \sin z$. Then g is partially differentiable in $\mathbb{R}^3$ and again we have, for each point $(x, y, z) \in \mathbb{R}^3$:

$$\frac{\partial g(x, y, z)}{\partial x} = \quad , \quad \frac{\partial g(x, y, z)}{\partial y} = \quad , \quad \frac{\partial g(x, y, z)}{\partial z} = \quad .$$

For a better understanding of what is actually being calculated here, let's take a closer look at the illustration in the box of Definition 10.11. In this case, $n = 2$ holds, so that the graph of $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^1$ forms a surface in $\mathbb{R}^{2+1} = \mathbb{R}^3$:

Visualization of partial derivatives for $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^1$

The curved surface G in the figure on the right shows the graph of a scalar field

$$f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^1.$$

The two points

$$\mathbf{x}_0 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \in D \subset \mathbb{R}^2$$

(here in $\mathbb{R}^3$ with $x_3 = 0$) and

$$\mathbf{p} = \begin{pmatrix} \mathbf{x}_0 \\ f(\mathbf{x}_0) \end{pmatrix} \in \mathbb{R}^3$$

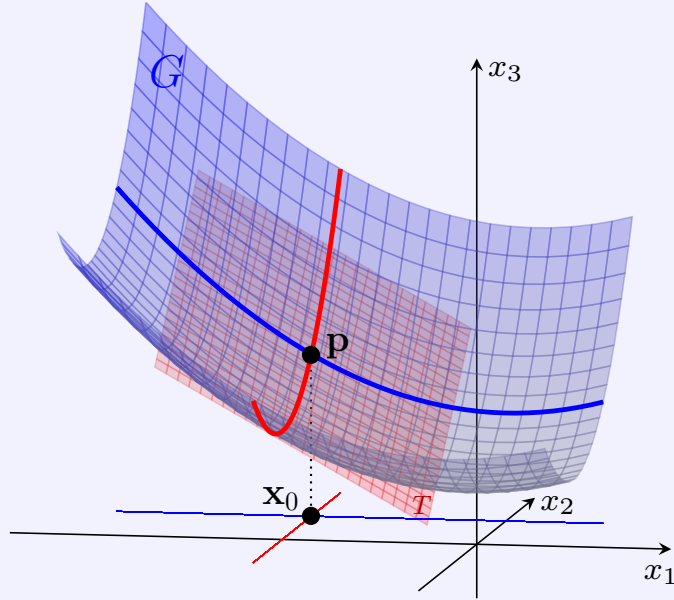
are highlighted. The point $\mathbf{p}$ lies on the graph of f , i.e. on the surface G .

Going through $\mathbf{x}_0$, we get the thin blue line when only varying the x_1 -coordinate of $\mathbf{x}_0$. These are the points of the form $\mathbf{x}_0 + t\mathbf{e}_1$ with $t \in \mathbb{R}$. Analogously, we get the thin red line if we just vary the x_2 -coordinate of $\mathbf{x}_0$, i.e., all points of the form $\mathbf{x}_0 + t\mathbf{e}_2$ with $t \in \mathbb{R}$. The thick blue and red curves are then the images of the thin lines, respectively, with respect to the function f . They hence lie on the surface G , where they intersect in $\mathbf{p}$.

The slope of the blue/red curve in the point $\mathbf{p}$ is what the two partial derivatives

$$\frac{\partial f(\mathbf{x}_0)}{\partial x_1} \quad \text{and} \quad \frac{\partial f(\mathbf{x}_0)}{\partial x_2} \quad (10.10)$$

indicate. The tangent line in $\mathbf{p}$ of the blue curve spans, together with the tangent line of the red curve, the tangential plane T (shown in bright red) of the surface G at the point $\mathbf{p}$. The slope of the plane T in x_1 - resp. x_2 -direction is thus given by the two values in (10.10).



If we understand all of that, the step to $n > 2$ is just a formality.

We come back to our Jacobian matrix (10.7) for scalar fields $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$, whose entries j_k turned out to be the partial derivatives of f in the k -th variable in (10.9). Wrapping up, the Jacobian matrix of f in $\mathbf{x}_0 \in D$ is

$$J_f = \left(\begin{array}{cccc} & & & \\ & & & \\ & & & \\ & & & \end{array} \right) \in \mathbb{R}^{1 \times n}. \quad (10.11)$$

⚠ Attention! Be careful with equality (10.11)!

Just because the partial derivatives $\frac{\partial f(\mathbf{x}_0)}{\partial x_i}$ on the right-hand side of (10.11) exist, the row vector formed from them is not necessarily the Jacobian matrix J_f at the point $\mathbf{x}_0$. It is indeed possible that f is only partially differentiable and not (totally) differentiable at $\mathbf{x}_0$; see Section 10.5 below. In that case, the Jacobian matrix J_f would not exist there at all. However, if f is (totally) differentiable at $\mathbf{x}_0$, then both sides of (10.11) exist and the equality holds as it is stated there.

Practically the same vector (10.11), only in transposed form, appears just as frequently in applications and therefore receives its own name:

Definition 10.13. gradient

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a scalar field and let f be partially differentiable in $\mathbf{x}_0 \in D$. Then we call the column vector

$$\begin{pmatrix} \frac{\partial f(\mathbf{x}_0)}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \end{pmatrix}$$

the gradient of f in $\mathbf{x}_0$ and denote it by $\nabla f(\mathbf{x}_0)$ or $\text{grad } f(\mathbf{x}_0)$.

The symbol ∇ is pronounced "nabla". It is not a letter of any alphabet.

⚠ Attention! The gradient only exists for scalar fields

While the concept of the Jacobian matrix also exists for vector fields, see (10.6), the gradient $\nabla f(\mathbf{x}_0)$ exists only for scalar fields!

The gradient plays an important role as a vector in $\mathbb{R}^n$. Therefore, it must be a column vector. The Jacobian matrix J_f , on the other hand, must be a row vector such that the matrix multiplication in (10.3) and (10.4) works as intended. Some books only use one of the two symbols and switch to the other via the transpose sign.

Example 10.14. gradient and Jacobian matrix (scalar field)

- a) The function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^1$ with $f(x, y, z) = 3x + 4y + 5z$ from Example 10.12 a) has in (x, y, z) the gradient

$$\nabla f(x, y, z) = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \quad \text{and the Jacobian matrix} \quad J_f = \begin{pmatrix} 3 & 4 & 5 \end{pmatrix}.$$

- b) The function $g : \mathbb{R}^3 \rightarrow \mathbb{R}^1$ with $g(x, y, z) = x^2 e^y + \sin z$ from Example 10.12 b) has in (x, y, z) the gradient

$$\nabla g(x, y, z) = \begin{pmatrix} 2xy e^y \\ x^2 e^y \\ \cos z \end{pmatrix} \quad \text{and the Jacobian matrix} \quad J_g = \begin{pmatrix} 2xy e^y & x^2 e^y & \cos z \end{pmatrix}.$$

- c) Let $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ with $h(\mathbf{x}) = \|\mathbf{x}\|$, i.e., $h(x, y) = \sqrt{x^2 + y^2}$ is the euclidean distance of $\mathbf{x}$ from the origin. h is partially differentiable in $(x, y) \in \mathbb{R}^2 \setminus \{\mathbf{o}\}$.

and therefore

$$\nabla h(x, y) = \frac{1}{\|\mathbf{x}\|} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\|\mathbf{x}\|} \mathbf{x} \quad \text{and} \quad J_h = \frac{1}{\|\mathbf{x}\|} \mathbf{x}^\top.$$

We notice that $\nabla h : \mathbb{R}^2 \setminus \{\mathbf{o}\} \rightarrow \mathbb{R}^2$ is the vector field from Example 10.6 b) outside of the discontinuity point $\mathbf{o}$.

10.4.2 Jacobian matrix of a vector field

It remains for us to close the circle of this section by getting back to the Jacobian matrix of a vector field $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$. So let us decompose f into its components, the scalar fields $f_1, \dots, f_m : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$, and combine (10.6) with (10.11). Then we get at the point $\mathbf{x}_0 \in D$

$$J_f = \begin{pmatrix} - J_{f_1} - \\ \vdots \\ - J_{f_m} - \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1(\mathbf{x}_0)}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{x}_0)}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m(\mathbf{x}_0)}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{x}_0)}{\partial x_n} \end{pmatrix} \in \mathbb{R}^{m \times n}. \quad (10.12)$$

Example 10.15. Jacobian matrix of a vector field

We glue both the scalar fields $f, g : \mathbb{R}^3 \rightarrow \mathbb{R}^1$ from Example 10.14 a) and b) together, leading to the vector field $h = \begin{pmatrix} f \\ g \end{pmatrix} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, i.e.,

$$h(x, y, z) = \begin{pmatrix} f(x, y, z) \\ g(x, y, z) \end{pmatrix} = \begin{pmatrix} \\ \end{pmatrix} \quad x, y, z \in \mathbb{R}.$$

Then the Jacobian matrix of h at the point (x, y, z) is equal to

$$J_h = \begin{pmatrix} - J_f - \\ - J_g - \end{pmatrix} = \begin{pmatrix} \\ \end{pmatrix}$$

10.5 Partial vs. total differentiability

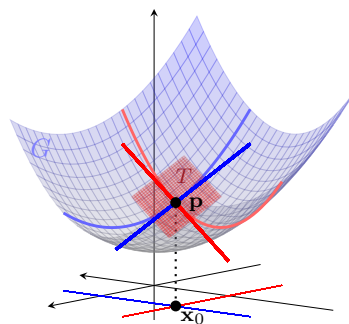
For a comparison between total and partial differentiability, we go back to scalar fields, since partial differentiability was defined only there.

🧐 Nerd box 10.16. partial differentiability of vector fields

Many books introduce partial differentiability also for vector fields and call f partially differentiable if each of its components $f_1, \dots, f_m$ is a partially differentiable scalar field. The partial derivative of f is then the vector composed of the partial derivatives of $f_1, \dots, f_m$.

Geometrically speaking: In case $\mathbb{R}^2 \rightarrow \mathbb{R}^1$...

- ...total differentiability in $\mathbf{x}_0$ means that the graph G of f has a tangential plane T in p ,
- while partial differentiability in $\mathbf{x}_0$ only means that the blue and red curve on G each have a tangential line in p .



Here is the analytical argument for arbitrary scalar fields $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$:

The implication

$$\text{total differentiability} \implies \text{partial differentiability} \quad (10.13)$$

holds since the existence of the limit in (10.8) implies that in (10.9).

The converse direction of (10.13) does not hold. Our geometric intuition in the case of $\mathbb{R}^2 \rightarrow \mathbb{R}^1$ leads us to a very simple counterexample:

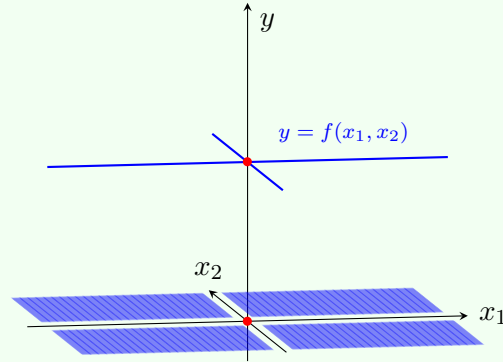
Example 10.17. partially but not totally differentiable

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ be given by

$$f(x_1, x_2) = \begin{cases} 1, & x_1 = 0, \\ 1, & x_2 = 0, \\ 0 & \text{otherwise.} \end{cases}$$

i.e., $f(\mathbf{x})$ is equal to 1 if $\mathbf{x}$ lies on one of the two axes, and $f(\mathbf{x})$ is otherwise equal to 0.

At the point $\mathbf{x}_0 = \mathbf{o}$, both partial derivatives exist, $\frac{\partial f(\mathbf{o})}{\partial x_1}$ and $\frac{\partial f(\mathbf{o})}{\partial x_2}$, but f is obviously discontinuous there and, by Proposition 10.9, not totally differentiable.



To get from partial to total differentiability, so more or less to get the converse direction in (10.13), we have to require a lot more than partial differentiability. For this we introduce the following terms:

Definition 10.18. continuous partial differentiability

Let $n \in \mathbb{N}$ and $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a scalar field. If f is partially differentiable in each point $\mathbf{x} \in D$, we call f partially differentiable, and we can treat the partial derivatives

$$\frac{\partial f(\mathbf{x})}{\partial x_1}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \in \mathbb{R}^1 \quad \text{or} \quad \frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \in \mathbb{R}^1$$

as functions

$$\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} : D \rightarrow \mathbb{R}^1$$

of $\mathbf{x} \in D$. If all these functions are continuous in $\mathbf{x}_0 \in D$, then we call f continuously partially differentiable in $\mathbf{x}_0$. Is this the case for each $\mathbf{x}_0 \in D$, then we call f continuously partially differentiable.

It's time for an example:

Example 10.19. continuous partial differentiability

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ be given by $f(\mathbf{x}) = \|\mathbf{x}\|$, i.e., $f(x, y) = \sqrt{x^2 + y^2}$. By Example

10.14 c), f is partially differentiable in $\mathbb{R}^2 \setminus \{\mathbf{o}\}$, where

$$\frac{\partial f}{\partial x}(x, y) = \frac{x}{\sqrt{x^2 + y^2}} \quad \text{and} \quad \frac{\partial f}{\partial y}(x, y) = \frac{y}{\sqrt{x^2 + y^2}}.$$

Since both functions, $\frac{\partial f}{\partial x}$ as well as $\frac{\partial f}{\partial y}$, are continuous on $\mathbb{R}^2 \setminus \{\mathbf{o}\}$, f is continuously partially differentiable there.

Now indeed, the continuous partial differentiability does the trick for us:

Proposition 10.20. converse direction holds for continuity of the partial derivative

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a partially differentiable scalar field and let $\mathbf{x}_0 \in D$.
If f is continuously partially differentiable in $\mathbf{x}_0$, f is totally differentiable in $\mathbf{x}_0$.

Proof: Using (10.5), we check componentwise that (10.12) plays the role of J_f in (10.3). For this, we split the transition from $\mathbf{x}$ to $\mathbf{x}_0$ into n steps, each parallel to one axis, each one of which leads to a step in the value of f equal to the step length times a partial derivative in a neighborhood of $\mathbf{x}_0$, via the mean value theorem. For details see the literature. ■

So in the end we have this chain of implications:

continuously partially differentiable $\implies$ totally differentiable $\implies$ partially differentiable

10.6 Differentiation rules

The differentiation rules from 1D calculus $\mathbb{R}^1 \rightarrow \mathbb{R}^1$ carry over to the multidimensional setting as follows. Their proofs also carry over, so that we do not have to repeat them.

Proposition 10.21. sum and scaling

Let $f, g : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be vector fields that are differentiable in $\mathbf{x}_0 \in D$. Let $\alpha \in \mathbb{R}$. Then also $f + g$ and αf are differentiable in $\mathbf{x}_0$ and

$$(f + g)'(\mathbf{x}_0) = f'(\mathbf{x}_0) + g'(\mathbf{x}_0) \quad \text{as well as} \quad (\alpha f)'(\mathbf{x}_0) = \alpha \cdot f'(\mathbf{x}_0)$$

hold.

At the level of Jacobian matrices, the last line corresponds to the sum of two $m \times n$ matrices or the product of a number α with an $m \times n$ matrix.

Next, we arrive at the product rule and quotient rule. For these, we need scalar fields, i.e., functions mapping into $\mathbb{R}^1$, in order to multiply and divide in the usual manner.

Proposition 10.22. product rule

Let $f, g : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be scalar fields that are differentiable in $\mathbf{x}_0 \in D$. Then $f \cdot g$ is also differentiable in $\mathbf{x}_0$ and we have

$$(f \cdot g)'(\mathbf{x}_0) = \underbrace{g(\mathbf{x}_0)}_{\in \mathbb{R}} \cdot f'(\mathbf{x}_0) + \underbrace{f(\mathbf{x}_0)}_{\in \mathbb{R}} \cdot g'(\mathbf{x}_0).$$

The equation expresses the linear map $(f \cdot g)'(\mathbf{x}_0)$ as a linear combination of $f'(\mathbf{x}_0)$ and $g'(\mathbf{x}_0)$. For the Jacobian matrix/the gradient in $\mathbf{x}_0$ we can write:

$$J_{fg}(\mathbf{x}_0) = g(\mathbf{x}_0)J_f(\mathbf{x}_0) + f(\mathbf{x}_0)J_g(\mathbf{x}_0) \text{ and } \nabla(f \cdot g)(\mathbf{x}_0) = g(\mathbf{x}_0) \cdot \nabla f(\mathbf{x}_0) + f(\mathbf{x}_0) \cdot \nabla g(\mathbf{x}_0).$$

We can understand the terms in the quotient rule in a similar way:

Proposition 10.23. quotient rule

Let $f, g : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be scalar fields that are differentiable in $\mathbf{x}_0 \in D$ and let $g(\mathbf{x}_0) \neq 0$. Then $\frac{f}{g}$ is also differentiable in $\mathbf{x}_0$ and we have

$$\left(\frac{f}{g}\right)'(\mathbf{x}_0) = \frac{g(\mathbf{x}_0) \cdot f'(\mathbf{x}_0) - f(\mathbf{x}_0) \cdot g'(\mathbf{x}_0)}{g(\mathbf{x}_0)^2}.$$

Example 10.24. of the product rule

We start with the scalar fields $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ with

$$f(x, y) = x + 2y \quad \text{and} \quad g(x, y) = 3x + 4y$$

and their product,

$$(f \cdot g)(x, y) =$$

before we come to its Jacobian matrix,

$$J_{fg}(x, y) =$$

It is claimed that the product rule comes to the same result:

$$J_{fg}(x, y) = g(x, y) \cdot J_f(x, y) + f(x, y) \cdot J_g(x, y)$$

$$=$$

$$=$$

$$=$$

True.

We can also generalize the chain rule if we are careful with the order of the terms:

Proposition 10.25. chain rule

Let f and g be vector fields that we can compose to $g \circ f$, i.e.,

$$f : D_f \subset \mathbb{R}^n \rightarrow \mathbb{R}^m \quad \text{and} \quad g : D_g \subset \mathbb{R}^m \rightarrow \mathbb{R}^k \quad \text{with} \quad f(D_f) \subset D_g.$$

If f is differentiable in $\mathbf{x}_0 \in D_f$ and g is differentiable in $f(\mathbf{x}_0) \in D_g$, then also $g \circ f : D_f \subset \mathbb{R}^n \rightarrow \mathbb{R}^k$ is differentiable in $\mathbf{x}_0$ and

$$\underbrace{(g \circ f)'(\mathbf{x}_0)}_{\mathbb{R}^k \leftarrow \mathbb{R}^n} = \underbrace{g'(f(\mathbf{x}_0))}_{\mathbb{R}^k \leftarrow \mathbb{R}^m} \circ \underbrace{f'(\mathbf{x}_0)}_{\mathbb{R}^m \leftarrow \mathbb{R}^n}.$$

The right hand side is the composition of two linear maps. We notice that the order of the terms must not be changed. The same is evident when writing the equation with Jacobian matrices:

$$\underbrace{J_{g \circ f}(\mathbf{x}_0)}_{\in \mathbb{R}^{k \times n}} = \underbrace{J_g(f(\mathbf{x}_0))}_{\in \mathbb{R}^{k \times m}} \cdot \underbrace{J_f(\mathbf{x}_0)}_{\in \mathbb{R}^{m \times n}}$$

For the chain rule we look at an example:

Example 10.26. of the chain rule

We compose the function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad f(r, \varphi) = \begin{pmatrix} r \cos \varphi \\ r \sin \varphi \end{pmatrix},$$

that maps polar coordinates to cartesian ones, and the function

$$g : \mathbb{R}^2 \rightarrow \mathbb{R}^1, \quad g(x, y) = x^2 + y^2$$

to get $g \circ f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$. We are looking for the derivative of $g \circ f$.

1st Approach: We calculate the composed function first,

$$(g \circ f)(r, \varphi) = g(f(r, \varphi)) =$$

and calculate its Jacobian matrix then:

$$J_{g \circ f}(r, \varphi) = \left(\frac{\partial(g \circ f)(r, \varphi)}{\partial r} \quad \frac{\partial(g \circ f)(r, \varphi)}{\partial \varphi} \right) =$$

2nd Approach: Using the chain rule and the Jacobian matrices of both functions:

$$J_{g \circ f}(r, \varphi) = J_g(f(r, \varphi)) J_f(r, \varphi)$$

=

If we formalize the last result for general compositions of the form

$$\begin{pmatrix} r \\ \varphi \end{pmatrix} \xrightarrow{f} \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{g} z \quad \text{i.e.} \quad z = z(x, y) = z(x(r, \varphi), y(r, \varphi)) =: \hat{z}(r, \varphi),$$

we get

$$\begin{pmatrix} \frac{\partial \hat{z}}{\partial r} & \frac{\partial \hat{z}}{\partial \varphi} \end{pmatrix} = J_{g \circ f}(r, \varphi) = J_g(f(r, \varphi)) J_f(r, \varphi) = \begin{pmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \varphi} \end{pmatrix}$$

and therefore

$$\frac{\partial \hat{z}}{\partial r} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial r} \quad \text{and} \quad \frac{\partial \hat{z}}{\partial \varphi} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial \varphi} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial \varphi}.$$

A bit simpler but still quite useful is the following general situation:

$$t \xrightarrow{f} \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{g} z \quad \text{i.e.} \quad z = z(x, y) = z(x(t), y(t)) =: \hat{z}(t).$$

Then we have

$$\frac{d\hat{z}}{dt}(t) = J_{g \circ f}(t) = J_g(f(t)) J_f(t)$$

and hence (in short):

$$\frac{d\hat{z}}{dt} = \begin{pmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

Note the mix of partial and 1D-derivatives.

10.7 Directional derivatives and level sets

This section deals exclusively with scalar fields $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$, i.e.,

$$J_f(\mathbf{x}_0) = \left(\frac{\partial f}{\partial x_1}(\mathbf{x}_0) \quad \cdots \quad \frac{\partial f}{\partial x_n}(\mathbf{x}_0) \right) = (\nabla f(\mathbf{x}_0))^\top.$$

We want to rely on geometric intuition here even more than in the previous sections and therefore we start with functions $\mathbb{R}^2 \rightarrow \mathbb{R}^1$ before generalizing to $\mathbb{R}^n \rightarrow \mathbb{R}^1$.

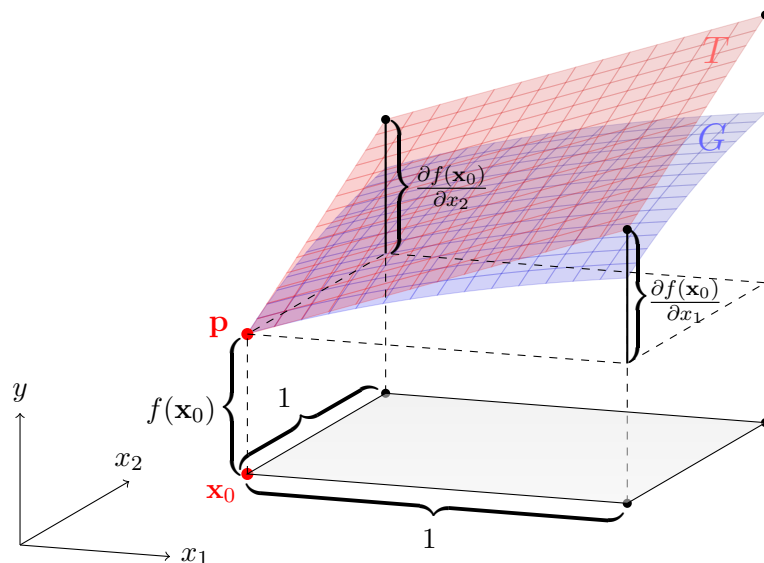
We start with the graph G of a function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^1.$$

Let f be differentiable in $\mathbf{x}_0 \in \mathbb{R}^2$ and let T be the corresponding tangent plane.

We now move around on the plane T , starting at the point

$$\mathbf{p} = \begin{pmatrix} \mathbf{x}_0 \\ f(\mathbf{x}_0) \end{pmatrix}.$$



With each step to the east (i.e. +1 in the x_1 direction), our position on T increases by $\frac{\partial f(\mathbf{x}_0)}{\partial x_1}$, and with each step to the north (i.e. +1 in the x_2 direction), our position on T increases by $\frac{\partial f(\mathbf{x}_0)}{\partial x_2}$.

So if we walk h_1 steps to the east and h_2 steps to the north, our height on T changes by

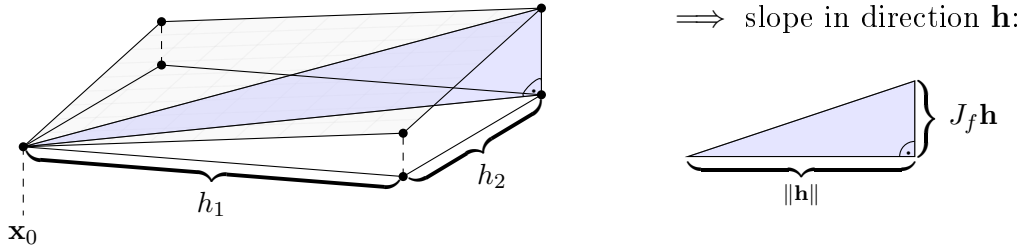
$$h_1 \cdot \frac{\partial f(\mathbf{x}_0)}{\partial x_1} + h_2 \cdot \frac{\partial f(\mathbf{x}_0)}{\partial x_2} = \underbrace{\begin{pmatrix} \frac{\partial f(\mathbf{x}_0)}{\partial x_1} & \frac{\partial f(\mathbf{x}_0)}{\partial x_2} \end{pmatrix}}_{J_f} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = J_f(\mathbf{x}_0) \mathbf{h}, \quad \text{where } \mathbf{h} = \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}.$$

At this point we have a look at (10.1):

$$\Delta f = J_f \Delta \mathbf{x} + r(\mathbf{x}),$$

i.e. the change in height Δf is obtained by multiplying the Jacobian matrix J_f by the change in position $\Delta \mathbf{x} = \mathbf{h}$, plus adding an error term $r(\mathbf{x})$, which describes the difference between the curved surface G and the straightened surface T (tangential plane) above the point $\mathbf{x} = \mathbf{x}_0 + \mathbf{h}$.

We easily obtain the slope of the tangent plane in the direction of $\mathbf{h} = \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$:



The aforementioned slope in the direction of $\mathbf{h}$ is therefore equal to

$$\frac{J_f \mathbf{h}}{\|\mathbf{h}\|} = J_f \frac{\mathbf{h}}{\|\mathbf{h}\|} = \left\langle \nabla f(\mathbf{x}_0), \frac{\mathbf{h}}{\|\mathbf{h}\|} \right\rangle.$$

The angle brackets $\langle \cdot, \cdot \rangle$ denote the standard inner product in $\mathbb{R}^n$. For an analytical derivation of the same expression, one looks at the derivative of the function

$$g : \mathbb{R}^1 \rightarrow \mathbb{R}^1, \quad \text{where} \quad g : t \mapsto f\left(\mathbf{x}_0 + t \frac{\mathbf{h}}{\|\mathbf{h}\|}\right) \quad (10.14)$$

with respect to t and determines $g'(0)$. If f is differentiable, the chain rule yields

$$g'(0) = J_f(\mathbf{x}_0) \frac{\mathbf{h}}{\|\mathbf{h}\|} = \left\langle \nabla f(\mathbf{x}_0), \frac{\mathbf{h}}{\|\mathbf{h}\|} \right\rangle. \quad (10.15)$$

Definition 10.27. directional derivative

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a scalar field, $\mathbf{x}_0 \in D$, $\mathbf{h} \in \mathbb{R}^n \setminus \{\mathbf{o}\}$ and let g denote the function from (10.14). Then the number

$$\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{h}} := g'(0)$$

is called directional derivative of f in the direction $\mathbf{h}$ at the point $\mathbf{x}_0$.

If f is differentiable then the directional derivative $\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{h}}$ is hence given by (10.15).

Example 10.28. directional derivative

$$\mathbf{h} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \quad \Rightarrow \quad \frac{\partial f}{\partial \mathbf{h}} = \frac{3 \cdot \frac{\partial f}{\partial x_1} - 2 \cdot \frac{\partial f}{\partial x_2}}{\sqrt{3^2 + (-2)^2}} = \frac{3}{\sqrt{13}} \cdot \frac{\partial f}{\partial x_1} - \frac{2}{\sqrt{13}} \cdot \frac{\partial f}{\partial x_2}$$

⚠ Attention! f need not be differentiable in Definition 10.27

The directional derivative $\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{h}} = g'(0)$ can exist (even in all directions $\mathbf{h}$) despite f being not differentiable. ↗ tutorials

Each partial derivative $\frac{\partial f}{\partial x_j}$ is a special case of a directional derivative $\frac{\partial f}{\partial \mathbf{h}}$ with $\mathbf{h} = \mathbf{e}_j$.

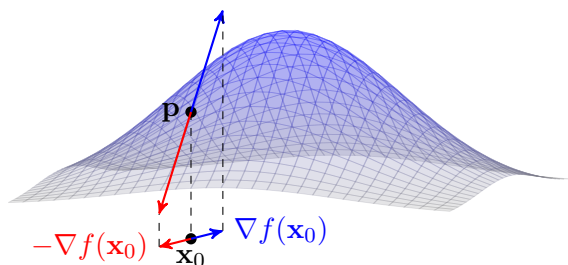
We know from linear algebra that $\langle \mathbf{a}, \mathbf{b} \rangle = \|\mathbf{a}\| \cdot \|\mathbf{b}\| \cdot \cos \angle(\mathbf{a}, \mathbf{b})$ holds and thus

$$\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{h}} = \left\langle \nabla f(\mathbf{x}_0), \frac{\mathbf{h}}{\|\mathbf{h}\|} \right\rangle =$$

=

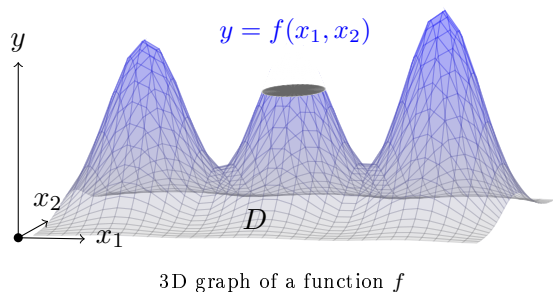
So if you are walking on the graph of f in the point $\mathbf{p} = \begin{pmatrix} \mathbf{x}_0 \\ f(\mathbf{x}_0) \end{pmatrix}$ and are looking for the direction $\mathbf{h}$ with the steepest slope uphill, $\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{h}}$, the cosine must be maximized. This means $\mathbf{h}$ must have the same direction as the gradient $\nabla f(\mathbf{x}_0)$, i.e.,

$\nabla f(\mathbf{x}_0)$ points in the direction of the steepest ascent of f in $\mathbf{x}_0$
and $-\nabla f(\mathbf{x}_0)$ points in the direction of the steepest descent of f in $\mathbf{x}_0$.

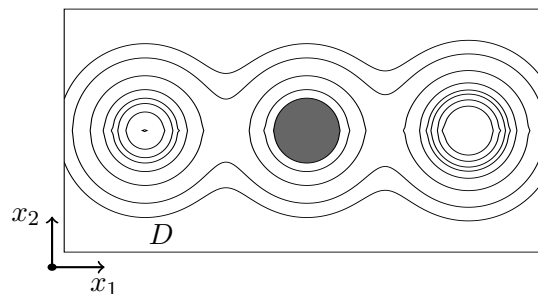


$\Rightarrow$ strategy for finding
local maxima / minima of f
numerically („gradient method“)

Classical 2D maps contain information about the elevation in the form of contour lines:



3D graph of a function f



contour lines ("level sets")

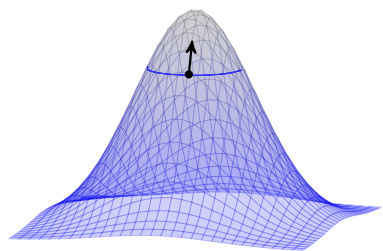
Definition 10.29. level sets

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a scalar field and let $c \in \mathbb{R}$ be fixed. Then

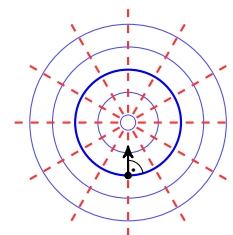
$$N_c := \{ \mathbf{x} \in D : f(\mathbf{x}) = c \}$$

is called the level set of f for the level c .

If you follow a level curve when hiking, you walk around the mountain at a constant altitude. However, if you want to change direction and climb to the top or walk to the valley, your new path appears to be at a right angle to the level curve:



level set
("circular walk" with constant altitude c)



level sets and
gradient or field lines
(steepest ascent and descent)

$$\text{gradient} \perp \text{level set}$$

Indeed, here is the mathematical confirmation: Let $K = \varphi([a, b])$ be a curve (the image of a C^1 -path $\varphi : [a, b] \rightarrow \mathbb{R}^n$), which is contained in one level set N_c . Then we have

$$f(\varphi(t)) \equiv c, \quad t \in [a, b].$$

Differentiation on both sides with respect to t yields, by the chain rule,

i.e., the gradient $\nabla f(\varphi(t))$ is orthogonal to the velocity vector $\varphi'(t)$ of the path φ and thus to the curve K at any time t .

10.8 Higher order derivatives and Taylors formula

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a partially differentiable scalar field and let $i \in \{1, \dots, n\}$. The mapping

$$f \mapsto \frac{\partial f}{\partial x_i},$$

which turns the function f into the function of its partial derivative with respect to x_i is denoted by

$$\frac{\partial}{\partial x_i} \quad \text{or just by} \quad D_i,$$

such that

$$\frac{\partial f}{\partial x_i} = \frac{\partial}{\partial x_i} f = D_i f. \quad (10.16)$$

10.8.1 Higher order derivatives and Schwarz's theorem

If the function $g = (10.16) : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ is partially differentiable, then its partial derivatives

$$\frac{\partial}{\partial x_j} g = \frac{\partial}{\partial x_j} \frac{\partial}{\partial x_i} f =: \frac{\partial \partial f}{\partial x_j \partial x_i} =: \frac{\partial^2 f}{\partial x_j \partial x_i} \quad \text{or} \quad D_j g = D_j (D_i f) = D_j D_i f$$

exist for $j = 1, \dots, n$ and are called partial derivatives of second order of f .

Analogously: partial derivatives of the third, fourth, ... order.

Definition 10.30. k times continuously partially differentiable

For a scalar field $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$, we write $f \in C^k(D)$ and we say that f is k times continuously partially differentiable if all partial derivatives up to the k -th order exist and if all these partial derivatives are continuous.

🧐 Nerd box 10.31. Higher order derivatives for total differentiation

In the process of total differentiation of $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, i.e.

$$f \mapsto f',$$

the object changes as follows:

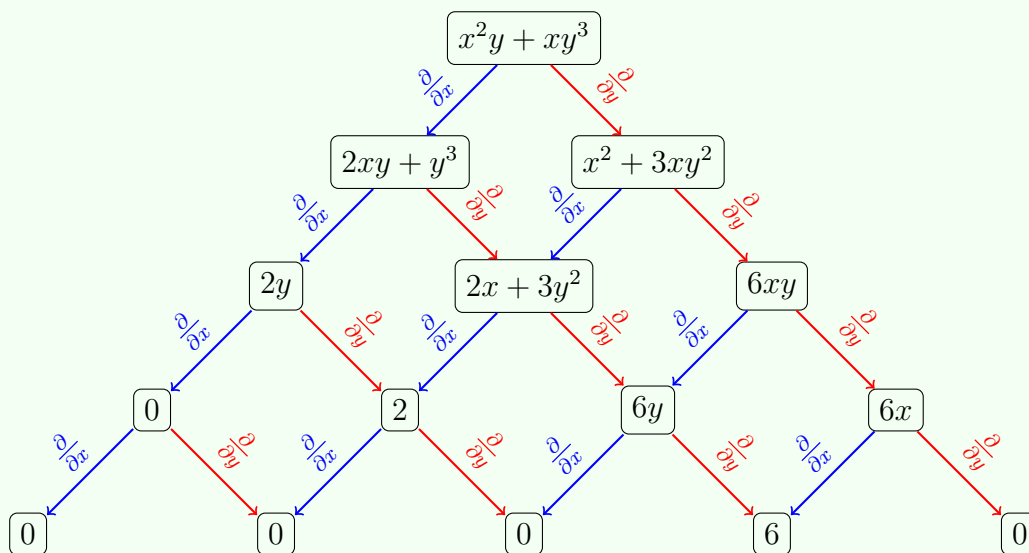
While f maps points $\mathbf{x} \in D \subset \mathbb{R}^n$ to vectors $f(\mathbf{x}) \in \mathbb{R}^m$, the derivative f' maps points $\mathbf{x} \in D \subset \mathbb{R}^n$ to linear maps $f'(\mathbf{x}) \cong J_f(\mathbf{x}) \in \mathbb{R}^{m \times n}$. Putting $k := m \cdot n$, we have $\mathbb{R}^{m \times n} \cong \mathbb{R}^{m \cdot n} = \mathbb{R}^k$, whence we can regard f' as a function $f' : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^k$.

One can then look at the total differentiability and the total derivative of f' again, and f is called k times continuously differentiable if the total derivatives up to order k exist and the k -th derivative is continuous. Sometimes the notation $C^k(D)$ is used for the set of functions $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ that are k times continuously (totally) differentiable in this sense. However, the equivalence of continuous total differentiability and continuous partial differentiability (see Proposition 10.20) ensures the consistency of the notation despite different definitions.

It's time for an example:

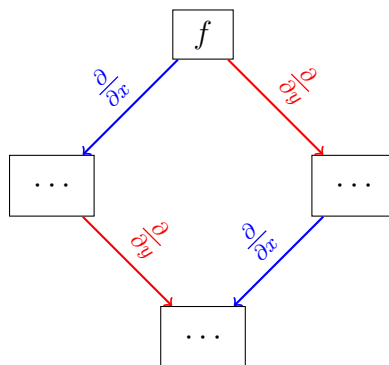
Example 10.32. higher order partial derivatives

Take $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ with $f(x, y) = x^2y + xy^3$. Then the partial derivatives up to order four are given by the following terms:



One can see that the partial derivatives of fifth, sixth, ... order also exist and are equal to the zero function.

The diagram in Example 10.32 contains the following piece



and also many other pieces of the same form.

Does the equality

$$\frac{\partial}{\partial x} \frac{\partial}{\partial y} f = \frac{\partial}{\partial y} \frac{\partial}{\partial x} f$$

always hold?

The same question in the general context $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ for $i, j \in \{1, \dots, n\}$:

Does $D_i D_j f = D_j D_i f$ always hold ?

Generally not:

Example 10.33. $D_i D_j f \neq D_j D_i f$

For the function

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases}$$

the following holds ($\nearrow$ tutorials)

$$D_1 D_2 f(0, 0) = 1 \neq -1 = D_2 D_1 f(0, 0).$$

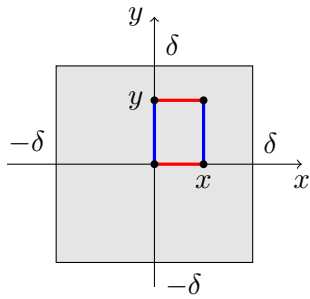
However:

Proposition 10.34. Schwarz's theorem (Hermann Amandus Schwarz, 1843–1921)

Let $D \subset \mathbb{R}^n$ and take $i, j \in \{1, \dots, n\}$. For $f \in C^2(D)$, i.e. if the second order partial derivatives are continuous, for every $\mathbf{x}_0 \in D \subset \mathbb{R}^n$, the equality of mixed partial derivatives holds:

$$D_i D_j f(\mathbf{x}_0) = D_j D_i f(\mathbf{x}_0)$$

Proof: It is sufficient to consider the case $n = 2$ with $i = 1$ and $j = 2$. For the sake of simplicity, let $\mathbf{x}_0 = (0, 0) \in D$. Choose $\delta > 0$ such that $(-\delta, \delta)^2 \subset D$.



For $x, y \in (-\delta, \delta)$ define

$$F_y(x) := f(x, y) - f(x, 0).$$

Then

$$\begin{aligned} f(x, y) - f(x, 0) - f(0, y) + f(0, 0) &= (f(x, y) - f(x, 0)) - (f(0, y) - f(0, 0)) \\ &= F_y(x) - F_y(0) \stackrel{\text{MVT-Diff}}{=} F'_y(\xi) \cdot x \\ &= (D_1 f(\xi, y) - D_1 f(\xi, 0)) \cdot x \\ &\stackrel{\text{MVT-Diff}}{=} D_2 D_1 f(\xi, \eta) \cdot x \cdot y \end{aligned}$$

for some $\xi \in (0, x)$ and $\eta \in (0, y)$.

Using $G_x(y) := f(x, y) - f(0, y)$, one can show analogously

$$f(x, y) - f(x, 0) - f(0, y) + f(0, 0) = D_1 D_2 f(\tilde{\xi}, \tilde{\eta}) \cdot x \cdot y$$

for some $\tilde{\xi} \in (0, x)$ and $\tilde{\eta} \in (0, y)$. Therefore we have

$$\begin{array}{ccc} D_1 D_2 f(\tilde{\xi}, \tilde{\eta}) & = & D_2 D_1 f(\xi, \eta) \\ \downarrow & & \downarrow \\ D_1 D_2 f(0, 0) & & D_2 D_1 f(0, 0) \end{array} \quad \text{as } (x, y) \rightarrow (0, 0)$$

due to the continuity of $D_1 D_2 f$ and $D_2 D_1 f$. ■

Corollary 10.35. commuting k partial derivatives

For $f \in C^k(D)$, the order of differentiation is not relevant for partial derivatives up to order k .

And what do we do with all these derivatives?

10.8.2 Taylor's theorem in several variables

Proposition 10.36. Taylor polynomial (and Lagrange's remainder) in 2 variables

Let $D \subset \mathbb{R}^2$, $\ell \in \mathbb{N}$, $f \in C^{\ell+1}(D)$ and let the two points $\mathbf{x}_0 = (x_0, y_0)$ and $\mathbf{x}_0 + \mathbf{h} = (x_0 + \Delta x, y_0 + \Delta y)$ in D be such that also the line segment

$$S := \{(x_0, y_0) + t(\Delta x, \Delta y) : t \in [0, 1]\}$$

connecting $\mathbf{x}_0$ and $\mathbf{x}_0 + \mathbf{h}$ is still in D . Then there is a point (ξ, η) on S such that

$$\begin{aligned} f(x_0 + \Delta x, y_0 + \Delta y) &= f(x_0, y_0) + \frac{1}{1!} \left(f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y \right) \\ &+ \frac{1}{2!} \left(f_{xx}(x_0, y_0) \Delta x^2 + 2f_{xy}(x_0, y_0) \Delta x \Delta y + f_{yy}(x_0, y_0) \Delta y^2 \right) \\ &+ \dots \\ &+ \frac{1}{\ell!} \sum_{j=0}^{\ell} \binom{\ell}{j} f_{x^{\ell-j} y^j}(x_0, y_0) \Delta x^{\ell-j} \Delta y^j \\ &+ \frac{1}{(\ell+1)!} \sum_{j=0}^{\ell+1} \binom{\ell+1}{j} f_{x^{\ell+1-j} y^j}(\xi, \eta) \Delta x^{\ell+1-j} \Delta y^j. \end{aligned}$$

Here $f_{x^i y^j}$ denotes the partial derivative $(\frac{\partial}{\partial x})^i (\frac{\partial}{\partial y})^j f = \frac{\partial^{i+j}}{\partial x^i \partial y^j} f$ of f .

Proof: Applying Proposition 7.35 (Taylor's theorem in 1D) to the function

$$g : [0, 1] \rightarrow \mathbb{R}^1 \quad \text{with} \quad g : t \mapsto f(x_0 + t\Delta x, y_0 + t\Delta y) \in \mathbb{R}^1$$

with Taylor expansion at $t_0 = 0$, one gets

$$g(t) = \sum_{k=0}^n \frac{g^{(k)}(0)}{k!} (t-0)^k + \frac{g^{(n+1)}(\xi)}{(n+1)!} (t-0)^{n+1}, \quad t \in [0, 1].$$

Evaluating this at $t = 1$, yields the assertion. ■

Example 10.37. of a Taylor polynomial in 2 variables

Take $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ with

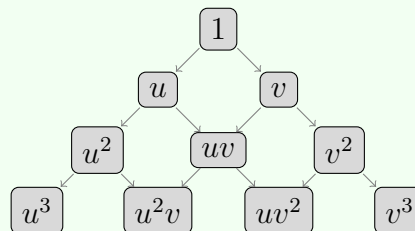
$$f(x, y) = e^{2x+3y}, \quad x, y \in \mathbb{R}.$$

We are looking for the Taylor polynomial p of order $\ell = 3$ of f with the center $\mathbf{x}_0 = \mathbf{o} \in \mathbb{R}^2$. If we set $\mathbf{h} = \begin{pmatrix} u \\ v \end{pmatrix}$ with $u, v \in \mathbb{R}$, then $p(\mathbf{h})$ is of the form

$$p(u, v) = \underbrace{\alpha}_{\text{degree 0}} + \underbrace{\beta u + \gamma v}_{\text{degree 1}} + \underbrace{\delta u^2 + \varepsilon uv + \varphi v^2}_{\text{degree 2}} + \underbrace{\kappa u^3 + \lambda u^2 v + \sigma uv^2 + \tau v^3}_{\text{degree 3}}$$

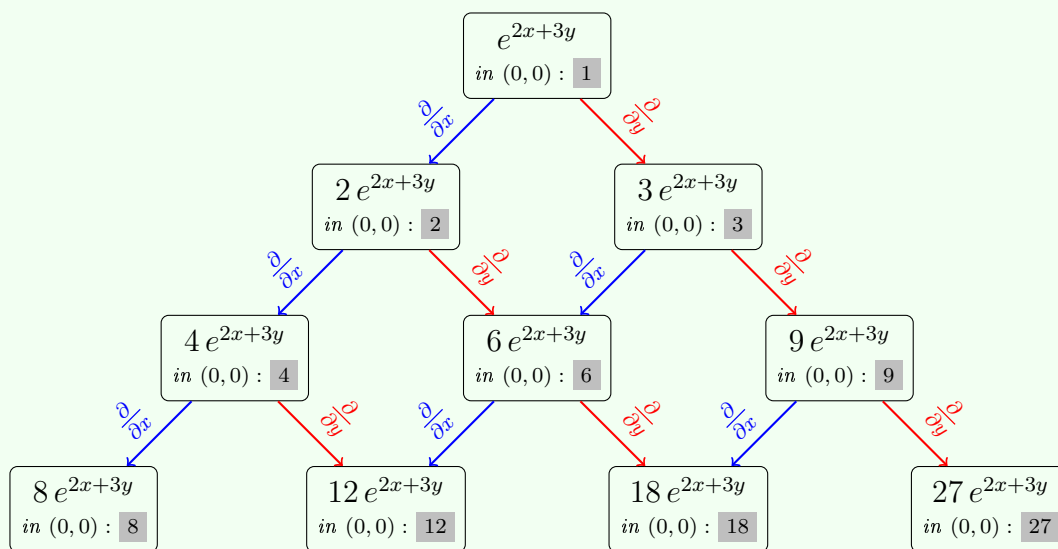
with coefficients $\alpha, \beta, \dots, \tau \in \mathbb{R}$ depending on f and its partial derivatives at the point $\mathbf{x}_0 = \mathbf{o}$.

In order to assign the above coefficients $\alpha, \beta, \dots, \tau$ correctly after we have determined them, we arrange the 2D monomials in the following scheme.



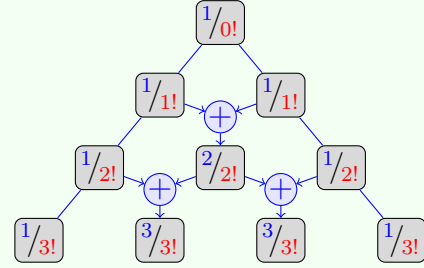
The rows of the diagram correspond (from top to bottom) to the 2D monomials of zeroth, first, second and third degree.

Now f comes into play: Similar to Example 10.32, we summarize the function terms of the partial derivatives of f up to the order $\ell = 3$ and insert the point $\mathbf{x}_0 = \mathbf{o}$, i.e. $(x, y) = (0, 0)$:



In the last step we regard both *the binomial-coefficients* $\binom{k}{j}$ from Proposition 10.36 and *the factor* $1/k!$ *for the term of k -th order*.

To do this, we divide *the entries of Pascal's triangle* by $k!$, where k is the considered monomial degree, i.e. the line number in the scheme.



It remains to multiply our three schemes pointwise and to add them up. Doing so, we obtain (line by line) $p = p_0 + p_1 + p_2 + p_3$ with

where p_k with $k \in \{0, 1, 2, 3\}$ combines the terms of degree k .

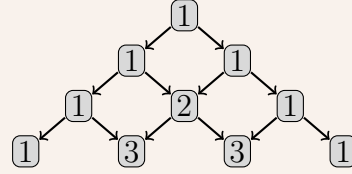
If the Lagrange remainder is also of interest, one has to carry all three schemes one line further. But for the partial derivatives of order $\ell + 1$, use the unknown intermediate value $(x, y) = (\xi, \eta)$ from Proposition 10.36 instead of $(x, y) = (0, 0)$.

🧐 Nerd box 10.38. followup on Example 10.37

a) In this example, the result is exactly the 1D Taylor polynomial

$$p(u, v) = \frac{(2u + 3v)^0}{0!} + \frac{(2u + 3v)^1}{1!} + \frac{(2u + 3v)^2}{2!} + \frac{(2u + 3v)^3}{3!}.$$

b) The binomial coefficients in Pascal's triangle (without division by $k!$) indicate how many different paths exist from the top of the diagram on the right to the considered node.



The formulation of Proposition 10.36 and Example 10.37 give little hope for the case of three or more variables. This is where so-called multi-indices come in for rescue: For $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$, we set

$$|\alpha| := \alpha_1 + \dots + \alpha_n, \quad D^\alpha f := D_{x_1}^{\alpha_1} \dots D_{x_n}^{\alpha_n} f, \quad \alpha! := \alpha_1! \dots \alpha_n!$$

$$\text{and} \quad (\Delta x_1, \dots, \Delta x_n)^\alpha := \Delta x_1^{\alpha_1} \dots \Delta x_n^{\alpha_n}.$$

Example 10.39. on multi-indices

Let $n = 2$, $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ and let $\alpha = (3, 2)$ be a multi-index with two components. Then

$$D^\alpha f = \left(\frac{\partial}{\partial x} \right)^3 \left(\frac{\partial}{\partial y} \right)^2 f = \frac{\partial^{3+2}}{\partial x^3 \partial y^2} f$$

is the partial derivative of f , taking three partial differentiations with respect to x and two with respect to y . It is therefore a partial derivative of the order

$$|\alpha| = 3 + 2 = 5.$$

If $f \in C^5(D)$, then it does not matter in which order the five partial differentiations are being applied. Furthermore,

$$\alpha! = 3! \cdot 2! = 12 \quad \text{and} \quad (\Delta x, \Delta y)^\alpha = (\Delta x)^3 \cdot (\Delta y)^2.$$

These notations allow for a perfect formulation of the multidimensional Taylor formula:

Proposition 10.40. Taylor polynomial in n variables

Let $D \subset \mathbb{R}^n$, $\ell \in \mathbb{N}$, let $f \in C^{\ell+1}(D)$ and take the two points $\mathbf{x}_0$ and $\mathbf{x}_0 + \mathbf{h}$ in D such that also the entire line segment

$$S := \{\mathbf{x}_0 + t\mathbf{h} : t \in [0, 1]\}$$

connecting $\mathbf{x}_0$ and $\mathbf{x}_0 + \mathbf{h}$ is in D . Then there exists a point $\boldsymbol{\xi}$ on S such that

$$f(\mathbf{x}_0 + \mathbf{h}) = \underbrace{\sum_{\substack{\boldsymbol{\alpha} \in \mathbb{N}_0^n \\ |\boldsymbol{\alpha}| \leq \ell}} \frac{1}{\boldsymbol{\alpha}!} D^{\boldsymbol{\alpha}} f(\mathbf{x}_0) \mathbf{h}^{\boldsymbol{\alpha}}}_{n\text{-dim. Taylor polynomial } p(\mathbf{h})} + \underbrace{\sum_{\substack{\boldsymbol{\alpha} \in \mathbb{N}_0^n \\ |\boldsymbol{\alpha}| = \ell+1}} \frac{1}{\boldsymbol{\alpha}!} D^{\boldsymbol{\alpha}} f(\boldsymbol{\xi}) \mathbf{h}^{\boldsymbol{\alpha}}}_{\text{Lagrange remainder}}.$$

Proof: As above. ■

 Nerd box 10.41. Is it really “as above”?

To see that Proposition 10.40 for $n = 2$ actually coincides with Proposition 10.36, we have to confirm, for $\boldsymbol{\alpha} = (i, j) \in \mathbb{N}_0^2$ with $i + j = k$, that

$$\frac{1}{k!} \cdot \binom{k}{j} = \frac{1}{k!} \cdot \frac{k!}{(k-j)!j!} = \frac{1}{(k-j)!j!} = \frac{1}{i!j!} = \frac{1}{\boldsymbol{\alpha}!}.$$

Under the weaker condition that $f \in C^{\ell-1}(U)$ for a neighbourhood U of $\mathbf{x}_0$ and that all partial derivatives of order ℓ exist in $\mathbf{x}_0$, it is already true that

$$f(\mathbf{x}_0 + \mathbf{h}) = p(\mathbf{h}) + r(\mathbf{x}_0 + \mathbf{h}) \quad \text{with} \quad \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|r(\mathbf{x}_0 + \mathbf{h})\|}{\|\mathbf{h}\|^\ell} = 0 \quad (10.17)$$

for the Taylor polynomial p from Proposition 10.40. For $\ell = 1$ this brings us back to (10.4).

10.9 Local extrema

In this section, we come back to our initial questions about the highest point on a graph landscape or about having noodles before the marathon. This section is also all about scalar fields.

Let $D \subset \mathbb{R}^n$ be an open set and let $f : D \rightarrow \mathbb{R}^1$ be a scalar field. As in Definition 7.13, we say that f has a local maximum at $\mathbf{x}_0 \in D$ if there is an ε -neighbourhood U of $\mathbf{x}_0$ with $U \subset D$ and

$$f(\mathbf{x}_0) \geq f(\mathbf{x}) \quad \text{for all} \quad \mathbf{x} \in U.$$

The term local minimum is defined analogously and both are referred to as a local extrema.

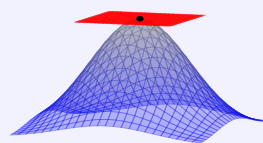
10.9.1 Necessary condition: Fermat's condition

As for $n = 1$, extreme value candidates $\mathbf{x}_0$ can be found using the derivative:

Proposition 10.42. Fermat's condition

If $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ is differentiable in $\mathbf{x}_0 \in D$ and has a local extremum there, then the following holds:

$$\nabla f(\mathbf{x}_0) = \mathbf{o}.$$



Proof: As f in $\mathbf{x}_0$, all functions $g_1, \dots, g_n : \mathbb{R} \rightarrow \mathbb{R}$

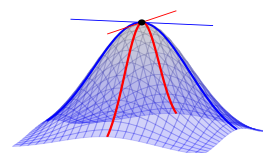
$$g_j : t \mapsto f(\mathbf{x}_0 + t\mathbf{e}_j), \quad j = 1, \dots, n,$$

are differentiable in $t = 0$ and each $g_1, \dots, g_n$ has a local extremum at $t = 0$ since f has one in $\mathbf{x}_0$.

By Proposition 7.14 (Fermat, 1D), we get that

$$0 = g'_j(0) = f'(\mathbf{x}_0 + 0\mathbf{e}_j) \frac{d}{dt}(\mathbf{x}_0 + t\mathbf{e}_j) = J_f(\mathbf{x}_0) \mathbf{e}_j = \frac{\partial f}{\partial x_j}(\mathbf{x}_0).$$

Thus, all partial derivatives $\frac{\partial f}{\partial x_1}(\mathbf{x}_0), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}_0)$ are equal to zero. ■



All curves that pass the summit have a horizontal tangent there. Therefore, the tangential plane at the summit is also horizontal.

Between us: picnic on the summit

When climbing a mountain, the gradient always points into the direction of the steepest ascent. And once you reach the summit, it says you don't need to go any further:
 $\nabla f(\mathbf{x}_0) = \mathbf{o}.$

Again, we get:

Necessary criterion for local extrema: For differentiable functions f

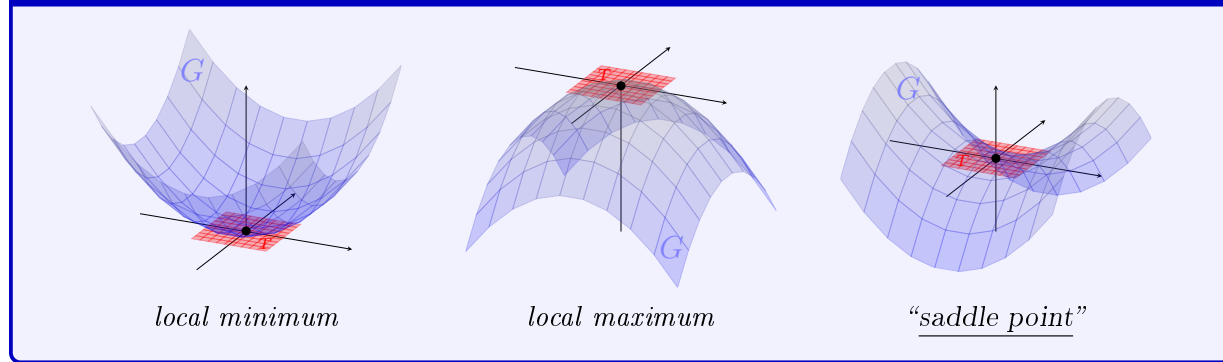
$$f \text{ has a local extremum in } \mathbf{x}_0 \implies \nabla f(\mathbf{x}_0) = \mathbf{o}.$$

In the case of $\nabla f(\mathbf{x}_0) = \mathbf{o}$, the point $\mathbf{x}_0$ is therefore referred to as an extreme value suspect or as a stationary point.

10.9.2 A sufficient condition and the Hessian matrix

Again, the converse does not apply: For the following three graphs of functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ (see Example 10.48) there is a point $\mathbf{x}_0$ with horizontal tangent plane, i.e. with $\nabla f(\mathbf{x}_0) = \mathbf{o}$, but in the third case f has neither a local minimum nor a local maximum at $\mathbf{x}_0$:

Just because $\nabla f(\mathbf{x}_0) = \mathbf{o}$, there does not have to be a local extremum!



A look back at (10.17) with $\ell = 2$ leads the way forward:

$$\underbrace{f(\mathbf{x}_0 + \mathbf{h})}_{\text{graph: } G} = \underbrace{f(\mathbf{x}_0) + \sum_{\substack{\alpha \in \mathbb{N}_0^n \\ |\alpha|=1}} \frac{1}{\alpha!} D^\alpha f(\mathbf{x}_0) \mathbf{h}^\alpha}_{\text{graph: } T} + \underbrace{\sum_{\substack{\alpha \in \mathbb{N}_0^n \\ |\alpha|=2}} \frac{1}{\alpha!} D^\alpha f(\mathbf{x}_0) \mathbf{h}^\alpha}_{p_2(\mathbf{h})} + \underbrace{r(\mathbf{x}_0 + \mathbf{h})}_{\substack{\text{is dominated by} \\ p_2 \text{ around } \mathbf{x}_0, \\ \text{see (10.17)}}}$$

where $p_k(\mathbf{h})$ denotes the part of the Taylor polynomial of f in $\mathbf{x}_0$ that contains exactly the terms of degree k , compare Example 10.37.

Box 10.43. interim conclusion

Therefore, the term $p_2(\mathbf{h})$ is responsible for whether, nearby $\mathbf{x}_0$, the graph G is

- above T ($\Rightarrow$ local minimum if T is horizontal) or
- below T ($\Rightarrow$ local maximum if T is horizontal)
- or partly above and partly below T ($\Rightarrow$ saddle point).

So let's take a closer look at $p_2(\mathbf{h})$: All terms have the form

$$\frac{1}{1! \cdot 1!} \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(\mathbf{x}_0) \cdot h_i h_j \quad \text{for} \quad i, j \in \{1, \dots, n\}, i \neq j \quad (10.18)$$

or

$$\frac{1}{2!} \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(\mathbf{x}_0) \cdot h_i h_j \quad \text{for} \quad i, j \in \{1, \dots, n\}, i = j, \quad (10.19)$$

where $h_1, \dots, h_n \in \mathbb{R}$ are the components of $\mathbf{h} \in \mathbb{R}^n$. Thus

$$p_2(\mathbf{h}) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) \cdot h_i h_j \quad (10.20)$$

$$= \frac{1}{2} (h_1 \quad \dots \quad h_n) \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}_0) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}_0) \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}_0) & \dots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}_0) \end{pmatrix} \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}, \quad (10.21)$$

because, by Schwarz's theorem, the terms of the form (10.18) all appear twice in (10.20), which is cancelled by the factor $\frac{1}{2}$ in (10.20). The terms of the form (10.19) appear only once, but they already have a coefficient $\frac{1}{2!} = \frac{1}{2}$.

The matrix in (10.21) is given its own name:

Definition 10.44. Hessian matrix

(Otto Hesse, 1811–1874)

For $D \subset \mathbb{R}^n$ and $f \in C^2(D)$ as well as $\mathbf{x}_0 \in D$, the matrix

$$H_f(\mathbf{x}_0) := \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}_0) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}_0) \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}_0) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}_0) \end{pmatrix} = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) \right)_{i,j=1}^n$$

is referred to as the Hessian matrix of f in $\mathbf{x}_0$.

According to Proposition 10.34 (Schwarz's theorem), $H_f(\mathbf{x}_0)$ is a symmetric matrix.

Example 10.45. Hessian matrix

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^1$ with $f(x, y, z) = x^2 e^y + \sin z$ be the function from Example 10.12 b) and Example 10.14. We know the first-order partial derivatives from the examples above. They are used for the Jacobian matrix

$$J_f(\mathbf{x}_0) = \left(\begin{array}{ccc} & & \end{array} \right)$$

in $\mathbf{x}_0 = (x, y, z)$. Accordingly, the Hessian matrix is

With (10.21) we get

$$p_2(\mathbf{h}) = \frac{1}{2} \mathbf{h}^\top H_f(\mathbf{x}_0) \mathbf{h} = \frac{1}{2} \left\langle \mathbf{h}, H_f(\mathbf{x}_0) \mathbf{h} \right\rangle$$

and, using Box 10.43 with $\nabla f(\mathbf{x}_0) = \mathbf{o}$, the function $f \in C^2(D)$ has

- a local minimum at $\mathbf{x}_0 \in D$ if $\langle \mathbf{h}, H_f(\mathbf{x}_0) \mathbf{h} \rangle > 0$ for all $\mathbf{h} \neq \mathbf{o}$,
- a local maximum at $\mathbf{x}_0 \in D$ if $\langle \mathbf{h}, H_f(\mathbf{x}_0) \mathbf{h} \rangle < 0$ for all $\mathbf{h} \neq \mathbf{o}$, and
- a saddle point at $\mathbf{x}_0 \in D$ if $\langle \mathbf{h}, H_f(\mathbf{x}_0) \mathbf{h} \rangle$ assumes both positive and negative values for $\mathbf{h} \neq \mathbf{o}$.

These cases remind us of the following fact from linear algebra:

Proposition 10.46. definite and indefinite matrices

Let $H = (a_{ij})_{i,j=1}^n \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then

a) The following statements are equivalent:

- H is positive definite,
- $\langle \mathbf{h}, H\mathbf{h} \rangle > 0$ holds for all $\mathbf{h} \in \mathbb{R}^n \setminus \{\mathbf{o}\}$ (this is usually the definition),
- all eigenvalues of H are positive (they are real because H is symmetric),
- $$a_{11}, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots, \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix} = \det H > 0,$$
- the LU decomposition $A = LR$ satisfies $r_{ii} > 0$ for all $i = 1, \dots, n$,
- $A = CC^\top$ holds for a regular triangular matrix C .

b) The following statements are equivalent:

- H is negative definite,
- $-H$ is positive definite,
- $\langle \mathbf{h}, H\mathbf{h} \rangle < 0$ holds for all $\mathbf{h} \in \mathbb{R}^n \setminus \{\mathbf{o}\}$
- all eigenvalues of H are negative (they are real because H is symmetric),
- for

$$D_k := \begin{vmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{vmatrix} \quad \text{mit} \quad k = 1, \dots, n$$

we get $(-1)^k D_k > 0$, i.e. $D_1 < 0$, $D_2 > 0$, $D_3 < 0$, $D_4 > 0, \dots$

c) The following statements are equivalent:

- H is indefinite,
- there exist $\mathbf{h}, \tilde{\mathbf{h}} \in \mathbb{R}^n \setminus \{\mathbf{o}\}$ such that $\langle \mathbf{h}, H\mathbf{h} \rangle > 0$ and $\langle \tilde{\mathbf{h}}, H\tilde{\mathbf{h}} \rangle < 0$
- H has both positive and negative eigenvalues.

The vectors $\mathbf{h}$ and $\tilde{\mathbf{h}}$ from c) can be chosen as eigenvectors for an eigenvalue $\lambda > 0$ or $\tilde{\lambda} < 0$ of the indefinite matrix H . They correspond to directions in which G is above or below T .

The list (positive definite, negative definite, indefinite) from Proposition 10.46 is not complete. The following two cases are still missing

- all eigenvalues are ≥ 0 (the matrix is then called positive semidefinite) but some of them are equal to zero (so that the matrix is not positive definite)

as well as

- all eigenvalues are ≤ 0 (the matrix is then called negative semidefinite) but some of them are equal to zero (so the matrix is not negative definite).

In short: A matrix with the eigenvalues 1 and 5 is positive definite, one with the eigenvalues 0 and 5 is only positive semidefinite, and one with the eigenvalues -1 and 5 is indefinite.

Now we summarize everything and get the following result:

Proposition 10.47. sufficient condition for local extrema

Let $D \in \mathbb{R}^n$ be open, $f \in C^2(D)$ and suppose $\mathbf{x}_0 \in D$ with $\nabla f(\mathbf{x}_0) = \mathbf{0}$.

- a) If $H_f(\mathbf{x}_0)$ is positive definite, then f has a local minimum at $\mathbf{x}_0$,
- b) if $H_f(\mathbf{x}_0)$ is negative definite, then f has a local maximum at $\mathbf{x}_0$,
- c) if $H_f(\mathbf{x}_0)$ is indefinite, then f has a saddle point at $\mathbf{x}_0$,
- d) However, if $H_f(\mathbf{x}_0)$ is (positive or negative) semidefinite but not definite, then two of the three situations above are possible for f at $\mathbf{x}_0$.

Proof: a), b) and c) follow directly from Box 10.43, (10.21) and Proposition 10.46. For d) see Example 10.49 ■

Example 10.48. about the statements a), b) and c) of Proposition 10.47

Again, we consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$, $\mathbf{x}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, and we write $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =: \begin{pmatrix} x \\ y \end{pmatrix}$.

- a) For $f(x, y) = x^2 + y^2$, we have $\nabla f(x, y) =$ and therefore $\nabla f(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
Furthermore,

$$H_f(0, 0) =$$

eigenvalues:

$\Rightarrow$

- b) For $f(x, y) = -x^2 - y^2$, we have $\nabla f(x, y) =$ and therefore $\nabla f(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
Furthermore,

$$H_f(0, 0) =$$

eigenvalues:

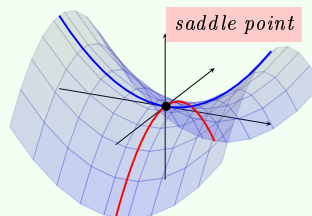
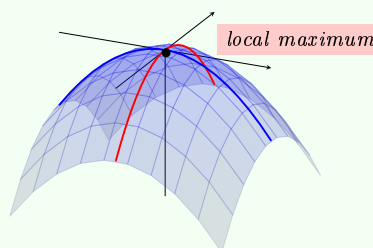
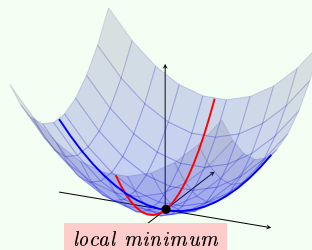
$\Rightarrow$

- c) For $f(x, y) = x^2 - y^2$, we have $\nabla f(x, y) =$ and therefore $\nabla f(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
Furthermore,

$$H_f(0, 0) =$$

eigenvalues:

$\Rightarrow$



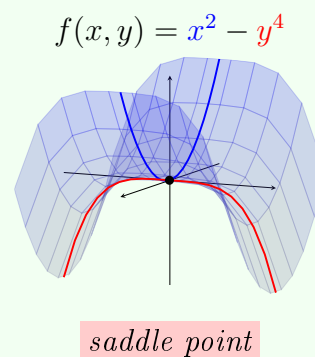
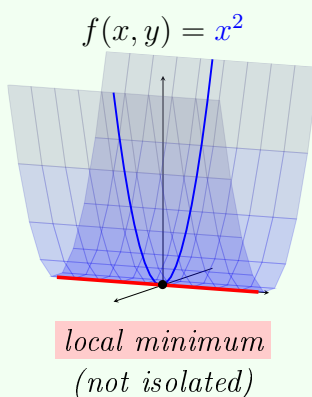
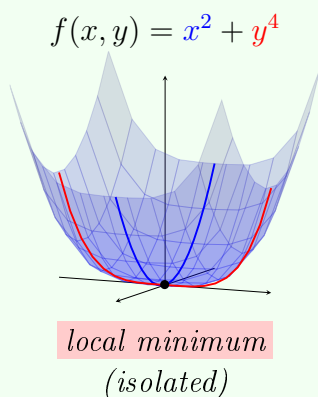
The following example shows that no conclusion is possible in the case of $\nabla f(\mathbf{x}_0) = \mathbf{0}$ with a semi-definite Hessian matrix:

Example 10.49. about d) of Proposition 10.47

The following three functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ satisfy $\nabla f(0,0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ with a positive semidefinite (but not positive definite) Hessian matrix

$$H_f(0,0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

A local maximum can be excluded by the positive eigenvalue 2. However, a local minimum and a saddle point are both possible:



The term y^4 grows so slowly at $y = 0$ that its first three derivatives are equal to 0 there and thus, unnoticed by the Hessian matrix $H_f(0,0)$, it causes the graph to rise in the direction of the y -axis.

10.9.3 Examples of extrema in domains with or without boundary points

Then we are now prepared for searching for local extrema in examples:

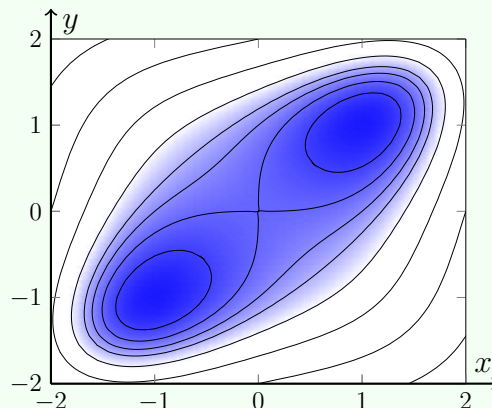
Example 10.50. local extrema in open set (without boundary)

We consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ with

$$f(x, y) = x^4 + y^4 - 4xy.$$

We are looking for local extrema of f in $D = \mathbb{R}^2$.

We start our search by determining all stationary points, i.e. all points (x, y) with $\nabla f(x, y) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:



$$\left. \begin{aligned} 0 &\stackrel{!}{=} \frac{\partial}{\partial x} f(x, y) = \\ 0 &\stackrel{!}{=} \frac{\partial}{\partial y} f(x, y) = \end{aligned} \right\} \Rightarrow$$

This means that either $x = 0$ or $x^8 = 1$. The latter equation has eight solutions (roots of unity) in $\mathbb{C}$, but only two of them are real: $x = -1$ and $x = 1$.

We therefore have three stationary points:

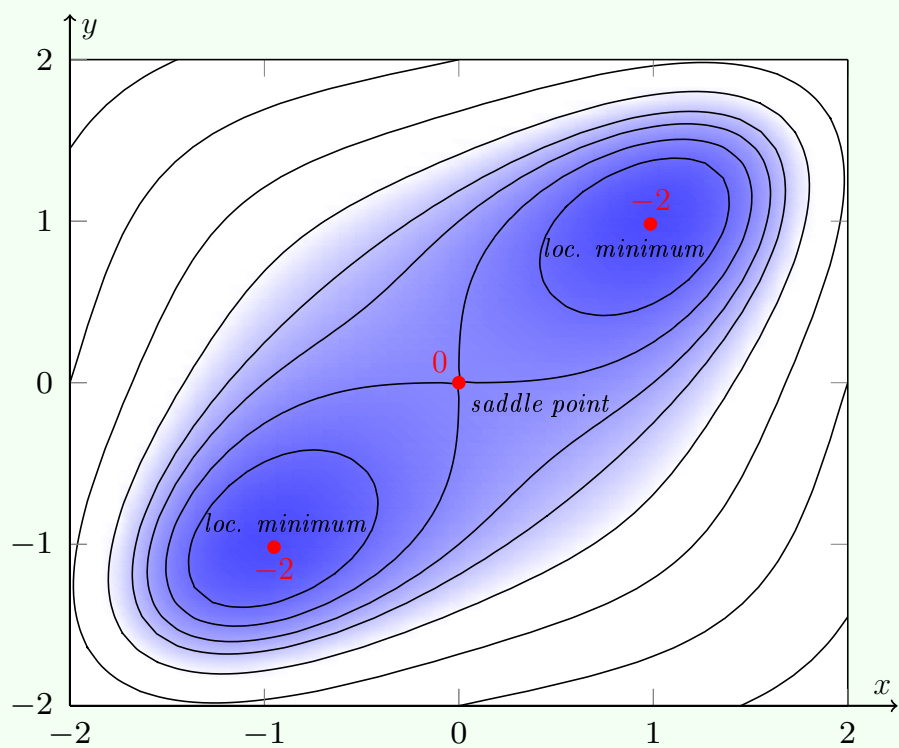
$$(x, y) = (0, 0) \quad \text{and} \quad (x, y) = (-1, -1) \quad \text{and} \quad (x, y) = (1, 1)$$

In order to find out which of the three points is a local extremum (and of which type), we look at the Hessian matrix at the respective point:

$$H_f(x, y) = \begin{pmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{pmatrix} = \quad , \quad (x, y) \in \mathbb{R}^2.$$

- $H_f(0, 0) =$

- $H_f(1,1) = H_f(-1,-1) =$



When searching for the extrema of a function $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ on a set D with boundary points (i.e. not open), all boundary points must also be examined:

- search on $\text{int } D$ (as above),
- search in the boundary points $\mathbf{x} \in D \setminus \text{int } D$ } comparison $\implies$ global extrema

Example 10.51. global extrema (in a domain with boundary)

We restrict the function from the previous example to $D = [0, 2]^2$, i.e. $f : D \rightarrow \mathbb{R}^1$ with

$$f(x, y) = x^4 + y^4 - 4xy.$$

We are looking for all local extrema of f in D .

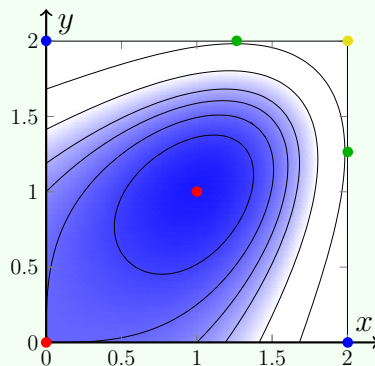
In Example 10.50 we already found the stationary points using $\nabla f(x, y) \stackrel{!}{=} \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Out of the three stationary points from Example 10.50, only $(0, 0)$ and $(1, 1)$ are in D . The corresponding function values are

$$f(0, 0) = 0 \quad \text{and} \quad f(1, 1) = -2.$$

In the next step we look at the boundary lines of D .

- left boundary: $x = 0$ and $y \in [0, 2]$.



- upper boundary: $x \in [0, 2]$ and $y = 2$.

- Because of $f(x, y) = f(y, x)$, the right boundary shows the same values as the top boundary, and the bottom boundary shows the same as the left boundary.

The extrema of the possible candidates thus provide the global extrema of f :

- the minimum is -2 , being assumed at $(x, y) = (1, 1)$ only,
- the maximum is 16 , being assumed at $(0, 2)$, $(2, 0)$ and $(2, 2)$.

10.10 Divergence and curl of a vector field

Remember that we sometimes like to write $\frac{\partial f}{\partial x}$ in place of $\frac{\partial}{\partial x}f$. One then speaks of $\frac{\partial}{\partial x}$ as the differential operator that maps f to another function: its derivative with respect to x . In the same spirit, the gradient ∇f of $f : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ is sometimes written as

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix} f, \quad \text{whence we abbreviate:} \quad \nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix}.$$

In the general case, $f : \mathbb{R}^n \rightarrow \mathbb{R}^1$, ∇ is the column vector of all $\frac{\partial}{\partial x_i}$ with $i = 1, \dots, n$.

Using this formal interpretation of ∇ as a vector of differential operators, two further important operators on vector fields $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be elegantly expressed as follows:

Definition 10.52. divergence of a vector field

For $n \in \mathbb{N}$ and a vector field $f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, the scalar field

$$\operatorname{div} f := \langle \nabla, f \rangle = \left\langle \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{pmatrix}, \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} \right\rangle = \frac{\partial}{\partial x_1} f_1 + \dots + \frac{\partial}{\partial x_n} f_n$$

is referred to as the divergence of f . In that case, $\operatorname{div} f : \mathbb{R}^n \rightarrow \mathbb{R}^1$.

If f describes the velocity of a flowing fluid at every point then $\operatorname{div} f(\mathbf{x})$ indicates how much fluid is created or lost at the point $\mathbf{x}$ – depending on the sign of $\operatorname{div} f(\mathbf{x})$. One also speaks of so-called sources and sinks of the vector field f , and refers to $\operatorname{div} f$ as the source density of f .

Definition 10.53. curl of a vector field

For a vector field $f = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the vector field

$$\operatorname{rot} f := \nabla \times f = \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \frac{\partial}{\partial x_3} \end{pmatrix} \times \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_2} f_3 - \frac{\partial}{\partial x_3} f_2 \\ \frac{\partial}{\partial x_3} f_1 - \frac{\partial}{\partial x_1} f_3 \\ \frac{\partial}{\partial x_1} f_2 - \frac{\partial}{\partial x_2} f_1 \end{pmatrix}$$

is referred to as the curl of f . One then has $\operatorname{rot} f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$.

For the velocity field f of a flowing fluid, $\operatorname{rot} f(\mathbf{x})$ indicates how fast and around which axis a particle moving with the flow rotates at the point $\mathbf{x}$. A vector field f with $\operatorname{rot} f(\mathbf{x}) = \mathbf{0}$ at every point $\mathbf{x}$ is called irrotational or vortex-free.

⚠ Attention! curl is only defined for $n = 3$

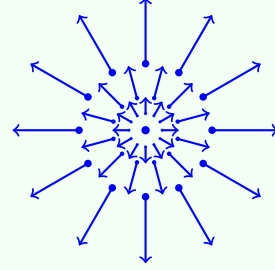
In contrast to $\operatorname{div} f$, the curl $\operatorname{rot} f$ is only defined for vector fields $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $n = 3$ (cf. scalar product vs. cross product). Exception: By attaching a zero as the third field component, $\operatorname{rot} f$ can also be studied when $n = 2$, see Example 10.54.

Example 10.54. divergence and curl

- a) Let $f(x, y) = \begin{pmatrix} x \\ y \end{pmatrix}$ be the 2D vector field from Example 10.6 a). At every point we have

$$\operatorname{div} f(x, y) = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y = 1 + 1 = 2.$$

In fact, not only the origin is a source but at every point $(x, y) \in \mathbb{R}^2$, the intensity of the field grows.



The rotation can only be calculated after introducing a third variable z into f (which does not appear anywhere in the function) on the input side and by adding a zero as the third component on the output:

$$\hat{f}(x, y, z) := \begin{pmatrix} f(x, y) \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}, \quad x, y, z \in \mathbb{R}$$

The visualisation of the new field $\hat{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in the (x, y) plane and in every of its parallel planes is exactly as that for f , see above. And now

$$\operatorname{rot} \hat{f}(x, y, z) = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial y}0 - \frac{\partial}{\partial z}y \\ \frac{\partial}{\partial z}x - \frac{\partial}{\partial x}0 \\ \frac{\partial}{\partial x}y - \frac{\partial}{\partial y}x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The field $\hat{f}$ that was lifted to $\mathbb{R}^3$ is irrotational. By the way:

$$\operatorname{div} \hat{f}(x, y, z) =$$

That $\operatorname{div} \hat{f}(x, y, z) = \operatorname{div} f(x, y)$ follows directly from $\hat{f}(x, y, z) := \begin{pmatrix} f(x, y) \\ 0 \end{pmatrix}$.

- b) Next, we lift the tornado f from Example 10.6 c) into $\mathbb{R}^3$, using the same lifting technique as above, that is

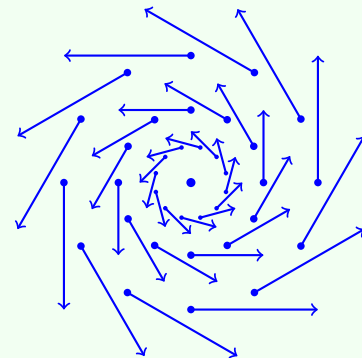
$$\hat{f}(x, y, z) := \begin{pmatrix} f(x, y) \\ 0 \end{pmatrix} = \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix}, \quad x, y, z \in \mathbb{R}.$$

Then

$$\operatorname{div} \hat{f} =$$

which agrees with $\operatorname{div} f$ again. Moreover,

$$\operatorname{rot} \hat{f} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} =$$



The field rotates at a constant angular velocity in the positive direction about the vector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. It is worth noting that this applies not only to points on the z -axis but everywhere in $\mathbb{R}^3$.

c) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $f(x, y, z) = \begin{pmatrix} z - y \\ x - z \\ y - x \end{pmatrix}$. Clearly,

$$\operatorname{div} f = \frac{\partial(z-y)}{\partial x} + \frac{\partial(x-z)}{\partial y} + \frac{\partial(y-x)}{\partial z} = \quad \text{and}$$

$$\operatorname{rot} f = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \times \begin{pmatrix} z-y \\ x-z \\ y-x \end{pmatrix} =$$

Indeed, this field is just a rotated version of the field $\hat{f}$ from Example b).

Just like the gradient, divergence and curl also feature in numerous practical problems – and not just in the case of flowing fluids:

- The equation

$$\frac{\partial f}{\partial t} = \operatorname{div}_{\mathbf{x}} \operatorname{grad}_{\mathbf{x}} f$$

describes both heat conduction and the mixing/diffusion of chemical concentrations

$f(x, y, z, t)$ at the spatial point $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ at time t . It is therefore referred to as the heat equation or the diffusion equation. Divergence and gradient are defined here only with respect to the spatial variables x, y, z .

- The equation

$$\frac{\partial^2 f}{\partial t^2} = v^2 \operatorname{div}_{\mathbf{x}} \operatorname{grad}_{\mathbf{x}} f$$

describes the propagation of waves at a speed v in a three-dimensional medium and is therefore referred to as the wave equation. Divergence and gradient are again defined only in terms of the spatial variables x, y, z . The combination $\operatorname{div} \operatorname{grad} f$ is also called the Laplace operator and is denoted by Δf .

- The famous Maxwell's equations describe an electric field E and a magnetic field B in $\mathbb{R}^3$ by means of conditions on

$$\operatorname{div} E, \quad \operatorname{div} B, \quad \operatorname{rot} E, \quad \operatorname{rot} B,$$

where $\operatorname{rot} B$ and $\operatorname{rot} E$ are connected to the time derivative of the other field.

All of the above examples are so-called partial differential equations, as they involve functions f, E, B of several variables x, y, z, t and relationships between various kinds of their partial derivatives. For more on this topic, see Mathematics IV.

Summary

- The total derivative $f'(\mathbf{x}_0)$ of $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\mathbf{x}_0 \in D$ is, among all linear maps $\ell : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the one that best approximates f nearby $\mathbf{x}_0$.

So we have

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \ell(\mathbf{h}) + r(\mathbf{x}_0 + \mathbf{h})$$

with an error of $r(\mathbf{x}_0 + \mathbf{h})$, such that

$$\frac{r(\mathbf{x}_0 + \mathbf{h})}{\|\mathbf{h}\|} \rightarrow \mathbf{0}$$

holds as $\mathbf{h} \rightarrow \mathbf{0}$.

- If such a linear map $\ell : \mathbb{R}^n \rightarrow \mathbb{R}^m$ exists, then it is unique and is denoted by $\ell = f'(\mathbf{x}_0)$. Then f is called (totally) differentiable at $\mathbf{x}_0$.
- For $m = n = 1$, we get exactly the concept of differentiability from Chapter 7.
- For the above characterization, one needs the norm $\|\cdot\|$ and convergence $\rightarrow$ for vectors in $\mathbb{R}^n$ and $\mathbb{R}^m$.
 - As the norm, we use the Euclidean norm, $\|\mathbf{h}\| = \sqrt{h_1^2 + \cdots + h_n^2}$,
 - and by convergence we mean componentwise convergence.
- Totally differentiable functions are continuous.
- A scalar field is a function that maps every point in $D \subset \mathbb{R}^n$ to a real number, for example a temperature.
- A vector field f maps each point in $D \subset \mathbb{R}^n$ to a vector in $\mathbb{R}^m$, for example the local wind. One can also think of f as a vector of m scalar fields $D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$, which we call the component functions (or simply components) of f .
- A vector field is continuous if and only if all of its components are continuous.
- A vector field is differentiable if and only if all of its components are differentiable.
 - This is why we can focus on scalar fields for the most part in this chapter.
- The Jacobian matrix $J_f(\mathbf{x}_0)$ is the representation matrix of the linear map $\ell = f'(\mathbf{x}_0)$ in the standard bases of $\mathbb{R}^n$ and $\mathbb{R}^m$. If it suffices, we just write J_f .
- For $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix J_f is a $m \times n$ -matrix.

- If we write f as a vector of its scalar component functions, $f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$, then the i -th row of the Jacobian matrix J_f is exactly the Jacobian matrix of f_i .
- The Jacobian matrix of a scalar field $g : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ is therefore of size $1 \times n$, i.e., a row vector.
- The same vector as a column vector is called the gradient of the scalar field g , written as $\nabla g(\mathbf{x}_0)$.
- The entries of the Jacobian matrix or of the gradient of a scalar field $g : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ are the first partial derivatives of g with respect to the n variables $x_1, \dots, x_n$.
- When partially differentiating a scalar field $g : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$, one of the n variables, $x_1, \dots, x_n$, is selected and differentiation is performed with respect to it, while all other variables are treated as constants. If x_j is the variable with respect to which differentiation is performed and $\mathbf{x}_0 \in D$ is the point where only x_j varies, then we denote the partial derivative by

$$\frac{\partial g(\mathbf{x}_0)}{\partial x_j} \quad \text{or} \quad \frac{\partial g}{\partial x_j}(\mathbf{x}_0) \quad \text{or} \quad \frac{\partial}{\partial x_j} g(\mathbf{x}_0) \quad \text{or} \quad D_j g(\mathbf{x}_0) \quad \text{or} \quad g_{x_j}(\mathbf{x}_0).$$

- If a scalar field g is totally differentiable (in $\mathbf{x}_0$), then its first-order partial derivatives also exist, and we call g partially differentiable (in $\mathbf{x}_0$).
 - Total differentiability means that there is a tangent plane (if $n > 2$: hyperplane) T to the graph at $\mathbf{x}_0$.
 - The partial derivative $\frac{\partial g(\mathbf{x}_0)}{\partial x_j}$ then measures the slope of T at $\mathbf{x}_0$ in the x_j -direction.
 - If we perform this for all $j = 1, \dots, n$, we have determined T .
 - In exactly this sense, the Jacobian matrix, $J_g(\mathbf{x}_0)$, and the gradient, $\nabla g(\mathbf{x}_0)$, are determined by the partial derivatives $\frac{\partial g(\mathbf{x}_0)}{\partial x_1}, \dots, \frac{\partial g(\mathbf{x}_0)}{\partial x_n}$.
- Conversely: If a scalar field g is partially differentiable in $\mathbf{x}_0$, it does not have to be totally differentiable in $\mathbf{x}_0$ (not even continuous). However, if in addition the partial derivatives $\frac{\partial g}{\partial x_1}, \dots, \frac{\partial g}{\partial x_n}$ are continuous in $\mathbf{x}_0$, then total differentiability of g in $\mathbf{x}_0$ follows.
- If f and g are differentiable vector fields that fit together so that $f + g$, resp. $f \circ g$, can be formed, then $f + g$, resp. $f \circ g$, as well as αf (with $\alpha \in \mathbb{R}$) is also differentiable, and matrix versions of the known differentiation rules apply.
- The same is true for scalar fields, $f \cdot g$ and $\frac{f}{g}$.
- The directional derivative $\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{h}} = \langle \nabla f(\mathbf{x}_0), \frac{\mathbf{h}}{\|\mathbf{h}\|} \rangle$ calculates the slope of the tangent plane of the scalar field $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ in $\mathbf{x}_0$ in the direction $\mathbf{h} \in \mathbb{R}^n$.
- The gradient is orthogonal to the current level set and it points in the direction of the steepest ascent.

- Let f be a scalar field $D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$. If the function $\frac{\partial f}{\partial x_j} : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ is partially differentiable in x_i , we call

$$\frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) (\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_i \partial x_j} (\mathbf{x}_0) = D_i D_j f(\mathbf{x}_0)$$

the second-order partial derivative of f in $\mathbf{x}_0$.

- If all partial derivatives of second-order, $D_i D_j f$ with $i, j \in \{1, \dots, n\}$, exist and are continuous in $\mathbf{x}_0$, then we have $D_i D_j f(\mathbf{x}_0) = D_j D_i f(\mathbf{x}_0)$.
- Applying this rule multiple times we get, e.g.,

$$D_1 D_2 D_3 f(\mathbf{x}_0) = D_2 D_1 D_3 f(\mathbf{x}_0) = D_2 D_3 D_1 f(\mathbf{x}_0) = D_3 D_2 D_1 f(\mathbf{x}_0),$$

in case all partial derivatives (now of third-order) are continuous in $\mathbf{x}_0$.

- Just like the equation of the tangent plane (being an n -dimensional polynomial of first degree) can be formed from the function value and the first-order partial derivatives of f in $\mathbf{x}_0$, one can form the n -dimensional Taylor polynomial of degree ℓ using the partial derivatives up to order ℓ .
- We can again describe the error by a Lagrange error term, and for $\mathbf{x} \rightarrow \mathbf{x}_0$ it is of smaller order than $\|\mathbf{x} - \mathbf{x}_0\|^\ell$.
- If the scalar field $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$ is differentiable at $\mathbf{x}_0 \in D$, then at a local extremum of f at $\mathbf{x}_0$, we always have $\nabla f(\mathbf{x}_0) = \mathbf{0}$.
- For the converse direction, one again needs an additional condition: this time the positive or negative definiteness of the Hessian matrix $H_f(\mathbf{x}_0) = (D_i D_j f(\mathbf{x}_0))_{i,j=1}^n$.
- Boundary points of D must be examined separately. There, extremality can also occur without $\nabla f(\mathbf{x}_0) = \mathbf{0}$.
- Introducing the vectorial differential operator $\nabla := \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)^\top$, we get that, for every scalar field $f : \mathbb{R}^n \rightarrow \mathbb{R}^1$ and every vector field $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$, the expression ∇f is the gradient of f and $\langle \nabla, g \rangle$ is the divergence aka source density, $\operatorname{div} g$, of g . In case $n = 3$, one can also study the vector product $\nabla \times g$, which is then the rotation, $\operatorname{rot} g$, of g .

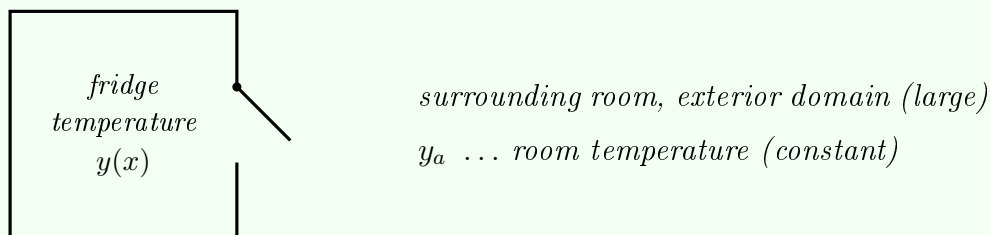
Chapter 11

Outlook: Ordinary Differential Equations (ODEs)

Sometimes it is easiest to start with an example:

Example 11.1. Defrosting the fridge

The fridge is frosted over. So we turn off the power and open the door:



x time (often denoted by t rather than x)

$y(x)$ fridge temperature at the time x

$y(0) = y_0$ initial fridge temperature at time $x_0 = 0$

physics: door open $\implies$ heat flow that is proportional to the current temperature difference (inside vs. outside)

initial value problem (IVP)

$y'(x) = c \cdot (y_a - y(x))$ differential equation (wanted: the function y)

$y(x_0) = y_0$ initial condition at time $x = x_0$

11.1 First order ODEs

Definition 11.2. ordinary differential equation of first order

Take $D \subset \mathbb{R}^2$ and let $f : D \rightarrow \mathbb{R}$ be continuous. Then we call

$$y' = f(x, y) \quad (11.1)$$

an ordinary differential equation of first order for the function y .

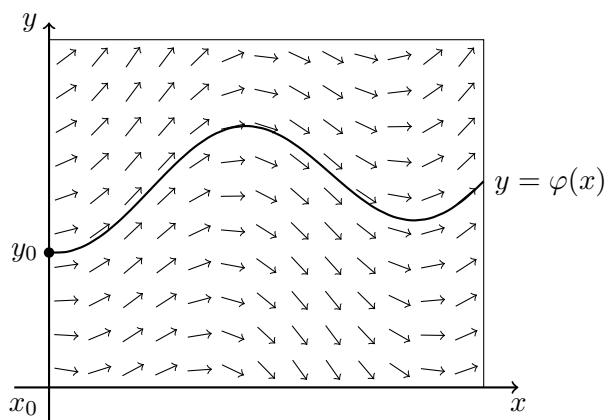
By a solution of (11.1) we mean a C^1 function $\varphi : I \rightarrow \mathbb{R}$ with

$$\varphi'(x) = f(x, \varphi(x)), \quad x \in I.$$

Here $I \subset \mathbb{R}$ denotes an interval and $\{(x, \varphi(x)) : x \in I\} \subset D$.

If, moreover, a point $(x_0, y_0) \in D$ is given and a solution of $y' = f(x, y)$ is to be found, where also $y(x_0) = y_0$ holds, one speaks of an initial value problem.

Geometric Interpretation:



$f \hat{=}$ “field of directions” on $D \subset \mathbb{R}^2$

Wanted:

graph of a function passing through (x_0, y_0) whose slope coincides with the field f everywhere

Between us: Imagine a grain of sand in the wind that someone lets go at the point (x_0, y_0) .

Idea: Integrate the slopes $\implies$ solution

$$\varphi(x) = y_0 + \int_{x_0}^x f(t, \underbrace{\varphi(t)}_{\text{Problem!}}) dt \quad (11.2)$$

Euler’s method (approximate numerical method)

$$\begin{aligned} \varphi(x_0) &:= y_0 \\ \varphi(x_n) &:= \varphi(x_{n-1}) + f(x_{n-1}, \varphi(x_{n-1})) \cdot \Delta x_n, \quad n = 1, 2, \dots \end{aligned}$$

If f does not depend on its second component then the integration of (11.2) yields an antiderivative φ of f , so it naturally solves this very simple form of (11.1):

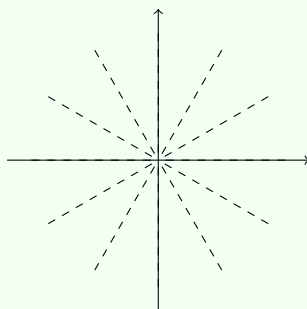
$$y'(x) = f(x)$$

$$y(x_0) = y_0.$$

Sometimes you can guess the solutions y just by looking at the direction field f :

Example 11.3. Solve with your eyes first.

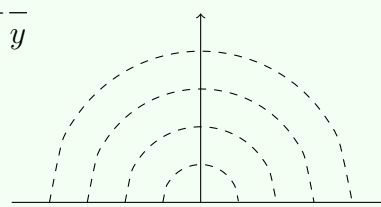
a) $y' = \frac{y}{x}$



$\Rightarrow$ straight lines?

Indeed:

b) $y' = -\frac{x}{y}$



$\Rightarrow$ semi-circles?

Indeed:

The method of solution used in Example 11.3 a) and b) is known as “separation of variables”, and it works when $f(x, y)$ can be written as $g(x) \cdot h(y)$.

$$(11.1) \quad \implies \quad y'(x) = f(x, y) = g(x) \cdot h(y) \quad \implies \quad \int \frac{y'(x)}{h(y)} dx = \int g(x) dx \quad (11.3)$$

In the special case when $h(y)=y$

...so that

$$y'(x) = f(x, y) = g(x) \cdot y, \quad (11.4)$$

one speaks of a homogeneous linear differential equation.

Solution:

$$(11.3) \quad \implies \quad \ln y(x) = \int_{x_0}^x g(t) dt + C \quad \xRightarrow{\exp} \quad y(x) = \exp \left(\int_{x_0}^x g(t) dt \right) \cdot \underbrace{e^C}_D$$

Example 11.3 a) is exactly of this form. Sometimes a so-called inhomogeneity is added to the right-hand side of (11.4):

In the special case when $f(x, y) = g(x) \cdot y + i(x)$

...so that

$$y'(x) = f(x, y) = g(x) \cdot y + i(x), \quad (11.5)$$

one speaks of an inhomogeneous linear differential equation.

Method of solution: “variation of the constant”, see Example 11.4.

We saw the structure (11.5) earlier when modelling the open fridge (Example 11.1):

Example 11.4. “variation of the constant”: the fridge is open again

The starting point for the open fridge model was this initial value problem:

$$y'(x) = c \cdot (y_a - y(x)), \quad (\text{DiffEq})$$

$$y(0) = y_0. \quad (\text{IC})$$

(DiffEq) can be rewritten as

$$y'(x) = \quad (11.6)$$

which is of the form (11.5): an inhomogeneous linear differential equation.

Now here comes the “cooking recipe” that we neither motivate nor prove:

- First pass to the homogeneous version of (11.6),

$$y'(x) =$$

and solve this as in Example 11.3 a) or using the solution formula for (11.4):

$$\ln y(x) + \hat{C} = \int \frac{y'(x)}{y(x)} dx = \int -c dx = -cx + \tilde{C} \implies \ln y(x) = -cx + C$$

whence

$$y(x) = \exp(-cx + C) = e^{-cx} \cdot \overbrace{e^C}^D = D \cdot e^{-cx}. \quad (11.7)$$

- The inhomogeneous problem (11.6) is now solved by replacing the constant D by the function $D(x)$ in (11.7) (“variation of the constant”):

$$\begin{aligned} y(x) &= D(x) e^{-cx} \\ \implies y'(x) &= \end{aligned} \quad (11.8)$$

- The last two expressions for $y(x)$ and $y'(x)$ can now be substituted into (11.6):

- We gratefully acknowledge the cancellations, reorder for $D'(x)$ and integrate:

$$D'(x) = c y_a e^{cx} \xrightarrow{\int} D(x) = y_a e^{cx} + E.$$

- With that expression for $D(x)$ we go back to (11.8):

$$y(x) = D(x) e^{-cx} = \quad (11.9)$$

- Finally we match this new constant E with our initial condition (IC)

$$y_0 \stackrel{(\text{IC})}{=} y(0) \stackrel{(11.9)}{=} \implies E =$$

- ...and then bring it back to (11.9):

$$\underline{\underline{y(x) = \quad .}}$$

The constants c, y_a and y_0 did already show up in (DiffEq) and (IC) so that it is okay (and actually important) to see them pop up in the solution also.

Test: It is easy to check that our result for $y(x)$ indeed satisfies both the differential equation (DiffEq) and the initial condition (IC) and hence our initial value problem.

The theory of ordinary first-order differential equations can be generalised:

11.2 Systems of first-order differential equations

Let's look at a very much simplified predator-prey model: We consider the size $f(x)$ of a fox population and $h(x)$ of a hare population over time x . Foxes hunt hares; the more hares there are, the better off the foxes are; the more foxes there are, the worse off the hares are:

$$\begin{aligned} f'(x) &= -\frac{1}{8}f(x) + \frac{1}{4}h(x), \\ h'(x) &= -2f(x) + 4h(x). \end{aligned}$$

This is referred to as a linear 2×2 differential equation system. Similar systems involving, e.g., ozone, nitrogen, CO_2 , temperature, etc. are found in climate models. As in linear algebra, we will now move on to vectors and a matrix:

$$\begin{pmatrix} f \\ h \end{pmatrix}' = \begin{pmatrix} & \\ & \end{pmatrix} \begin{pmatrix} f \\ h \end{pmatrix} \quad (11.10)$$

Denoting the vector $\begin{pmatrix} f(x) \\ h(x) \end{pmatrix}$ by $y(x)$ and the 2×2 matrix by A , (11.10) simply reads $y' = Ay$: a homogeneous first-order linear differential equation – just that now $y(x)$ is a vector and A is a matrix of according size.

One can show that also $y' = Ay$ is uniquely solvable with solution

$$y(x) = \exp(xA) y_0 = \left((xA)^0 + \frac{(xA)^1}{1!} + \frac{(xA)^2}{2!} + \dots \right) y_0,$$

where $y_0 = y(0) = \begin{pmatrix} f(0) \\ h(0) \end{pmatrix}$ denotes the initial state of the system.

If the matrix A is diagonalisable (as in our example)¹, $A = VDV^{-1}$ with $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, then $(xA)^k = (xVDV^{-1})^k = x^k VD^k V^{-1}$, so that

$$\exp(xA) = \sum_{k=0}^{\infty} \frac{(xA)^k}{k!} = \sum_{k=0}^{\infty} \frac{x^k VD^k V^{-1}}{k!} = V \sum_{k=0}^{\infty} \frac{x^k D^k}{k!} V^{-1} = V \begin{pmatrix} e^{x\lambda_1} & 0 \\ 0 & e^{x\lambda_2} \end{pmatrix} V^{-1},$$

which means that the solution

$$y(x) = \exp(xA) y_0 = V \begin{pmatrix} e^{x\lambda_1} & 0 \\ 0 & e^{x\lambda_2} \end{pmatrix} V^{-1} y_0$$

is easily written down once we have the eigenvalues and eigenvectors of A .

¹Otherwise use the Jordan decomposition $A = V(D + E)V^{-1}$, having a bit more trouble with the powers $(D + E)^k$ than with D^k . On the bright side, by the structure of E , one has $E^n = 0$ for $A \in \mathbb{C}^{n \times n}$.

11.3 Linear differential equations of order n

Even in the case of homogeneous linear differential equations of the second, third or higher order (with constant coefficients), we can fall back on the theory of first-order linear differential equations. Take this example:

$$y'''(x) = 2y''(x) + 7y'(x) - 5y(x) \quad (11.11)$$

With this simple trick

$$\begin{pmatrix} y'' \\ y' \\ y \end{pmatrix}' = \begin{pmatrix} y''' \\ y'' \\ y' \end{pmatrix} = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \begin{pmatrix} y'' \\ y' \\ y \end{pmatrix}$$

we transfer the equation (11.11) of third order into the 3×3 system of order one. Solving the latter is then again a matter of diagonalisation or Jordan decomposition of the system matrix and the formulas from Section 11.2.

As an alternative approach to (11.11), one tries to work with $y(x) = e^{\lambda x}$. Substituting this into (11.11), we get

The zeros $\lambda_1, \lambda_2, \lambda_3$ give us three solutions $y_k(x) = e^{\lambda_k x}$ with $k = 1, 2, 3$ for (11.11). Indeed, the set of solutions of the linear equation (11.11) is a three-dimensional linear space, so that, by the linear independence of (y_1, y_2, y_3) , that solution space must be $\text{Span}(y_1, y_2, y_3)$. How can two approaches that are so different arrive at the same solution?

The polynomial $p(\lambda) = \lambda^3 - 2\lambda^2 - 7\lambda + 5$ and the matrix $A = \begin{pmatrix} 2 & 7 & -5 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

are very close relatives: A is the so-called companion matrix of p , and p is the characteristic polynomial of A . That's why the eigenvalues of A are exactly the zeros of p : λ_1, λ_2 and λ_3 .

Two other things that also seem to fit unexpectedly well are

Linear Algebra and Analysis.

How handy that you now know both.

The best of luck for the rest of your studies!

Index

— Symbols —

a^x	95
e	91
C^∞	131
C^n	122
$\arccos x$	101
$\arcsin x$	101
$\arctan x$	101
$\cos x$	98
$\cot x$	100
$\exp$	91
$\exp_a$	94
$\ln$	93
$\log_a$	96
π	99
$\sin x$	98
$\tan x$	100
ε -neighbourhood	45, 207
k times continuously partially differentiable	237
n times continuously differentiable ...	122
n times differentiable	122
p -norm	209
IVP	265

— A —

absolutely convergent	36, 197
accumulation point	27
alternating harmonic series	35, 37
alternating series	35
analytic at x_0	132
antiderivative	140
äquivalente Normen	209

arcus cosine	101
arcus sine	101
arcus tangens	101
average	162

— B —

bisection	71
boundary	46
boundary point	46
bounded	10, 52
bounded from above	10, 52
bounded from below	10, 52

— C —

Cantor set	51
Cauchy criterion for series	36
Cauchy principal value	180, 183
Cauchy sequence	20
Cauchy-Hadamard formula	87
center	90
center of gravity	173
chain rule	229
chain rule	111
closed set	48
closure	46
compact	53
complete	21
components	211
componentwise decomposition	211
concave	125
conditionally convergent	36
continuous	64
continuous extension	64

continuous in 64
 continuously extendable 64
 continuously partially differentiable .. 225
 continuously partially differentiable in $\mathbf{x}_0$

225

convergent 7
 converges 7
 converges pointwise 188
 converges uniformly 189
 convex 125
 cosine 98
 cotangent 100
 curl 258
 curve 168

— D —

degree 79
 derivative 106, 214
 difference quotient 106
 differentiable 106, 214
 differentiable in $\mathbf{x}_0 \in D$ 214
 diffusion equation 260
 directional derivative 233
 Dirichlet function 157
 disc of convergence 89
 disjoint union 46
 divergence 258
 divergent 7

— E —

euclidean distance 206
 euclidean norm 206
 Euler's number 91
 exponential function 91
 Exponential series 85
 exterior 46
 exterior point 46
 extreme value suspect 115, 247

— F —

fineness 154
 function series 196

— G —

geometric series 31

gradient 221

— H —

harmonic series 33
 heat equation 260
 Hessian matrix 249
 homogeneous linear differential equation
 268

— I —

improper integral 177, 181
 indefinite 250
 indefinite integral 142
 infimum 24, 52
 inflection point 126
 inhomogeneous linear differential
 equation 268
 initial value problem 266
 integration by parts 143, 166
 integration by substitution 166
 interior 46
 interior point 46
 irrotational 258
 isolated point 48

— J —

Jacobian matrix 214

— L —

Lagrange remainder 130
 Laplace operator 260
 lateral surface of a solid of revolution 175
 left-continuous 65
 left-sided limit 59
 level set 235
 limit 7, 58
 limit point 48
 linear substitutions 149
 Lipschitz continuous 70
 local extrema 246
 local extremum 114
 local maximum 114, 246
 local minimum 114, 246
 locally constant function 142
 logarithm to the base a 96

lower bound 52
 lower bounds 10
 lower integral 156
 lower sum 155

— M —

mass 173
 maximum 24, 52
 maximum norm 209
 Maxwell's equations 260
 measure zero 159
 member of the sequence 5
 minimum 24, 52
 monomial 77
 monotonic 21, 73
 monotonically decreasing 21, 73
 monotonically increasing 21, 73
 moving the center 90
 multi-indices 244

— N —

natural logarithm 93
 negative definite 250
 negative semidefinite 251
 null sequence 13
 number sequence 5

— O —

open set 48
 ordinary differential equation of first
 order 266

— P —

parametrization 168
 partial derivative 218
 partial derivatives of second order 236
 partial fraction decomposition 151
 partial fractions of the 1st kind 151
 partial fractions of the 2nd kind 152
 partial integration 143
 partial limit 27
 partial sum 30
 partially differentiable 218, 225
 partition 154

path 168
 period 100
 periodic 100
 point of expansion 90
 pointwise convergence 188
 pointwise limit 188
 pole 83
 polynomial 79
 positive definite 250
 positive semidefinite 251
 power series 85
 power series centered at 90
 principal branches 101
 product rule 109, 227

— Q —

quotient rule 109, 227

— R —

radius of convergence 86, 89
 rational function 83
 region of convergence 85
 Riemann integrable 157
 Riemann integral 157
 right-continuous 65
 right-sided limit 59

— S —

saddle point 247
 scalar field 210
 secant 106
 second derivative 122
 sector formula 169
 sequence 5
 series 30
 sine 98
 sinks 258
 solid of revolution 175
 source density 258
 sources 258
 stationary point 115, 247
 strictly concave 125
 strictly convex 125
 strictly decreasing 21

strictly increasing 21
 strictly monotonic 21, 73
 strictly monotonically decreasing 73
 strictly monotonically increasing 73
 subsequence 25
 substitution 143
 supremum 24, 52

— **T** —

tangent 100
 Taylor coefficients 128
 Taylor series 131
 Taylor-Polynom 128
 total differentiability 214
 triangle inequality 9

— **U** —

unbounded 10, 52
 uniform limit 189
 uniformly continuous 70

uniformly convergent 197
 upper bound 52
 upper bounds 10
 upper integral 156
 upper sum 155

— **V** —

variation of the constant 268
 vector field 210
 velocity vector 171
 volume of a solid of revolution 175
 vortex-free 258

— **W** —

wave equation 260
 weight function 162
 weighted average 162

— **Z** —

zero function 188